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Risk Management for PSPs

Risk Management for PSPs is a solution designed for acquirers, gateways, and Payments Service Providers (PSPs), which face the challenge of detecting abnormal payment behavior in order to build trust with their clients as well as to save costs. At the same time, this solution implements a scheduled behavioral analysis over PSPs' complete merchant portfolio, which spots abnormal merchant behavior.

Get started

Markdown page example

You don't need React to write simple standalone pages.

- monitoring
    - getHealth Check

- payments
    - postPayment
    - getResponses
    - getPayment
    - postInfo
    - putInfo
    - putFeedback
    - putSettlement

- Ad-Hoc Investigation API
    - postInvestigation
    - putFeedback
    - putInfo

- Merchant Provisioning API
    - putMerchant Provisioning
    - patchMerchant Provisioning

[API docs by Redocly]

Analytics Dashboards' Fields

This page includes all the relevant fields you can use in each `index-pattern` to refine the scope of your analysis.

`rmpsp-pulse-cm-events` â

- @timestamp
- date
- system_decision
- alerted
- feedback.fraud.history.realOutcome
- feedback.fraud.history.type.reportInstant
- feedback.fraud.latest.realOutcome
- feedback.fraud.latest.reportInstant

- pulse.rules_names
- pulse.id
- pulse.instant
- pulse.outcomes.decision.outcomeDecision
- pulse.outcomes.decision.score
- pulse.outcomes.fraud.outcomeDecision
- pulse.outcomes.fraud.score
- pulse.outcomes.sca_required.outcomeDecision
- pulse.outcomes.results.enrich.createdAt
- pulse.outcomes.results.enrich.createdBy
- pulse.outcomes.results.enrich.description
- pulse.outcomes.results.enrich.isSelfServiceRule
- pulse.outcomes.results.enrich.name
- pulse.outcomes.results.enrich.originalId
- pulse.outcomes.results.enrich.ruleId
- pulse.outcomes.results.enrich.ruleProjectId
- pulse.outcomes.results.enrich.ruleProjectSnapshotId
- pulse.outcomes.results.enrich.scoreEnabled
- pulse.outcomes.results.enrich.targetVarValue
- pulse.outcomes.results.enrich.workflowRuleBlockName
- pulse.outcomes.result.outcome
- pulse.outcomes.result.ruleId
- pulse.outcomes.result.scores
- pulse.outcomes.result.targetOutcome
- pulse.outcomes.result.targetOutcomeLabel
- pulse.outcomes.result.originalRuleId
- pulse.triggeredActions.name
- pulse.triggeredActions.nestedSourceId
- pulse.triggeredActions.sourceId
- pulse.alerted
- pulse.channel_id
- fields.amount_unified_currency
- fields.event_type
- fields.info_type
- fields.feedback_type
- fields.lifecycle_id
- fields.merchant_id
- fields.merchant_mcc
- fields.merchant_country_code
- fields.merchant_name
- fields.merchant_url
- fields.payment_3ds_status_code
- fields.payment_auth_status_code
- fields.payment_card_country_code
- fields.payment_method_type
- fields.terminal_ip_country_code
- fields.terminal_latitude
- fields.terminal_longitude
- fields.device_ip_country_code
- fields.terminal_ip_location
- fields.device_ip_location
- fields.amount_decimal
- fields.customer_region
- fields.alert_id
- fields.payment_region
- fields.tenant_alert
- fields.channel
- fields.merchant_parent_merchant_id

- fields.card_presence_flag
- cm.alert
- cm.versionTimestamp
- cm.assignedAt
- cm.decisionUserId
- cm.decisionUsername
- cm.decisionAt
- cm.decisionTimeSinceCreated
- cm.decisionTimeSinceAssigned
- cm.first_assignee_user_id
- cm.first_assignee_username
- cm.channel
- cm.eventStates.mm_status.stateId
- cm.eventStates.mm_analyst_decision.stateId
- cm.eventStates.rmpsp_operational_status.stateId
- cm.eventStates.status.stateId
- cm.eventStates.status_payments.stateId
- cm.eventStates.rmpsp_status.stateId
- cm.id
- cm.reasons
- cm.states
- cm.status
- cm.timestamp
- cm.userId
- cm.username
- cm.versionId
- cm.fraud_label
- cm.analyst_decision
- cm.state

`rmpsp-triggered-rules` â

- @timestamp
- enrich.createdAt
- enrich.createdBy
- enrich.description
- enrich.isSelfServiceRule
- enrich.name
- enrich.originalId
- enrich.ruleId
- enrich.ruleProjectId
- enrich.ruleProjectSnapshotId
- enrich.scoreEnabled
- enrich.targetVarValue
- enrich.workflowRuleBlockName
- fields.alert_id
- fields.amount_decimal
- fields.amount_unified_currency
- fields.channel
- fields.customer_region
- fields.device_ip_country_code
- fields.event_type
- fields.ip_location
- fields.lifecycle_id
- fields.merchant_country_code
- fields.merchant_id
- fields.merchant_mcc
- fields.merchant_name

- fields.merchant_parent_merchant_id
- fields.merchant_url
- fields.payment_3ds_status_code
- fields.payment_card_country_code
- fields.payment_method_type
- fields.terminal_ip_country_code
- fields.terminal_ip_location
- fields.device_ip_location
- fields.payment_region
- fields.card_presence_flag
- pulse.channel_id
- pulse.alerted
- pulse.id
- pulse.instant
- pulse.outcomes.decision.outcomeDecision
- pulse.outcomes.decision.score
- pulse.outcomes.fraud.outcomeDecision
- pulse.outcomes.fraud.score
- pulse.result.result.originalRuleId
- pulse.result.result.ruleId
- pulse.result.result.targetOutcome
- pulse.result.result.targetOutcomeLabel
- pulse.result.result.outcome
- pulse.result.sourceId
- pulse.result.weight
- pulse.sourceId@timestamp
- enrich.createdAt
- enrich.createdBy
- enrich.description
- enrich.isSelfServiceRule
- enrich.name
- enrich.originalId
- enrich.ruleId
- enrich.ruleProjectId
- enrich.ruleProjectSnapshotId
- enrich.scoreEnabled
- enrich.targetVarValue
- enrich.workflowRuleBlockName
- fields.alert_id
- fields.amount_decimal
- fields.amount_unified_currency
- fields.channel
- fields.customer_region
- fields.device_ip_country_code
- fields.event_type
- fields.ip_location
- fields.lifecycle_id
- fields.merchant_country_code
- fields.merchant_id
- fields.merchant_mcc
- fields.merchant_name
- fields.merchant_parent_merchant_id
- fields.merchant_url
- fields.payment_3ds_status_code
- fields.payment_card_country_code
- fields.payment_method_type
- fields.terminal_ip_country_code
- fields.terminal_ip_location

- fields.device_ip_location
- fields.payment_region
- fields.card_presence_flag
- pulse.channel_id
- pulse.alerted
- pulse.id
- pulse.instant
- pulse.outcomes.decision.outcomeDecision
- pulse.outcomes.decision.score
- pulse.outcomes.fraud.outcomeDecision
- pulse.outcomes.fraud.score
- pulse.result.result.originalRuleId
- pulse.result.result.ruleId
- pulse.result.result.targetOutcome
- pulse.result.result.targetOutcomeLabel
- pulse.result.result.outcome
- pulse.result.sourceId
- pulse.result.weight
- pulse.sourceId

`rmpsp-cm-cases` â

- created_at
- channel
- closed_at
- timeSinceCreated
- case_source_type
- state
- updatedAt
- assignee_id
- assignee_username
- id.identifier
- case_source_id

`rmpsp-cm-report-1` â

- timestamp
- rule_name
- rule_original_id
- user
- rule_status

`rmpsp-cm-report-2` â

- @timestamp
- app_id
- rules.app
- rules.comment
- rules.createdAt
- rules.createdBy
- rules.desc
- rules.enabled
- rules.id
- rules.name
- rules.rulesProjectId
- rules.scoreEnabled
- rules.targetVarValue

- rules.updatedAt
- rules.updatedBy
- rules.weight

FAQ

How does Bin enrichment work?â

As of version 1.4.0, Pulse enriches payments with card-related information, if given a valid card bin. Fields sent via API will never be overwritten by its enriched counterparts. The enriched fields are as follows:

- `payment_card_brand`
- `payment_card_type`
- `payment_card_country_code`
- `payment_card_category`
- `payment_card_issuing_org_name`

What does the threshold value mean if the MM rule compares two ratios? â□□

1. **Simple ratio** means percentage, as follows:
    - In the past week, the merchant received 3 chargebacks (i.e. disputed transactions) over a total of 200 processed transactions.
        - ratio = 0.015 (or 1.5%).
2. **Relation between two ratios** means X times greater, as follows:
    - In the past 3 weeks, the merchant received 5 chargebacks over a total of 800 processed transactions:
        - ratio = 0.00625 (or 0.625%).
    - In the past week, the merchant received 3 chargebacks over a total of 200 processed transactions:
        - ratio = 0.015 (or 1.5%).

The relation between these two ratios is `0,015 / 0,00625 = 2.4`.

This means that, in the past week, the percentage of chargebacks is 2.4 times greater than the previous 3 weeks.

What do the `payment_auth_status` statuses mean?â□□

The `payment_auth_status` field is used to transmit the authorization status at the issuer bank after Feedzai scores the payment. This is an open field that accepts the **Accepted** and **Declined** statuses, as well as the following ones:

- **Verified**: Zero Dollar Authorization ($0 auths), also known as Verification transaction, provides merchants with an effective way of validating account number and other elements such as Card Verification Value (CVV) and Address Verification Service (AVS) without charging the card.
- **Pending**: Used by some regions/issuers to process payments asynchronously.

What is Non-Delivery Exposure?â□□

Non-Delivery Exposure (NDX) is a business metric that indicates the actual risk exposure for the PSP regarding each merchant, i.e. it represents the potential financial losses for the PSP in case of fraudulent merchant activity.

`NDX risk = SUM (TPV-1 : TPV-ADD)`

- TPV = total payment volume
- ADD = average delivery days
- TPV-1 = previous dayâ□□s TPV
- TPV-ADD = TPV of the last day included in the ADD range

Example:

- MCC 7829 has ADD = 5
- TPV for merchant (displayed in the table)
- NDX = 245 201,00 USD

Days	TPV
Day -5	3 444,00
Day -4	2 421,00
Day -3	235 346,00
Day -2	434,00
Day -1	3 556,00
Sum	245 201,00

tip

Note that the MCC is reference data and needs to be sent in the merchant reference data file for the following reasons:

- There may not be transactions for specific days. Therefore, the transaction monitoring's merchant data cannot be used.
- The MCC may be changed by an agent. This means that part of the transactions may be different from the rest.

Which fields are enriched by Reference Data?â

All the fields from entities are enriched by Reference Data. Check the Merchant Provisioning endpoint to see the list of enriched fields.

Which fields are enriched by third-party providers?â

As an extension of the out-of-the-box enrichment, it is possible to enrich IP, BIN, and currency exchange fields by using enrichment functions. Check Pulse's documentation for more information.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/overview

Risk Management for PSPs

About Risk Management for PSPs

Learn the basics to have a solid understanding of how the solution works

Admin Reference

Understand how Risk Management for PSPs manages accesses and dashboards

-
- About Risk Management for PSPs
- How it Works

How it Works

This is how RMPSP works:

1. A Payment Service Provider (PSP) calls Feedzai's Payments API, which consumes real-time transactions and subsequent updates to transactions, such as chargebacks. Refer to the Transaction Life Cycle to learn more about each transaction's stage.
2. Once a transaction is sent to Feedzai, it enters Feedzai's Transaction Fraud engine where itâs processed via workflow. As it passes through this workflow, the transaction will go through a series of rule sets and models to receive an Approve, Decline, or Challenge response. Once transactions have received a response for their status, they are sent into Feedzai's alert management tool - Case Manager - for further review.
3. Case Manager provides all the relevant data that analysts need to make effective decisions. Information related to the transaction - clear explanations as to why models or rules have applied the relevant recommendation, and visual link analysis - allow fraud, risk, and financial crime analysts to quickly analyze patterns and trends at a transactional, merchant, or portfolio level.

4. Case Manager contains a set of dashboards to obtain visibility into areas such as payment scheme compliance, geographical trends per merchant category code (MCC) / merchant, and unusual merchant behavior such as increased refunds / chargebacks / volumes to help manage potential credit risk issues.
5. Feedzai enables acquirers and PSPs to send additional Merchant's reference data to help manage and monitor merchantsâ□□regulatory and payment scheme compliance. Once the additional data is received, a batch processor component continuously sends this data to the Merchant Monitoring workflow. This component runs the merchant's reference data on a daily basis to evaluate both current and new merchant data. In the second case, the new merchant data is updated when PSPs call the *Merchant Provisioning* endpoint.
6. Once the merchant's reference data is sent to the Merchant Monitoring workflow, Pulse calculates metrics on specific entities over a time period (i.e. profiles). The reference data is then combined with the Transaction Fraud Engine workflow data for greater insight into the merchantâ□□s transaction behavior.
7. If suspicious behavior or compliance risks are flagged by Pulse, the events are sent to Case Manager so that all the events classified as alerts can be reviewed by the acquirer or payment service provider. This allows for settlements to be stopped, and for further investigation as required. Case Manager allows alerts to be escalated to senior team members for review.
8. The decisions made within Case Manager automatically feedback into both Transaction Fraud and Merchant Monitoring workflows, allowing rules, models, and negative or positive lists to be updated automatically. This helps the Pulse application identify future fraudulent/risky merchants with higher precision.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/about/overview

About Risk Management for PSPs

Risk Management for Payment Service Providers (RMPSP) consists of an integration of two use cases - Transaction Fraud for Payment Service Providers (TFP) and Merchant Monitoring (MM) - and is used to prevent fraud at the merchant level. This solution is designed for acquirers, gateways, and Payments Service Providers (PSPs), which face the challenge of detecting abnormal payment behavior in order to build trust with their clients as well as to save costs.

RMPSP entails the following use cases:

- **Transaction Fraud for PSPs:** a real-time fraud detection solution that detects different fraud scenarios related to card transactions and alternative payments initiated at the merchant's site/store.
- **Merchant Monitoring:** an account monitoring solution for acquirers and PSPs to prevent and detect fraud at the merchant level. By implementing a scheduled (batch) behavioral analysis over the merchant portfolio's transactional data, acquirers can spot abnormal merchant behavior, risk of bankruptcy or high chargeback levels, and thus act before further fraud and damages occur.

Manage Users

PSPs need to manage their merchants' accounts so that they can access Case Manager in order to investigate fraud. This page helps fraud prevention managers understand the steps they must take to successfully create users for their merchants to access Case Manager with the right permissions.

1. **Invoke the Merchant Provisioning endpoint**: Before a new merchant's first payment, you must invoke the *Merchant Provisioning* endpoint to add the new merchant entity to Feedzai.

2. **Create user**: To successfully create a user and associate it with the merchant, read the Manage Users page from the Pulse documentation. While creating the new user, make sure to assign the role and the tenant in the **Create User** form (see image below). Note that Risk Management for PSPs delivers out-of-the-box roles and their respective permissions, which you can check in the Roles and Permissions page.

Multi-Tenancy in Case Manager

Multi-tenancy is the ability to host and centrally manage multiple tenants in a single application. This functionality allows a large company (e.g. PSP) to manage multiple clients (e.g. merchants). On the other hand, clients can also use Feedzai's solution, but with restricted access to their own data.

This page describes the investigation differences between PSPs and merchants.

Investigation flowâ

The investigation flow for merchants is simpler than for PSP. Merchants can see all their event details (i.e. payment, refunds, OCTs, etc.), make searches to understand sales patterns, rules triggered and system decisions, create cases and SSRs.

Since a payment cannot be in two queues at the same time, having queues on the merchant's end restricts the PSP from using queues. Similarly, a payment cannot be assigned to two analysts at the same time, as having assignees on the merchant's end restricts the PSP from working with the **Assign to** functionality. As such, merchants can't perform any of these actions.

Merchants aren't able to use neither status transitions, nor the **Get Next** button.

PSP

Merchant

Alert flagâ

PSPs can see both the alerted Self Service Rules (SSRs) and out-of-the-box rules.

Merchants can create their own SSRs and decide to alert a payment without creating an alert for the PSP, and they won't see alerts from rules other than their own SSRs.

PSP

Merchant

Fraud / Not Fraud feedbackâ

On the PSPs' end, Case Manager displays both PSPs and merchants' feedback labels, **Fraud Label** and **Merchant Fraud Label**, respectively. The feedback from merchants provides PSPs with extra information for the decision process.

Note that PSPs' decisions are not visible to merchants. Showing PSPs' decisions to merchants presents a risk, as merchants may notice they are under fraud investigation.

PSP

Merchant

Rules explanation visibilityâ

By default, the Rules explanation in the **Rules Triggered** widget is visible to all users with generic permissions to see rules.

However, PSPs may not want their merchants to see certain triggered rules, due to compliance reasons or internal policies. Therefore, depending on the configuration you make, rules may or may not become available to merchants. Merchants will still be able to see the system's decision (`approve` , `decline`), and whether the payment has been alerted or not.

PSP

Merchant

To learn how to configure the rules visibility in Case Manager, refer to the **Rules visibility in Case Manager** section from the Work with Rules guide.

Notesâ

Notes and attachments left by the PSPs are not visible to merchants. However, any note left by merchants will be available to PSPs for their consideration during an investigation.

PSP

Merchant

Activity Logâ

The activity log may also include sensitive information, such as notes or reasons for a fraud decision. For this reason, merchants can only see their own activity log. On the other hand, PSPs are able to see their merchants' activity log, to better understand the reasoning behind merchants' feedback decision.

PSP

Merchant

-
- Admin Reference

Admin Reference

This documentation is intended for:

- Fraud prevention managers, who need to understand the multi-tenancy gap between PSPs (landlord) and merchants (tenants) - read the Multi-Tenancy in Case Manager and Manage Users to see the differences between these two entities in Case Manager, and to create users for merchants (to access Case Manager), respectively.
- System administrators, who provide and manage accesses to Pulse, Case Manager, and analytics dashboards - read the Roles and Permissions page to understand each role's allowed actions to perform their work, and to quickly check each role's permissions.

-
- Admin Reference
- Roles and Permissions

On this page

Roles and Permissions

Learn about the default roles and their associated permissions to understand how RMPSP manages the access within Pulse, Case Manager, and dashboards.

Overviewâ

The following list summarizes the out-of-the-box roles in RMPSP with high-level information about the allowed actions for each role:

Fraud Prevention Agent L1 - PSP

- Mark up payment and merchant monitoring events
- Add and remove files from alerts and cases
- Add entities to PosNeg lists - global and tenanted

- Create and modify Self-Service Rules (SSRs) - global and tenanted
- Access Genome

Fraud Prevention Agent L2 - PSP

- Mark up payment and merchant monitoring events
- Create and manage cases in Case Manager
- Add and remove files from alerts and cases
- Add entities to PosNeg lists - global and tenanted
- Create and modify Self-Service Rules (SSRs) - global and tenanted
- Access Genome

Fraud Prevention Manager - PSP

- Mark up payment and merchant monitoring events
- Create and manage cases in Case Manager
- Add and remove files from alerts and cases
- Add entities to PosNeg lists - global and tenanted
- Create and modify SSRs - global and tenanted
- Access Genome
- Override Fraud Prevention Agents' actions
- Access analytics dashboards
- Create and delete reasons
- Self Service Configuration (UI)

Fraud Strategist - PSP

- Maintain rules, lists, and thresholds in Pulse and Case Manager
- Read events and alerts

Fraud Strategist Manager - PSP

- Maintain rules, lists, and thresholds in Pulse and Case Manager
- Read events and alerts
- Override Fraud Strategists' actions
- Access analytics dashboards

Monitoring and Auditing

- Quality monitoring - selects a sample of alerts per agent in a given month and examines the investigation's accuracy (i.e. typically an internal auditor)
- Read alerts, cases, and SSRs from users with a reporting responsibility
- Access analytics dashboards

Fraud Prevention Agent L1 - Merchant

- See and act on payments from their own tenant
- Mark up payments
- Add and remove files from alerts

Fraud Prevention Agent L2 - Merchant

- See and act on payments from their own tenant
- Mark up payments
- Add and remove files from alerts
- Add entities to "by Merchant PosNeg" lists (tenanted)

Fraud Prevention Agent L3 - Merchant

- See and act on payments from their own tenant
- Mark up payments
- Add and remove files from alerts
- Add entities to "by Merchant PosNeg" lists (tenanted)
- Create and modify "by Merchant" SSRs

Merchant Monitoring

- View, assign, and manage Merchant Monitoring events
- View Merchant Monitoring channel

Integrator

- Manage API JWKs

caution

Users with any of the following roles must be added to the `rmpsp_landlord_resources` group and be granted the `landlord` role, when creating a user:

- `Fraud Prevention Agent L1 - PSP`
- `Fraud Prevention Agent L2 - PSP`
- `Fraud Prevention Manager - PSP`
- `Fraud Strategist - PSP`
- `Fraud Strategist Manager - PSP`
- `Merchant Monitoring`

Detailed viewâ

The following table shows a detailed view of each role's permissions:

Tool	Permissions	Fraud Prevention Agent L1 - PSP	Fraud Prevention Agent L2 - PSP	Fraud Prevention Manager - PSP	Fraud Strategist - PSP	Fraud Strategist Manager - PSP	Fraud Prevention Agent L1 - Merchant	Fraud Prevention Agent L2 - Merchant	Fraud Prevention Agent L3 - Merchant	Merchant Monitoring	MM Ad- Investig
Pulse	4-Eye Check - Create / Review				X	X					
	4-Eye Check - Review					X					
	Export Resources			X		X					
	Manage Application			X	X	X					
	Manage Custom Services			X	X	X					
	Manage Data Science			X	X	X					
	Manage History			X	X	X					
	Manage Jobs			X	X	X					
	Manage JWK										
	Manage List Items	X	X	X	X	X		X	X		
	Manage Lists			X	X	X					
	Manage Outcome Combinators			X	X	X					
	Manage Sandbox Evaluations			X	X	X					
	Manage SSRs			X	X	X			X		
	Manage Workflows			X	X	X					
	Publish SSRs			X	X	X			X		
	Read Application			X	X	X					
	Read Custom Services			X	X	X					
	Read Data Science			X	X	X					
	Read Data Science Rules Projects						X	X	X		
	Read History			X	X	X					
	Read Jobs			X	X	X					
	Read Lists	X	X	X	X	X	X	X	X		

Tool	Permissions	Fraud Prevention Agent L1 - PSP	Fraud Prevention Agent L2 - PSP	Fraud Prevention Manager - PSP	Fraud Strategist - PSP	Fraud Strategist Manager - PSP	Fraud Prevention Agent L1 - Merchant	Fraud Prevention Agent L2 - Merchant	Fraud Prevention Agent L3 - Merchant	Merchant Monitoring	MM Ad-Investig
	Read Outcome Combinators			X	X	X					
	Read Sandbox Evaluations			X	X	X					
	Read SSRs			X	X	X			X		
	Read Users										
	Read Workflows			X	X	X	X	X	X		
	Login			X	X	X					
	Manage Users			X		X					
Case Manager	Access Entity Notes			X						X	X
	Assign Events	X	X								
	Assign Ad-Hoc Investigations Events										X
	Assign Merchant Monitoring Events									X	
	Landlord	X	X	X	X	X					
	Login	X	X	X	X	X	X	X	X		
	Manage Automation Rules			X							
	Manage Cases		X	X			X	X	X		
	Manage Ad-Hoc Investigations Events										X
	Manage Events - Base	X	X	X			X	X	X		
	Manage Events - Landlord	X	X	X							
	Manage Merchant Monitoring Events									X	
							X	X	X		

Tool	Permissions	Fraud Prevention Agent L1 - PSP	Fraud Prevention Agent L2 - PSP	Fraud Prevention Manager - PSP	Fraud Strategist - PSP	Fraud Strategist Manager - PSP	Fraud Prevention Agent L1 - Merchant	Fraud Prevention Agent L2 - Merchant	Fraud Prevention Agent L3 - Merchant	Merchant Monitoring	MM Ad-Investig
	Manage Events - Tenant										
	Manage List Items	X	X	X	X	X		X	X		
	Manage Queues			X							
	Manage Reports			X							
	Manage SLAs			X							
	Manage SSRs			X	X	X			X		
	Manage User Queues	X	X	X							
	Publish SSRs			X	X	X			X		
	Read / Create Reasons			X							
	Read / Delete Reasons			X							
	Reveal Encrypted Values		X	X		X		X	X		
	Review Next	X	X	X	X						
	Self-Service Config			X							
	Read / Write Merchant Monitoring Ad-Hoc Investigations from Tenants										X
	Read / Write Merchant Monitoring from Tenants									X	
	Read / Write Payments from Tenants	X	X	X	X	X	X	X	X		
	Test SSRs			X	X						
	Read Automation Rules			X							
	Read Cases		X	X		X	X	X	X		
	Read Dashboards			X		X					

Tool	Permissions	Fraud Prevention Agent L1 - PSP	Fraud Prevention Agent L2 - PSP	Fraud Prevention Manager - PSP	Fraud Strategist - PSP	Fraud Strategist Manager - PSP	Fraud Prevention Agent L1 - Merchant	Fraud Prevention Agent L2 - Merchant	Fraud Prevention Agent L3 - Merchant	Merchant Monitoring	MM Ad-Investig
	Read Entity Files			X						X	X
	Read Entity Notes						X	X	X		
	Read Ad-Hoc Investigations Events										X
	Read Events - Base	X	X	X	X	X	X	X	X		
	Read Events - Landlord	X	X	X	X	X					
	Read Merchant Monitoring Events									X	
	Read Events - Tenant						X	X	X		
	Read Merchants	X	X	X	X	X					
	Read Queues			X	X	X					
	Read Reports			X							
	Read SLAs			X							
	Read SSRs			X	X	X			X		

-
- Merchant Monitoring

Merchant Monitoring

About Merchant Monitoring

Learn the basics to have a solid understanding of how the solution works

Operational Work

Explore MM's practical guides to help data scientists and fraud prevention agents use the software

Out-of-the-Box Resources

Learn about the resources that come with the solution, namely rules and lists

Transaction Fraud for PSPs

About Transaction Fraud for PSPs

Learn the basics to have a solid understanding of how the solution works

Operational Work

Explore TFP's practical guides to help data scientists and analysts use the software

API Resources

Learn all the operations you can do with Feedzai's API and how to communicate with its multiple endpoints

Out-of-the-box Resources

Learn about the resources that come with the solution, namely rules and lists

About Merchant Monitoring

Merchant Monitoring is an account monitoring solution for Payment Service Providers (PSPs) to prevent and detect risk at the merchant level. By implementing a scheduled behavioral analysis over their complete merchants' portfolio, PSPs can spot abnormal merchant behavior and act before more fraud and damages occur. As a result, PSPs are able to prevent losses in money chargebacks and in case of merchant bankruptcy, as well as reduce the risk of reputational damage caused by risky merchants. Merchant Monitoring offers out-of-the-box functionalities to:

- Proactively look for patterns of risky activities and take preemptive action to reduce losses.
- Track merchants' performance and alerts so that PSPs can decide which merchants are worth keeping and which jeopardize PSPs' reputation.
- Ensure that merchants are compliant with regulations and policies regarding risk exposure.
- Provide an alert management tool (Case Manager) to analyze merchants' information and alerts.

End-to-end packageâ

These are the out-of-the-box capabilities that come with Merchant Monitoring:

- Standard configurations: Out-of-the-box set of standard configurations for Pulse, Case Manager, visual link analysis (Genome), and analytics dashboards.
- Reference data: List of merchants sent by the PSP, which is used as an input and enricher of the Merchant Monitoring workflow to detect risk and to provide analysts with additional information.
- Metrics and rules: Pre-packaged set of rules and metrics for different risky factors, and shared metrics with Transaction Fraud for PSPs for the same end.

Risk Scenarios

This page covers the risk scenarios and the detection strategy for merchant monitoring. The standard package delivers out-of-the-box rules that capture user patterns associated with each risk scenario.

Collusive / fraudulent / bust-out merchantâ

In a collusive, fraudulent, bust-out merchant scenario, fraudsters create a fake company that looks legitimate in the eyes of the acquirer. Once the acquirer makes an agreement with the company, fraudsters generate high amount transactions in a short time before disappearing completely. Usually, these criminals use stolen credit cards to process these transactions.

How Merchant Monitoring prevents collusive / fraudulent / bust-out merchantsâ

Here are some examples of rules that help detect a collusive / fraudulent / bust-out merchant:

- **Visa-VFMP-Early-Warning-FR5:** An alert is generated if, over the last month, the fraud to sales ratio of the Visa Fraud Monitoring Program is above a set threshold.
- **Fraud-Amount-Rate-FR11:** An alert is generated if, over the last month, the fraud amount rate is equal to or above a set threshold.
- **Rejected-Count-Rate-FR13:** An alert is generated if, over the last month, the rejected count rate is equal to or above a set threshold.

Bankruptcyâ

Merchant bankruptcy occurs when the merchant is unable to pay its debts. In this scenario, most commonly, merchants conceal assets so that creditors can only liquidate the assets listed by the merchant.

How Merchant Monitoring prevents bankruptcyâ

Here are some examples of rules that help detect bankruptcy:

- **Monthly-Amount-Decrease-BR1:** An alert is generated if the decrease in monthly total transaction amount is equal to or below a set threshold.
- **Monthly-Refund-Amount-BR5:** An alert is generated if, over the last month, the refund amount rate is equal to or above a set threshold.
- **Refund-Count-Increase-BR8:** An alert is generated if the refund count rate of 1 month compared to the previous 3 months is equal to or above a set threshold.

Money launderingâ

Money laundering perpetrated by merchants is a process in which legitimate goods and services are being processed on behalf of illicit ones.

How Merchant Monitoring prevents money launderingâ

Here are some examples of rules that help detect money laundering:

- **High-Amount-Count-ML1:** An alert is generated if the number of high amount transactions, over the last 24 hours, is equal to or above a set threshold.
- **High-Risk-Countries-ML3:** An alert is generated if the number of transactions from high-risk-country cards is equal to or above a set threshold.
- **Inactivity_Period_ML4:** An alert is generated if the merchant has been inactive for at least 60 days.

Chargeback riskâ

The risk of excessive chargebacks is especially concerning for merchants that base their business on e-commerce, where only card not present transactions happen. Card not present transactions increase the likelihood of transaction disputes while keeping the goods for free.

How Merchant Monitoring prevents chargeback riskâ

Here are some examples of rules that help detect chargeback risk:

- **Visa-VDMP-Early-Warning-CB1:** An alert is generated if, over the last month, the chargeback to sales ratio of the Visa Dispute Monitoring Program is equal to or above a set threshold.
- **Mastercard-ECM-CB4:** An alert is generated if, over the last month, the chargeback to sales ratio of the Mastercard Excessive Chargeback Program is equal to or above a set threshold.
- **Chargeback-Cnt-Increase-CB6:** An alert is generated if the chargeback count rate of 1 month, over the last 3 months, is equal to or above a set threshold.

Unusual order activityâ

This type of scenario occurs when a merchant shows unusual behavior, such as the increase of gift or prepaid card transactions, or manual card transactions. This unusual behavior may be related to fraudulent activities.

How Merchant Monitoring prevents unusual order activityâ

Here are some examples of rules that help detect unusual order activity:

- **Card-Manual-Increase-UAR8:** An alert is generated if the manually entered card payments' rate of 1 month, over the last 3 months, is equal to or above a set threshold.
- **Merch-Avg-Ticket-Size-UAR13:** An alert is generated if the merchant's daily average transaction value is unusually higher when compared to the daily average of the last 3 months.
- **Gift-Prepaid-Card-Rate-UAR14:** An alert is generated if the daily gift or prepaid card payments rate is equal to or above a set threshold.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/api-resources/additional-resources

Additional Resources

This section contains information about the following topics:

- Authentication

-
- [API Resources]
- [Additional Resources]
- Authentication

On this page

Authentication

Feedzai allows you to authenticate and protect your data through JSON Web Tokens (JWTs). A JSON Web Token (JWT) is a piece of information that is transmitted as a JSON object from clients to Feedzai, and its purpose is to ensure that client requests (e.g. transactions for processing) have not been compromised.

These tokens are generated using the clientâs private key, which can be verified with the corresponding public key. The public key must be shared with Feedzai beforehand to verify that the token sent with the request belongs to the client.

Overviewâ

JWTs protect against different attacks using the following mechanisms:

- **Encryption/signing** so that only authenticated users can access sensitive data.

- **Expiration dates** to protect against replay attacks.
- **JWT revocation** to block exposed JWTs as soon as possible.

You can see more details on integrating JWTs with your software in the sub-sections below, but in general you will:

1. Generate a private-public key pair to sign the JWT.
2. Provide the public key to Feedzai in JSON Web Key Set format.
3. Frequently generate new JWTs with a reduced validity period (e.g. 5 minutes). Each JWT is signed with your private key generated in step 1.
4. For each request to Feedzai, include a JWT in the `Authorization` header.

Given the recommended algorithm and parameters, you can choose the tool that best fits your system. A comprehensive list of tools is available at the libraries section of jwt.io.

danger

Third-party security vulnerabilities: When choosing a library, make sure it has no known security vulnerabilities.

JSON Web Key

A key pair is necessary to cryptographically sign and verify the JWTs sent with your requests.

- You will use the private key to sign your JWTs.
- Feedzai will use the public key to verify the signature, and therefore the authenticity of the requests.

To create a JWK set, you need to choose the signing algorithm and the JWK parameters. Feedzai recommends the `RS512` algorithm (â□□RSASSA-PKCS1-v1_5 using SHA-512â□□) with a 4096-bit key. See the sections below to learn how to generate a JWK set with standard tools.

danger

Beware of private key parameters: Construct the JWK set so that you **do not** send any of the private key parameters.

Your JWK set should look similar to this:

```
 {
    "keys": [
        {
            "e": "AQAB",
            "n":
"187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6xj9ZnqyyS
cle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374XNfLRb0s
F7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbOWgEPUKoR
BRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
            "kty": "RSA",
            "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
            "use": "sig"
        }
    ]
}

{
        "kid":"mykey2",
        "kty":"RSA",
        "alg":"RS512",
        "use":"sig",
        "e":"AQAB",
        "n":"k51ZPyu9gocn508aXzob_hSKaIAnJApY4pY3h1oym1Z15-6DN3WxB3YHUyXPRd-iBzF7177XjQAywTzAaz-
PKLZ4bC52noanLx7Ihyf3nQAOwhEv16zaJLI-
HqMEMp6JKPjKJ8LKanVnvxWNI0zUBwcBOwt9cxph0evyVO94cXA0RZxUZxa7C3FuSYkO2zWorMH0ZkC3harkJx0C9Q7nBiGW
PQmB8ZkLx04iWlm8ldyxDJPKl-PC4-
gS3y2mgPfZVAaK_VGod88PUmF4vFax7a03v2VLtn9wm8UWs9SXATxJ5cExgIIvFkdsZAcS3hWnhUe3Of4zAEbKh2Ah5DtfY9
xvtsU8FGlak7wLCwVnubWWglI2hmFvFU7Jsyzb_L-
jD7XCYrg7QuFV8mX2jVcTbsSxnd8GLRNbvmztvsjfaQHXEHraTRJFoGO1Y3JXDiY8Kj40OQXxBNXaFxHda9lBfzWfDkgWvcy
eiTPeWpKc6cS1zcDaQNRzZr3aZR-
gI4U8YTAXcIKDdom2GCQSx9WfmFq3uAj95duL5CgyGfw7L-02c6ckRr6kBAKebURkSsZXOyTO8bdbS8GzPMD59CrRL2SnJJe
qHyrEav8JCu0x3UCXbyqlfqQgzVsXNpFUMIS-sUhe3nyVQOeJ_WoYq7oAbjLpx_Bw5elQJf4zJbJd2Pk"
    }
```

In the aforementioned example, the following parameters are used:

Name	Extended name	Description
`e`	Exponent	RSA public exponent e.
`n`	Modulus	RSA public modulus n.
`kty`	Key Type	Should be `RSA` to symbolize you are using the RS512 algorithm.
`kid`	Key Id	Should specify a key identifier of your choice.
`use`	Public Key Use	Should be `sig` to note this key will be used to verify JWTs.

How to generate a key-pair and JWKâ

tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

Consider the following requirements if you're generating JWTs in a different way:

- Make sure you store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**

- Manage JWKs in Pulse.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

1.

info

Make sure you repeat steps 1 through 3 for a total of two private-public key pairs: one for `prod` and another one for `qa`.

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWK \
  --generateKeys \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --kid "key_id"
```

   The JWK Set JSON will be displayed on your standard output. Alternatively, you can use the `--jwk` parameter to specify a file where it should be stored instead.
5. Store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**
6. Manage JWKs in Pulse.

7. *This step is internal and focused on Customer Success teams. Be careful when sharing it with clients.*

Feedzai adds the generated JWK Set JSON to the following folders:

````
 ```bash
delivery/pulse/pulse-assembly-runtime/src/main/resources/[environment]/files/conf/jwk.json
```
````

tip

Note that `[environment]` is the client's cloud environment (e.g. `qa` and `prod`).

JSON Web Tokensâ

Once you have a private key, you can generate your JWTs. All JWTs are composed of a set of headers, a set of claims and a signature.

JWT Headersâ

Besides the algorithm (`alg`), the only mandatory header for JWTs is the `kid` (the key ID), which indicates which key was used to sign the token and **must** match a valid key in the JWK Set used for validation. The set of headers should be similar to:

```
{
  "kid": "keyID",
  "alg": "RS512"
}
```

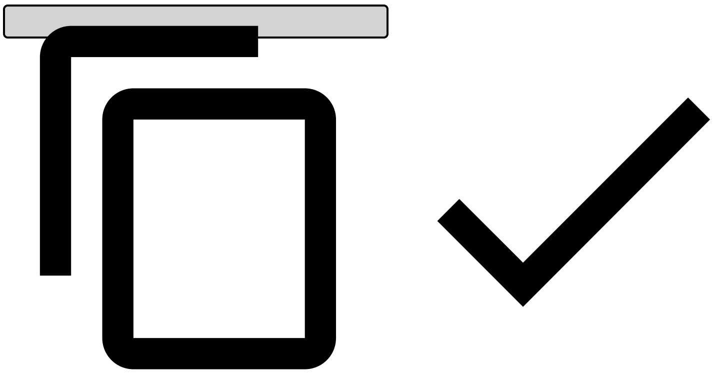
JWT Claimsâ

Typically, the following claims should be present on the JWTs:

Claim name	Extended name	Description
`iss`	Issuer name	Identifies the issuer of the JWT. This value will be provided by Feedzai, and it should match the tenant id, if applicable.
`exp`	Expiration Time	Date after which the JWT must be rejected.
`nbf`	Not before Time	Date before which the JWT must be rejected.
`jti`	JWT id	Identifier used to revoke JWTs.
`sub`	Subject	Identifies the user of the JWT.

Your final set of claims should be similar to:

```
{
    "iss": "feedzai",
    "exp": 1523022355,
    "nbf": 1523022055,
    "jti": "a00cd6dd-4e55-4694-aeb5-4050e7d063b6",
    "sub": "700e55f9-15b0-4f95-8094-960138a0d886"
}
```



How to generate a JWTâ

tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWT \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --iss "issuer" \
  --sub "subject" \
  --exp "$((EPOCHSECONDS + 300))" \
  --kid "key_id"
```

The JWT will be displayed on your standard output. Alternatively, you can use the `--jwt` parameter to specify a file where it should be stored instead.

How to send the signed JWTâ

danger

Note that you can only send the signed JWT after the corresponding JWK is uploaded to Pulse (see [Manage JWKs in Pulse]).

Send the signed JWT in the `Authorization` header of HTTP requests made to Feedzai's API, as follows:

```
Authorization: Bearer <your_jwt>
```

Manage JWKs in Pulseâ

Once you create the JWK set, the following JWK steps must be taken:

1. Log into Pulse and confirm the current list of Authorization keys. Before revoking any key, make sure you are no longer using it.
2. Add the new key to Pulse.
3. Start signing messages with JWTs based on the new key.
4. Revoke the old key.
5. Delete revoked Authorization keys.

List Authorization keysâ

To list the existing Authorization keys, access the **Administration** page and select the **Authorization Keys** tab on the sidebar.

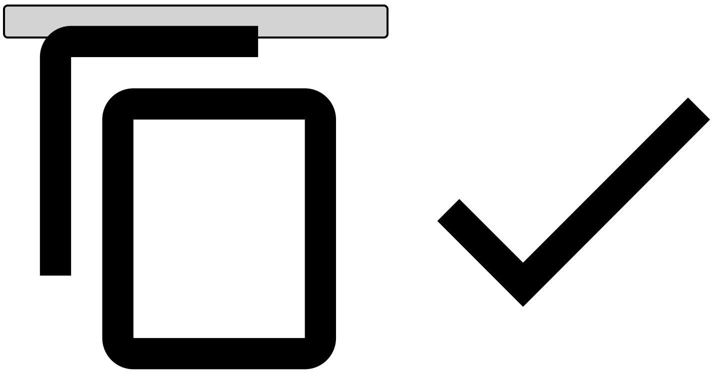
Select a label name or expand to see the Authorization keys associated with that label:

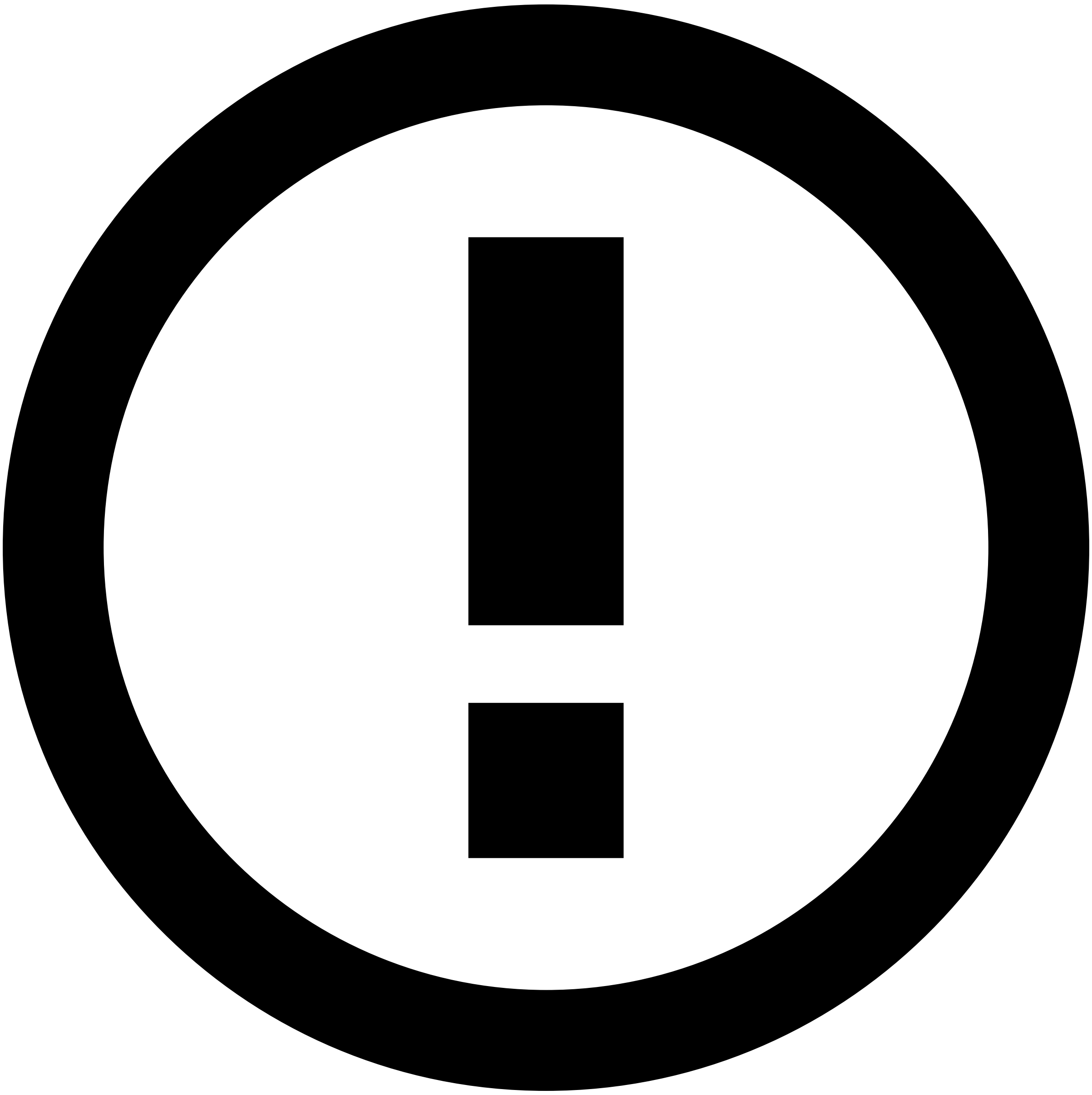
Add Authorization keysâ

To add a new Authorization key, select a label name and upload the JSON file that contains the new key:

The JSON file should be an array with the key object, using the following format:

```
 [
    {
        "e": "AQAB",
        "n":
"187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6xj9ZnqyyS
cle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374XNfLRb0s
F7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbOWgEPUKoR
BRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
        "kty": "RSA",
        "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
        "use": "sig"
    }
]
```





info

The new key must be associated with an existing label.

tip

Make sure you start signing the new messages with the new key.

Revoke Authorization keysâ

The following animation illustrates how to add a new key and revoke the current one.

danger

An Authorization key should only be revoked once a new Authorization key has been added (see previous step) and JWTs are already being signed with the new key.

Delete revoked Authorization keysâ

By deleting an Authorization key, you are permanently removing it from the list of keys. Only revoked keys can be deleted.

-
- [API Resources]
- Ad-Hoc Investigation Endpoints' Life Cycles

On this page

Ad-Hoc Investigation Endpoints' Life Cycles

This page explains the life cycles of the Investigations API's endpoints. The Investigations API includes the following endpoints:

- [Investigation `POST`]
- [Info `PUT`]
- [Feedback `PUT`]

Investigation (`POST`)â

To perform an ad-hoc investigation, take the following steps:

1. Send an `investigation` (`POST`) request to Feedzai. The request must contain the `merchant_id` to fetch the merchant data from the Reference Data Service and populate the merchant event in Case Manager. Consider the following payload when sending the request:

   ```
   {
       "event_external_id": "Tx_8448122",
       "lifecycle_id": "Iv_8448121",
       "injection_time": 1603207359060,
       "id": "unique_identifier",
       "reason_code": "string",
       "origin": "subpoenas"
   }
   ```

2. Go to Case Manager's Ad-Hoc Investigations channel and open the created event.

Info (`PUT`)â

The `info` endpoint allows you to update and add new information to a previously created investigation, e.g. signalling that the investigation can be closed.

Feedback (`PUT`)â

The `feedback` endpoint allows you to label merchants as **risky** (`fraud`) or **not risky** (i.e. `not fraud`). The labelling of an event is highly recommended, as it gives feedback to Feedzai's risk engine, which uses this information to learn and consolidate fraud patterns.

- API Resources

API Resources

This section provides you with the information you need to work with the Investigations API.

- [Ad-Hoc Investigation Endpoints' Life Cycles]
- [Ad-Hoc Investigations' Endpoints]
- [Additional Resources]

-
- Out-of-the-Box Resources
- Lists

Lists

Name	Description
default_threshold_list	
multi_tenanted_threshold_list	
terrorist_sanctions_list	Terrorist & Sactions Names List
track_list	
regulated_countries_3ds	List of countries with legal or regulatory requirements for strong cardholder authentication
exclusion_mid_list	Merchants to be excluded from inactivity rules (seasonal merchants, new onboarded merchants, etc)

Schemas

Schemas

Ad-hoc Investigations API Schema

Field Name	Field Type

injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING

merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
merchant_mcc	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
alert_id	STRING
event_external_id	STRING
lifecycle_id	STRING
event_type	STRING
info_type	STRING
event_received_at	LONG
feedback_label	STRING
feedback_value	BOOLEAN
feedback_type	STRING
reason_code	STRING
description	STRING
origin	STRING

Ad-hoc Investigations Extended API Schema

Field Name	Field Type
injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING

merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
merchant_mcc	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE

merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
alert_id	STRING
event_external_id	STRING
lifecycle_id	STRING
event_type	STRING
info_type	STRING
event_received_at	LONG
feedback_label	STRING
feedback_value	BOOLEAN
feedback_type	STRING
reason_code	STRING
description	STRING
origin	STRING
risky	BOOLEAN

Ad-hoc Investigations Plan Schema

Field Name	Field Type
injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING

merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
merchant_mcc	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
merchant_account_name	STRING

merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
alert_id	STRING
event_external_id	STRING
lifecycle_id	STRING
event_type	STRING
info_type	STRING
event_received_at	LONG
feedback_label	STRING
feedback_value	BOOLEAN
feedback_type	STRING
reason_code	STRING
description	STRING
origin	STRING

Extended Merchant Monitoring Schema

Field Name	Field Type
injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG

merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
risky	BOOLEAN
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
merchant_mcc	STRING
injection_time_prev_day	LONG
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG

Merchant Monitoring Schema

Field Name	Field Type
injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING

merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
merchant_mcc	STRING
injection_time_prev_day	LONG
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG

Merchant Monitoring Workflow Schema

Field Name	Field Type
injection_time	LONG
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING

merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
risky	BOOLEAN
merchant_mcc	STRING
injection_time_prev_day	LONG
month_of_year_string	STRING
prev_month_of_year_string	STRING
visa_cnt_cur_m	LONG
visa_cbk_cnt_cur_m	LONG
mc_cbk_cnt_cur_m	LONG
mc_fraud_cbk_cnt_cur_m	LONG
visa_amt_cur_m	LONG
visa_us_3ds_ok_amt_cur_m	LONG
visa_fraud_amt_cur_m	LONG
visa_us_3ds_ok_fraud_amt_cur_m	LONG
mc_fraud_cbk_amt_cur_m	LONG
mc_3ds_ok_amt_cur_m	LONG

mc_amt_cur_m	LONG
mc_cnt_prev_m	LONG
mc_ecom_cnt_prev_m	LONG
cnt_manual_3m_4w_dlay	LONG
avg_score_4w_4w_dlay	DOUBLE
cnt_4w_4w_dlay	LONG
std_score_4w_4w_dlay	DOUBLE
sum_amt_4w_4w_dlay	LONG
sum_amt_8w_8w_dlay	LONG
avg_score_1w_1w_dlay	DOUBLE
std_score_1w_1w_dlay	DOUBLE
sum_amt_1w_1d_dlay	LONG
cnt_6m_1d_dlay	LONG
avg_accpt_amt_90d_1d_dlay	DOUBLE
cnt_accpt_90d_1d_dlay	LONG
std_accpt_amt_90d_1d_dlay	DOUBLE
avg_accpt_amt_1d	DOUBLE
cnt_1d	LONG
aml_cnt_1d	LONG
rejct_cnt_1d	LONG
accpt_cnt_1d	LONG
prepaid_card_cnt_1d	LONG
sum_amt_1d	LONG
cnt_manual_1m	LONG
high_risk_cntries_1m	LONG
avg_score_1w	DOUBLE
cnt_1w	LONG
sum_amt_1w	LONG
sum_amt_10d	LONG
sum_amt_15d	LONG
sum_amt_2d	LONG
sum_amt_30d	LONG
avg_score_4w	DOUBLE
3ds_ko_cnt_4w	LONG
cnt_4w	LONG
rfnd_cnt_4w	LONG
3ds_ko_amt_4w	LONG
rfnd_amt_4w	LONG
sum_amt_4w	LONG
sum_amt_45d	LONG

sum_amt_5d	LONG
cnt_60d	LONG
sum_amt_8w	LONG
3ds_ko_cnt_3m_4w_dlay	LONG
cnt_3m_4w_dlay	LONG
rfnd_cnt_3m_4w_dlay	LONG
rejct_cnt_3m_4w_dlay	LONG
dist_cardcntry_3m_4w_dlay	LONG
dist_card_3m_4w_dlay	LONG
3ds_ko_amt_3m_4w_dlay	LONG
rfnd_amt_3m_4w_dlay	LONG
rejct_amt_3m_4w_dlay	LONG
sum_amt_3m_4w_dlay	LONG
cbk_cnt_4w_4w_dlay	LONG
fraud_amt_4w_4w_dlay	LONG
cbk_amt_4w_4w_dlay	LONG
cnt_3m_8w_dlay	LONG
cbk_cnt_3m_8w_dlay	LONG
fraud_amt_3m_8w_dlay	LONG
cbk_amt_3m_8w_dlay	LONG
sum_amt_3m_8w_dlay	LONG
cnt_8w_8w_dlay	LONG
rejct_cnt_3m_1d_dlay	LONG
cnt_3m_1d_dlay	LONG
rejct_cnt_4w	LONG
dist_cardcntry_4w	LONG
dist_card_4w	LONG
rejct_amt_4w	LONG
cnt_8w	LONG
mcc_avg_accpt_amt_90d_1d_dlay	DOUBLE
mcc_std_accpt_amt_90d_1d_dlay	DOUBLE
rt_cbk_amt_4w_4w_dlay	DOUBLE
rt_cbk_amt_3m_8w_dlay	DOUBLE
rt_rejct_amt_4w	DOUBLE
rt_rejct_amt_3m_4w_dlay	DOUBLE
rt_rfnd_amt_4w	DOUBLE
rt_rfnd_amt_3m_4w_dlay	DOUBLE
rt_fraud_amt_4w_4w_dlay	DOUBLE
rt_fraud_amt_3m_8w_dlay	DOUBLE
rt_rejct_cnt_4w	DOUBLE

rt_rejct_cnt_3m_4w_dlay	DOUBLE
rt_rfnd_cnt_4w	DOUBLE
rt_rfnd_cnt_3m_4w_dlay	DOUBLE
rt_3ds_ko_amt_4w	DOUBLE
rt_3ds_ko_amt_3m_4w_dlay	DOUBLE
rt_3ds_ko_cnt_4w	DOUBLE
rt_3ds_ko_cnt_3m_4w_dlay	DOUBLE
rt_cbk_cnt_4w_4w_dlay	DOUBLE
rt_cbk_cnt_3m_8w_dlay	DOUBLE
rt_visa_cbk_cnt_cur_m	DOUBLE
rt_rejct_cnt_1d	DOUBLE
rt_rejct_cnt_3m_1d_dlay	DOUBLE
rt_manual_cnt_1m	DOUBLE
rt_manual_cnt_3m_4w_dlay	DOUBLE
rt_prepaid_card_cnt_1d	DOUBLE
rt_visa_fraud_amt_1m	DOUBLE
rt_visa_us_3ds_ok_fraud_amt_1m	DOUBLE
rt_mc_cbk_cnt_1m	DOUBLE
rt_mc_fraud_cbk_cnt_1m	DOUBLE
rt_mc_3ds_ok_amt_1m	DOUBLE
rtcomp_cbk_amt_1m_to_3m	DOUBLE
rtcomp_rejct_amt_1m_to_3m	DOUBLE
rtcomp_rfnd_amt_1m_to_3m	DOUBLE
rt_amt_1m_to_3m	DOUBLE
rt_amt_lastm_to_prevm	DOUBLE
rt_amt_last2m_to_prev2m	DOUBLE
rt_dist_card_1m_to_3m	DOUBLE
rtcomp_fraud_1m_to_3m	DOUBLE
rt_dist_cardcntry_1m_to_3m	DOUBLE
rtcomp_rejct_cnt_1m_to_3m	DOUBLE
rtcomp_rfnd_cnt_1m_to_3m	DOUBLE
rt_cnt_1m_to_3m	DOUBLE
rt_cnt_lastm_to_prevm	DOUBLE
rt_cnt_last2m_to_prev2m	DOUBLE
rtcomp_3ds_ko_amt_1m_to_3m	DOUBLE
rtcomp_3ds_ko_cnt_1m_to_3m	DOUBLE
rtcomp_cbk_cnt_1m_to_3m	DOUBLE
rtcomp_rejct_cnt_1d_to_3m	DOUBLE
rt_amt_1d_to_1w	DOUBLE
rtcomp_manual_cnt_1m_to_3m	DOUBLE

ndx_approx	DOUBLE
add_approx	DOUBLE
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
custom_fields	MAP
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
oct_rfd_amount_ref_rev_mid_30d	LONG
oct_rfd_amount_payment_mid_30d	LONG
oct_rfd_mid_count_6m_delay_30d	LONG
oct_rfd_count_mid_30d	LONG
oct_rfd_amount_mid_30d	LONG
oct_rfd_count_mid_7d	LONG
oct_rfd_sum_amt_24h	LONG
oct_rfd_sum_amt_mid_90d	LONG
oct_rfd_rt_amt_1m_to_3m	DOUBLE
oct_rfd_sum_amt_1w	LONG
oct_rfd_cnt_1w	LONG
oct_rfd_sum_amt_3m_4w_dlay	LONG
oct_rfd_sum_amt_4w	LONG
oct_rfd_rt_cnt_1m_to_3m	DOUBLE
oct_rfd_cnt_3m_4w_dlay	LONG
oct_rfd_cnt_4w	LONG
cnt_risky_ahi_30d	LONG

Payments Extended API Schema

Field Name	Field Type
fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING
domain	STRING

event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
info_type	STRING
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING

payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING

device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN

merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN
payment_bank_account_number	STRING
payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING

terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING
dispute_end_date	LONG
dispute_refunded	STRING
card_mti	STRING
card_processing_code	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
payment_card_pan	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN

recurring_type	STRING
is_scoring	BOOLEAN
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG

Pre Omnichannel Metrics Plan Schema

Field Name	Field Type
fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING
domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING

payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING
payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING

customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING
device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING

merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN

payment_bank_account_number	STRING
payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING
terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING

dispute_end_date	LONG
dispute_refunded	STRING
payment_card_pan	STRING
card_mti	STRING
card_processing_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
cbk_reason_code	STRING
aml_high_risk_coutries	BOOLEAN
norm_payment_card_brand	STRING
month_of_year_string	STRING
prev_month_of_year_string	STRING
us_domestic_3ds_payments	BOOLEAN
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
is_scoring	BOOLEAN
oct_rfd_count_card_mid_24h	LONG
oct_rfd_sum_amt_card_mid_24h	LONG
oct_rfd_count_card_mid_7d	LONG
oct_rfd_sum_amount_card_mid_7d	LONG
oct_rfd_sum_amt_card_mid_30d	LONG

oct_rfd_count_card_mid_30d	LONG
oct_rfd_count_mid_24h	LONG
oct_rfd_count_mid_7d	LONG
oct_rfd_count_mid_30d	LONG
oct_rfd_sum_amount_mid_24h	LONG
oct_rfd_sum_amount_mid_7d	LONG
oct_rfd_sum_amount_mid_30d	LONG
oct_rfd_count_card_24h	LONG
oct_rfd_count_card_7d	LONG
oct_rfd_count_card_30d	LONG
oct_rfd_sum_amount_card_24h	LONG
oct_rfd_sum_amount_card_7d	LONG
oct_rfd_sum_amount_card_30d	LONG
oct_rfd_count_bin_mid_24h	LONG
oct_rfd_count_bin_mid_7d	LONG
oct_rfd_count_bin_mid_30d	LONG
oct_rfd_amount_bin_mid_24h	LONG
oct_rfd_amount_bin_mid_7d	LONG
oct_rfd_amount_bin_mid_30d	LONG
oct_rfd_count_card_country_7d	LONG
oct_rfd_amount_card_country_7d	LONG
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG

Out-of-the-box resources

Merchant Monitoring offers out-of-the-box resources, such as lists, schemas, and rules projects. These resources are included in the solution standard package to help Payment Service Providers monitor risky merchants.

This section contains the following reference documentation on those resources:

- Lists
- Schemas
- Rules Projects
  - Bankruptcy Risk Rules
  - Chargeback Rules
  - Fraud Rules
  - Money Laundering Rules
  - Unusual Activity Rules
  - Refund and OCT Rules
- Analytics Dashboards' Fields

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/overview

Operational Work

Data Scientist & Fraud Strategist

Fraud Prevention Agent

Fraud Prevention Manager

Integration Engineer

Auth and Capture Scenarios

Typically, authorization and capture requests take place simultaneously via the `authorization_type` field. The authorization allows merchants to ensure customers have sufficient funds for the product / service they're purchasing. The capture allows merchants to collect those funds.

In some cases, however, the capture may take place some time after the authorization has occurred. For example, let's consider a customer who subscribes to a streaming platform using a credit card. After the customer submits the card information, the merchant verifies if the card is valid and then authorizes the purchase, thus allowing the customer to use the service. A few days / weeks later, the merchant moves the funds (i.e. capture occurs at a later time).

Now that you have a high-level understanding of how auth and capture requests work, take a look at the following scenarios to understand how they impact metrics when calculating total amounts in profiles:

Info endpoint scenario	Event flow
Payment with authorization and capture taking place simultaneously in a single event	1. PSP sends payment event to Feedzai with `authorization_type` = "auth-capture".

Info endpoint scenario	Event flow
	2. No further capture event is expected in this case, the final transaction amount considered will be the payment event amount. 3. If the `authorization_type` field is null, Feedzai assumes its value to be "auth-capture", and therefore, the amount to be considered is the payment event amount.
Payment with capture taking place after authorization, i.e. two separate events	1. PSP sends payment event to Feedzai with `authorization_type` = "auth". 2. At a later stage, when capturing the amount, the PSP sends a specific info event to Feedzai with `info_type` = "capture". 3. The amount to be considered for metrics will be the info event amount where `info_type` = "capture".
Payment whose card is verified or tokenized before any capture of funds takes place	1. PSP sends the first payment event to Feedzai with: Â Â Â Â a. `authorization_type` = "verify". Â Â Â Â b. Amount expected to be 0 (or, e.g. 0.1â□¬ if 0 is not allowed). 2. Subsequent payments from the registered card are expected to take place.

- About Transaction Fraud for PSPs
- Chargeback Life Cycle

Chargeback Life Cycle

Learn about TFP's chargeback life cycle.

1. The merchant delivers goods to the customer.
2. The customer receives the goods.
3. Funds are settled.
4. The customer initiates a chargeback.
5. The Payment Service Provider (PSP) informs Feedzai about the chargeback that is taking place through the Payments API.

info

Endpoint: The PSP invokes the *Feedback* PUT endpoint. This request is registered in the `feedback_type` with the `chargeback` value.

6. Feedzai's risk engine processes the feedback to update metrics.
7. The updated metrics are used in Merchant Monitoring's rules and analytics dashboards, thus making fraudulent chargeback prevention more accurate.

-
- [About Transaction Fraud for PSPs]
- Fraud Scenarios

On this page

Fraud Scenarios

This page covers the fraud scenarios and the detection strategy for fraudulent customers. The standard package delivers out-of-the-box rules that capture user patterns associated with each fraud scenario.

Card-not-Present fraudâ

Card-not-present (CNP) fraud is prevalent in transactions made over the internet, but may also take place over the phone. In the CNP scenario, fraudulent customers are able to purchase goods without a physical card, since they have already managed to access the victim's card credentials via hacking, phishing, or skimming schemes.

How TFP prevents CNP fraudâ

Here are some examples of rules that help detect a CNP fraud scenario:

- **Cards-Behavior-FR17:** Alert payment if multiple transactions with the same amount have been approved by the same merchant over the last 24 hours.
- **Cards-Behavior-FR18:** Decline payment and block customer email if the number of distinct rejected cards per customer email, over the last 7 days, is above a set threshold.
- **Cards-CNP-FR27:** Alert payment if the card has made multiple CNP transactions in a high number of unique merchants over the last 5 minutes.

Cardingâ

Carding is a term used for fraudulent customers who attempt to access credit or debit card credentials for their personal interest. Fraudulent customers may use methods such as manipulating bots and botnets to steal a person's card details in order to sell their information on the dark web.

How TFP prevents cardingâ

Here are some examples of rules that help detect a carding scenario:

- **Cards-Mismatch-FR28:** Alert payment for listed merchants if there is a mismatch between card and merchant countries.
- **Cards-Behavior-FR20:** Decline payment if, over the last 24 hours, the number of distinct emails from the same card and merchant is above a set threshold.

Skimmingâ

Skimming is a type of fraud that consists in stealing a person's card details and using them in a fraudulent card, i.e. electronic copying card information to make cloned cards. Fraudulent customers may use several methods to do this. For instance, they may use a skimming device that swipes the victims' card information, or simply use the victim's receipts to extract their card details. They then proceed to create "fake plastic" cards to store the stolen victim's information.

How TFP prevents skimmingâ

Here is an example of a rule that helps detect a skimming scenario:

- **MagStripe-Fallback-CP1:** Decline a fallback payment if the amount is above a set threshold. This rule will check if a transaction's Card Entry Mode is "Magnetic Stripe" and if the authentication mode is "PIN Bypassed".

Card testingâ

A bot is a software or configurable script that runs scripts over the internet and attacks e-commerce merchants and financial institutions. This is usually performed by botnets, a group of internet devices running one or more bots. Fraudsters can easily rent bots and adapt botnets (for different malicious purposes) to deploy them fast. Dozens of purchases with the same card in a short period of time (e.g. within 5 minutes) is a good example of a bot attack.

How TFP prevents card testingâ

Here are some examples of rules that help detect a card testing scenario:

- **Cards-Velocity-FR14:** Decline payment if, over the last minute, the number of distinct cards with failed payments per merchant is above a set threshold.
- **Cards-MCC-FR24:** Decline payment if, over the last 15 minutes, the number of small amount card transactions per merchant with listed merchant category codes (MCC) is above a set threshold (e.g. digital goods, in-app purchases).
- **Cards-CNP-FR27:** Alert payment if, over the last 5 minutes, the card has made multiple CNP transactions in a high number of unique merchants.

BIN attackâ

A bank identification number (BIN) corresponds to the first 6 or 8 digits of a credit card number. A BIN attack occurs when a fraudster uses those digits and runs software to generate the rest of the numbers. These auto-generated card numbers will then be used in fraudulent transactions.

How TFP prevents BIN attackâ

Here is an example of a rule that helps detect a BIN attack scenario:

- **Cards-BinAttack-FR29:** Decline payment if, over the last 10 minutes, the ratio of distinct cards and distinct expiration dates per BIN and merchant is below a set threshold.

Lost & stolenâ

The lost & stolen scenario occurs when a person is a victim of physical theft, i.e. a criminal accesses the victim's pocket or purse and uses the card for unauthorized transactions. These individuals use the card until the cardholder reports to the issuing bank, which blocks the account.

How TFP prevents lost & stolenâ

Here is an example of a rule that helps detect a lost & stolen scenario:

- **Cards-Behavior-FR15:** Decline payment if, over the last 24 hours, the number of transactions per Card ID and merchant is above a set threshold.
- **Cards-Behavior-FR16:** Decline payment if, over the last 24 hours, the amount of transactions per Card ID and merchant is above a set threshold.
- **Cards-Behavior-FR17:** Alert payment if, over the last 24 hours, multiple transactions with the same amount have been approved by the same merchant.

Unusual activityâ

Unusual activity includes transactions that are not part of a customer's usual behavior, i.e. they start spending large amounts, or making several transactions in a short time period.

How TFP prevents unusual activityâ

Here are some examples of rules that help detect unusual activity:

- **All-Payments-Velocity-FR1:** Decline payment if, over the last 24 hours, the number of transactions per customer ID and merchant is above a set threshold.
- **All-Payments-Velocity-FR7**: Decline payments if, over the last 24 hours, the number of transactions per Shipping Address and merchant is above a set threshold.
- **Velocity-APM1:** Alert payment if, over the last 24 hours, the total amount per alternative payment method (APM) - e.g Paypal, Neteller -, merchant, and customer email is above a set threshold.

About Transaction Fraud for PSPs

Every merchant that processes payments with a Payment Service Provider (PSP) poses a threat to fraud and subsequent losses. As a result, Feedzai's Transaction Fraud for PSPs (TFP) offers real-time fraud prevention to detect and stop risky payment transactions. By understanding what each customer's behavior looks like, TFP creates contextual awareness that allows for accurate fraud detection. Rules and machine learning models receive this information to score each payment transaction for PSPs to make accept/decline decisions.

TFP evaluates payment transactions over their entire life cycle through the Payments API. The following sections focus on those specifics.

Paymentsâ

The integration with the Payments API allows PSPs to score payment transactions. It provides a RESTful API that allows customers to send information at different stages of the payment's life cycle. Payments API is built based on ISO 8583 and

ISO 20022 international standards, and gives customers the ability to consume payment-related data via a unified API for all payment types.

New payment types can be easily and quickly integrated into the Payments API, thus ensuring customers remain agile in the evolving payment landscape. A unified channel prevents issues caused by multiple payment channels.

Feedzai's Payments API consumes real-time transactions as well as subsequent updates to transactions, such as chargebacks.

End-to-end packageâ

These are the out-of-the-box capabilities that respond to the use case end to end:

- REST API: Sends information via REST API at different stages of the transaction life cycle.
- Reference data: Sends merchant information upfront via a standard API. This information is then used in the payment's scoring process.
- Pre-configured risk engine: Processes transactions, generates real-time accept/decline decisions, as well as challenge recommendations that require manual decision.
- Pre-configured fraud rules: Detect and prevent the most sophisticated fraud typologies.
- Pre-configured alert and case management interface: Reviews suspicious payments that require manual decision. This interface includes visual link analysis and graph technology (i.e. Genome) to find and visualize complex financial crime patterns.
- Analytics dashboards: Display information about fraud prevention agent and rule performance.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/about/refund-oct-life-cycle

-
- [About Transaction Fraud for PSPs]
- Refund / OCT Life Cycle

Refund / OCT Life Cycle

Learn about Transaction Fraud for PSPs' refund / original credit transaction (OCT) life cycle and understand what happens in each refund's stage.

For the sake of the refund / OCT use case, the life cycle begins after the customer paid for the goods.

1. The merchant delivers goods to the customer.
2. The customer receives the goods.
3. Funds are settled.
4. The customer requests a refund (e.g. the good hasn't been delivered to their address) or an OCT (e.g. the merchant passes gambling winnings back to the card used to place the bet).
5. The Payment Service Provider (PSP) calls Feedzai's Payments API to score the refund / OCT.

info

Endpoint: The PSP makes this by invoking the *Info* POST endpoint. This request is registered in the `info_type` with the `refund` or `oct` value.

caution

PUT vs POST: Note that the Info endpoint accepts two methods: POST and PUT. Any INFO event with `info_type=refund/oct` sent as a POST will return a decision outcome and alert flag, whereas an INFO event sent as a PUT will merely ingest the event into Feedzai's system for displaying purposes and metric calculations.

6. Feedzai's risk engine processes and scores the refund / OCT in real time, and sends the recommendation (Approve/Decline) to the PSP. Note that the event can be alerted regardless of the system decision's recommendation (see step 10).
7. The PSP makes an automated decision based on the risk engine recommendation (auto-approve or auto-decline).
8. The merchant receives the `approve` or `decline` message.
9. If the request is approved, the customer receives the refund / OCT.
10. Fraud prevention agents review alerted refunds / OCTs at any time as a post-mortem analysis to provide feedback on fraud/not fraud decisions and improve scoring of subsequent refunds / OCTs.

info

Endpoint: This decision is registered in the `feedback_value` field as true or false for `fraud` and `not_fraud`, respectively.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/about/transaction-life-cycle

- About Transaction Fraud for PSPs
- Payment Life Cycle

Payment Life Cycle

Learn about Transaction Fraud for PSPs' payment life cycle and understand what happens in each payment's stage.

1. The customer uses a card or payment app to purchase a product in the merchant's physical or online store.
2. The merchant processes the order.
3. The transaction is redirected to the Payment Service Provider (e.g. Adyen, Stripe).
4. The Payment Service Provider (PSP) calls Feedzai's Payments API to score the transaction.

Info and *Merchant Provisioning* endpoints

The PSP calls Feedzai's Payment API by invoking the *Payment* endpoint. Alternatively, the PSP can send the transaction through the *Info* endpoint in case they don't want to request a scoring.

Before a new merchant's first payment, you must invoke the *Merchant Provisioning* endpoint to add the new merchant entity to Feedzai.

5. Feedzai's risk engine processes and scores the transaction in real time, and sends the recommendation (Approve/Decline) to the PSP.
6. The PSP makes an automated decision based on the risk engine recommendation.
7. The PSP processes the payment if the transaction is approved. Note that, if the transaction is declined by the PSP, the `decline` communication is sent to the merchant (jump to step 10).

Info and *Feedback* endpoints

The PSP should invoke Feedzai's *Info* endpoint to communicate a different decision from Feedzai's risk engine recommendation (`do_not_honor`).

The PSP can also invoke the *Feedback* endpoint to notify Feedzai about a fraudulent transaction.

8. The processed payment is redirected to the acquirer/network, and ultimately reaches the issuer bank.
9. The PSP receives the notification that the acquirer/network and issuer bank authorized the transaction.

Declined *Infos*

Declined *Infos* are mandatory. By default, Feedzai assumes the payment is approved, unless it receives an *Info* saying it was declined. To ensure metrics are accurate, Feedzai must receive an *Info* on every `decline` event - regardless of who influenced the declining decision (i.e. the issuer bank, Feedzai, the PSP, etc).

Approved *Infos*

Approved *Infos* are optional and do not count for metrics, as they only add context for analysts (e.g. `do_not_honor`). Feedzai recommends not to send *Infos* for approved payments to minimize the number of events.

10. The merchant receives an `approve` message. Note that if, in step 7, the PSP rejects the transaction, the merchant receives a `decline` message.
11. If the transaction has been approved, the merchant accepts the order and the customer receives the product.
12. Later, funds are settled between the PSP and the merchant.

Settlement endpoint

Settlement events provide information about money moves and not fraud insights. They provide analysts with context, and are only a requirement for compliance use cases, such as Anti-Money Laundering. The Transaction Fraud for PSPs' standard integration does not send settlements.

13. Fraud analysts review alerted transactions (see step 5) at any time as a post-mortem analysis to provide feedback on fraud/not fraud decisions and improve scoring of subsequent transactions.

Additional Resources

This section contains information about the following topics:

- Schema Mapping
- Authentication
- Responses
- Generated Fields
- Asynchronous Operation
- Custom Fields

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/api-resources/asynchronous-operation

- API Resources
- Additional Resources
- Asynchronous Operation

Asynchronous Operation

The Payments API connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations, such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations), are executed immediately. Any invalid request fails immediately, as in the synchronous operation mode.

Once validations are successfully completed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will eventually be processed. Since the request has not been fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the *Payment* endpoint will not be present in this response.

Requests are consumed by the message broker at a configurable rate and execute the same operations as if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one produced in synchronous operation mode. You can use the *Responses Get* endpoint to retrieve the responses from the message broker.

The responses obtained from the *Responses Get* endpoint can be correlated to the original request using the `event_external_id` response field, which will have the same value that was sent in the equivalent request field.

-
- API Resources
- Additional Resources
- Authentication

On this page

Authentication

Feedzai allows you to authenticate and protect your data through JSON Web Tokens (JWTs). A JSON Web Token (JWT) is a piece of information that is transmitted as a JSON object from clients to Feedzai, and its purpose is to ensure that client requests (e.g. transactions for processing) have not been compromised.

These tokens are generated using the clientâ□□s private key, which can be verified with the corresponding public key. The public key must be shared with Feedzai beforehand to verify that the token sent with the request belongs to the client.

Overviewâ□□

JWTs protect against different attacks using the following mechanisms:

- **Encryption/signing** so that only authenticated users can access sensitive data.

- **Expiration dates** to protect against replay attacks.
- **JWT revocation** to block exposed JWTs as soon as possible.

You can see more details on integrating JWTs with your software in the sub-sections below, but in general you will:

1. Generate a private-public key pair to sign the JWT.
2. Provide the public key to Feedzai in JSON Web Key Set format.
3. Frequently generate new JWTs with a reduced validity period (e.g. 5 minutes). Each JWT is signed with your private key generated in step 1.
4. For each request to Feedzai, include a JWT in the `Authorization` header.

Given the recommended algorithm and parameters, you can choose the tool that best fits your system. A comprehensive list of tools is available at the libraries section of jwt.io.

danger

Third-party security vulnerabilities: When choosing a library, make sure it has no known security vulnerabilities.

JSON Web Keyâ

A key pair is necessary to cryptographically sign and verify the JWTs sent with your requests.

- You will use the private key to sign your JWTs.
- Feedzai will use the public key to verify the signature, and therefore the authenticity of the requests.

To create a JWK set, you need to choose the signing algorithm and the JWK parameters. Feedzai recommends the `RS512` algorithm (â□□RSASSA-PKCS1-v1_5 using SHA-512â□□) with a 4096-bit key. See the sections below to learn how to generate a JWK set with standard tools.

danger

Beware of private key parameters: Construct the JWK set so that you **do not** send any of the private key parameters.

Your JWK set should look similar to this:

```
 {
    "keys": [
        {
            "e": "AQAB",
            "n":
"187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6xj9ZnqyyS
cle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374XNfLRb0s
F7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbOWgEPUKoR
BRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
            "kty": "RSA",
            "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
            "use": "sig"
        }
    ]
}

{
        "kid":"mykey2",
        "kty":"RSA",
        "alg":"RS512",
        "use":"sig",
        "e":"AQAB",
        "n":"k51ZPyu9gocn508aXzob_hSKaIAnJApY4pY3h1oym1Z15-6DN3WxB3YHUyXPRd-iBzF7177XjQAywTzAaz-
PKLZ4bC52noanLx7Ihyf3nQAOwhEv16zaJLI-
HqMEMp6JKPjKJ8LKanVnvxWNI0zUBwcBOwt9cxph0evyVO94cXA0RZxUZxa7C3FuSYkO2zWorMH0ZkC3harkJx0C9Q7nBiGW
PQmB8ZkLx04iWlm8ldyxDJPKl-PC4-
gS3y2mgPfZVAaK_VGod88PUmF4vFax7a03v2VLtn9wm8UWs9SXATxJ5cExgIIvFkdsZAcS3hWnhUe3Of4zAEbKh2Ah5DtfY9
xvtsU8FGlak7wLCwVnubWWglI2hmFvFU7Jsyzb_L-
jD7XCYrg7QuFV8mX2jVcTbsSxnd8GLRNbvmztvsjfaQHXEHraTRJFoGO1Y3JXDiY8Kj40OQXxBNXaFxHda9lBfzWfDkgWvcy
eiTPeWpKc6cS1zcDaQNRzZr3aZR-
gI4U8YTAXcIKDdom2GCQSx9WfmFq3uAj95duL5CgyGfw7L-02c6ckRr6kBAKebURkSsZXOyTO8bdbS8GzPMD59CrRL2SnJJe
qHyrEav8JCu0x3UCXbyqlfqQgzVsXNpFUMIS-sUhe3nyVQOeJ_WoYq7oAbjLpx_Bw5elQJf4zJbJd2Pk"
    }
```

In the aforementioned example, the following parameters are used:

Name	Extended name	Description
`e`	Exponent	RSA public exponent e.
`n`	Modulus	RSA public modulus n.
`kty`	Key Type	Should be `RSA` to symbolize you are using the RS512 algorithm.
`kid`	Key Id	Should specify a key identifier of your choice.
`use`	Public Key Use	Should be `sig` to note this key will be used to verify JWTs.

How to generate a key-pair and JWKâ

tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

Consider the following requirements if you're generating JWTs in a different way:

- Make sure you store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**

- [Manage JWKs](#) in Pulse.

1. [Download Feedzai's JWT Generator](#).
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

1. [Download Feedzai's JWT Generator](#).

info

Make sure you repeat steps 1 through 3 for a total of two private-public key pairs: one for `prod` and another one for `qa`.

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWK \
  --generateKeys \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --kid "key_id"
```

The JWK Set JSON will be displayed on your standard output. Alternatively, you can use the `--jwk` parameter to specify a file where it should be stored instead.

5. Store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**
6. [Manage JWKs] in Pulse.

7. *This step is internal and focused on Customer Success teams. Be careful when sharing it with clients.*

Feedzai adds the generated JWK Set JSON to the following folders:

````
 ```bash
delivery/pulse/pulse-assembly-runtime/src/main/resources/[environment]/files/conf/jwk.json
```
````

tip

Note that `[environment]` is the client's cloud environment (e.g. `qa` and `prod`).

JSON Web Tokensâ

Once you have a private key, you can generate your JWTs. All JWTs are composed of a set of headers, a set of claims and a signature.

JWT Headersâ

Besides the algorithm (`alg`), the only mandatory header for JWTs is the `kid` (the key ID), which indicates which key was used to sign the token and **must** match a valid key in the JWK Set used for validation. The set of headers should be similar to:

```
{
  "kid": "keyID",
  "alg": "RS512"
}
```

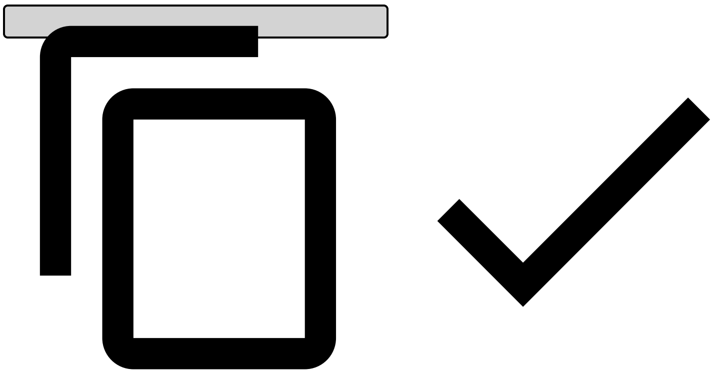
JWT Claimsâ

Typically, the following claims should be present on the JWTs:

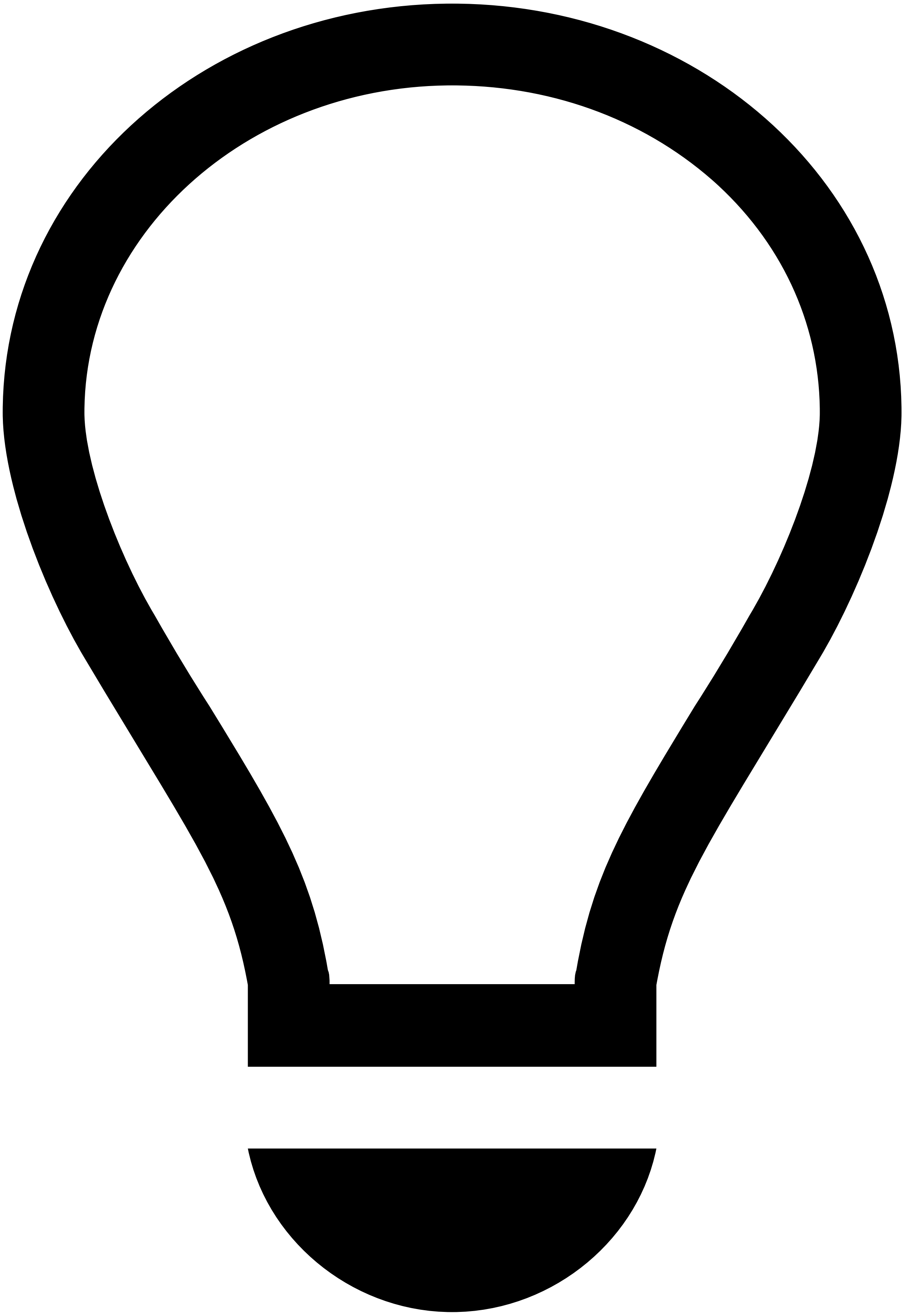
Claim name	Extended name	Description
`iss`	Issuer name	Identifies the issuer of the JWT. This value will be provided by Feedzai, and it should match the tenant id, if applicable.
`exp`	Expiration Time	Date after which the JWT must be rejected.
`nbf`	Not before Time	Date before which the JWT must be rejected.
`jti`	JWT id	Identifier used to revoke JWTs.
`sub`	Subject	Identifies the user of the JWT.

Your final set of claims should be similar to:

```
{
    "iss": "feedzai",
    "exp": 1523022355,
    "nbf": 1523022055,
    "jti": "a00cd6dd-4e55-4694-aeb5-4050e7d063b6",
    "sub": "700e55f9-15b0-4f95-8094-960138a0d886"
}
```



How to generate a JWTâ



tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWT \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --iss "issuer" \
  --sub "subject" \
  --exp "$((EPOCHSECONDS + 300))" \
  --kid "key_id"
```

The JWT will be displayed on your standard output. Alternatively, you can use the `--jwt` parameter to specify a file where it should be stored instead.

How to send the signed JWTâ

danger

Note that you can only send the signed JWT after the corresponding JWK is uploaded to Pulse (see [Manage JWKs in Pulse]).

Send the signed JWT in the `Authorization` header of HTTP requests made to Feedzai's API, as follows:

```
Authorization: Bearer <your_jwt>
```

Manage JWKs in Pulseâ

Once you create the JWK set, the following JWK steps must be taken:

1. Log into Pulse and confirm the current list of Authorization keys. Before revoking any key, make sure you are no longer using it.
2. Add the new key to Pulse.
3. Start signing messages with JWTs based on the new key.
4. Revoke the old key.
5. Delete revoked Authorization keys.

List Authorization keysâ

To list the existing Authorization keys, access the **Administration** page and select the **Authorization Keys** tab on the sidebar.

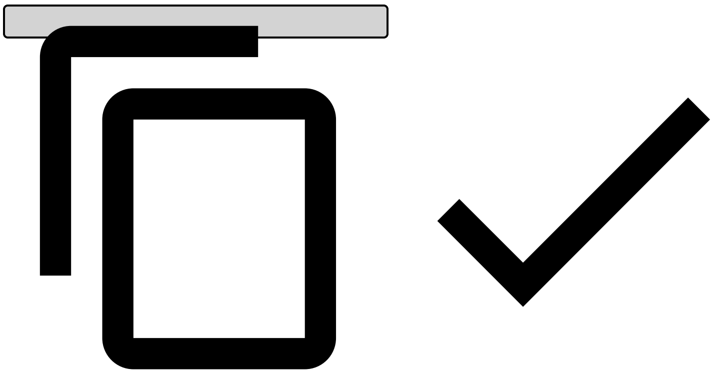
Select a label name or expand to see the Authorization keys associated with that label:

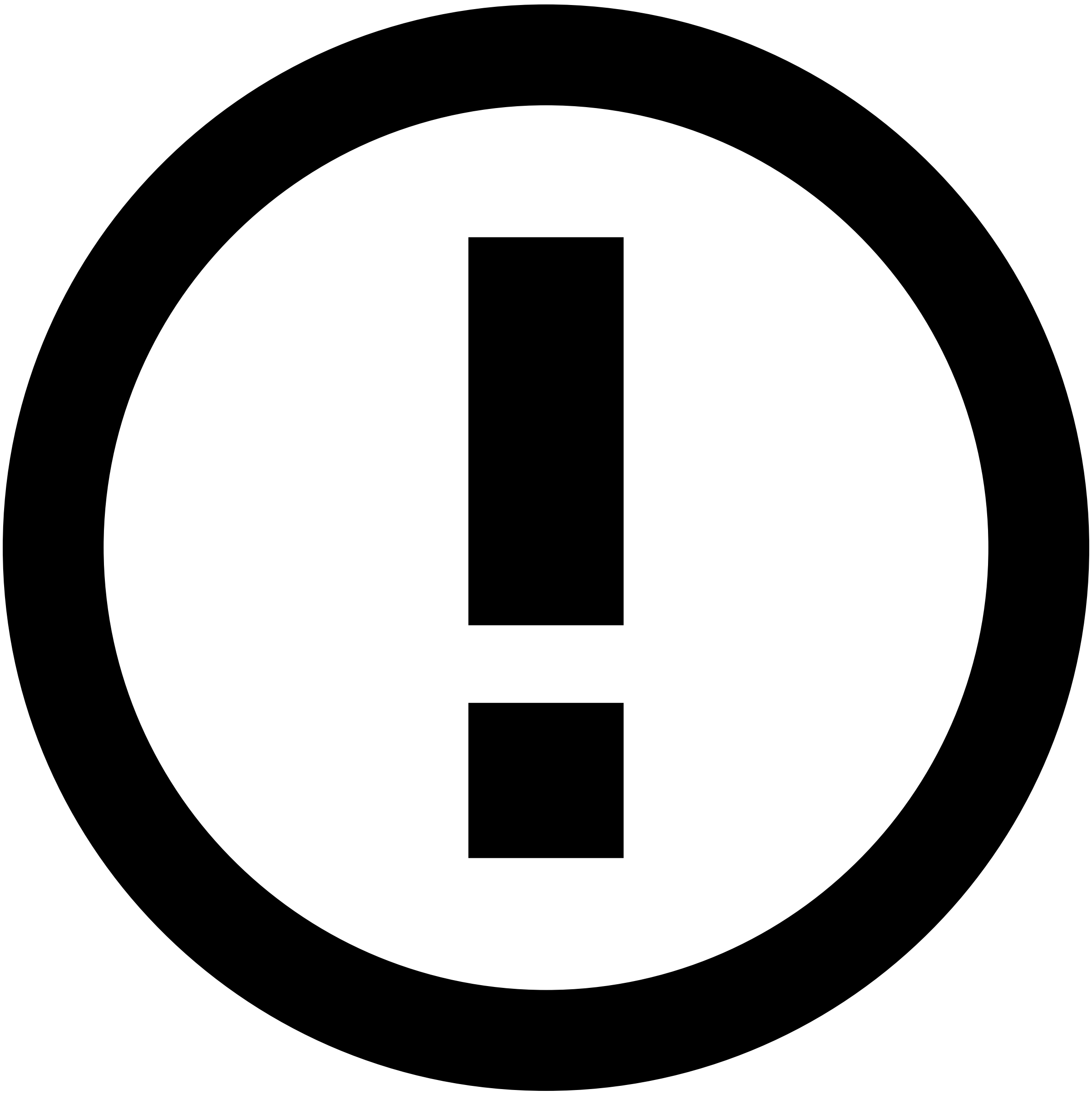
Add Authorization keysâ

To add a new Authorization key, select a label name and upload the JSON file that contains the new key:

The JSON file should be an array with the key object, using the following format:

```
 [
    {
        "e": "AQAB",
        "n":
"187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6xj9ZnqyyScle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374XNfLRb0sF7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbOWgEPUKoRBRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
        "kty": "RSA",
        "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
        "use": "sig"
    }
]
```





info

The new key must be associated with an existing label.

tip

Make sure you start signing the new messages with the new key.

Revoke Authorization keysâ

The following animation illustrates how to add a new key and revoke the current one.

danger

An Authorization key should only be revoked once a new Authorization key has been added (see previous step) and JWTs are already being signed with the new key.

Delete revoked Authorization keysâ

By deleting an Authorization key, you are permanently removing it from the list of keys. Only revoked keys can be deleted.

Custom Fields

The `custom_fields` is a complex field that allows you to use custom fields without making changes to Pulse and Case Manager's schemas. This field is a JSON object that supports key values of type string.

info

Only the `custom_fields` field is under a strict interface as an optional JSON object with optional properties of type string. The internal properties won't be under a strict interface and only the type string will be validated. Note that all the integrations must be carefully done and should respect the defined names for the custom fields.

The following example helps you understand how you can add two custom fields (`risk_score` and `risk_level`) to the message payload:

```
 (...)
"custom_fields": {
    "risk_score": "7.3",
    "risk_level": "high"
},
(...)
```

Note that, even though the `risk_score` is a number, it must be sent as a string and handled as such when needed in the workflow.

These custom fields can be easily used in rules or displayed in Case Manager. User-Defined Functions (UDFs) allow you to use custom fields in rules. You can extract the desired value from the custom field and convert it to the desired type. The following image shows you an example of the aforementioned custom fields:

You can add them to Case Manager by using the **Edit** button from the Event page. Type the `event.fields.custom_fields` prefix (e.g. `event.fields.custom_fields.risk_score`) as follows:

Configurationâ

Custom fields support the following configurations:

- Limit: maximum number of custom fields in a single event.
  - Optional with default value `10`.
  - The affected endpoints have the `customFields.limit` configuration with value `10`
- Allowed fields list: if defined, only fields present in the list can be sent.
  - Optional and without default value.
  - The affected endpoints don't have the `customFields.allowedFieldsList` configuration defined.

Restrictionsâ

- The limit of custom fields must be defined and greater than `0`.
- The name field must be between 1 and 30 characters.
- The name field allows only lowercase letters, numbers, underscore and dash.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/api-resources/generated-fields

Generated Fields

The Payments API generates fields that will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Payment	payment
Info	info
Settlement	settlement

Endpoint	event_type
Feedback	feedback
Reponses	responses

event_received_atâ

The `event_received_at` represents the time the request reached the Payments API as a timestamp, in milliseconds.

-
- API Resources

API Resources

This section provides you with the information you need to work with the API.

Endpoints

Learn all the operations you can do with Feedzai's API and how to communicate with its multiple endpoints.

Additional Resources

Find information about schema mapping, authentication and other necessary resources to successfully work with the API.

Responses

Successful responseâ

Successful calls to Feedzai's API result in a response with HTTP 200 code. Please check each endpoint's documentation to see the structure of a successful response.

Warning responseâ

A warning response is a subtype of a successful response which is sent with the HTTP 200 code. The transaction is also processed; however, the response returns the `status` field "warning" and the `warnings` field explaining the reasons for this outcome.

Error responseâ

When a call to Feedzai's API results in an error, the response object returned always has the same fields. In particular, the `status` field is shared with the successful response objects of all endpoints. The **Error Response** object is defined as follows:

Name	Field Type	Description
`code`	String	An indicative code for the error. See possible values below.
`errors`	Array of strings	A list of free text explanations for the error.
`status`	String	Always with value `error` .

Generic Error codes - Possible Valuesâ

Error code	Description	HTTP Code
`authenticationError`	The request could not be authenticated.	`401`
`authorizationError`	The authenticated user does not have permission to access the endpoint.	`403`
`parseError`	The request could not be parsed.	`400`
`invalidParameterError`	The URL parameter provided is invalid.	`400`
`invalidTimestampError`	The timestamp provided in the request is invalid.	`400`
`nonexistentEndpoint`	The endpoint being called does not exist.	`404`
`unsupportedMethod`	The HTTP method to call this endpoint is not supported.	`405`
`tooManyRequests`	The allowed request rate was exceeded.	`429`
`internalError`	Feedzai's system produced an internal error.	`500`
`timeoutError`	Feedzai's system failed to produce a timely response.	`504`

Please check each [endpoint's documentation]() for specific error codes.

Responses for timestamps out of rangeâ

This subsection describes responses for requests with timestamps that are out of the valid range.

Endpoint	Timestamp	Response HTTP code	Error or Warning	Is transaction processed?
Authorization	`event_occurred_at` in the past	400	Error	No
Authorization	`event_occurred_at` in the future	400	Error	No
Info, Clearing, Feedback	`event_occurred_at` in the past	200	Warning	Yes
Info, Clearing, Feedback	`event_occurred_at` in the future	400	Error	No

Error Response Example:

```
{
    "status": "error",
    "code": "invalidTimestampError",
    "errors": [
        "Event has timestamp in the future: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:41:30.774Z'"
```

```
    ]
}
```

Warning response example:

```
 {
    "lifecycle_id": <LIFECYCLE_ID>,
    "event_external_id": <EVENT_EXTERNAL_ID>,
    "warnings": [
        "Event has timestamp in the past: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:40:40.522Z'"
    ],
    "status": "warning"
}
```

Schema Mapping

This page contains comma-separated value (CSV) files that allow you to confirm where API fields are used. Among other parameters, these CSVs contain rules, metrics, and variables from Payments.

As a business analyst, you need to understand the impact of missing data fields on your client's data. As such, the following files allow you to:

- Validate the API fields required to run each rule (e.g. fields in a rule's condition and metrics).
- Confirm where an API field is being used in the whole solution (e.g. in which metrics, rule conditions, or variables).

RulesÂ Â API Schema

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/oob-resources/overview

- Out-of-the-Box Resources

Out-of-the-box Resources

Transaction Fraud for Payment Service Providers (PSPs) offers out-of-the-box resources, such as lists, schemas, and rules projects. These resources are included in the solution standard package to help PSPs detect and prevent crime.

This section contains the following reference documentation on those resources:

- Lists
- Schemas
- Rules projects
  - Feedback Rules
  - Refund and OCT Rules
  - PosNeg Rules
  - Self-Service Rules
  - Fraud Rules
  - APM Rules
  - Card Present Rules
  - By Merchant PosNeg Rules

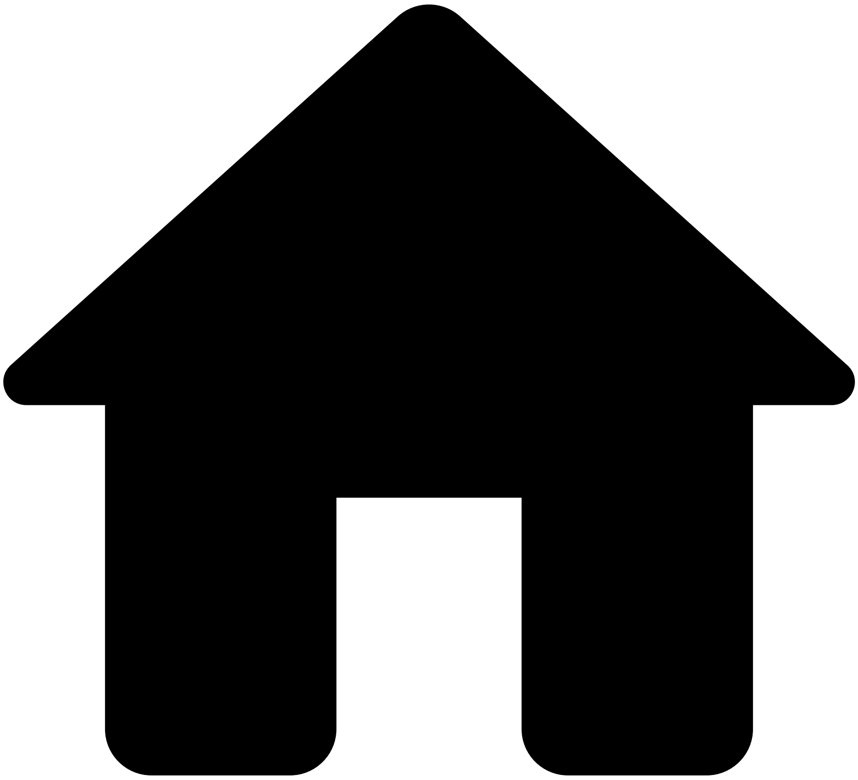
Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/oob-resources/tfp-lists

Lists

Name	Description
PosNeg Listed Countries	PosNeg List for country codes managed by the Landlord.
PosNeg Listed Cards	PosNeg List for cards by card_id managed by the Landlord.
PosNeg Terminal IP	PosNeg List for terminal IP by terminal_ip
aml_high_risk_countries	High-Risk Countries that must be monitored by AML rules
By Merchant Tenanted Listed PosNeg Shipping Address	List of PosNeg Listed Shipping Addresses per tenant that is managed by the Merchant.
By Merchant Tenanted PosNeg Listed Card BIN	List of PosNeg Listed Card BIN per tenant that is managed by the Merchant.
By Merchant Tenanted PosNeg Listed Card Country	List of PosNeg Listed Card Countries per tenant that is managed by the Merchant.
By Merchant Tenanted PosNeg Listed Cards	List of PosNeg Listed Cards per tenant that is managed by the Merchant

By Merchant Tenanted PosNeg Listed Customer Email	List of PosNeg Listed Customer Emails per tenant that is managed by the Merchant.
By Merchant Tenanted PosNeg Listed Device IP	List of PosNeg Listed Device IPs per tenant that is managed by the Merchant.
By Merchant Tenanted PosNeg Listed Devices	List of PosNeg Listed Device IDs per tenant that is managed by the Merchant
By Merchant Tenanted PosNeg Listed Terminals	List of PosNeg Listed Terminals per tenant that is managed by the Merchant
Card Testing MCC	List of MCCs used on carding/card testing fraud scenarios
Global Default - Rules Thresholds	Default thresholds used in rules regardless of the Merchant ID
Listed Merchants - International Payments	List os MIDs that are limited to accept international card payments
MCC Limits	List of daily amount limits per MCC and MID
Merchant ID BIN Attacks	Merchant IDs to be excluded from 'Bin Attack' Rule
Merchant Risk Levels - Rules Thresholds	Different thresholds per Merchant Risk Level
Tenanted PosNeg Listed Customer Email	List of PosNeg Listed Customer Emails per tenant that is managed by the Landlord
Tenanted Rules Thresholds	List of rules thresholds per tenant (Merchant ID) managed by the Landlord.
Tenanted PosNeg Listed Card Country	List of PosNeg Listed Card Countries per tenant that is managed by the Landlord
Tenanted PosNeg Listed Device IP	List of PosNeg Listed Device IPs per tenant that is managed by the Landlord
Tenanted PosNeg Listed Devices	List of PosNeg Listed Device IDs per tenant that is managed by the Landlord
Tenanted PosNeg Listed Terminals	List of PosNeg Listed Terminals per tenant that is managed by the Landlord
Tenanted Listed PosNeg Shipping Address	List of PosNeg Listed Shipping Addresses per tenant that is managed by the Landlord
Tenanted PosNeg Listed Card BIN	List of PosNeg Listed Card BIN per tenant that is managed by the Landlord
Tenanted PosNeg Listed Cards	List of PosNeg Listed Cards per tenant that is managed by the Landlord
PosNeg Listed Customer Email	PosNeg List for Customer Emails managed by the Landlord.
PosNeg Listed Device IP	PosNeg List for device IP by device_ip managed by the Landlord.
PosNeg Listed Ship Addr ZIP	PosNeg List for shipping address zip codes managed by the Landlord.
PosNeg Listed Card BIN	PosNeg List for Card BIN managed by the Landlord.
PosNeg Listed Card Country	PosNeg List for Card Country codes managed by the Landlord.
PosNeg Listed Devices	PosNeg List for Devices managed by the Landlord.
PosNeg Listed Merchants	PosNeg List for Merchant IDs managed by the Landlord.
PosNeg Listed Shipping Address	PosNeg List of Shipping Address managed by the Landlord.
PosNeg Listed Terminals	PosNeg List for Terminal IDs managed by the Landlord.
Restricted MCC per BIN	List of all restricted MCCs for a specific BIN
Dummy IPs	List of dummy IP addresses to be excluded from the Device IP rules
By Merchant Tenanted Country Thresholds	
Global Negative listed MCCs	List of Negative listed MCCs

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/oob-resources/tfp-schemas

- 

- Out-of-the-Box Resources
- Schemas

Schemas

Schemas

- Features Plan Schema
- Merchant Provisioning Workflow Schema (Empty)
- Payments API Schema
- Payments Extended API Schema
- Payments Workflow Schema
- Pre Omnichannel Metrics Plan Schema
- Reference Data Workflow Schema (Empty)
- TFP Scoring Model Schema

Features Plan Schema

Field Name	Field Type

fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING
domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING

payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING
payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING

device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING
device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING

merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN
payment_bank_account_number	STRING
payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING

shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING
terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING
dispute_end_date	LONG
dispute_refunded	STRING
payment_card_pan	STRING
card_mti	STRING
card_processing_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
cbk_reason_code	STRING
full_shipping_address	STRING
card_days_to_expire	LONG
norm_shipp_addr_line	STRING

posneg_card_value	STRING
posneg_device_ip_value	STRING
posneg_customer_email_value	STRING
posneg_mid_value	STRING
posneg_terminal_id_value	STRING
posneg_device_id_value	STRING
posneg_card_country_value	STRING
posneg_card_bin_value	STRING
posneg_full_ship_addr_value	STRING
posneg_norm_shipp_addr_line	STRING
small_amount_trn_value	DOUBLE
email_merch_sum_amt_7d	LONG
email_merch_cnt_7d	LONG
fail_email_merch_dist_card_7d	LONG
email_merch_sum_amt_1d	LONG
email_merch_cnt_1d	LONG
fail_email_dist_card_7d	LONG
cust_merch_sum_amt_7d	LONG
cust_merch_cnt_7d	LONG
cust_merch_sum_amt_1d	LONG
cust_merch_cnt_1d	LONG
devip_merch_cnt_1d	LONG
shipaddr_merch_cnt_7d	LONG
shipaddr_merch_cnt_1d	LONG
merch_bill_email_dist_card_1d	LONG
merch_device_dist_card_1d	LONG
merch_cardbin_cntDist_cardExpD	LONG
merch_cardbin_cntDist_card_10m	LONG
card_merch_sum_amt_1d	LONG
fail_dist_card_merch_1min	LONG
merch_small_amt_cnt_15min	LONG
appr_card_merch_amtstr_cnt_1d	LONG
card_merch_dist_email_1d	LONG
card_merch_cnt_1d	LONG
fail_card_cnt_1d	LONG
cnp_card_dist_merch_5min	LONG
apm_merch_cust_email_sum_24h	LONG
apm_merch_cust_email_cnt_24h	LONG
merch_cardbin_cur7	BOOLEAN
aml_high_risk_coutries	BOOLEAN

norm_payment_card_brand	STRING
month_of_year_string	STRING
prev_month_of_year_string	STRING
us_domestic_3ds_payments	BOOLEAN
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
is_dummy_device_ip	BOOLEAN
is_dummy_terminal_ip	BOOLEAN
is_dummy_merchant_ip	BOOLEAN
is_dummy_merchant_registration_ip	BOOLEAN
fail_ip_dist_card_1d	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
dayOfMonth_event_occurred_at	INT
dayOfWeek_event_occurred_at	INT
emailName_billing_email	STRING
norm_billing_email	STRING
hour_event_occurred_at	INT
log_amount_unified_currency	DOUBLE
month_event_occurred_at	INT
cos_dayOfWeek_event_occurred_at	DOUBLE
cos_hour_event_occurred_at	DOUBLE
merchant_id_avg_amount_unified_currency_1_hour	DOUBLE
merchant_id_avg_amount_unified_currency_1_day	DOUBLE
merchant_id_avg_log_amount_unified_currency_1_hour	DOUBLE
merchant_id_avg_log_amount_unified_currency_1_day	DOUBLE
merchant_id_countDistinct_customer_id_1_hour	LONG
merchant_id_countDistinct_merchant_mcc_1_hour	LONG
merchant_id_countDistinct_payment_card_id_1_hour	LONG
merchant_id_countDistinct_payment_card_bin_1_hour	LONG
merchant_id_countDistinct_billing_email_1_hour	LONG
merchant_id_countDistinct_customer_id_1_day	LONG
merchant_id_countDistinct_merchant_mcc_1_day	LONG
merchant_id_countDistinct_payment_card_id_1_day	LONG
merchant_id_countDistinct_payment_card_bin_1_day	LONG
merchant_id_countDistinct_billing_email_1_day	LONG
merchant_id_count_1_hour	LONG

merchant_id_count_1_day	LONG
merchant_id_max_amount_unified_currency_1_hour	LONG
merchant_id_max_amount_unified_currency_1_day	LONG
merchant_id_stdDev_amount_unified_currency_1_hour	DOUBLE
merchant_id_stdDev_amount_unified_currency_1_day	DOUBLE
merchant_id_stdDev_log_amount_unified_currency_1_hour	DOUBLE
merchant_id_stdDev_log_amount_unified_currency_1_day	DOUBLE
merchant_id_sum_amount_unified_currency_1_hour	LONG
merchant_id_sum_amount_unified_currency_1_day	LONG
merchant_id_timeSincePrev_event_occurred_at_1_hour	LONG
merchant_id_timeSincePrev_event_occurred_at_1_day	LONG
merchant_id_vel_1_hour	DOUBLE
merchant_id_vel_1_day	DOUBLE
merchant_id_avg_amount_unified_currency_1_month	DOUBLE
merchant_id_avg_log_amount_unified_currency_1_month	DOUBLE
merchant_id_count_1_month	LONG
merchant_id_max_amount_unified_currency_1_month	LONG
merchant_id_stdDev_amount_unified_currency_1_month	DOUBLE
merchant_id_stdDev_log_amount_unified_currency_1_month	DOUBLE
merchant_id_sum_amount_unified_currency_1_month	LONG
merchant_id_timeSincePrev_event_occurred_at_1_month	LONG
merchant_id_vel_1_month	DOUBLE
probability_log_amount_unified_currency_merchant_id_avg_log_amount_unified_currency_1_month	DOUBLE
probability_log_amount_unified_currency_merchant_id_avg_log_amount_unified_currency_1_hour	DOUBLE
probability_log_amount_unified_currency_merchant_id_avg_log_amount_unified_currency_1_day	DOUBLE
ratio_merchant_id_avg_amount_unified_currency_1_day_merchant_id_avg_amount_unified_currency_1_month	DOUBLE
ratio_merchant_id_avg_amount_unified_currency_1_hour_merchant_id_avg_amount_unified_currency_1_month	DOUBLE
ratio_merchant_id_avg_amount_unified_currency_1_hour_merchant_id_avg_amount_unified_currency_1_day	DOUBLE
ratio_merchant_id_countDistinct_merchant_mcc_1_hour_merchant_id_countDistinct_merchant_mcc_1_day	DOUBLE
ratio_merchant_id_countDistinct_billing_email_1_hour_merchant_id_countDistinct_billing_email_1_day	DOUBLE
ratio_merchant_id_countDistinct_customer_id_1_hour_merchant_id_countDistinct_customer_id_1_day	DOUBLE
ratio_merchant_id_count_1_day_merchant_id_count_1_month	DOUBLE
ratio_merchant_id_countDistinct_payment_card_bin_1_hour_merchant_id_countDistinct_payment_card_bin_1_day	DOUBLE
ratio_merchant_id_count_1_hour_merchant_id_count_1_day	DOUBLE
ratio_merchant_id_count_1_hour_merchant_id_count_1_month	DOUBLE
ratio_merchant_id_countDistinct_payment_card_id_1_hour_merchant_id_countDistinct_payment_card_id_1_day	DOUBLE
ratio_merchant_id_max_amount_unified_currency_1_hour_merchant_id_max_amount_unified_currency_1_day	DOUBLE
ratio_merchant_id_max_amount_unified_currency_1_hour_merchant_id_max_amount_unified_currency_1_month	DOUBLE
ratio_merchant_id_max_amount_unified_currency_1_day_merchant_id_max_amount_unified_currency_1_month	DOUBLE
ratio_merchant_id_sum_amount_unified_currency_1_hour_merchant_id_sum_amount_unified_currency_1_day	DOUBLE

ratio_merchant_id_sum_amount_unified_currency_1_hour_merchant_id_sum_amount_unified_currency_1_month	DOUBLE
ratio_merchant_id_sum_amount_unified_currency_1_day_merchant_id_sum_amount_unified_currency_1_month	DOUBLE
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
is_scoring	BOOLEAN
oct_rfd_count_card_mid_24h	LONG
oct_rfd_sum_amt_card_mid_24h	LONG
oct_rfd_count_card_mid_7d	LONG
oct_rfd_sum_amount_card_mid_7d	LONG
oct_rfd_sum_amt_card_mid_30d	LONG
oct_rfd_count_card_mid_30d	LONG
oct_rfd_count_mid_24h	LONG
oct_rfd_count_mid_7d	LONG
oct_rfd_count_mid_30d	LONG
oct_rfd_sum_amount_mid_24h	LONG
oct_rfd_sum_amount_mid_7d	LONG
oct_rfd_sum_amount_mid_30d	LONG
oct_rfd_count_card_24h	LONG
oct_rfd_count_card_7d	LONG
oct_rfd_count_card_30d	LONG
oct_rfd_sum_amount_card_24h	LONG
oct_rfd_sum_amount_card_7d	LONG
oct_rfd_sum_amount_card_30d	LONG
oct_rfd_count_bin_mid_24h	LONG
oct_rfd_count_bin_mid_7d	LONG
oct_rfd_count_bin_mid_30d	LONG
oct_rfd_amount_bin_mid_24h	LONG
oct_rfd_amount_bin_mid_7d	LONG
oct_rfd_amount_bin_mid_30d	LONG

oct_rfd_count_card_country_7d	LONG
oct_rfd_amount_card_country_7d	LONG
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
payment_method_subtype	STRING
payment_account_reference	STRING
refund_status	STRING
refund_type	STRING
external_refund_id	STRING
region_id	STRING

Merchant Provisioning Workflow Schema (Empty)

Field Name	Field Type

Payments API Schema

Field Name	Field Type
event_external_id	STRING
lifecycle_id	STRING
domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING

payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_pan	STRING
payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING
payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING

customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING
device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING

billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG

merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
cross_border_merchant	BOOLEAN
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING
terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN

feedback_label	STRING
dispute_end_date	LONG
dispute_refunded	STRING
card_mti	STRING
card_processing_code	STRING
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
payment_bank_account_number	STRING
payment_bank_identifier	STRING
cbk_reason_code	STRING
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG

is_scoring	BOOLEAN
payment_method_subtype	STRING
payment_account_reference	STRING
refund_status	STRING
refund_type	STRING
external_refund_id	STRING
region_id	STRING

Payments Extended API Schema

Field Name	Field Type
fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING
domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_country_code	STRING

payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING
payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING

customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING
device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING

merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN
payment_bank_account_number	STRING

payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING
terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING
dispute_end_date	LONG

dispute_refunded	STRING
card_mti	STRING
card_processing_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
cbk_reason_code	STRING
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
payment_card_pan	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
is_scoring	BOOLEAN
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
payment_method_subtype	STRING
payment_account_reference	STRING
refund_status	STRING
refund_type	STRING
external_refund_id	STRING
region_id	STRING

Payments Workflow Schema

Field Name	Field Type
fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING
domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING

arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING
payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING

device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING
device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN

merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN
merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN
payment_bank_account_number	STRING
payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING

shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING
terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING
dispute_end_date	LONG
dispute_refunded	STRING
payment_card_pan	STRING
card_mti	STRING
card_processing_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
cbk_reason_code	STRING

full_shipping_address	STRING
card_days_to_expire	LONG
norm_shipp_addr_line	STRING
posneg_card_value	STRING
posneg_device_ip_value	STRING
posneg_customer_email_value	STRING
posneg_mid_value	STRING
posneg_terminal_id_value	STRING
posneg_device_id_value	STRING
posneg_card_country_value	STRING
posneg_card_bin_value	STRING
posneg_full_ship_addr_value	STRING
posneg_norm_shipp_addr_line	STRING
small_amount_trn_value	DOUBLE
email_merch_sum_amt_7d	LONG
email_merch_cnt_7d	LONG
fail_email_merch_dist_card_7d	LONG
email_merch_sum_amt_1d	LONG
email_merch_cnt_1d	LONG
fail_email_dist_card_7d	LONG
cust_merch_sum_amt_7d	LONG
cust_merch_cnt_7d	LONG
cust_merch_sum_amt_1d	LONG
cust_merch_cnt_1d	LONG
devip_merch_cnt_1d	LONG
shipaddr_merch_cnt_7d	LONG
shipaddr_merch_cnt_1d	LONG
merch_bill_email_dist_card_1d	LONG
merch_device_dist_card_1d	LONG
merch_cardbin_cntDist_cardExpD	LONG
merch_cardbin_cntDist_card_10m	LONG
card_merch_sum_amt_1d	LONG
fail_dist_card_merch_1min	LONG
merch_small_amt_cnt_15min	LONG
appr_card_merch_amtstr_cnt_1d	LONG
card_merch_dist_email_1d	LONG
card_merch_cnt_1d	LONG
fail_card_cnt_1d	LONG
cnp_card_dist_merch_5min	LONG
apm_merch_cust_email_sum_24h	LONG

apm_merch_cust_email_cnt_24h	LONG
merch_cardbin_cur7	BOOLEAN
aml_high_risk_coutries	BOOLEAN
norm_payment_card_brand	STRING
month_of_year_string	STRING
prev_month_of_year_string	STRING
us_domestic_3ds_payments	BOOLEAN
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING
merchant_declared_annual_revenue	LONG
is_dummy_device_ip	BOOLEAN
is_dummy_terminal_ip	BOOLEAN
is_dummy_merchant_ip	BOOLEAN
is_dummy_merchant_registration_ip	BOOLEAN
fail_ip_dist_card_1d	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
is_scoring	BOOLEAN
oct_rfd_count_card_mid_24h	LONG
oct_rfd_sum_amt_card_mid_24h	LONG
oct_rfd_count_card_mid_7d	LONG
oct_rfd_sum_amount_card_mid_7d	LONG
oct_rfd_sum_amt_card_mid_30d	LONG
oct_rfd_count_card_mid_30d	LONG
oct_rfd_count_mid_24h	LONG

oct_rfd_count_mid_7d	LONG
oct_rfd_count_mid_30d	LONG
oct_rfd_sum_amount_mid_24h	LONG
oct_rfd_sum_amount_mid_7d	LONG
oct_rfd_sum_amount_mid_30d	LONG
oct_rfd_count_card_24h	LONG
oct_rfd_count_card_7d	LONG
oct_rfd_count_card_30d	LONG
oct_rfd_sum_amount_card_24h	LONG
oct_rfd_sum_amount_card_7d	LONG
oct_rfd_sum_amount_card_30d	LONG
oct_rfd_count_bin_mid_24h	LONG
oct_rfd_count_bin_mid_7d	LONG
oct_rfd_count_bin_mid_30d	LONG
oct_rfd_amount_bin_mid_24h	LONG
oct_rfd_amount_bin_mid_7d	LONG
oct_rfd_amount_bin_mid_30d	LONG
oct_rfd_count_card_country_7d	LONG
oct_rfd_amount_card_country_7d	LONG
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
payment_method_subtype	STRING
payment_account_reference	STRING
refund_status	STRING
refund_type	STRING
external_refund_id	STRING
region_id	STRING

Pre Omnichannel Metrics Plan Schema

Field Name	Field Type
fraud	BOOLEAN
fraud_score	DOUBLE
decision	STRING
event_external_id	STRING
lifecycle_id	STRING

domain	STRING
event_occurred_at	LONG
event_received_at	LONG
event_type	STRING
transaction_id	STRING
transaction_created_at	LONG
channel	STRING
authorization_type	STRING
capture_mode	STRING
amount	LONG
currency_code	STRING
amount_unified_currency	LONG
amount_conv_rate	DOUBLE
cross_border_transaction	BOOLEAN
payment_method_type	STRING
payment_method_id	STRING
payment_apm_token	STRING
payment_card_name	STRING
payment_card_bin	STRING
payment_card_last4	STRING
payment_card_token	STRING
payment_card_expiration	STRING
payment_card_id	STRING
payment_card_country_code	STRING
payment_card_country_risky	BOOLEAN
payment_card_brand	STRING
payment_card_type	STRING
payment_card_category	STRING
payment_card_tier	STRING
payment_card_issuing_org_name	STRING
payment_card_issuing_org_site	STRING
payment_card_issuing_org_phone	STRING
arn	STRING
payment_auth_status	STRING
payment_auth_status_code	STRING
payment_auth_status_scheme	STRING
payment_cvv_status	STRING
payment_cvv_status_code	STRING
payment_cvv_status_scheme	STRING
payment_avs_status	STRING

payment_avs_status_code	STRING
payment_avs_status_scheme	STRING
payment_3ds_status	STRING
payment_3ds_status_code	STRING
payment_3ds_status_scheme	STRING
eci	STRING
acquirer_id	STRING
acquirer_country	STRING
installments_no	STRING
dispute_date	LONG
dispute_reason	STRING
dispute_reason_code	STRING
dispute_status	STRING
dispute_status_code	STRING
customer_id	STRING
customer_name	STRING
customer_email	STRING
customer_email_domain	STRING
customer_email_free	BOOLEAN
customer_email_disposable	BOOLEAN
customer_phone	STRING
customer_addr_line1	STRING
customer_addr_line2	STRING
customer_zip	STRING
customer_city	STRING
customer_region	STRING
customer_country_code	STRING
customer_country_risky	BOOLEAN
customer_dob	STRING
customer_created_at	LONG
customer_tier	STRING
device_ip	STRING
device_ip_latitude	DOUBLE
device_ip_longitude	DOUBLE
device_ip_addr_line1	STRING
device_ip_addr_line2	STRING
device_ip_zip	STRING
device_ip_city	STRING
device_ip_region	STRING
device_ip_country_code	STRING

device_ip_country_risky	BOOLEAN
device_ip_scanning_host	BOOLEAN
device_ip_malicious_host	BOOLEAN
device_ip_command_control	BOOLEAN
device_ip_proxy	BOOLEAN
device_ip_tor_node	BOOLEAN
device_id	STRING
device_info_session_id	STRING
billing_name	STRING
billing_email	STRING
billing_email_domain	STRING
billing_email_free	BOOLEAN
billing_email_disposable	BOOLEAN
billing_phone	STRING
billing_addr_line1	STRING
billing_addr_line2	STRING
billing_zip	STRING
billing_city	STRING
billing_region	STRING
billing_country_code	STRING
billing_country_risky	BOOLEAN
billing_descriptor	STRING
merchant_id	STRING
merchant_domain	STRING
merchant_name	STRING
merchant_account_beneficiary_name	STRING
merchant_url	STRING
merchant_ip	STRING
merchant_email	STRING
merchant_email_domain	STRING
merchant_email_free	BOOLEAN
merchant_email_disposable	BOOLEAN
merchant_phone	STRING
merchant_addr_line1	STRING
merchant_addr_line2	STRING
merchant_zip	STRING
merchant_city	STRING
merchant_region	STRING
merchant_country_code	STRING
merchant_country_risky	BOOLEAN

merchant_mcc	STRING
merchant_created_at	LONG
merchant_payout_bank_account	STRING
merchant_payout_currency	STRING
merchant_payout_frequency	STRING
merchant_payout_bank_country_code	STRING
merchant_payout_bank_account_issuing_bank_name	STRING
merchant_payout_bank_account_beneficiary_name	STRING
merchant_payout_bank_account_date_added	LONG
merchant_go_live	LONG
merchant_registration_ip	STRING
merchant_registration_ip_country_code	STRING
merchant_timezone	STRING
merchant_type	STRING
merchant_max_ticket_amount	LONG
merchant_max_monthly_volume	LONG
merchant_max_monthly_volume_unified_currency	LONG
merchant_add	DOUBLE
merchant_ubo_names	STRING
merchant_tax_id	STRING
merchant_account_status	STRING
merchant_account_code	STRING
cross_border_merchant	BOOLEAN
payment_bank_account_number	STRING
payment_bank_identifier	STRING
items	LIST
shipping_type	STRING
shipping_name	STRING
shipping_phone	STRING
shipping_addr_line1	STRING
shipping_addr_line2	STRING
shipping_zip	STRING
shipping_city	STRING
shipping_region	STRING
shipping_country_code	STRING
shipping_country_risky	BOOLEAN
terminal_id	STRING
card_entry_mode	STRING
terminal_latitude	STRING
terminal_longitude	STRING

terminal_ip	STRING
terminal_ip_latitude	DOUBLE
terminal_ip_longitude	DOUBLE
terminal_ip_addr_line1	STRING
terminal_ip_addr_line2	STRING
terminal_ip_zip	STRING
terminal_ip_city	STRING
terminal_ip_region	STRING
terminal_ip_country_code	STRING
terminal_ip_country_risky	BOOLEAN
terminal_ip_scanning_host	BOOLEAN
terminal_ip_malicious_host	BOOLEAN
terminal_ip_command_control	BOOLEAN
terminal_ip_proxy	BOOLEAN
terminal_ip_tor_node	BOOLEAN
fallback_flag	BOOLEAN
card_presence_flag	BOOLEAN
authentication_mode	STRING
pos_read_cap	STRING
atm_country_code	STRING
feedback_type	STRING
feedback_value	BOOLEAN
feedback_label	STRING
dispute_end_date	LONG
dispute_refunded	STRING
payment_card_pan	STRING
card_mti	STRING
card_processing_code	STRING
merchant_registration_longitude	DOUBLE
merchant_registration_latitude	DOUBLE
merchant_fiscal_number	STRING
cbk_reason_code	STRING
aml_high_risk_coutries	BOOLEAN
norm_payment_card_brand	STRING
month_of_year_string	STRING
prev_month_of_year_string	STRING
us_domestic_3ds_payments	BOOLEAN
alert_id	STRING
info_type	STRING
merchant_risk_rating	STRING

merchant_declared_annual_revenue	LONG
merchant_declared_atv	DOUBLE
merchant_parent_merchant_id	STRING
merchant_cryptocurrencyindicator	STRING
customer_bank_account	STRING
customer_bank_id	STRING
customer_bank_country_code	STRING
merchant_branch_id	STRING
custom_fields	MAP
sign_count	LONG
sign_amount	LONG
payment_auth_status_desc	STRING
payment_reason	STRING
frictionless	BOOLEAN
liability_shift	BOOLEAN
request_exemption	BOOLEAN
recurring_type	STRING
is_scoring	BOOLEAN
customer_ssn	STRING
merchant_account_name	STRING
merchant_terminated_reason	STRING
merchant_statement_descriptor	STRING
merchant_mid_status	STRING
merchant_legal_entity_created_at	LONG
payment_method_subtype	STRING
payment_account_reference	STRING
refund_status	STRING
refund_type	STRING
external_refund_id	STRING
region_id	STRING

Reference Data Workflow Schema (Empty)

Field Name	Field Type

TFP Scoring Model Schema

Field Name	Field Type
fraud	STRING

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/overview

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Operational Work

Data Scientist & Fraud Strategist

Learn how to fight fraud with rules and models in Pulse.

Fraud Prevention Agent

Learn how to work with alerts and/or cases using Case Manager.

Fraud Prevention Manager

View real-time insights on agent performance and rule monitoring.

Integration Engineer

Learn how to test rules, and ensure the system is properly tested / running properly.

-

- Out-of-the-Box Resources
- Rules Projects
- Ad-Hoc Investigations Rules

On this page

Ad-Hoc Investigations Rules

Descriptionâ

The Ad-Hoc Investigations rules project includes rules that use information from the ad-hoc investigations' channel, e.g. alerts a merchant if there is at least one ad-hoc investigation marked as fraud over the course of 30 days.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

AHIR1

Comment

Actions

- Alert
- [AHIR1]

Variables

N/A

Condition

```
event.cnt_risky_ahi_30d >= 1
```

Explanation

> Merchant ({event.merchant_id}) has recorded {event.cnt_risky_ahi_30d} investigations marked as 'Risky' in the last 30 days. Please review the Investigations channel for further details.

Mark As

N/A

-
- [Out-of-the-Box Resources]
- Rules Projects
- Bankruptcy Risk Rules

On this page

Bankruptcy Risk Rules

Descriptionâ

Bankruptcy risk rules account for merchants who are at risk of or are concealing their inability to ensure payment of their debt.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Monthly-Amount-Decrease-BR1

Comment

The decrease in monthly total transaction amount is equal to or below a set threshold

Actions

- Alert
- [Monthly-Amount-Decrease-BR1]

Variables

```
 threshold = multi_tenanted_threshold_list["amount_decrease_1m"].threshold ??
default_threshold_list["amount_decrease_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_amt_lastm_to_prevm * 100;
```

Condition

```
 // Filters: (event_type == "info") and (payment_auth_status == "accepted")
// Total payment amount last month over the previous month (delayed over 4w)
event.rt_amt_lastm_to_prevm > 0 and event.rt_amt_lastm_to_prevm <= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) decrease in total transaction amount: last month is {localvar.formated_ratio} % of the previous month, which is equal/below the limit of{localvar.formated_threshold}%.

Mark As

N/A

Monthly-Count-Decrease-BR2

Comment

The decrease in monthly total transaction count is equal to or below a set threshold

Actions

- Alert
- [Monthly-Count-Decrease-BR2]

Variables

```
 threshold = multi_tenanted_threshold_list["count_decrease_1m"].threshold ??
default_threshold_list["count_decrease_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
```

```
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_cnt_lastm_to_prevm * 100;
```

Condition

```
 // (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
// Total payment count last month over the previous month (delayed over 4w)
event.rt_cnt_lastm_to_prevm > 0 and event.rt_cnt_lastm_to_prevm <= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) decrease in total transaction count: last month is {localvar.formated_ratio} % of the previous month, which is equal/below the limit of{localvar.formated_threshold}%).

Mark As

N/A

BiMonthly-Amt-Decrease-BR3

Comment

The decrease in bi-monthly total transaction amount is equal to or below a set threshold

Actions

- [BiMonthly-Amt-Decrease-BR3]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["amount_decrease_2m"].threshold ??
default_threshold_list["amount_decrease_2m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_amt_last2m_to_prev2m * 100;
```

Condition

```
 // (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
// Total payment amount last 2 month over the previous 2 month (delayed over 8w)
event.rt_amt_last2m_to_prev2m > 0 and event.rt_amt_last2m_to_prev2m <= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) decrease in total transaction amount: last 2 months are {localvar.formated_ratio} % of the previous 2 months, which is equal/below the limit of {localvar.formated_threshold}%).

Mark As

N/A

BiMonthly-Count-Decrease-BR4

Comment

The decrease in bi-monthly total transaction count is equal to or below a set threshold

Actions

- Alert
- [BiMonthly-Count-Decrease-BR4]

Variables

```
 threshold = multi_tenanted_threshold_list["count_decrease_2m"].threshold ??
default_threshold_list["count_decrease_2m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_cnt_last2m_to_prev2m * 100;
```

Condition

```
 // (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
// Total payment count last 2 months over the previous 2 months (delayed over 8w)
event.rt_cnt_last2m_to_prev2m > 0 and event.rt_cnt_last2m_to_prev2m <= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) decrease in total transaction amount: last 2 months are {localvar.formated_ratio} % of the previous 2 months, which is equal/below the limit of {localvar.formated_threshold}%).

Mark As

N/A

Monthly-Refund-Amount-BR5

Comment

Refund amount rate, in 1 month, is equal to or above a set threshold

Actions

- [Monthly-Refund-Amount-BR5]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["refund_amount_limit_1m"].threshold ??
default_threshold_list["refund_amount_limit_1m"].threshold;
```

```
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_rfnd_amt_4w * 100;
```

Condition

```
 // (Filters = (event_type == "info") and (info_type == "refund"))
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
// Total refund amount over the total payment amount last month
event.rt_rfnd_amt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) refund rate of one month is {localvar.formated_ratio} %, which is equal/ below the limit of {localvar.formated_threshold}%).

Mark As

N/A

Monthly-Refund-Count-BR6

Comment

Refund count rate, in 1 month, is equal to or above a set threshold

Actions

- [Monthly-Refund-Count-BR6]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["refund_count_limit_1m"].threshold ??
default_threshold_list["refund_count_limit_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_rfnd_cnt_4w * 100;
```

Condition

```
 // Total refund count over the total payment count last month
// (Filters = (event_type == "info") and (info_type == "refund"))
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.rt_rfnd_cnt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) refund count rate of one month is {localvar.formated_ratio} %, which is equal/below the limit of {localvar.formated_threshold}%).

Mark As

N/A

Refund-Amount-Increase-BR7

Comment

Refund amount rate of 1 month over 3 months is equal or above a set threshold

Actions

- Alert
- [Refund-Amount-Increase-BR7]

Variables

```
 threshold = multi_tenanted_threshold_list["refund_amount_1m_3m_list"].threshold ??
default_threshold_list["refund_amount_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Refund rate of one month over the refund rate of three months (delayed over 4w)
// (Filters = (event_type == "info") and (info_type == "refund"))
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.rtcomp_rfnd_amt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) refund amount rate of one month is event.rtcomp_rfnd_amt_1m_to_3m x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Refund-Count-Increase-BR8

Comment

Refund count rate of 1 month over 3 months is equal or above a set threshold

Actions

- [Refund-Count-Increase-BR8]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["refund_count_1m_3m_list"].threshold ??
default_threshold_list["refund_count_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Refund count rate of one month over the refund rate of three months (delayed over 4w)
// (Filters = (event_type == "info") and (info_type == "refund"))
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.rtcomp_rfnd_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

> Merchant ({event.merchant_id}) refund count rate of one month is {event.rtcomp_rfnd_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

-
- Out-of-the-Box Resources
- Rules Projects
- Chargeback Rules

On this page

Chargeback Rules

Descriptionâ

Chargeback rules account for merchants with abnormal customer transaction activity, which may lead to higher volume and risk of reported transaction chargebacks.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Visa-VDMP-Early-Warning-CB1

Comment

Visa Dispute Monitoring Program - Early Warning (VDMP) - Chargeback to sales ratio, in one month.

Actions

- [Visa-VDMP-Early-Warning-CB1]
- Alert

Variables

```
 min_vdmp_early_rat_threshold = default_threshold_list["min_vdmp_early_threshold"].threshold; //
0.65%
max_vdmp_early_rat_threshold = default_threshold_list["max_vdmp_early_threshold"].threshold; //
0.9%
min_vdmp_early_cnt_threshold =
default_threshold_list["min_vdmp_early_cnt_threshold"].threshold; // 75
max_vdmp_early_cnt_threshold =
default_threshold_list["max_vdmp_early_cnt_threshold"].threshold; // 100
//For Rule Explanation
visa_cbk_to_sales_ratio = event.rt_visa_cbk_cnt_cur_m*100;
```

Condition

```
 //Visa ratio of disputed payments to all payments
// (Filter: feedback_type == "chargeback")
// (Filter: event_type == "info" and payment_auth_status == 'accepted')
(event.rt_visa_cbk_cnt_cur_m >= min_vdmp_early_rat_threshold
 and event.rt_visa_cbk_cnt_cur_m < max_vdmp_early_rat_threshold)
and
// The total number of Visa payments that have been disputed
(event.visa_cnt_cur_m >= min_vdmp_early_cnt_threshold
 and event.visa_cnt_cur_m < max_vdmp_early_rat_threshold)
```

Explanation

Visa Dispute Monitoring Program - Early Warning (VDMP) - The chargeback to sales ratio, in one month, is {localvar.visa_cbk_to_sales_ratio} and the number of CBKs is {event.visa_cbk_cnt_cur_m}

Mark As

N/A

Visa-VDMP-Standard-CB2

Comment

Visa Dispute Monitoring Program - Standard (VDMP) - Chargeback to sales ratio, in one month.

Actions

- [Visa-VDMP-Standard-CB2]
- Alert

Variables

```
 min_vdmp_standard_rat_threshold =
default_threshold_list["min_vdmp_standard_rat_threshold"].threshold; // 0.9%
max_vdmp_standard_rat_threshold =
default_threshold_list["max_vdmp_standard_rat_threshold"].threshold; // 1.8%
min_vdmp_standard_cnt_threshold =
default_threshold_list["min_vdmp_standard_cnt_threshold"].threshold; // 100
max_vdmp_standard_cnt_threshold =
default_threshold_list["max_vdmp_standard_cnt_threshold"].threshold; // 1000
//For Rule Explanation
visa_cbk_to_sales_ratio = event.rt_visa_cbk_cnt_cur_m*100;
```

Condition

```
 //Visa ratio of disputed payments to all payments
// (Filter: feedback_type == "chargeback")
// (Filter: event_type == "info" and payment_auth_status == 'accepted')
(event.rt_visa_cbk_cnt_cur_m >= min_vdmp_standard_rat_threshold
 and event.rt_visa_cbk_cnt_cur_m < max_vdmp_standard_rat_threshold)
and
// The total number of Visa payments that have been disputed
(event.visa_cnt_cur_m >= min_vdmp_standard_cnt_threshold
 and event.visa_cnt_cur_m < max_vdmp_standard_cnt_threshold)
```

Explanation

Visa Dispute Monitoring Program - Standard (VDMP) - The chargeback to sales ratio, in one month, is {localvar.visa_cbk_to_sales_ratio} and the number of CBKs is {event.visa_cnt_cur_m}

Mark As

N/A

Visa-VDMP-Excessive-CB3

Comment

Visa Dispute Monitoring Program - Excessive (VDMP) - Chargeback to sales ratio, in one month.

Actions

- [Visa-VDMP-Excessive-CB3]
- Alert

Variables

```
 min_vdmp_excessive_threshold =
default_threshold_list["min_vdmp_excessive_threshold"].threshold; // 1.8%
min_vdmp_excessive_cnt_threshold =
default_threshold_list["min_vdmp_excessive_cnt_threshold"].threshold; // 1000
//For Rule Explanation
visa_cbk_to_sales_ratio = event.rt_visa_cbk_cnt_cur_m*100;
```

Condition

```
 //Visa ratio of disputed payments to all payments
// (Filter: feedback_type == "chargeback")
// (Filter: event_type == "info" and payment_auth_status == 'accepted')
event.rt_visa_cbk_cnt_cur_m >= min_vdmp_excessive_threshold
and
```

```
// The total number of Visa payments that have been disputed
event.visa_cbk_cnt_cur_m >= min_vdmp_excessive_cnt_threshold
```

Explanation

Visa Dispute Monitoring Program - Excessive (VDMP) - The chargeback to sales ratio, in one month, is {localvar.visa_cbk_to_sales_ratio} and the number of CBKs is {event.visa_cnt_cur_m}

Mark As

N/A

Mastercard-ECM-CB4

Comment

Mastercard Excessive Chargeback Program (ECM) - Chargeback to sales ratio, in one month.

Actions

- [Mastercard-ECM-CB4]
- Alert

Variables

```
 min_mc_ecm_rat_threshold = default_threshold_list["min_mc_ecm_rat_threshold"].threshold; // 1.5%
max_mc_ecm_rat_threshold = default_threshold_list["max_mc_ecm_rat_threshold"].threshold; // 2.99%
min_mc_ecm_cnt_threshold = default_threshold_list["min_mc_ecm_cnt_threshold"].threshold; // 100
max_mc_ecm_cnt_threshold = default_threshold_list["max_mc_ecm_cnt_threshold"].threshold; // 299
//For Rule Explanation
mc_cbk_to_sales_ratio = event.rt_mc_cbk_cnt_1m*100;
```

Condition

```
 //Mastercard ratio of disputed payments of the current month to all payments processed in the
previous month
// (Filter: feedback_type == "chargeback")
// (Filter: event_type == "info" and payment_auth_status == 'accepted')
(event.rt_mc_cbk_cnt_1m >= min_mc_ecm_rat_threshold
 and event.rt_mc_cbk_cnt_1m < max_mc_ecm_rat_threshold)
and
// The total number of Mastercard payments that have been disputed in the current month
(event.mc_cbk_cnt_cur_m >= min_mc_ecm_cnt_threshold
 and event.mc_cbk_cnt_cur_m < max_mc_ecm_cnt_threshold)
```

Explanation

Mastercard Excessive Chargeback Program (ECM) - The chargeback to sales ratio, in one month, is {localvar.mc_cbk_to_sales_ratio} and the number of CBKs is {event.mc_cbk_cnt_cur_m}

Mark As

N/A

Mastercard-HECM-CB5

Comment

Mastercard High Excessive Chargeback Program (HECM) - Chargeback to sales ratio, in one month.

Actions

- [Mastercard-HECM-CB5]
- Alert

Variables

```
min_mc_hecm_rat_threshold = default_threshold_list["min_mc_hecm_rat_threshold"].threshold; // 3%
min_mc_hecm_cnt_threshold = default_threshold_list["min_mc_hecm_cnt_threshold"].threshold; // 300
//For Rule Explanation
mc_cbk_to_sales_ratio = event.rt_mc_cbk_cnt_1m*100;
```

Condition

```
//Mastercard ratio of disputed payments of the current month to all payments processed in the
previous month
// (Filter: feedback_type == "chargeback")
// (Filter: event_type == "info" and payment_auth_status == 'accepted')
event.rt_mc_cbk_cnt_1m >= min_mc_hecm_rat_threshold
and
// The total number of Mastercard payments that have been disputed in the current month
event.mc_cbk_cnt_cur_m >= min_mc_hecm_cnt_threshold
```

Explanation

Mastercard High Excessive Chargeback Program (HECM) - The chargeback to sales ratio, in one month, is {localvar.mc_cbk_to_sales_ratio} and the number of CBKs is {event.mc_cbk_cnt_cur_m}

Mark As

N/A

Chargeback-Count-Increase-CB6

Comment

Chargeback count rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Chargeback-Cnt-Increase-CB6]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["chargeback_count_1m_3m_list"].threshold ??
default_threshold_list["chargeback_count_1m_3m_list"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
// Refund count rate of one month (delayed over 4w) over the refund rate of three months (delayed
over 8w)
// (Filter: feedback_type == "chargeback")
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.rtcomp_cbk_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Chargeback count rate of one month is event.rtcomp_cbk_cnt_1m_to_3m x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Chargeback-Amount-Increase-CB7

Comment

Chargeback amount rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Chargeback-Amt-Increase-CB7]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["chargeback_amount_1m_3m_list"].threshold ??
default_threshold_list["chargeback_amount_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

// Rule Explanation
formated_threshold = threshold * 100;
formated_cgbck_ratio = event.rtcomp_cbk_amt_1m_to_3m * 100;
```

Condition

```
 // Refund amount rate of one month (delayed over 4w) over the refund rate of three months
(delayed over 8w)
// (Filter: feedback_type == "chargeback")
// (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.rtcomp_cbk_amt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Chargeback amount rate of one month is event.rtcomp_cbk_amt_1m_to_3m x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

-
- Out-of-the-Box Resources
- Rules Projects
- Fraud Rules

On this page

Fraud Rules

Descriptionâ

Fraud rules prevent various fraud schemes that are generally associated with collusive/fraudulent merchants. A collusive merchant scheme occurs when a merchant's owner(s) and/or their employees conspire to commit fraud using their customers' (cardholder) accounts and/or personal information. Moreover, the scheme encompasses any abnormal behavior, which may indicate illicit activity or illegitimate gains by the merchant.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Failed-3DS-Count-Rate-FR1

Comment

Failed 3ds count rate, in 1 month, is equal to or above a set threshold

Actions

- [Failed-3DS-Count-Rate-FR1]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["count_3ds_limit_1m"].threshold ??
default_threshold_list["count_3ds_limit_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//Rule Explanation
formated_threshold = threshold * 100;
formated_3ds_count_ratio = event.rt_3ds_ko_cnt_4w * 100;
```

Condition

```
 // number of failed 3DS payments over the total number of payments in one month
// (Filter: payment_3ds_status_code defined and not in ("","y","Y"))
event.rt_3ds_ko_cnt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) failed 3DS count rate of one month is {localvar.formated_3ds_count_ratio} % and is equal/above the limit of {localvar.formated_threshold} %.

Mark As

N/A

Failed-3DS-Amount-Rate-FR2

Comment

Failed 3ds amount rate, in 1 month, is equal to or above a set threshold

Actions

- [Failed-3DS-Amount-Rate-FR2]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["amount_3ds_limit_1m"].threshold ??
default_threshold_list["amount_3ds_limit_1m"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
```

```
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_3ds_amount_ratio = event.rt_3ds_ko_amt_4w * 100;
```

Condition

```
 // total amount of failed 3DS payments over the total amount of payments in one month
// (Filter: payment_3ds_status_code defined and not in ("","y","Y"))
event.rt_3ds_ko_amt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) failed 3DS amount rate of one month is {localvar.formated_3ds_amount_ratio} % and is equal/above the limit of {localvar.formated_threshold} %.

Mark As

N/A

Failed-3DS-Cnt-Increase-FR3

Comment

Failed 3ds count rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Failed-3DS-Cnt-Increase-FR3]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["count_3ds_1m_3m_list"].threshold ??
default_threshold_list["count_3ds_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // failed 3DS rate of one month over the failed 3DS rate of three months (delayed over 4w)
// (Filter: payment_3ds_status_code defined and not in ("","y","Y"))
event.rtcomp_3ds_ko_cnt_1m_to_3m >=  threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) failed 3DS count rate of one month is {event.rtcomp_3ds_ko_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Failed-3DS-Amt-Increase-FR4

Comment

Failed 3ds amount rate of 1 month over 3 months is equal to or above a set threshold

Actions

- Alert
- [Failed-3DS-Amt-Increase-FR4]

Variables

```
 threshold = multi_tenanted_threshold_list["amount_3ds_1m_3m_list"].threshold ??
default_threshold_list["amount_3ds_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // failed 3DS amt rate of one month over the failed 3DS rate of three months (delayed over 4w)
// (Filter: payment_3ds_status_code != "Y"')
event.rtcomp_3ds_ko_amt_1m_to_3m >=  threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) failed 3DS amount rate of one month is {event.rtcomp_3ds_ko_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Visa-VFMP-Early-Warning-FR5

Comment

Visa Fraud Monitoring Program - Early Warning (VFMP) - Fraud to sales ratio, in one month.

Actions

- [Visa-VFMP-Early-Warning-FR5]
- Alert

Variables

```
 min_vfmp_early_rat_threshold =
default_threshold_list["min_vfmp_early_rat_threshold"].threshold; // 0.65%
max_vfmp_early_rat_threshold =
default_threshold_list["max_vfmp_early_rat_threshold"].threshold; // 0.9%
min_vfmp_early_amt_threshold =
default_threshold_list["min_vfmp_early_amt_threshold"].threshold; // USD 50,000
max_vfmp_early_amt_threshold =
default_threshold_list["max_vfmp_early_amt_threshold"].threshold; // USD 75,000
//For Rule Explanation
```

```
visa_fraud_to_sales_ratio = event.rt_visa_fraud_amt_1m*100;
visa_fraud_amount = event.visa_fraud_amt_cur_m/100;
```

Condition

```
 //The amount ratio of Visa fraudulent payments to all payments (feedback_label == 'fraud' and
feedback_value == true)
(event.rt_visa_fraud_amt_1m >= min_vfmp_early_rat_threshold
 and event.rt_visa_fraud_amt_1m < max_vfmp_early_rat_threshold)
and
// The total amount of Visa payments that are fraudulent
(event.visa_fraud_amt_cur_m>= min_vfmp_early_amt_threshold
 and event.visa_fraud_amt_cur_m < max_vfmp_early_amt_threshold)
```

Explanation

Visa Fraud Monitoring Program - Early Warning (VFMP) - The fraud to sales ratio, in one month, is {localvar.visa_fraud_to_sales_ratio} and the amount of fraudulent payments is {localvar.visa_fraud_amount}

Mark As

N/A

Visa-VFMP-Standard-FR6

Comment

Visa Fraud Monitoring Program - Standard (VFMP) - Fraud to sales ratio, in one month.

Actions

- [Visa-VFMP-Standard-FR6]
- Alert

Variables

```
 min_vfmp_standard_rat_threshold =
default_threshold_list["min_vfmp_standard_rat_threshold"].threshold; // 0.9%
max_vfmp_standard_rat_threshold =
default_threshold_list["max_vfmp_standard_rat_threshold"].threshold; // 1.8%
min_vfmp_standard_amt_threshold =
default_threshold_list["min_vfmp_standard_amt_threshold"].threshold; // USD 75,000
max_vfmp_standard_amt_threshold =
default_threshold_list["max_vfmp_standard_amt_threshold"].threshold; // USD 250,000
//For Rule Explanation
visa_fraud_to_sales_ratio = event.rt_visa_fraud_amt_1m *100;
visa_fraud_amount = event.visa_fraud_amt_cur_m/100;
```

Condition

```
 //The amount ratio of Visa fraudulent payments to all payments (feedback_label == 'fraud' and
feedback_value == true)
(event.rt_visa_fraud_amt_1m >= min_vfmp_standard_rat_threshold
 and event.rt_visa_fraud_amt_1m < max_vfmp_standard_rat_threshold)
and
// The total amount of Visa payments that are fraudulent
(event.visa_fraud_amt_cur_m>= min_vfmp_standard_amt_threshold
 and event.visa_fraud_amt_cur_m < max_vfmp_standard_amt_threshold)
```

Explanation

Visa Fraud Monitoring Program - Standard (VFMP) - The fraud to sales ratio, in one month, is {localvar.visa_fraud_to_sales_ratio} and the amount of fraudulent payments is {localvar.visa_fraud_amount}

Mark As

N/A

Visa-VFMP-Excessive-FR7

Comment

Visa Fraud Monitoring Program - Excessive (VFMP) - Fraud to sales ratio, in one month.

Actions

- Alert
- [Visa-VFMP-Excessive-FR7]

Variables

```
 min_vfmp_excessive_rat_threshold =
default_threshold_list["min_vfmp_excessive_rat_threshold"].threshold; // 1.8%
min_vfmp_excessive_amt_threshold =
default_threshold_list["min_vfmp_excessive_amt_threshold"].threshold; // USD 250,000
//For Rule Explanation
visa_fraud_to_sales_ratio = event.rt_visa_fraud_amt_1m*100;
visa_fraud_amount = event.visa_fraud_amt_cur_m/100;
```

Condition

```
 //The amount ratio of Visa fraudulent payments to all payments (feedback_label == 'fraud' and
feedback_value == true)
event.rt_visa_fraud_amt_1m >= min_vfmp_excessive_rat_threshold
and
// The total amount of Visa payments that are fraudulent
event.visa_fraud_amt_cur_m >= min_vfmp_excessive_amt_threshold
```

Explanation

Visa Fraud Monitoring Program - Early Warning (VFMP) - The fraud to sales ratio, in one month, is {localvar.visa_fraud_to_sales_ratio} and the amount of fraudulent payments is {localvar.visa_fraud_amount}

Mark As

N/A

Visa-VFMP-3DS-Early-FR8

Comment

Visa Fraud Monitoring Program - 3DS - Early Warning (VFMP) - 3DS Fraud to sales ratio, in one month. (US only)

Actions

- Alert
- [Visa-VFMP-3DS-Early-FR8]

Variables

```
 min_vfmp_3ds_early_rat_threshold =
default_threshold_list["min_vfmp_3ds_early_rat_threshold"].threshold; // 0.5%
max_vfmp_3ds_early_rat_threshold =
default_threshold_list["max_vfmp_3ds_early_rat_threshold"].threshold; // 0.75%
min_vfmp_3ds_early_amt_threshold =
default_threshold_list["min_vfmp_3ds_early_amt_threshold"].threshold; // USD 5,000
max_vfmp_3ds_early_amt_threshold =
default_threshold_list["max_vfmp_3ds_early_amt_threshold"].threshold; // USD 7,500
//For Rule Explanation
visa_3ds_fraud_to_sales_ratio = event.rt_visa_fraud_amt_1m*100;
visa_3ds_fraud_amount = event.visa_fraud_amt_cur_m/100;
```

Condition

```
 //The amount ratio of Visa fraudulent US domestic 3DS payments to all US 3DS domestic payments
(event.rt_visa_us_3ds_ok_fraud_amt_1m >= min_vfmp_3ds_early_rat_threshold
 and event.rt_visa_us_3ds_ok_fraud_amt_1m < max_vfmp_3ds_early_rat_threshold)
and
// The total amount of Visa US domestic 3DS payments that are fraudulent
(event.visa_us_3ds_ok_fraud_amt_cur_m >= min_vfmp_3ds_early_amt_threshold
 and event.visa_us_3ds_ok_fraud_amt_cur_m < max_vfmp_3ds_early_amt_threshold)
```

Explanation

Visa Fraud Monitoring Program - 3DS - Early Warning (VFMP) - The 3DS fraud to sales ratio, in one month, is {localvar.visa_3ds_fraud_to_sales_ratio} and the amount of 3DS fraudulent payments is {localvar.visa_3ds_fraud_amount}

Mark As

N/A

Visa-VFMP-3DS-Standard-FR9

Comment

Visa Fraud Monitoring Program - 3DS - Standard (VFMP) - 3DS Fraud to sales ratio, in one month. (US only)

Actions

- [Visa-VFMP-3DS-Standard-FR9]
- Alert

Variables

```
 min_vfmp_3ds_standard_rat_threshold =
default_threshold_list["min_vfmp_3ds_standard_rat_threshold"].threshold; // 0.75%
min_vfmp_3ds_standard_amt_threshold =
default_threshold_list["min_vfmp_3ds_standard_amt_threshold"].threshold; // USD 7,500
//For Rule Explanation
visa_3ds_fraud_to_sales_ratio = event.rt_visa_fraud_amt_1m*100;
visa_3ds_fraud_amount = event.visa_fraud_amt_cur_m/100;
```

Condition

```
 //The amount ratio of Visa fraudulent US domestic 3DS payments to all US 3DS domestic payments
event.rt_visa_us_3ds_ok_fraud_amt_1m >= min_vfmp_3ds_standard_rat_threshold
and
```

```
// The total amount of Visa US domestic 3DS payments that are fraudulent
event.visa_us_3ds_ok_fraud_amt_cur_m >= min_vfmp_3ds_standard_amt_threshold
```

Explanation

Visa Fraud Monitoring Program - 3DS - Standard (VFMP) - The 3DS fraud to sales ratio, in one month, is {localvar.visa_3ds_fraud_to_sales_ratio} and the amount of 3DS fraudulent payments is {localvar.visa_3ds_fraud_amount}

Mark As

N/A

Mastercard-EFM-FR10

Comment

Mastercard Excessive Fraud Merchant Compliance Program (EFM)

Actions

- [Mastercard-EFM-3DS-FR10]
- Alert

Variables

```
 mc_efm_ecom_sales_cnt_threhsold =
default_threshold_list["mc_efm_ecom_sales_cnt_threhsold"].threshold; // 1000
mc_efm_fraud_cbk_amt_threshold =
default_threshold_list["mc_efm_fraud_cbk_amt_threshold"].threshold; // USD 50,000
mc_efm_fraud_cbk_rate_threshold =
default_threshold_list["mc_efm_fraud_cbk_rate_threshold"].threshold; // 0.5%
mc_efm_3ds_amt_rate_non_reg_threshold =
default_threshold_list["mc_efm_3ds_amt_rate_non_reg_threshold"].threshold; // 10%
mc_efm_3ds_amt_rate_reg_threshold =
default_threshold_list["mc_efm_3ds_amt_rate_reg_threshold"].threshold; // 50%

//For Rule Explanation
mc_fraud_cbk_amount = event.mc_fraud_cbk_amt_cur_m/100;
rt_mc_fraud_cbk_cnt_1m = event.rt_mc_fraud_cbk_cnt_1m*100;
```

Condition

```
 //The number of E-Comm Mastercard sales in the previous month
event.mc_ecom_cnt_prev_m >= mc_efm_ecom_sales_cnt_threhsold
and
// The total amount of Mastercard chargebacks that are fraud related (reason code 4837) in the
current month
event.mc_fraud_cbk_amt_cur_m >= mc_efm_fraud_cbk_amt_threshold
and
// The number of Mastercard fraud CBK in the current month over the number of E-Comm sales of the
previous month
event.rt_mc_fraud_cbk_cnt_1m >= mc_efm_fraud_cbk_rate_threshold
and
// The Mastercard 3DS volume rate for Regulated and Non-Regulated Countries
((!regulated_countries_3ds.contains(event.merchant_country_code) and event.rt_mc_3ds_ok_amt_1m <=
mc_efm_3ds_amt_rate_non_reg_threshold)
or
(regulated_countries_3ds.contains(event.merchant_country_code) and event.rt_mc_3ds_ok_amt_1m <=
mc_efm_3ds_amt_rate_reg_threshold))
```

Explanation

Mastercard Excessive Fraud Merchant Compliance Program (EFM) - The Fraud CBK amount is {localvar.mc_fraud_cbk_amount} and the Fraud CBK rate is {localvar.rt_mc_fraud_cbk_cnt_1m}

Mark As

N/A

Fraud-Amount-Rate-FR11

Comment

Fraud amount rate, in one month, is equal to or above a set threshold

Actions

- [Fraud-Amount-Rate-FR11]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["fraud_amount_limit_1m"].threshold ??
default_threshold_list["fraud_amount_limit_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_fraud_amt_4w_4w_dlay * 100;
```

Condition

```
 // total amount of fraudulent payments over the total amount of payments in one month (delayed
over 4w)
// (Filter: feedback_type == 'fraud' and feedback_value == 'true')
event.rt_fraud_amt_4w_4w_dlay >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) fraud rate of one month is {localvar.formated_ratio} % and is equal/above the limit of {localvar.formated_threshold} %.

Mark As

N/A

Fraud-Amount-Increase-FR12

Comment

Fraud amount rate of, 1 month vs 3 months, is equal to or above a set threshold

Actions

- Alert
- [Fraud-Amount-Increase-FR12]

Variables

```
 threshold = multi_tenanted_threshold_list["fraud_amount_1m_3m_list"].threshold ??
default_threshold_list["fraud_amount_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Fraud amount rate of one month (delayed over 4w) over the fraud amount rate of three months
(delayed over 8w)
// (Filter: feedback_type == 'fraud' and feedback_value == 'true')
event.rtcomp_fraud_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Fraud amount rate of one month is event.rtcomp_fraud_1m_to_3m x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Rejected-Count-Rate-FR13

Comment

Rejected count rate, in one month, is equal to or above a set threshold

Actions

- Alert
- [Rejected-Count-Rate-FR13]

Variables

```
 threshold = multi_tenanted_threshold_list["reject_count_limit_1m"].threshold ??
default_threshold_list["reject_count_limit_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

formated_threshold = threshold * 100;
formated_ratio = event.rt_rejct_cnt_4w * 100;
```

Condition

```
 // The number of rejected payments over the total number of payments in one month
// (Filter: event_type == "info" and payment_auth_status == "declined")
event.rt_rejct_cnt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) rejected count rate of one month is {localvar.formated_ratio} % and is equal/above the limit of {localvar.formated_threshold} %.

Mark As

N/A

Rejected-Amount-Rate-FR14

Comment

Rejected amount rate, in one month, is equal to or above a set threshold

Actions

- Alert
- [Rejected-Amount-Rate-FR14]

Variables

```
 threshold = multi_tenanted_threshold_list["reject_amount_limit_1m"].threshold ??
default_threshold_list["reject_amount_limit_1m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

// For Rule Explanation
formated_threshold = threshold * 100;
formated_ratio = event.rt_rejct_amt_4w * 100;
```

Condition

```
 // The amount of rejected payments over the total amount of payments in one month
// (Filter: event_type == "info" and payment_auth_status == "declined")
event.rt_rejct_amt_4w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) rejected amount rate of one month is {localvar.formated_ratio} % and is equal/above the limit of {localvar.formated_threshold} %.

Mark As

N/A

Rejected-Count-Increase-FR15

Comment

Rejected count rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Rejected-Count-Increase-FR15]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["reject_count_1m_3m_list"].threshold ??
default_threshold_list["reject_count_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Rejected rate of one month over the rejected rate of three months (delayed over 4w)
// (Filter: event_type == "info" and payment_auth_status == "declined")
event.rtcomp_rejct_cnt_1m_to_3m >=  threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

> Merchant ({event.merchant_id}) rejected count rate of one month is {event.rtcomp_rejct_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Rejected-Amt-Increase-FR16

Comment

> Rejected amount rate of 1 month over 3 months is equal to or above a set threshold

Actions

- Alert
- [Rejected-Amt-Increase-FR16]

Variables

```
 threshold = multi_tenanted_threshold_list["reject_amount_1m_3m_list"].threshold ??
default_threshold_list["reject_amount_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // rejected amount rate of one month over the rejected amount rate of three months (delayed over
4w)
// (Filter: event_type == "info" and payment_auth_status == "declined")
event.rtcomp_rejct_amt_1m_to_3m >=  threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

> Merchant ({event.merchant_id}) rejected amount rate of one month is event.rtcomp_rejct_amt_1m_to_3m x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Rejected-Daily-Increase-FR17

Comment

Rejected count rate of 1 day over 3 months is equal to or above a set threshold

Actions

- Alert
- [Rejected-Daily-Increase-FR17]

Variables

```
 threshold = multi_tenanted_threshold_list["reject_count_1d_3m_list"].threshold ??
default_threshold_list["reject_count_1d_3m_list"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count_3m_1d_dlay"].threshold ??
default_threshold_list["min_trx_count_3m_1d_dlay"].threshold;
```

Condition

```
 // Rejected rate of one day over the rejected rate of three months (delayed over 1day)
// (Filter: event_type == "info" and payment_auth_status == "declined")
event.rtcomp_rejct_cnt_1d_to_3m >=  threshold
and event.cnt_3m_1d_dlay >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) rejected count rate of one day is {event.rtcomp_rejct_cnt_1d_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

- Out-of-the-Box Resources
- Rules Projects
- Money Laundering Rules

On this page

Money Laundering Rules

Descriptionâ

Merchants with abnormal activity indicating potential business fronts used to process transactions for products and/or services that are unknown and/or illegal.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

High-Amount-Count-ML1

Comment

The number of high amount transactions, in one day, is equal to or above a set threshold.

Actions

- [High-Amount-Count-ML1]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["aml_count_list"].threshold ??
default_threshold_list["aml_count_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // The number of payments in the last day with amount >= 15 000
event.aml_cnt_1d >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) number of payments in the last day with amount above 15 000 ({event.aml_cnt_1d} payments) is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Sanctions-List-ML2

Comment

Merchant name or merchant UBO names are part of a sanctions list.

Actions

- Auto-Decline-MID
- [Sanctions-List-ML2]
- Alert

Variables

```
 name=event.merchant_name;
ubos=event.merchant_ubo_names;
match_name=terrorist_sanctions_list.contains(name);
match_ubos=terrorist_sanctions_list.contains(ubos);
```

Condition

```
 match_name or match_ubos
```

Explanation

Merchant name ({event.merchant_name}) or merchant beneficiary name(s) ({event.merchant_ubo_names}) are part of a sanctions list.

Mark As

N/A

High-Risk-Countries-ML3

Comment

Number of transactions using cards from high-risk countries, in this merchant, is equal or above a set threshold.

Actions

- [High-Risk-Countries-ML3]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["high_risk_card_country_count"].threshold ??
default_threshold_list["high_risk_card_country_count"].threshold;
```

Condition

```
event.high_risk_cntries_1m >= threshold
```

Explanation

Merchant ({event.merchant_id}) processed {event.high_risk_cntries_1m} transactions with cards from high-risk countries, in one month, which is equal/above the limit of {localvar.threshold}

Mark As

N/A

Inactivity_Period_ML4

Comment

Merchant has been inactive for at least 60 days

Actions

- Alert
- [Inactivity_Period_ML4]

Variables

N/A

Condition

```
event.cnt_60d <= 0
  and (event.merchant_id ?? '' != '') and !exclusion_mid_list.contains(event.merchant_id)
```

Explanation

Merchant ({event.merchant_id}) has been inactive for at least 60 days.

Mark As

N/A

First_Trx_6months_ML5

Comment

First transaction was processed after the merchant has been inactive for over 6 months.

Actions

- [First_Trx_6months_ML5]
- Alert

Variables

N/A

Condition

```
event.cnt_6m_1d_dlay <= 0
 and event.cnt_1d > 0
 and (event.merchant_id ?? '' != '') and !exclusion_mid_list.contains(event.merchant_id)
```

Explanation

Merchant ({event.merchant_id}) had his first transaction after being innactive for 6 months.

Mark As

N/A

Unusual Activity Rules

Descriptionâ

Contains rules that look for suspicious activity that is not yet associated with a particular fraud scenario but requires close monitoring. Subsequent transactions performed by alerted merchants may evolve into a specific fraud type such as bankruptcy, fraud or money laundering.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Score-Deviation-Weekly-UAR1

Comment

Average score from last week deviates by at least n standard deviation

Actions

- [Score-Deviation-Weekly-UAR1]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["weekly_fraud_score_limit"].threshold ??
default_threshold_list["weekly_fraud_score_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

max_deviation = (threshold * event.std_score_1w_1w_dlay) + event.avg_score_1w_1w_dlay; // (n *
std dev of previous week fraud score) + avg score of previous week
```

Condition

```
event.avg_score_1w >= max_deviation // Average fraud score of last week // aggregated model score
per merchant from TF channel
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) current weekly average score of {event.avg_score_1w} deviates by at least {localvar.threshold} standard deviations compared to previous weekly average score of {event.avg_score_1w_1w_dlay}.

Mark As

N/A

Score-Deviation-Monthly-UAR2

Comment

Average score from last month deviates by at least n standard deviation

Actions

- [Score-Deviation-Monthly-UAR2]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["monthly_fraud_score_limit"].threshold ??
default_threshold_list["monthly_fraud_score_limit"].threshold;
threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
```

```
default_threshold_list["min_trx_count"].threshold;

max_deviation = (threshold * event.std_score_4w_4w_dlay) + event.avg_score_4w_4w_dlay;  // (n * std dev of previous week fraud score) + avg score of previous week
```

Condition

```
 event.avg_score_4w >= max_deviation // Average fraud score of last month // aggregated model score per merchant from TF channel
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) current monthly average score of {event.avg_score_4w} deviates by at least {localvar.threshold} standard deviations compared to previous monthly average score of {event.avg_score_4w_4w_dlay}.

Mark As

N/A

MID-Track-List-UAR3

Comment

Merchant ID is in a track list

Actions

- Alert
- [MID-Track-List-UAR3]

Variables

```
 listed = track_list.contains(event.merchant_id??"");
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 listed
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) is in a track list.

Mark As

N/A

NDX-Limit-UAR4

Comment

NDX (non-delivery exposure) limit is equal to or above a set threshold

Actions

- [NDX-Limit-UAR4]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["ndx_limit"].threshold ??
default_threshold_list["ndx_limit"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;

// For Rule Explanation
formated_threshold = threshold / 100;
formated_ndx = event.ndx_approx / 100;
```

Condition

```
 event.ndx_approx >= threshold // if Merchant ADD == x days then NDX Apprx = Merchant Sum Amt x
days
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Non Delivery Exposure is {localvar.formated_ndx} and is equal/above the limit of {localvar.formated_threshold} .

Mark As

N/A

ADD-Limit-UAR5

Comment

ADD (average delivery days) limit is equal to or above a set threshold

Actions

- [ADD-Limit-UAR5]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["add_limit"].threshold ??
default_threshold_list["add_limit"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 event.merchant_add >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) Average Delivery Days is {event.merchant_add} and is equal/above the hard limit of {localvar.threshold}.

Mark As

N/A

Distinct-Cards-Increase-UAR6

Comment

Number of distinct cards in 1 month has grown abnormally compared to the last 3 months

Actions

- [Distinct-Cards-Increase-UAR6]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["distinct_cards_1m_3m"].threshold ??
default_threshold_list["distinct_cards_1m_3m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Number of distinct cards one month over the number of distinct of three months (delayed over
4w)
event.rt_dist_card_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) distinct cards number of one month is {event.rt_dist_card_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Dist-Country-Increase-UAR7

Comment

Number of distinct card countries in 1 month has grown abnormally compared to the last 3 months

Actions

- Alert
- [Dist-Country-Increase-UAR7]

Variables

```
 threshold = multi_tenanted_threshold_list["distinct_cardcountry_1m_3m"].threshold ??
default_threshold_list["distinct_cardcountry_1m_3m"].threshold;
```

```
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Number of distinct card country one month over the number of distinct card country of three
months (delayed over 4w)
event.rt_dist_cardcntry_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) distinct card country number of one month is {event.rt_dist_cardcntry_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Card-Manual-Increase-UAR8

Comment

Card manually entered payments rate of 1 month over 3 months is equal to or above a set threshold

Actions

- [Card-Manual-Increase-UAR8]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["card_manual_count_1m_3m_list"].threshold ??
default_threshold_list["card_manual_count_1m_3m_list"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // Card manual payments rate of one month over the card manual rate of three months (delayed over
4w)
// Filters: card_entry_mode == 'manual'
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rtcomp_manual_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) card manually entered payments rate of one month is {event.rtcomp_manual_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Monthly-Count-Increase-UAR9

Comment

Number of payments in 1 month has grown abnormally compared to the last 3 months

Actions

- Alert
- [Monthly-Count-Increase-UAR9]

Variables

```
 threshold = multi_tenanted_threshold_list["count_increase_1m_3m"].threshold ??
default_threshold_list["count_increase_1m_3m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // total coount of approved payments one month over the total amount of approved payments of
three months (delayed over 4w)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_cnt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) number of payments one month is {event.rt_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Monthly-Amt-Increase-UAR10

Comment

Total amount of payments in 1 month has grown abnormally compared to the last 3 months

Actions

- [Monthly-Amt-Increase-UAR10]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["amount_increase_1m_3m"].threshold ??
default_threshold_list["amount_increase_1m_3m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // total amount of approved payments one month over the total amount of approved payments of
three months (delayed over 4w)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_amt_1m_to_3m >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) total amount of payments one month is {event.rt_amt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Daily-Amount-Increase-UAR11

Comment

Total amount of payments in 1 day has grown abnormally compared to the last week

Actions

- [Daily-Amount-Increase-UAR11]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["amount_increase_1d_1w"].threshold ??
default_threshold_list["amount_increase_1d_1w"].threshold;
 threshold_amount = multi_tenanted_threshold_list["min_avg_amount"].threshold ??
default_threshold_list["min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["min_trx_count"].threshold ??
default_threshold_list["min_trx_count"].threshold;
```

Condition

```
 // total amount of approved payments one day over the total amount of approved payments of 1w
(delayed over 1day)
// Filters: (event_type == "info") and (payment_auth_status == "accepted")
event.rt_amt_1d_to_1w >= threshold
and (event.sum_amt_1w/7.0) >= threshold_amount
and event.cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) total amount of payments one day is {event.rt_amt_1d_to_1w}x higher than one week and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

Peer-Avg-Ticket-Size-UAR12

Comment

Merchant daily average transaction value deviates by at least n Standard Deviations of the MCC 90 day average transaction value

Actions

- Alert
- [Peer-Avg-Ticket-Size-UAR12]

Variables

```
 threshold = multi_tenanted_threshold_list["mcc_avg_ticket_size_limit"].threshold ??
default_threshold_list["mcc_avg_ticket_size_limit"].threshold;
threshold_count = multi_tenanted_threshold_list["min_count_succ_90d"].threshold ??
default_threshold_list["min_count_succ_90d"].threshold;

max_deviation = (threshold * event.mcc_std_accpt_amt_90d_1d_dlay) +
event.mcc_avg_accpt_amt_90d_1d_dlay; // (n * std dev of MCC succ trx amount last 90 days) + avg
MCC succ trx amount of last 90 days
```

Condition

```
 // (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.avg_accpt_amt_1d >= max_deviation // Average succ trx amount last day >= Max Deviation
  and event.cnt_accpt_90d_1d_dlay >= threshold_count // min 1 succ transaction per day
```

Explanation

Merchant ({event.merchant_id}) daily average ticket size is {event.avg_accpt_amt_1d} and deviates by at least {localvar.threshold} standard deviation compared to previous 90 days MCC average ticket size of {event.mcc_avg_accpt_amt_90d_1d_dlay}.

Mark As

N/A

Merch-Avg-Ticket-Size-UAR13

Comment

Merchant daily average transaction value deviates by at least n Standard Deviations of the Merchant 90 day average transaction value

Actions

- Alert
- [Merch-Avg-Ticket-Size-UAR13]

Variables

```
 threshold = multi_tenanted_threshold_list["merch_avg_ticket_size_limit"].threshold ??
default_threshold_list["merch_avg_ticket_size_limit"].threshold;
threshold_count = multi_tenanted_threshold_list["min_count_succ_90d"].threshold ??
default_threshold_list["min_count_succ_90d"].threshold;

max_deviation = (threshold * event.std_accpt_amt_90d_1d_dlay) +
event.avg_accpt_amt_90d_1d_dlay; // (n * std dev of succ trx amount last 90 days) + avg succ trx
```

```
amount of last 90 days
```

Condition

```
 // (Filters: (event_type == "info") and (payment_auth_status == "accepted"))
event.avg_accpt_amt_1d >= max_deviation // Average succ trx amount last day >= Max Deviation
  and event.cnt_accpt_90d_1d_dlay >= threshold_count // min 1 succ transaction per day
```

Explanation

Merchant ({event.merchant_id}) daily average ticket size is {event.avg_accpt_amt_1d} and deviates by at least {localvar.threshold} standard deviation compared to previous 90 days average ticket size of {event.avg_accpt_amt_90d_1d_dlay}.

Mark As

N/A

Gift-Prepaid-Card-Rate-UAR14

Comment

Daily gift or prepaid card payments rate is equal or above a set threshold

Actions

- [Gift-Prepaid-Card-Rate-UAR14]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["gift_prepaid_card_limit"].threshold ??
default_threshold_list["gift_prepaid_card_limit"].threshold;

//For Rule Explanation
formated_threshold = threshold * 100;
formated_rate = event.rt_prepaid_card_cnt_1d * 100;
```

Condition

```
 // Approved Gift or prepaid payments rate of one day
// Filters: payment_auth_status == 'accepted' and payment_card_type == 'giftcard' or 'prepaid'
event.rt_prepaid_card_cnt_1d >= threshold
```

Explanation

Merchant ({event.merchant_id}) gift/prepaid card payments rate of one day is {localvar.formated_rate} and it's equal/above the limit of {localvar.formated_threshold}

Mark As

N/A

-
- Out-of-the-Box Resources
- Rules Projects
- Refund and OCT Rules

On this page

Refund and OCT Rules

Descriptionâ

A payment refund is the process in which a payment made to the merchant account for a specific transaction is sent back to the customer.

The Original Credit Transaction (OCT) is a transaction that can be used to send or "push" funds to an eligible card-based account, resulting in a credit of funds to the cardholder's account. The "push" payment framework speeds up the funds delivery. It differs from the traditional method, in which funds are credited to the receiver's bank account. These rules monitor the volume and trends of refunds and OCTs for merchants.

Rules Project Metadata

- *Target Variable:* risky
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

OCT-MM-1

Comment

Alert when OCT/Refunds to sales (amount) ratio exceeds threshold over 30 days

Actions

- Alert
- [OCT-MM-1]

Variables

```
ratio_threshold = multi_tenanted_threshold_list["oct_rfd_sales_ratio_mid_30d_th"].threshold ??
default_threshold_list["oct_rfds_sales_ratio_mid_30d_th"].threshold ?? 0.01;
```

Condition

```
(event.oct_rfd_amount_mid_30d-event.oct_rfd_amount_ref_rev_mid_30d)/
event.oct_rfd_amount_payment_mid_30d  > ratio_threshold
```

Explanation

OCT/Refunds to sales ratio exceeds threshold ({localvar.ratio_threshold}) over 30 days

Mark As

N/A

OCT-MM-2

Comment

Alert when OCT/Refunds count by merchant, in 7 days frame exceeds threshold

Actions

- Alert
- [OCT-MM-2]

Variables

```
count_threshold = multi_tenanted_threshold_list["oct_rfd_count_mid_7d_th"].threshold ??
default_threshold_list["oct_rfd_count_mid_7d_th"].threshold ?? 10;
```

Condition

```
event.oct_rfd_count_mid_7d >= count_threshold
```

Explanation

OCT/Refunds count ({event.oct_rfd_count_mid_7d}) by Merchant ({event.merchant_id}) in 7 days frame

Mark As

N/A

OCT-MM-3

Comment

Alert OCT/Refund growth rate exceeds threshold, compared to 90 day average

Actions

- [OCT-MM-3]
- Alert

Variables

```
rate_threshold = multi_tenanted_threshold_list["oct_rfd_amt_mid_90d_24h_rate_th"].threshold ??
default_threshold_list["oct_rfd_amt_mid_90d_24h_rate_th"].threshold ?? 2;
```

Condition

```
event.oct_rfd_sum_amt_24h / ( event.oct_rfd_sum_amt_mid_90d / 90 ) >= rate_threshold
```

Explanation

OCT/Refunds growth rate exceeds thresholds ({localvar.rate_threshold}) compared to 90 day averag

Mark As

N/A

OCT-MM-4

Comment

Total amount of Refunds/OCTs in 1 month has grown abnormally compared to the last 3 months

Actions

- [OCT-MM-4]
- Alert

Variables

```
threshold = multi_tenanted_threshold_list["oct_rfd_amount_increase_1m_3m"].threshold ??
default_threshold_list["oct_rfd_amount_increase_1m_3m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["oct_rfd_min_avg_amount"].threshold ??
default_threshold_list["oct_rfd_min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["oct_rfd_min_trx_count"].threshold ??
default_threshold_list["oct_rfd_min_trx_count"].threshold;
```

Condition

```
// total amount of approved Refunds/OCTs one month over the total amount of approved Refunds/OCTs
of three months (delayed over 4w)
// Filters: (event_type == "info") and (info_type == "refund" or info_type == "oct")
event.oct_rfd_rt_amt_1m_to_3m >= threshold
and (event.oct_rfd_sum_amt_1w/7.0) >= threshold_amount
and event.oct_rfd_cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) total amount of OCT/Refunds one month is {event.oct_rfd_rt_amt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

OCT-MM-5

Comment

Number of Refunds/OCTs in 1 month has grown abnormally compared to the last 3 months

Actions

- [OCT-MM-5]
- Alert

Variables

```
 threshold = multi_tenanted_threshold_list["oct_rfd_count_increase_1m_3m"].threshold ??
default_threshold_list["oct_rfd_count_increase_1m_3m"].threshold;
 threshold_amount = multi_tenanted_threshold_list["oct_rfd_min_avg_amount"].threshold ??
default_threshold_list["oct_rfd_min_avg_amount"].threshold;
 threshold_count = multi_tenanted_threshold_list["oct_rfd_min_trx_count"].threshold ??
default_threshold_list["oct_rfd_min_trx_count"].threshold;
```

Condition

```
 // total coount of approved Refunds/OCTs one month over the total amount of approved Refunds/OCTs
of three months (delayed over 4w)
// Filters: (event_type == "info") and (info_type == "refund" or info_type == "oct")
event.oct_rfd_rt_cnt_1m_to_3m >= threshold
and (event.oct_rfd_sum_amt_1w/7.0) >= threshold_amount
and event.oct_rfd_cnt_1w >= threshold_count
```

Explanation

Merchant ({event.merchant_id}) number of OCT/Refunds one month is {event.oct_rfd_rt_cnt_1m_to_3m}x higher than three months and is equal/above the limit of {localvar.threshold}.

Mark As

N/A

OCT-MM-6

Comment

Alert OCT/refunds count or amount exceeds threshold per MID over 30days when account has been inactive/dormant for at least 6 months

Actions

- [OCT-MM-6]
- Alert

Variables

```
threshold_amount = multi_tenanted_threshold_list["oct_rfd_refund_amount_mid_30d_th"].threshold ??
default_threshold_list["oct_rfd_refund_amount_mid_30d_th"].threshold;
 threshold_count = multi_tenanted_threshold_list["oct_rfd_refund_count_mid_30d_th"].threshold ??
default_threshold_list["oct_rfd_refund_count_mid_30d_th"].threshold;
 converted_amount_threshold = threshold_amount / 100;
 converted_amount = event.oct_rfd_amount_mid_30d / 100;
```

Condition

```
event.oct_rfd_mid_count_6m_delay_30d == 0 and
 event.oct_rfd_count_mid_30d >= threshold_count and
  event.oct_rfd_amount_mid_30d >= threshold_amount
```

Explanation

Merchant ID is dormant for at least 6 months and count ({event.oct_rfd_count_mid_30d}) and total amount ({localvar.converted_amount}) exceeds threshold ({localvar.threshold_count} and {localvar.converted_amount_threshold}, respectively)

Mark As

N/A

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/data-scientist/overview

- Operational Work
- Data Scientist & Fraud Strategist

Data Scientist & Fraud Strategist

Pulse Guide

Learn how to detect risky merchants using rules and models in Pulse.

Fraud Prevention Agent

Case Manager for Fraud Prevention Agents

Learn how to work with alerts and/or cases using Case Manager.

- Operational Work
- Fraud Prevention Manager

Fraud Prevention Manager

Case Manager for Fraud Prevention Managers

Learn how to create reasons or queues using Case Manager.

Analytics Dashboards Guide

View real-time insights on merchant portfolio and temporal trends.

- Operational Work
- Integration Engineer

Integration Engineer

Integration Guide

Learn how to test rules, and ensure the system is properly tested / running properly.

- Out-of-the-Box Resources
- Rules Projects
- APM Rules

On this page

APM Rules

Descriptionâ

Alternative Payment Method (APM) rules look for payments whose number or total amount of transactions per APM is above the threshold.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Velocity-APM1

Comment

Alert payment if the total amount per APM (e.g Paypal, Neteller), MID and Customer Email, in the last 24h, is above a set threshold

Actions

- [Velocity-APM1]
- Alert

Variables

```
 amount_threshold =
  tenanted_rules_thresholds['velocity_apm1_apm_merch_cust_email_sum_24h'].threshold
  ??
merchant_risk_levels__rules_thresholds['velocity_apm1_apm_merch_cust_email_sum_24h'].threshold
  ?? global_default__rules_thresholds['velocity_apm1_apm_merch_cust_email_sum_24h'].threshold
  ?? 0;
```

Condition

```
 //Filters
amount_threshold > 0 and (event.payment_method_id ?? '' != '')  and (event.merchant_id ?? '' !=
'')
// Limits
and event.apm_merch_cust_email_sum_24h >= amount_threshold
```

Explanation

Total amount spent using this APM ({event.payment_method_id}) in this MID ({event.merchant_id}) with the Email {event.customer_email}, in the last 24h is {event.apm_merch_cust_email_sum_24h}, and is equal/above limit of {localvar.amount_threshold} .

Mark As

N/A

Velocity-APM2

Comment

Alert payment if the number of transactions per APM (e.g Paypal, Neteller), MID and Customer Email, in the last 24h, is above a set threshold

Actions

- [Velocity-APM2]
- Alert

Variables

```
 count_threshold =
  tenanted_rules_thresholds['velocity_apm2_apm_merch_cust_email_cnt_24h'].threshold
  ??
merchant_risk_levels__rules_thresholds['velocity_apm2_apm_merch_cust_email_cnt_24h'].threshold
```

```
  ?? global_default__rules_thresholds['velocity_apm2_apm_merch_cust_email_cnt_24h'].threshold
  ?? 0;
```

Condition

```
 //Filters
count_threshold > 0 and (event.payment_method_id ?? '' != '') and (event.merchant_id ?? '' !=
'')  and (event.customer_email ?? '' != '')
// Limits
and event.apm_merch_cust_email_cnt_24h >= count_threshold
```

Explanation

Total number of transactions using this APM ({event.payment_method_id}), in this MID ({event.merchant_id}) and Email ({event.customer_email}), in last 24h is {event.apm_merch_cust_email_cnt_24h} and is equal/above limit of {localvar.count_threshold} .

Mark As

N/A

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/oob-resources/rules-projects/rmpsp-by-merchant-posneg-rules

-

- [Out-of-the-Box Resources]
- Rules Projects
- By Merchant PosNeg Rules

On this page

By Merchant PosNeg Rules

Descriptionâ

By Merchant PosNeg Rules allow the Merchant or the PSP to mark transactions that have entities in PosNeg Lists as "approved" or "declined" for a given Merchant. By default, the solution contains rules that mark the transaction as "declined". These rules act only on Payment Initiation events. For each rule, check the filter inside the condition code block to know which entities are being evaluated.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Card-Decline-PN1

Comment

Decline payment if Card ID is listed to be auto declined for this MID.

Actions

- [Card-Decline-PN1]

Variables

```
card_id_listed_value = by_merchant_tenanted_posneg_listed_cards [event.payment_card_id ??
''].value;
```

Condition

```
//Filters
(event.payment_card_id ?? '') != ''
and card_id_listed_value == 'decline'
```

Explanation

Card ID ({event.payment_card_id}) is listed by the Merchant to be auto decline.

Mark As

The outcome of this rule is: **decline**

Card-Approve-PN2

Comment

Approve payment if Card ID is listed to be auto-approved for this MID.

Actions

- [Card-Approve-PN2]

Variables

```
card_id_listed_value = by_merchant_tenanted_posneg_listed_cards [event.payment_card_id ??
''].value;
```

Condition

```
//Filters
(event.payment_card_id ?? '') != ''
and card_id_listed_value == 'approve'
```

Explanation

Card ID ({event.payment_card_id}) is listed by the Merchant to be auto-approve.

Mark As

The outcome of this rule is: **approve**

Card-Review-PN3

Comment

Alert payment if Card ID is listed to be limited for this MID.

Actions

- Tenant Alert
- [Card-Review-PN3]

Variables

```
 card_id_listed_value = by_merchant_tenanted_posneg_listed_cards  [event.payment_card_id ??
''].value;
```

Condition

```
 //Filters
(event.payment_card_id ?? '') != ''
and
(card_id_listed_value ?? '') == 'review'
```

Explanation

Card ID ({event.payment_card_id}) is listed by the Merchant to be limited.

Mark As

N/A

Email-Decline-PN4

Comment

Decline payment if Customer Email is listed to be auto declined for this MID.

Actions

- [Email-Decline-PN4]

Variables

```
 email_listed_value = by_merchant_tenanted_posneg_listed_customer_email [event.customer_email ??
''].value;
```

Condition

```
 //Filters
(event.customer_email ?? '') != ''
and
email_listed_value == 'decline'
```

Explanation

Customer Email ({event.customer_email}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Email-Approve-PN5

Comment

Approve payment if Customer Email is listed to be auto-approved per MID.

Actions

- [Email-Approve-PN5]

Variables

```
email_listed_value = by_merchant_tenanted_posneg_listed_customer_email  [event.customer_email ??
''].value;
```

Condition

```
//Filters
(event.customer_email ?? '') != ''
and
email_listed_value == 'approve'
```

Explanation

Customer Email ({event.customer_email}) is listed by the Merchant to be auto-approved.

Mark As

The outcome of this rule is: **approve**

Device-IP-Decline-PN6

Comment

Decline payment if Device IP is listed to be auto declined for this MID.

Actions

- [Device-IP-Decline-PN6]

Variables

```
device_ip_listed_value = by_merchant_tenanted_posneg_listed_device_ip  [event.device_ip ??
''].value;
```

Condition

```
(event.device_ip ?? '') != ''
and
device_ip_listed_value == 'decline'
```

Explanation

Device IP ({event.device_ip}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Terminal-ID-Decline-PN7

Comment

Decline payment if Terminal ID is listed to be auto declined for this MID.

Actions

- [Terminal-ID-Decline-PN7]

Variables

```
terminal_id_listed_value = by_merchant_tenanted_posneg_listed_terminals [event.terminal_id ??
''].value;
```

Condition

```
(event.terminal_id ?? '') != ''
and
terminal_id_listed_value == 'decline'
```

Explanation

Terminal ID ({event.terminal_id}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Terminal-ID-Approve-PN8

Comment

Approve payment if Terminal ID is listed to be auto-approved for this MID.

Actions

- [Terminal-ID-Approve-PN8]

Variables

```
terminal_id_listed_value = by_merchant_tenanted_posneg_listed_terminals  [event.terminal_id ??
''].value;
```

Condition

```
(event.terminal_id ?? '') != ''
and
terminal_id_listed_value == 'approve'
```

Explanation

Terminal ID ({event.terminal_id}) is listed by the Merchant to be auto-approve.

Mark As

The outcome of this rule is: **approve**

Device-ID-Decline-PN9

Comment

Decline payment if Device ID is listed to be auto declined for this MID.

Actions

- [Device-ID-Decline-PN9]

Variables

```
device_id_listed_value = by_merchant_tenanted_posneg_listed_devices  [event.device_id ??
''].value;
```

Condition

```
(event.device_id ?? '') != ''
and
device_id_listed_value == 'decline'
```

Explanation

Device ID ({event.device_id}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Device-ID-Approve-PN10

Comment

Approve payment if Device ID is listed to be auto-approved for this MID.

Actions

- [Device-ID-Approve-PN10]

Variables

```
device_id_listed_value = by_merchant_tenanted_posneg_listed_devices  [event.device_id ??
''].value;
```

Condition

```
(event.device_id ?? '') != ''
and
device_id_listed_value == 'approve'
```

Explanation

Device ID ({event.device_id}) is listed by the Merchant to be auto-approved.

Mark As

The outcome of this rule is: **approve**

Device-ID-Review-PN11

Comment

Alert payment if Device ID is listed to be limited for this MID.

Actions

- Tenant Alert
- [Device-ID-Review-PN11]

Variables

```
 device_id_listed_value = by_merchant_tenanted_posneg_listed_devices  [event.device_id ??
''].value;
```

Condition

```
 (event.device_id ?? '') != ''
and
device_id_listed_value == 'review'
```

Explanation

Device ID ({event.device_id}) is listed by the Merchant to be limited.

Mark As

N/A

Card-Country-Decline-PN12

Comment

Decline payment if Card Country is listed to be auto declined for this MID.

Actions

- [Card-Country-Decline-PN12]

Variables

```
 card_country_listed_value = by_merchant_tenanted_posneg_listed_card_country
[event.payment_card_country_code ?? ''].value;
```

Condition

```
 (event.payment_card_country_code ?? '') != ''
and
card_country_listed_value == 'decline'
```

Explanation

Card Country ({event.payment_card_country_code}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Card-BIN-Decline-PN13

Comment

Decline payment if Card BIN is listed to be auto declined for this MID.

Actions

- [Card-BIN-Decline-PN13]

Variables

```
card_bin_listed_value = by_merchant_tenanted_posneg_listed_card_bin  [event.payment_card_bin ??
''].value;
```

Condition

```
(event.payment_card_bin ?? '') != ''
and
card_bin_listed_value == 'decline'
```

Explanation

Card BIN ({event.payment_card_bin}) is listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

Ship-Addr-Decline-PN14

Comment

Decline payment if full Shipping Address is listed to be auto declined for this MID.

Actions

- [Ship-Addr-Decline-PN14]

Variables

```
norm_shipp_addr_listed_value = by_merchant_tenanted_listed_posneg_shipping_address
[event.norm_shipp_addr_line ?? ''].value;
full_shipp_addr_listed_value = by_merchant_tenanted_listed_posneg_shipping_address
[event.full_shipping_address ?? ''].value;
```

Condition

```
((event.norm_shipp_addr_line != '') and norm_shipp_addr_listed_value == 'decline') // Norm Shipp
Addr Line is equal to the following normalized and concateneted fields: addr_line1 and addr_line2
or
((event.full_shipping_address != '') and full_shipp_addr_listed_value == 'decline') // Full
Shipping Address is equal to the following normalized and concateneted fields: addr_line1,
addr_line2, ZIP, City, Region and Country
```

Explanation

Normalized Shipp Addr Lines ({event.norm_shipp_addr_line}) or Full Ship Address ({event.full_shipping_address}) are listed by the Merchant to be auto declined.

Mark As

The outcome of this rule is: **decline**

-

- [Out-of-the-Box Resources]
- Rules Projects
- By Merchant Self-Service Rules

On this page

By Merchant Self-Service Rules

Descriptionâ

Standard rules written in Pulse are normally developed and tested in staging environments to ensure that the work in development does not affect the work in production. These rules are developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production. At the same time, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one. Self-service rules (SSR) allow you to fill this gap, and have an active rule in production in a few minutes. With this capability, you can write a rule in Case Manager, test and publish it directly to production. As an example, you can use SSRs to create a rule with a different threshold than the one already defined in a standard rule and publish it in production to have it quickly live. Therefore, in a new installation, this rules project is empty, and is used in the Pulse workflow to receive the SSRs written in Case Manager. By Merchant SSRs allow Merchants and PSPs to write SSRs that affect a single given merchant. Note that any rule in this block is shared between the merchant and the PSP, and any of them can apply changes (according to permissions), so they must be seen as shared resources.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_DISTINCT

-
- Out-of-the-Box Resources
- Rules Projects
- Card Present Rules

On this page

Card Present Rules

Descriptionâ

Contains rule(s) specific to fraud patterns happening in Card Present transactions, and using data elements related to cardsâ magnetic stripe or chip+PIN attempts.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

MagStripe-Fallback-CP1

Comment

Decline a magstripe fallback payment to signature if the amount is above a set threshold

Actions

- [MagStripe-Fallback-CP1]

Variables

```
 magstripe_amt_threshold =
global_default__rules_thresholds['magstripe_fallback_cp1_amount'].threshold ?? 0;
```

Condition

```
 //Filters
magstripe_amt_threshold > 0
and (event.card_entry_mode == 'magstripe' or event.card_entry_mode == 'contactless-magstripe')
and event.authentication_mode == 'pin_bypassed'
//Limits
and event.amount_unified_currency ?? 0 >= magstripe_amt_threshold
```

Explanation

Magstripe ({event.card_entry_mode}) and PIN Bypassed payment ({event.authentication_mode}) with amount ({event.amount_unified_currency}) equal or above limit [{localvar.magstripe_amt_threshold}]

Mark As

The outcome of this rule is: **decline**

Decision Rules

Descriptionâ

Decision Rules are triggered based on the score value produced by a machine learning model. If your application workflow does not have a model, the `fraud_score` field remains empty, and thus the rules inside the Decision Rules project are not triggered.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Score-Decline-DR1

Comment

Decline a payment if it scored above a set threshold.

Actions

- [Score-Decline-DR1]

Variables

```
 HIGH_SCORE_THRESHOLD = global_default__rules_thresholds["model_high_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) >= HIGH_SCORE_THRESHOLD
```

Explanation

Model score ({localvar.score}) was equal or above the threshold ({localvar.HIGH_SCORE_THRESHOLD}).

Mark As

The outcome of this rule is: **decline**

Score-Review-DR2

Comment

Alert a payment if it scored between a set threshold.

Actions

- [Score-Review-DR2]
- Alert

Variables

```
 LOW_SCORE_THRESHOLD = global_default__rules_thresholds["model_low_score"].threshold;
HIGH_SCORE_THRESHOLD = global_default__rules_thresholds["model_high_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) > LOW_SCORE_THRESHOLD
  and (score ?? 0) < HIGH_SCORE_THRESHOLD
```

Explanation

Model score ({localvar.score}) was between the low score threshold ({localvar.LOW_SCORE_THRESHOLD}) and the high score threshold ({localvar.HIGH_SCORE_THRESHOLD}).

Mark As

N/A

Feedback Rules

Descriptionâ

Feedback Rules allow you to apply a set of actions to events that were confirmed as fraud or not fraud. For example, adding an entity to a PosNeg list over a period of time. Take the following example: A fraud analyst reviews an alert in Case Manager and marks it as fraud. In this case, an event is automatically sent to Pulse with the `fraud_label` as 'true' (i.e. fraud) and âevent_subtypeâfield as 'cm_feedback'. A Feedback rule will process this feedback event and will put the card into a PosNeg list to auto-decline for the next 5 days.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Auto-Decline-Card-FB1

Comment

Flag card to be auto declined if the transaction is marked as 'Fraud'. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Auto-Decline-Card
- [Auto-Decline-Card-FB1]

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == true
and
// Filters
(event.payment_card_id ?? '' != '')
```

Explanation

Transaction is marked as 'Fraud'. Card ID ({event.payment_card_id}) flagged for auto decline.

Mark As

N/A

Auto-Approve-Card-FB2

Comment

Flag card to be auto-approved if the transaction is marked as 'Not Fraud'. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Auto-Approve-Card-FB2]
- Auto-Approve-Card

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == false
and
```

```
// Filters
(event.payment_card_id ?? '' != '')
```

Explanation

Transaction is marked as 'Not Fraud'. Card ID ({event.payment_card_id}) flagged for auto approve.

Mark As

N/A

Auto-Decline-Email-FB3

Comment

Flag a customer email to be auto declined if the transaction is marked as 'Fraud'. Depends on 'List Management Service' to put the customer email in the configured list.

Actions

- Auto-Decline-Email
- [Auto-Decline-Email-FB3]

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == true
and
// Filters
(event.customer_email ?? '' != '') and (event.card_presence_flag == false)
```

Explanation

Transaction is marked as 'Fraud'. Customer Email ({event.customer_email}) flagged for auto decline.

Mark As

N/A

Auto-Approve-Email-FB4

Comment

Flag a customer email to be auto-approved if the transaction is marked as 'Not Fraud'. Depends on 'List Management Service' to put the customer email in the configured list.

Actions

- Auto-Approve-Email
- [Auto-Approve-Email-FB4]

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == false
and
// Filters
(event.customer_email ?? '' != '') and (event.card_presence_flag == false)
```

Explanation

Transaction is marked as 'Not Fraud'. Customer Email ({event.customer_email}) flagged for auto decline.

Mark As

N/A

- Out-of-the-Box Resources
- Rules Projects
- Fraud Rules

On this page

Fraud Rules

Descriptionâ

Contains rules that look for standard fraud patterns. These rules are used to approve/decline a payment and optionally mark the event as an alert which can then be further investigated by fraud prevention agents in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

All-Payments-Velocity-FR1

Comment

Decline payment if the number of transactions per Customer ID and MID, in the last day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR1]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_count_1d"].thre
shold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_co
unt_1d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr1_max_customer_id_merchant_id_count_1d"
].threshold
  ?? 0;
```

Condition

```
 // Limits
  COUNT_LIMIT > 0
  and
(event.cust_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}) in the last day ({event.cust_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR2

Comment

Decline payment if the number of transactions per Customer ID and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR2]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_2_max_customer_id_merchant_id_count_7d"].thresh
old
  ??
```

```
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr2_max_customer_id_merchant_id_co
unt_7d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr2_max_customer_id_merchant_id_count_7d"
].threshold
  ?? 0;
```

Condition

```
 COUNT_LIMIT > 0
  and
(event.cust_merch_cnt_7d ?? 0) >= COUNT_LIMIT
```

Explanation

> Number of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}) in the last week ({event.cust_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR3

Comment

> Decline payment if the number of transactions per Customer Email and MID, in the past day, is above a set threshold

Actions

- [All-Payments-Velocity-FR3]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_count_1d"].t
hreshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_
count_1d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr3_max_customer_email_merchant_id_count_
1d"].threshold
  ?? 0;
```

Condition

```
 COUNT_LIMIT > 0
 and
(event.email_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

> Number of transactions for the same Customer Email ({event.customer_email}) and MID ({event.merchant_id}) in the last day ({event.email_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR4

Comment

Decline payment if the number of transactions per Customer Email and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR4]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr4_max_customer_email_merchant_id_count_7d"].t
hreshold
  ?? merchant_risk_levels__rules_thresholds
["all_payments_velocity_fr4_max_customer_email_merchant_id_count_7d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr4_max_customer_email_merchant_id_count_
7d"].threshold
  ?? 0;
```

Condition

```
COUNT_LIMIT > 0
  and
(event.email_merch_cnt_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions for the same Customer Email ({event.customer_email}) and MID ({event.merchant_id}) in the last week ({event.email_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR5

Comment

Decline payment if the number of transactions per Device IP and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR5]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].
threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr5_max_ip_merchant_id_count_1d"].thresh
```

```
old
  ?? 0;
```

Condition

```
  COUNT_LIMIT > 0
  and
  (event.devip_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

> Number of transactions for the same device ip ({event.device_ip}) and MID ({event.merchant_id}), in the last day ({event.devip_merch_cnt_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR7

Comment

> Decline payments if the number of transactions per Shipping Address and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR7]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merchant_id_count_
1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merc
hant_id_count_1d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr7_max_full_shipping_address_merchant_i
d_count_1d"].threshold
  ?? 0;
```

Condition

```
 COUNT_LIMIT > 0
  and
(event.shipaddr_merch_cnt_1d ?? 0) >= COUNT_LIMIT
// Full Shipping Address - Normalized and concatenated fields: shipping_addr_line1,
shipping_addr_line2, shipping_zip, shipping_city, shipping_region, shipping_country_code
```

Explanation

> Number of transactions per Shipping Address ({event.full_shipping_address}) and MID ({event.merchant_id}) in the last day ({event.shipaddr_merch_cnt_1d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR8

Comment

Decline payments if the number of transactions per Shipping Address and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR8]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merchant_id_count_
7d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merc
hant_id_count_7d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr8_max_full_shipping_address_merchant_i
d_count_7d"].threshold
  ?? 0;
```

Condition

```
 COUNT_LIMIT > 0
  and
(event.shipaddr_merch_cnt_7d ?? 0) >= COUNT_LIMIT
// Full Shipping Address - Normalized and concatenated fields: shipping_addr_line1,
shipping_addr_line2, shipping_zip, shipping_city, shipping_region, shipping_country_code
```

Explanation

Number of transactions for the same Shipping Address ({event.full_shipping_address}) and MID ({event.merchant_id}) in the last week ({event.shipaddr_merch_cnt_7d}) is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR9

Comment

Decline payment if the amount of transactions for the same Customer ID and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR9]

Variables

```
 AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_sum_amount_unif
ied_currency_1d"].threshold
  ??
```

```
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_su
m_amount_unified_currency_1d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr9_max_customer_id_merchant_id_sum_amou
nt_unified_currency_1d"].threshold
  ?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.cust_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;
```

Condition

```
 AMOUNT_LIMIT > 0
  and
(event.cust_merch_sum_amt_1d ?? 0) >= AMOUNT_LIMIT
```

Explanation

Amount of transactions for the same Customer ID ({event.customer_id}) and MID ({event.merchant_id}), in the last day ({localvar.AMOUNT_CONVERTED} {event.currency_code}), is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR10

Comment

Decline payment if the amount of transactions for the same Customer ID and MID, in the past week, is above a set threshold

Actions

- [All-Payments-Velocity-FR10]

Variables

```
 AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_sum_amount_uni
fied_currency_7d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_s
um_amount_unified_currency_7d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr10_max_customer_id_merchant_id_sum_amo
unt_unified_currency_7d"].threshold
  ?? 0;
//for explanation
AMOUNT_CONVERTED = ((event.cust_merch_sum_amt_7d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0) /100;
```

Condition

```
 AMOUNT_LIMIT > 0
and
event.cust_merch_sum_amt_7d ?? 0 >= AMOUNT_LIMIT
```

Explanation

Amount of transactions for the same Customer ID and MID ({event.merchant_id}) in the last week ({localvar.AMOUNT_CONVERTED}) is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR11

Comment

Decline payment if the amount of transactions for the same Customer Email and MID, in the past day, is above a set threshold.

Actions

- [All-Payments-Velocity-FR11]

Variables

```
AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_FR11_max_customer_email_merchant_id_sum_amount_unified_currency_1d"].threshold
  ?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.email_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;
```

Condition

```
AMOUNT_LIMIT > 0
and
(event.email_merch_sum_amt_1d ?? 0) >= AMOUNT_LIMIT
```

Explanation

Amount of transactions for the same Customer Email and MID ({event.merchant_id}) in the last day ({localvar.AMOUNT_CONVERTED} {event.currency_code}) is equal or above limit ({localvar.AMOUNT_LIMIT} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

All-Payments-Velocity-FR12

Comment

Decline payment if the amount of transactions for the same Customer Email and MID, in the past week, is above a set threshold.

Actions

- [All-Payments-Velocity-FR12]

Variables

```
 AMOUNT_LIMIT =

tenanted_rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
  ??
merchant_risk_levels__rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
  ??
global_default__rules_thresholds["all_payments_velocity_fr12_max_customer_email_merchant_id_sum_amount_unified_currency_7d"].threshold
  ?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.email_merch_sum_amt_7d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0)/ 100;
```

Condition

```
 AMOUNT_LIMIT > 0
  and
(event.email_merch_sum_amt_7d ?? 0) >= AMOUNT_LIMIT
```

Explanation

> Amount of transactions for the same Customer Email and MID ({event.merchant_id}) in the last week ({localvar.AMOUNT_CONVERTED} {event.currency_code}) is equal or above limit ({localvar.AMOUNT_LIMIT} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR13

Comment

> Decline payment if the card has a number of failed transactions, in the last 24 hours, above a set threshold.

Actions

- [Cards-Velocity-FR13]

Variables

```
 COUNT_LIMIT =
  tenanted_rules_thresholds["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
  ?? merchant_risk_levels__rules_thresholds
["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
  ??
global_default__rules_thresholds["cards_velocity_fr13_max_failed_payment_card_id_count_1d"].threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_card_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Card ({event.payment_card_id}) failed transactions ({event.fail_card_cnt_1d}), in the last day, are equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR14

Comment

Decline payment if the number of distinct cards with failed payments per MID, in the last minute, is above a set threshold

Actions

- [Cards-Velocity-FR14]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ??
global_default__rules_thresholds["cards_velocity_fr14_max_fail_dist_card_merch_1min"].threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_dist_card_merch_1min?? 0) >= COUNT_LIMIT
```

Explanation

Distinct failed Card transactions per MID ({event.fail_dist_card_merch_1min}), in the last minute, is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR15

Comment

Decline payment if the card has a number of transactions, in the last day, above a set threshold.

Actions

- [Cards-Velocity-FR15]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_count_1d"].thresh
old
  ??
merchant_risk_levels__rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_coun
t_1d"].threshold
  ??
global_default__rules_thresholds["cards_velocity_fr15_max_payment_card_id_merchant_id_count_1d"].
threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.card_merch_cnt_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of transactions ({event.card_merch_cnt_1d}) for the same Card ID ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day, is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-FR16

Comment

Decline payment if the amount of transactions per Card ID and MID, in the past day, is above a set threshold.

Actions

- [Cards-Velocity-FR16]

Variables

```
AMOUNT_LIMIT =
  tenanted_rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
  ??
```

```
merchant_risk_levels__rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
  ?? global_default__rules_thresholds["cards_velocity_fr16_card_merch_sum_amt_1d"].threshold
  ?? 0;

//for explanation
AMOUNT_CONVERTED = ((event.card_merch_sum_amt_1d ?? 0) * 1.0) /100;
AMOUNT_LIMIT_CONVERTED = ((AMOUNT_LIMIT) * 1.0) /100;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
AMOUNT_LIMIT > 0
and
(event.card_merch_sum_amt_1d?? 0) >= AMOUNT_LIMIT
```

Explanation

> Total amount transactions ({event.card_merch_cnt_1d}) for the same Card ID ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day, is equal or above limit ({localvar.AMOUNT_LIMIT_CONVERTED}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR17

Comment

> Alert payment and flag card to be limited if multiple approved transactions with the same amount, in the same MID, in the last day. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Behavior-FR17]
- [Limit/Review_Card]
- Alert

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["cards_behavior_fr17_max_appr_auth_payment_card_id_merchant_id_amount_
as_str_count_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_behavior_fr17_max_appr_auth_payment_card_id_mercha
nt_id_amount_as_str_count_1d"].threshold
  ??
global_default__rules_thresholds["cards_behavior_1_max_appr_auth_payment_card_id_merchant_id_amo
unt_as_str_count_1d"].threshold
  ?? 0;
AMOUNT_LIMIT =

global_default__rules_thresholds["cards_behavior_fr17_max_amount_unified_currency"].threshold ??
0;
//for explanation
AMOUNT_CONVERTED = ((event.amount_unified_currency ?? 0) * 1.0) /100;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and (event.payment_card_id ?? '' != '')
and COUNT_LIMIT > 0
and AMOUNT_LIMIT > 0
// Limits
and (event.appr_card_merch_amtstr_cnt_1d ?? 0) >= COUNT_LIMIT
and (event.amount_unified_currency ?? 0) >= AMOUNT_LIMIT // current transaction amount >= 20
```

Explanation

Card transactions with the same amount of {localvar.AMOUNT_CONVERTED} {event.currency_code}, in the same merchant ({event.appr_card_merch_amtstr_cnt_1d}), in the last 24h, are equal or above limit ({localvar.COUNT_LIMIT}). Card ID flagged to be limited.

Mark As

N/A

Cards-Behavior-FR18

Comment

Decline payment and block Customer Email if the number of distinct rejected cards per Customer Email, in the last 7 days, is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Behavior-FR18]
- [Auto_Decline_Email]

Variables

```
 COUNT_LIMIT =
global_default__rules_thresholds["cards_behavior_fr18_max_failed_customer_email_countdist_paymen
t_card_id_7d"].threshold ?? 0;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_email_dist_card_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct cards ({event.fail_email_dist_card_7d}) with failed transaction per Customer Email, in the last week, is equal or above limit ({localvar.COUNT_LIMIT}). Customer Email ({event.customer_email}) is flagged to be auto declined

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR19

Comment

Decline and block Device IP if the number of distinct rejected cards per Device IP, in the last day, is above a set threshold. Depends on 'List Management Service' to put the IP in the configured list.

Actions

- [Auto_Decline_IP]
- [Cards-Behavior-FR19]

Variables

```
COUNT_LIMIT =
global_default__rules_thresholds["cards_behavior_fr19_max_failed_ip_countdist_payment_card_id_1d"
].threshold ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_ip_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Cards with failed transactions per Device IP ({event.device_ip}), in the last day ({event.fail_ip_dist_card_1d}), are equal or above limit {localvar.COUNT_LIMIT}. Device IP ({event.device_ip}) is flagged to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR20

Comment

Decline payment if the number of distinct emails for the same card and MID, in the last day, is above a set threshold.

Actions

- [Cards-Behavior-FR20]

Variables

```
COUNT_LIMIT =

tenanted_rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_countdist_custome
r_email_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_coun
tdist_customer_email_1d"].threshold
  ??
```

```
global_default__rules_thresholds["cards_behavior_fr20_max_payment_card_id_merchant_id_countdist_customer_email_1d"].threshold
  ?? 0;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.card_merch_dist_email_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Customer Email per card ({event.payment_card_id}) and MID ({event.merchant_id}), in the last day ({event.card_merch_dist_email_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR21

Comment

Decline payment if the number of distinct failed card transactions per Customer Email and MID, in the last week, is above a set threshold

Actions

- [Cards-Behavior-FR21]

Variables

```
 COUNT_LIMIT =

tenanted_rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
  ??
global_default__rules_thresholds["cards_behavior_fr21_max_failed_customer_email_merchant_id_countdist_payment_card_id_7d"].threshold
  ?? 0;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.fail_email_merch_dist_card_7d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of cards with failed transactions per Email ({event.customer_email}) and MID ({event.merchant_id}), in the last 24h ({event.fail_email_merch_dist_card_7d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-FR22

Comment

Alert payment if the number of distinct cards for the same Device ID and MID, in the last day, is above a set threshold.

Actions

- [Limit/Review_Device]
- Alert
- [Cards-Behavior-FR22]

Variables

```
 COUNT_LIMIT =
  tenanted_rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold

  ?? global_default__rules_thresholds["cards_behavior_fr22_merch_device_dist_card_1d"].threshold
  ?? 0;
```

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.device_id ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.merch_device_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Cards per Device ID ({event.device_id}) and MID ({event.merchant_id}), in the last day ({event.merch_device_dist_card_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-Behavior-FR23

Comment

Alert payment if the number of distinct cards for the same Billing Email and MID, in the last day, is above a set threshold.

Actions

- Alert
- [Cards-Behavior-FR23]

Variables

```
COUNT_LIMIT =
  tenanted_rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].threshold
  ??
merchant_risk_levels__rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].thre
shold
  ??
global_default__rules_thresholds["cards_behavior_fr23_merch_bill_email_dist_card_1d"].threshold
  ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.billing_email ?? '' != '') and (event.merchant_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
(event.merch_bill_email_dist_card_1d ?? 0) >= COUNT_LIMIT
```

Explanation

Number of distinct Cards per Billing Email ({event.billing_email}) and MID ({event.merchant_id}), in the last day ({event.merch_bill_email_dist_card_1d}), is equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-MCC-FR24

Comment

Decline payment if the number of small amount card transactions per MID with listed MCCs, in the last 15min, is above a set threshold.

Actions

- [Cards-MCC-FR24]

Variables

```
COUNT_LIMIT
=global_default__rules_thresholds["cards_behavior_fr24_merch_small_amt_cnt_15min"].threshold ??
0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.merchant_id ?? '' != '') and (event.merchant_mcc ?? '' != '')
and
// Limits
```

```
COUNT_LIMIT > 0
and (event.merch_small_amt_cnt_15min ?? 0) >= COUNT_LIMIT // trx amount <= 10
and card_testing_mcc.contains(event.merchant_mcc) // e.g Digital Goods, TaxiCabs, Streaming
Services
```

Explanation

MCC is listed ({event.merchant_mcc}) and multiple small amount ({event.amount_unified_currency}) card payments per MID ({event.merchant_id}), in the last 15m ({event.merch_small_amt_cnt_15min}), are equal or above limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-MCC-FR25

Comment

Decline payment if MCC is restricted for this BIN

Actions

- [Cards-MCC-FR25]

Variables

N/A

Condition

```
 // Filters
event.payment_method_type == 'card'
and
(event.payment_card_bin ?? '' != '') and (event.merchant_mcc ?? '' != '')
and
// Limits
restricted_mcc_per_bin.contains(event.merchant_mcc)
```

Explanation

Card payment with MCC ({event.merchant_mcc}) that is restricted for this BIN ({event.payment_card_bin})

Mark As

The outcome of this rule is: **decline**

Cards-MCC-FR26

Comment

Decline payment if the transaction amount is above MCC limit.

Actions

- [Cards-MCC-FR26]

Variables

```
MCC_AMOUNT_LIMIT = mcc_limits[event.merchant_mcc].amt_limit ?? 0;

//for explanations
AMOUNT_CONVERTED = ((event.amount_unified_currency ?? 0) * 1.0) /100;
MCC_AMOUNT_LIMIT_CONVERTED = ((MCC_AMOUNT_LIMIT) * 1.0) / 100;
```

Condition

```
MCC_AMOUNT_LIMIT > 0
  and
(event.amount_unified_currency ?? 0) >= MCC_AMOUNT_LIMIT
```

Explanation

> Transaction amount ({localvar.AMOUNT_CONVERTED}) is equal or above this MCC limit ({localvar.MCC_AMOUNT_LIMIT_CONVERTED}) (MCC {event.merchant_mcc}).

Mark As

The outcome of this rule is: **decline**

Cards-CNP-FR27

Comment

> Alert payment and flag a card to be limited if it has made multiple CNP transactions in a high number of unique merchants, in the last 5 minutes. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Limit/Review_Card]
- Alert
- [Cards-CNP-FR27]

Variables

```
COUNT_LIMIT =
global_default__rules_thresholds["cards_cnp_fr28_max_auth_cnp_payment_card_id_countdist_merchant_
id_5m"].threshold ?? 0;
```

Condition

```
// Filters
event.payment_method_type == 'card'
and
(event.payment_card_id ?? '' != '')
and
// Limits
COUNT_LIMIT > 0
and
// Limits
(event.cnp_card_dist_merch_5min ?? 0) >= COUNT_LIMIT
```

Explanation

> Card CNP transactions in unique merchants, in the last 5 minutes ({event.cnp_card_dist_merch_5min}), are equal or over the limit ({localvar.COUNT_LIMIT}). Card ({event.payment_card_id}) flagged to be limited.

Mark As

N/A

Cards-Mismatch-FR28

Comment

Alert a card payment for listed MIDs if card country mismatch merchant country

Actions

- [Cards-Mismatch-FR28]
- Alert

Variables

N/A

Condition

```
 // Filters
event.payment_method_type == 'card'
and (event.payment_card_country_code ?? '' != '') and (event.merchant_country_code ?? '' != '')
and (event.merchant_id ?? '' != '')
//Limits
and listed_merchants__international_payments.contains(event.merchant_id)
and event.payment_card_country_code != event.merchant_country_code
```

Explanation

Mismatch between Card Country ({event.payment_card_country_code}) and Merchant Country ({event.merchant_country_code}) and MID ({event.merchant_id}) is limited to international card payments

Mark As

N/A

Cards-BinAttack-FR29

Comment

Ratio of distinct card_id and distinct expiry dates per BIN and Merchant in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-BinAttack-FR29]

Variables

```
 // curve7 (th7) is expected to have better results then the others (good compromise between
number of alerts and detection rate)
binAttackThreshold = 7;

// merch_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 7 then return event.merch_cardbin_cur7;
  end
end
```

Condition

```
 thresholdDecision
and !merchant_id_bin_attacks.contains(event.merchant_id ?? '') // Whitelisted Merchants to be
excluded from this rule
```

Explanation

The ratio between distinct card_id ({event.merch_cardbin_cntdist_card_10m}) and distinct exp_date ({event.merch_cardbin_cntdist_cardexpd}), per BIN/Merchant on the last 10m, is below limit { {localvar.binAttackThreshold} }

Mark As

The outcome of this rule is: **decline**

-
- [Out-of-the-Box Resources]
- Rules Projects
- Refund and OCT Rules

On this page

Refund and OCT Rules

Descriptionâ

A payment refund is the process in which a payment made to the merchant account for a specific transaction is sent back to the customer.

The Original Credit Transaction (OCT) is a transaction that can be used to send or "push" funds to an eligible card-based account, resulting in a credit of funds to the cardholder's account. The "push" payment framework speeds up the funds delivery. It differs from the traditional method, in which funds are credited to the receiver's bank account.

Rules Project Metadata

- *Target Variable:* decision

- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

OCT-Velocity-1

Comment

Decline OCT/Refunds if count and total amount exceeds threshold per Card & MID over 24h

Actions

- [OCT-Velocity-1]
- Alert

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_mid_24h_th"].threshold ?? 5000;
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_card_mid_24h_th"].threshold ?? 10;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amt_card_mid_24h / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_sum_amt_card_mid_24h >= amount_threshold
 and event.oct_rfd_count_card_mid_24h >= count_threshold
```

Explanation

Count({event.oct_rfd_count_card_mid_24h}) and amount({localvar.converted_amount}) exceeds threshold({localvar.count_threshold} and {localvar.converted_amount_threshold}) per Card({event.payment_card_id})/MID({event.merchant_id}) over 24h

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-2

Comment

Decline OCT/ Refunds if count and total amount exceeds per card & MID over 7 days

Actions

- [OCT-Velocity-2]
- Alert

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_mid_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_mid_7d_th"].threshold ?? 10000;
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_mid_7d_th"].threshold ??
```

```
global_default__rules_thresholds["oct_rfd_count_card_mid_7d_th"].threshold ?? 10;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amount_card_mid_7d / 100;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_sum_amount_card_mid_7d >= amount_threshold
and event.oct_rfd_count_card_mid_7d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_card_mid_7d}) and amount ({localvar.converted_amount}) exceeds threshold ({localvar.count_threshold} and {localvar.converted_amount_threshold}) per Card ({event.payment_card_id}) & MID ({event.merchant_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-3

Comment

Decline OCT/ Refunds if count and total amount exceeds per card & MID over 30 days

Actions

- [OCT-Velocity-3]
- Alert

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_mid_30d_th"].threshold ?? 30000;
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_card_mid_30d_th"].threshold ?? 30;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amt_card_mid_30d / 100;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_sum_amt_card_mid_30d >= amount_threshold
and event.oct_rfd_count_card_mid_30d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_card_mid_30d}) and total amount ({localvar.converted_amount}) exceeds threshold ({localvar.count_threshold} and {localvar.converted_amount_threshold}) per Card ({event.payment_card_id}) & MID ({event.merchant_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-4

Comment

Decline OCT/ Refunds exceed count threshold per CardId for country over 7 days

Actions

- [OCT-Velocity-4]
- Alert

Variables

```
count_threshold =
by_merchant_tenanted_country_thresholds[event.payment_card_country_code].count_threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_count_card_country_7d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_card_country_7d}) exceeds threshold ({localvar.count_threshold}) per Card ({event.payment_card_id}) per country ({event.payment_card_country_code}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-5

Comment

Decline OCT/ Refunds exceed amount threshold per CardId for country over 7 days

Actions

- [OCT-Velocity-5]
- Alert

Variables

```
amount_threshold =
by_merchant_tenanted_country_thresholds[event.payment_card_country_code].amount_threshold ?? 10;
converted_amount_threshold = amount_threshold / 100;
converted_amount = event.oct_rfd_amount_card_country_7d / 100;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_amount_card_country_7d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold {localvar.converted_amount_threshold} per Card ({event.payment_card_id}) per country ({event.payment_card_country_code}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Scheme-6

Comment

Decline OCT/ Refunds for Visa, where MCC is listed to be declined

Actions

- Alert
- [OCT-Scheme-6]

Variables

```
is_decline_listed=global_negative_listed_mccs.contains( event.merchant_mcc );
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.norm_payment_card_brand == 'visa'
 and is_decline_listed == true
```

Explanation

Card brand is {event.norm_payment_card_brand} and {event.merchant_mcc} MCC is listed to be declined

Mark As

The outcome of this rule is: **decline**

OCT-Scheme-7

Comment

Decline OCT/ Refunds for Mastercard, where MCC is listed to be declined

Actions

- Alert
- [OCT-Scheme-7]

Variables

```
is_decline_listed=global_negative_listed_mccs.contains( event.merchant_mcc );
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.norm_payment_card_brand == 'mastercard'
 and is_decline_listed == true
```

Explanation

Card brand is {event.norm_payment_card_brand} and {event.merchant_mcc} MCC is listed to be declined

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-8

Comment

Decline OCT/Refunds count exceeds threshold per Merchant Id over 24 hours

Actions

- Alert
- [OCT-Velocity-8]

Variables

```
count_threshold = tenanted_rules_thresholds["oct_rfd_count_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_mid_24h_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_count_mid_24h >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_mid_24h}) exceeds threshold ({localvar.count_threshold}) per Merchant ID ({event.merchant_id}) over 24 hours

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-9

Comment

Decline OCT/Refunds count exceeds threshold per Merchant Id over 7 days

Actions

- Alert
- [OCT-Velocity-9]

Variables

```
count_threshold = tenanted_rules_thresholds["oct_rfd_count_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_mid_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_mid_7d_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_count_mid_7d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_mid_7d}) exceeds threshold ({localvar.count_threshold}) per Merchant ID ({event.merchant_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-10

Comment

Decline OCT/Refunds count exceeds threshold per Merchant Id over 30 days

Actions

- [OCT-Velocity-10]
- Alert

Variables

```
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_mid_30d_th"].threshold ?? 30;
```

Condition

```
 event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_mid_30d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_mid_30d}) exceeds threshold ({localvar.count_threshold}) per Merchant ID ({event.merchant_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-11

Comment

Decline OCT/Refunds total amount exceeds threshold per MID over 24 hours

Actions

- [OCT-Velocity-11]
- Alert

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_mid_24h_th"].threshold ?? 500000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amount_mid_24h / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_sum_amount_mid_24h >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Merchant ID ({event.merchant_id}) over 24 hours

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-12

Comment

Decline OCT/Refunds total amount exceeds threshold per MID over 7 days

Actions

- [OCT-Velocity-12]
- Alert

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_mid_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_mid_7d_th"].threshold ?? 1000000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amount_mid_7d / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_sum_amount_mid_7d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Merchant ID ({event.merchant_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-13

Comment

Decline OCT/Refunds total amount exceeds threshold per MID over 30 days

Actions

- [OCT-Velocity-13]
- Alert

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_mid_30d_th"].threshold ?? 3000000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amount_mid_30d / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_sum_amount_mid_30d >= amount_threshold
```

Explanation

> Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Merchant ID ({event.merchant_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-14

Comment

> Decline OCT/Refunds count exceeds threshold per Card Id over 24 hours

Actions

- [OCT-Velocity-14]
- Alert

Variables

```
count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_card_24h_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_card_24h >= count_threshold
```

Explanation

> Count ({event.oct_rfd_count_card_24h}) exceeds threshold ({localvar.count_threshold}) per Card ID ({event.payment_card_id}) over 24 hours

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-15

Comment

> Decline OCT/Refunds count exceeds threshold per Card Id over 7 days

Actions

- [OCT-Velocity-15]
- Alert

Variables

```
count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_card_7d_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_card_7d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_card_7d}) exceeds threshold ({localvar.count_threshold}) per Card ID ({event.payment_card_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-16

Comment

Decline OCT/Refunds count exceeds threshold per Card Id over 30 days

Actions

- [OCT-Velocity-16]
- Alert

Variables

```
count_threshold = tenanted_rules_thresholds["oct_rfd_count_card_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_card_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_card_30d_th"].threshold ?? 30;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_card_30d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_card_30d}) exceeds threshold ({localvar.count_threshold}) per Card ID ({event.payment_card_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-17

Comment

Decline OCT/Refunds total amount exceeds threshold per Card Id over 24 hours

Actions

- Alert
- [OCT-Velocity-17]

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_24h_th"].threshold ?? 500000;
converted_amount_threshold = amount_threshold / 100;
converted_amount = event.oct_rfd_sum_amount_card_24h / 100;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_sum_amount_card_24h >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Card ID ({event.payment_card_id}) over 24 hours

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-18

Comment

Decline OCT/Refunds total amount exceeds threshold per Card Id over 7 days

Actions

- Alert
- [OCT-Velocity-18]

Variables

```
amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_7d_th"].threshold ?? 1000000;
converted_amount_threshold = amount_threshold / 100;
converted_amount = event.oct_rfd_sum_amount_card_7d / 100;
```

Condition

```
event.event_type == 'info' and
(event.info_type == 'refund' or event.info_type == 'oct')
and event.oct_rfd_sum_amount_card_7d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Card ID ({event.payment_card_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-19

Comment

Decline OCT/Refunds total amount exceeds threshold per Card Id over 30 days

Actions

- Alert
- [OCT-Velocity-19]

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_card_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_card_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_card_30d_th"].threshold ?? 3000000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_sum_amount_card_30d / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_sum_amount_card_30d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per Card ID ({event.payment_card_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-20

Comment

Decline OCT/ Refunds count exceed threshold per BIN/Merchant Id over 24h

Actions

- [OCT-Velocity-20]
- Alert

Variables

```
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_bin_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_bin_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_bin_mid_24h_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_bin_mid_24h >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_bin_mid_24h}) exceeds threshold ({localvar.count_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 24h

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-21

Comment

Decline OCT/ Refunds count exceed threshold per BIN/Merchant Id over 7d

Actions

- [OCT-Velocity-21]
- Alert

Variables

```
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_bin_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_bin_mid_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_bin_mid_7d_th"].threshold ?? 10;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_bin_mid_7d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_bin_mid_7d}) exceeds threshold ({localvar.count_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-22

Comment

Decline OCT/ Refunds count exceed threshold per BIN/Merchant Id over 30d

Actions

- [OCT-Velocity-22]
- Alert

Variables

```
 count_threshold = tenanted_rules_thresholds["oct_rfd_count_bin_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_count_bin_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_count_bin_mid_30d_th"].threshold ?? 30;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_count_bin_mid_30d >= count_threshold
```

Explanation

Count ({event.oct_rfd_count_bin_mid_30d}) exceeds threshold ({localvar.count_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-23

Comment

Decline OCT/ Refunds total amount exceed threshold per BIN/Merchant Id 24h

Actions

- [OCT-Velocity-23]
- Alert

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_bin_mid_24h_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_bin_mid_24h_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_bin_mid_24h_th"].threshold ?? 500000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_amount_bin_mid_24h / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_amount_bin_mid_24h >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 24h

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-24

Comment

Decline OCT/ Refunds total amount exceed threshold per BIN/ Merchant Id 7d

Actions

- [OCT-Velocity-24]
- Alert

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_bin_mid_7d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_bin_mid_7d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_bin_mid_7d_th"].threshold ?? 1000000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_amount_bin_mid_7d / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_amount_bin_mid_7d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 7 days

Mark As

The outcome of this rule is: **decline**

OCT-Velocity-25

Comment

Decline OCT/ Refunds total amount exceed threshold per BIN/ Merchant Id 30d

Actions

- [OCT-Velocity-25]
- Alert

Variables

```
 amount_threshold = tenanted_rules_thresholds["oct_rfd_amount_bin_mid_30d_th"].threshold ??
merchant_risk_levels__rules_thresholds["oct_rfd_amount_bin_mid_30d_th"].threshold ??
global_default__rules_thresholds["oct_rfd_amount_bin_mid_30d_th"].threshold ?? 3000000;
 converted_amount_threshold = amount_threshold / 100;
 converted_amount = event.oct_rfd_amount_bin_mid_30d / 100;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and event.oct_rfd_amount_bin_mid_30d >= amount_threshold
```

Explanation

Total amount ({localvar.converted_amount}) exceeds threshold ({localvar.converted_amount_threshold}) per BIN({event.payment_card_bin}) and Merchant ID ({event.merchant_id}) over 30 days

Mark As

The outcome of this rule is: **decline**

OCT-PosNeg-26

Comment

Decline OCT/refund if Card Id is in a PosNeg/Sanctioned list

Actions

- Alert
- [OCT-PosNeg-26]

Variables

```
card_id_listed_value = by_merchant_tenanted_posneg_listed_cards[event.payment_card_id].value;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and (event.payment_card_id ?? '') != ''
  and card_id_listed_value == 'decline'
```

Explanation

Card ID ({event.payment_card_id}) is listed by the Tenant to be auto decline.

Mark As

The outcome of this rule is: **decline**

OCT-PosNeg-27

Comment

Approve OCT/refund if Card Id is listed to be auto-approved globally or per MID.

Actions

- [OCT-PosNeg-27]
- Alert

Variables

```
card_id_listed_value = by_merchant_tenanted_posneg_listed_cards[event.payment_card_id].value;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and (event.payment_card_id ?? '') != ''
  and card_id_listed_value == 'approve'
```

Explanation

Card ID ({event.payment_card_id}) is listed by the Tenant to be auto-approved.

Mark As

The outcome of this rule is: **decline**

OCT-PosNeg-28

Comment

Decline OCT/refund if Merchant Id is in a PosNeg/Sanctioned list

Actions

- [OCT-PosNeg-28]
- Alert

Variables

```
merchant_id_listed_value = by_merchant_tenanted_posneg_listed_cards[event.merchant_id].value;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and (event.payment_card_id ?? '') != ''
  and merchant_id_listed_value == 'decline'
```

Explanation

Card ID ({event.merchant_id}) is listed by the Tenant to be auto decline.

Mark As

The outcome of this rule is: **decline**

OCT-PosNeg-29

Comment

Approve OCT/refund if Merchant Id is listed to be auto-approved.

Actions

- [OCT-PosNeg-29]
- Alert

Variables

```
merchant_id_listed_value = by_merchant_tenanted_posneg_listed_cards[event.merchant_id].value;
```

Condition

```
event.event_type == 'info' and
 (event.info_type == 'refund' or event.info_type == 'oct')
 and (event.payment_card_id ?? '') != ''
  and merchant_id_listed_value == 'approve'
```

Explanation

Card ID ({event.merchant_id}) is listed by the Landlord to be auto-approve.

Mark As

The outcome of this rule is: **decline**

PosNeg Rules

Descriptionâ

PosNeg Rules allow you to mark transactions that have entities in PosNeg Lists as "fraud" or "not fraud". By default, the solution contains rules that mark transactions as âdeclinedâ. These rules act only on Payment Initiation events. For each rule, check the filter inside the condition code block to know which entities are being tested.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* MULTI_TENANT_SHARED

Rules in this rules project:

Card-Decline-PN1

Comment

Decline payment if Card ID is listed to be auto declined globally or per MID

Actions

- [Card-Decline-PN1]

Variables

```
card_id_listed_value = (event.posneg_card_value ??
tenanted_posneg_listed_cards[event.payment_card_id ?? ''].value) ?? '';
```

Condition

```
//Filters
(event.payment_card_id ?? '') != ''
and card_id_listed_value == 'decline'
```

Explanation

Card ID {event.payment_card_id} is listed by the Landlord to be auto decline.

Mark As

The outcome of this rule is: **decline**

Card-Approve-PN2

Comment

Approve payment if Card ID is listed to be auto-approved globally or per MID.

Actions

- [Card-Approve-PN2]

Variables

```
card_id_listed_value = (event.posneg_card_value ??
tenanted_posneg_listed_cards[event.payment_card_id ?? ''].value) ?? '';
```

Condition

```
//Filters
(event.payment_card_id ?? '') != ''
and card_id_listed_value == 'approve'
```

Explanation

Card ID {event.payment_card_id} is listed by the Landlord to be auto-approve.

Mark As

The outcome of this rule is: **approve**

Card-Review-PN3

Comment

Alert payment if Card ID is listed to be limited globally or per MID.

Actions

- Alert
- [Card-Review-PN3]

Variables

```
card_id_listed_value = (event.posneg_card_value ??
tenanted_posneg_listed_cards[event.payment_card_id ?? ''].value) ?? '';
```

Condition

```
//Filters
(event.payment_card_id ?? '') != ''
and
(card_id_listed_value ?? '') == 'review'
```

Explanation

Card ID {event.payment_card_id} is listed by the Landlord to be limited.

Mark As

N/A

Email-Decline-PN4

Comment

Decline payment if Customer Email is listed to be auto declined globally or per MID.

Actions

- [Email-Decline-PN4]

Variables

```
email_listed_value = (event.posneg_customer_email_value ??
tenanted_posneg_listed_customer_email[event.customer_email ?? ''].value) ?? '';
```

Condition

```
//Filters
(event.customer_email ?? '') != ''
and
email_listed_value == 'decline'
```

Explanation

Customer Email ({event.customer_email}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

Email-Approve-PN5

Comment

Approve payment if Customer Email is listed to be auto-approved globally or per MID.

Actions

- [Email-Approve-PN5]

Variables

```
email_listed_value = (event.posneg_customer_email_value ?? tenanted_posneg_listed_customer_email
[event.customer_email ?? ''].value) ?? '';
```

Condition

```
//Filters
(event.customer_email ?? '') != ''
and
email_listed_value == 'approve'
```

Explanation

Customer Email ({event.customer_email}) is listed by the Landlord to be auto-approved.

Mark As

The outcome of this rule is: **approve**

Device-IP-Decline-PN6

Comment

Decline payment if Device IP is listed to be auto declined globally or per MID.

Actions

- [Device-IP-Decline-PN6]

Variables

```
device_ip_listed_value = (event.posneg_device_ip_value ?? tenanted_posneg_listed_device_ip
[event.device_ip ?? ''].value) ?? '';
```

Condition

```
(event.device_ip ?? '') != ''
and
device_ip_listed_value == 'decline'
```

Explanation

Device IP ({event.device_ip}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

MID-Decline-PN7

Comment

Decline payment if Merchant ID is listed to be auto declined.

Actions

- [MID-Decline-PN7]

Variables

N/A

Condition

```
 (event.merchant_id ?? '') != ''
and
(event.posneg_mid_value ?? '') == 'decline'
```

Explanation

Merchant ID ({event.merchant_id}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

MID-Approve-PN8

Comment

Approve payment if Merchant ID is listed to be auto-approved.

Actions

- [MID-Approve-PN8]

Variables

N/A

Condition

```
 (event.merchant_id ?? '') != ''
and
(event.posneg_mid_value ?? '') == 'approve'
```

Explanation

Merchant ID ({event.merchant_id}) is listed by the Landlord to be auto-approve.

Mark As

The outcome of this rule is: **approve**

Terminal-ID-Decline-PN9

Comment

Decline payment if Terminal ID is listed to be auto declined globally or per MID.

Actions

- [Terminal-ID-Decline-PN9]

Variables

```
terminal_id_listed_value = (event.posneg_terminal_id_value ??
tenanted_posneg_listed_terminals[event.terminal_id ?? ''].value) ?? '';
```

Condition

```
(event.terminal_id ?? '') != ''
and
terminal_id_listed_value == 'decline'
```

Explanation

Terminal ID ({event.terminal_id}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

Terminal-ID-Approve-PN10

Comment

Approve payment if Terminal ID is listed to be auto-approved globally or per MID.

Actions

- [Terminal-ID-Approve-PN10]

Variables

```
terminal_id_listed_value = (event.posneg_terminal_id_value ?? tenanted_posneg_listed_terminals
[event.terminal_id ?? ''].value) ?? '';
```

Condition

```
(event.terminal_id ?? '') != ''
and
terminal_id_listed_value == 'approve'
```

Explanation

Terminal ID ({event.terminal_id}) is listed by the Landlord to be auto-approve.

Mark As

The outcome of this rule is: **approve**

Device-ID-Decline-PN11

Comment

Decline payment if Device ID is listed to be auto declined globally or per MID.

Actions

- [Device-ID-Decline-PN11]

Variables

```
 device_id_listed_value = (event.posneg_device_id_value ?? tenanted_posneg_listed_devices
[event.device_id ?? ''].value) ?? '';
```

Condition

```
 (event.device_id ?? '') != ''
and
device_id_listed_value == 'decline'
```

Explanation

Device ID ({event.device_id}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

Device-ID-Approve-PN12

Comment

Approve payment if Device ID is listed to be auto-approved globally or per MID.

Actions

- [Device-ID-Approve-PN12]

Variables

```
 device_id_listed_value = (event.posneg_device_id_value ?? tenanted_posneg_listed_devices
[event.device_id ?? ''].value) ?? '';
```

Condition

```
 (event.device_id ?? '') != ''
and
device_id_listed_value == 'approve'
```

Explanation

Device ID ({event.device_id}) is listed by the Landlord to be auto-approved.

Mark As

The outcome of this rule is: **approve**

Device-ID-Review-PN13

Comment

Alert payment if Device ID is listed to be limited globally or per MID.

Actions

- [Device-ID-Review-PN13]
- Alert

Variables

```
device_id_listed_value = (event.posneg_device_id_value ?? tenanted_posneg_listed_devices
[event.device_id ?? ''].value) ?? '';
```

Condition

```
(event.device_id ?? '') != ''
and
device_id_listed_value == 'review'
```

Explanation

Device ID ({event.device_id}) is listed by the Landlord to be limited.

Mark As

N/A

Card-Country-Decline-PN14

Comment

Decline payment if Card Country is listed to be auto declined globally or per MID.

Actions

- [Card-Country-Decline-PN14]

Variables

```
card_country_listed_value = (event.posneg_card_country_value ??
tenanted_posneg_listed_card_country [event.payment_card_country_code ?? ''].value) ?? '';
```

Condition

```
(event.payment_card_country_code ?? '') != ''
and
card_country_listed_value == 'decline'
```

Explanation

Card Country ({event.payment_card_country_code}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

Card-BIN-Decline-PN15

Comment

Decline payment if Card BIN is listed to be auto declined globally or per MID.

Actions

- [Card-BIN-Decline-PN15]

Variables

```
card_bin_listed_value = (event.posneg_card_bin_value ?? tenanted_posneg_listed_card_bin
[event.payment_card_bin ?? ''].value) ?? '';
```

Condition

```
(event.payment_card_bin ?? '') != ''
and
card_bin_listed_value == 'decline'
```

Explanation

Card BIN ({event.payment_card_bin}) is listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

Ship-Addr-Decline-PN16

Comment

Decline payment if full Shipping Address is listed to be auto declined globally or per MID.

Actions

- [Ship-Addr-Decline-PN16]

Variables

```
norm_shipp_addr_listed_value = (event.posneg_norm_shipp_addr_line ??
tenanted_listed_posneg_shipping_address [event.norm_shipp_addr_line ?? ''].value) ?? '';
full_shipp_addr_listed_value = (event.posneg_full_ship_addr_value ??
tenanted_listed_posneg_shipping_address [event.full_shipping_address ?? ''].value) ?? '';
```

Condition

```
((event.norm_shipp_addr_line != '') and norm_shipp_addr_listed_value == 'decline') // Norm Shipp
Addr Line is equal to the following normalized and concateneted fields: addr_line1 and addr_line2
or
((event.full_shipping_address != '') and full_shipp_addr_listed_value == 'decline') // Full
Shipping Address is equal to the following normalized and concateneted fields: addr_line1,
addr_line2, ZIP, City, Region and Country
```

Explanation

Normalized Shipp Addr Lines ({event.norm_shipp_addr_line}) or Full Ship Address ({event.full_shipping_address}) are listed by the Landlord to be auto declined.

Mark As

The outcome of this rule is: **decline**

MID-Review-PN17

Comment

Alert payment if Merchant ID is listed to be limited.

Actions

- [MID-Review-PN17]
- Alert

Variables

N/A

Condition

```
 (event.merchant_id ?? '') != ''
and
(event.posneg_mid_value ?? '') == 'review'
```

Explanation

Merchant ID ({event.merchant_id}) is listed by the Landlord to be limited.

Mark As

N/A

Self-Service Rules

Descriptionâ

Standard rules written in Pulse are normally developed and tested in staging environments to ensure that the work in development does not affect the work in production. These rules are developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production. At the same time, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one. Self-service rules (SSR) allow you to fill this gap, and have an active rule in production in a few minutes. With this capability, you can write a rule in Case Manager, test and publish it directly to production. As an example, you can use SSRs to create a rule with a different threshold than the one already defined in a standard rule and publish it in production to have it quickly live. Therefore, this rules project is empty, and is used in the Pulse workflow to receive the SSRs written in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

- Operational Work
- Analytics Dashboards Guide

Analytics Dashboards Guide

The analytics dashboards include metrics that provide you with real-time charts with information about fraud prevention agents' performance and rule monitoring. Based on these insights, you can improve workload management, or write self-service rules to quickly block a fraudulent behavior.

This guide provides you with the following information:

- Agent performance - alerts dashboard
- Agent performance - cases dashboard
- Rule monitoring dashboard

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/analytics-dashboards/view-agent-performance-dashboard-alerts

-

- Operational Work
- Fraud Prevention Manager
- Analytics Dashboards Guide
- Agent Performance Review - Alerts

On this page

Agent Performance Review - Alerts

Fraud prevention managers need to regularly manage and measure their fraud prevention agents' performance on an individual level. The **Agent Performance Review - Alerts** dashboard measures fraud prevention agents' performance, such as the alert's average resolution time. The Transaction Fraud for PSPs' Analyst Performance dashboard addresses this need, and consequently allows managers to make decisions that enhance their agents' performance more efficiently.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by fields. For instance, the `cm.first_assignee_username` field allows you to filter by a specific agent name.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the field you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. To finalize the creation of your condition, enter a value.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

The **Analyst Performance Review - Alerts** dashboard provides other filters to refine your search such as:

- **Analyst**
- **Fraud label**
- **Fraud reason**
- **Status**
- **Analyst decision (MM only)**
- **Event type**
- **Feedzai channel**

Alertsâ

- **Number of alerts**: The total number of generated alerts.
- **Decision time since assigned**: The average decision time since alert was assigned.
- **Decision time since created**: The average decision time since alert was created.
- **Closed - events**: The total number of closed events.

- **Unreviewed - events**: The total number of unreviewed events.
- **In progress - events**: The tota number of in progress events.
- **Alerts by status**: The average decision time per agent (in seconds).
- **Unique count of merchants and number of alerts by analyst**: The total number of merchants assigned per agent, and the corresponding number of generated alerts.

caution

Note that selecting bars works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Alerts** dashboard.

- **Average decision time over the selected time range**: The average decision time (in seconds) over the defined interval. Hover over each bar to see the average decision time for each decision.
- **Average decision time by analyst**: The average decision time per analyst over a 12-hour window. By hovering over each dot, you can see which agent made the decision and how long it took them in seconds to reach a decision.

caution

Note that selecting bars or line dots works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Alerts** dashboard.

- **Average decision time by analyst**: The distribution of workload per agent over a 12-hour window.
- **Alert status over time**: The distribution of Closed, Unreviewed, and In Progress alerts over a 12-hour window.

- Operational Work
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- Agent Performance Review - Cases

On this page

Agent Performance Review - Cases

Fraud prevention managers need to regularly manage and measure their fraud prevention agents' performance on an individual level. The **Agent Performance Review - Cases** dashboard measures fraud prevention agents' performance, such as the case's average resolution time. The Transaction Fraud for PSPs' Analyst Performance dashboard addresses this need, and consequently allows managers to make decisions that enhance their agents' performance more efficiently.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by fields. For instance, the `cm.first_assignee_username` field allows you to filter by a specific agent name.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the field you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. To finalize the creation of your condition, enter a value.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

The **Analyst Performance Review - Cases** dashboard provides other filters to refine your search such as:

- **Analyst**
- **Source type**
- **Case state**: Closed, Pending review, or Review in progress

Casesâ

- **Cases by assignee**: The total number of created cases.
- **Decision time since created**: The average decision time since the case was created.
- **Average decision time over the selected time range**: The distribution of the average decision time within a 24-hour period.

- **Cases by assignee**: The number of cases per assignee.

- **Cases by status**: The number of cases in status **Closed**, **Pending**, and/or **Review in progress** over a 24-hour window.

caution

Note that selecting bars or line dots works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

- **Average decision time by analyst**: The average decision time per analyst over a 24-hour window.

caution

Note that selecting line dots works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Rule Monitoring

The rule monitoring dashboard identifies which rules are being triggered by the risk engine platform and which are not. It also provides information about rules hit rate and precision. Based on this information, you can write self-service rules to control and improve fraud prevention.

Understanding if the rules behave as expected, or quickly discovering which need to be adjusted, or even deleted, is critical to improve your fraud prevention system.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by fields. For instance, the `enrich.name: allpaymentsvelocityfr1` field allows you to filter by a specific rule.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An [index pattern] selects the data to use and allows you to define properties of fields.
2. Add the [field] you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. A **Value** field appears after selecting the operator. Enter a value to finalize the creation of your condition.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

The **Rule Monitoring** dashboard provides other filters to refine your search such as:

- **Rule block**
- **Rule owner**
- **Is self-service rule?**
- **Name**
- **Customer region**
- **Event type**: `info`, `merchant monitoring`, `payment`
- **Feedzai channel**: `mmchannel`, `payments`
- **Merchant ID**
- **Merchant name**
- **Merchant MCC**
- **Payment region**

Rule KPIsâ

- **Number of triggered rules**: The total number of rules triggered.
- **Number of transactions with triggered rules**: The total number of transactions that triggered rules.
- **Overall rule(s) false positives**: The false positive rate.
- **Overall rule(s) true positives**: The true positive rate.

- **Total transactions amount by rule name distributed over time**: The total transaction amount and the corresponding triggered rule(s) in the selected interval over a defined period (e.g. 7 hours).
- **Number of transactions by rule name distributed over time**: The number of transactions and the corresponding triggered rule(s) in the selected interval over a defined period (e.g. 7 hours).

- **Total transaction amount by rule name**: The total transaction amount and the corresponding triggered rule.
- **Number of transactions by rule name**: The number of transactions that triggered a specific rule.

caution

Note that selecting bars or line dots works like a filter, and therefore this action results in changes to all the results of the **Rule Monitoring** dashboard.

Rule detailsâ

The following table includes the top rules triggered and their respective transaction counts over a defined period.

Drill down by merchantâ

The table below shows the top triggered rules and their corresponding transaction counts per merchant over a defined period.

tip

- These are sortable tables. You can arrange data into ascending or descending order according to a column.
- By hovering over each **rule name**, **rule description**, **is SSR?**, **merchant ID**, or **merchant name**, you can use the **magnifying glass** icon to either filter out or filter by a specific value.
- You can also export the table by selecting **Raw** or **Formatted**.

Rules triggered by terminal locationâ

The following geographical map shows the regions where specific rules triggered. Select a location and use the pagination to see all triggered rules for that specific region. Note that selecting a specific rule will result in changes to all the results of the **Rule Monitoring** dashboard.

- Operational Work
- Case Manager Guide

Case Manager Guide

PSPs

Merchants

overview-import

Case Manager is an alert management tool that enables fraud prevention agents to review transactions that fall within a risk threshold and raise alerts. Case Manager includes out-of-the-box resources tailored to the Transaction Fraud for PSPs use case, which automate event processing and organize teams' workload efficiently.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/integration-guide/overview

-
- Operational Work
- Integration Guide

Integration Guide

This guide provides you with the relevant steps to successfully test the system.

- User Acceptance Testing: Learn how to test rules, and ensure the system is well configured / running properly.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/pulse/pulse-guide

Pulse Guide

The Pulse guide leads you on the recommended tasks you must perform in Pulse throughout the complete life cycle of your project. It contains instructions on what you should do to perform each task and how you should keep moving forward, often complemented with examples, and always providing more context and conceptual details.

Pulse focuses on a democratization of Data Science, which allows both data scientists and fraud prevention agents to use the platform. All of the the tasks in this journey will be about writing or improving the risk strategy for the transaction fraud use case, heavily influenced by Data Science and Machine Learning models, providing an understanding on how to work with rules and models.

This guide provides you with the following steps:

- Work with rules
- Build and use models

Rule Metrics

The following table lists the schema fields that are required for the PSP Rules Plan:

Field Name	Field Type	Description
amount	LONG	The total transaction amount, in cents. For currencies that don't have cents representation, multiply the value by 100 in order to guarantee processing consistency.
amount_unified_currency	LONG	Amount converted to cents for a given unified currency.
billing_email	STRING	The email from the billing information associated with this transaction.

card_presence_flag	BOOLEAN	Indicates if the transaction is CP or CNP.
customer_email	STRING	The email address of the end customer.
customer_id	STRING	ID of the end customer initiating the operation in the client's system.
device_id	STRING	Customer's device unique ID.
device_ip	STRING	The IP of the device used to perform the transaction. Typically from the end customer.
event_occurred_at	TIMESTAMP	Timestamp corresponding to the moment the current event occurred (i.e. not the original event in Info and Settlement events, with the exception of Pre-decided Info events; in milliseconds since the Unix epoch).
event_type	STRING	N/A
info_type	STRING	Indicates the sub-type of the info event. Possible values: dispute, refund, capture, void, not_honor, pre_decided, additional_info.
merchant_id	STRING	The ID of the transaction's recipient in the client's system.
payment_auth_status	STRING	The status of the authorization for post-authorization scoring. Pending only applied to 3D Secure payments.
payment_card_bin	STRING	The BIN (first 6 digits) of the card number.
payment_card_country_code	STRING	The country (ISO 3166 alpha-2 code) of the card's issuer.
payment_card_expiration	STRING	Specifies the card's expiration date (YYMM).
payment_card_id	STRING	A best effort identifier of this card.
payment_method_id	STRING	The ID of this payment method within the transaction. E.g. Klarna, PayPal, ApplePay, GPay.
payment_method_type	STRING	The payment method registered for the transaction at the time of the event. E.g. card, apm.
shipping_addr_line1	STRING	The first line of this shipping address.
shipping_addr_line2	STRING	The second line of this shipping address.
shipping_city	STRING	The city of this shipping address.
shipping_country_code	STRING	The country (ISO 3166 alpha-2 code) of this shipping address.
shipping_region	STRING	The region of this shipping address.
		The ZIP code of this shipping address.

shipping_zip	STRING	
terminal_id	STRING	Unique ID of the terminal that processed the transaction.

The following table shows the features in the standard plan together with the fields needed to calculate each of them:

Plan Field	**Field Type**	**Schema fields used**	**Description**
amount_as_str	STRING	amount	Payment amount written as a string.
full_shipping_address	STRING	shipping_addr_line1 shipping_addr_line2 shipping_city shipping_country_code shipping_regionÂ shipping_zip	Concatenation of all the fields composing the shipping address in a single string.
card_days_to_expire	LONG	event_occurred_at payment_card_expiration	Number of remaining days for the card to expire until the event date.
norm_shipp_addr_line	STRING	shipping_addr_line1 shipping_addr_line2	Concatenation of the first and second line of the shipping address in a single string.
sign_count	LONG	event_type info_type payment_auth_status	
posneg_card_value	STRING	payment_card_id	Value defined by a Posneg list indicating the action to be taken by the rules according to the payment_card_id used in the payment.
posneg_device_ip_value	STRING	device_ip	Value defined by a Posneg list indicating the action to be taken by the rules according to the device_ip used in the payment.
posneg_customer_email_value	STRING	customer_email	Value defined by a Posneg list indicating the action to be taken by the rules according to the customer_email used in the payment.
posneg_mid_value	STRING	merchant_id	Value defined by a Posneg list indicating the action to be taken by the rules according to the merchant_id involved in the payment.
posneg_terminal_id_value	STRING	terminal_id	Value defined by a Posneg list indicating the action to be taken by the rules according to the terminal_id used in the payment.

posneg_device_id_value	STRING	device_id	Value defined by a Posneg list indicating the action to be taken by the rules according to the device_id used in the payment.
posneg_card_country_value	STRING	payment_card_country_code	Value defined by a Posneg list indicating the action to be taken by the rules according to the payment_card_country_code.
posneg_card_bin_value	STRING	payment_card_bin	Value defined by a Posneg list indicating the action to be taken by the rules according to the payment_card_bin used in the payment.
posneg_full_ship_addr_value	STRING	full_shipping_address	Value defined by a Posneg list indicating the action to be taken by the rules according to the full_shipping_address used in the payment.
posneg_norm_shipp_addr_line	STRING	norm_shipp_addr_line	Value defined by a Posneg list indicating the action to be taken by the rules according to the norm_shipp_addr_line used in the payment.
small_amount_trn_value	DOUBLE	-	Small amount transaction value defined in the Global Default - Rules Thresholds.
email_merch_sum_amt_7d	LONG	amount_unified_currency customer_email event_type merchant_id	Sum amount involved in payments performed by a given customer_email to a merchant_id over the last 7 days.
email_merch_cnt_7d	LONG	customer_email event_type merchant_id	Number of payments performed by a customer_email to a merchant_id over the last 7 days.
fail_email_merch_dist_card_7d	LONG	customer_email event_type merchant_id payment_auth_status payment_card_id	Number of distinct payment_card_ids involved in failed payment attempts performed by a given customer_email to a given merchant_id over the last 7 days.
email_merch_sum_amt_1d	LONG	amount_unified_currency customer_email event_type merchant_id	Sum amount involved in payments performed by a given customer_email to a merchant_id over the past day.

email_merch_cnt_1d	LONG	customer_email event_type merchant_id	Number of payments performed by a customer_email to a merchant_id over the past day.
fail_email_dist_card_7d	LONG	event_type customer_email payment_auth_status payment_card_id	Number of distinct payment_card_ids involved in failed payment attempts performed by a given customer_email over the last 7 days.
cust_merch_sum_amt_7d	LONG	amount_unified_currency customer_id event_type merchant_id	Sum amount involved in payments performed by a given customer_id to a merchant_id over the last 7 days.
cust_merch_cnt_7d	LONG	customer_id event_type merchant_id	Number of payments performed by a customer_id to a merchant_id over the last 7 days.
cust_merch_sum_amt_1d	LONG	amount_unified_currency customer_id event_type merchant_id	Sum amount involved in payments performed by a given customer_id to a merchant_id over the past day.
cust_merch_cnt_1d	LONG	customer_id event_type merchant_id	Number of payments performed by a customer_id to a merchant_id over the past day.
devip_merch_cnt_1d	LONG	device_ip event_type merchant_id	Number of payments performed through a device_ip to a merchant_id over the past day.
fail_ip_dist_card_1d	LONG	device_ip event_type merchant_id payment_auth_status payment_card_id	Number of distinct payment_card_ids involved in failed payment attempts performed by a given device_ip over the past day.
shipaddr_merch_cnt_7d	LONG	event_type full_shipping_address merchant_id	Number of payments to a merchant_id having the same full_shipping_address over the last 7 days.

shipaddr_merch_cnt_1d	LONG	event_type full_shipping_address merchant_id	Number of payments to a merchant_id having the same full_shipping_address over the past day.
merch_bill_email_dist_card_1d	LONG	billing_email event_type merchant_id payment_card_id	Number of distinct payment_card_ids used by a billling_email in payments to merchant_id over the past day.
merch_device_dist_card_1d	LONG	device_id event_type merchant_id payment_card_id	Number of distinct payment_card_ids used by a device_id in payments to merchant_id over the past day.
merch_cardbin_cntDist_cardExpD	LONG	card_days_to_expire merchant_id payment_card_bin payment_card_expiration payment_card_id	Number of distinct payment_card_expirations associated to events involving a given merchant_id and a payment_card_bin over the last 10 minutes.
merch_cardbin_cntDist_card_10m	LONG	card_days_to_expire merchant_id payment_card_bin payment_card_expiration payment_card_id	Number of distinct payment_card_ids associated with events involving a given merchant_id and a payment_card_bin over the last 10 minutes.
card_merch_sum_amt_1d	LONG	amount_unified_currency event_type merchant_id payment_card_id	Sum amount involved in payments performed by a given payment_card_id to a merchant_id over the past day.
fail_dist_card_merch_1min	LONG	event_type merchant_id payment_auth_status payment_card_id	Number of distinct payment_card_ids involved in failed payment attempts to a given merchant_id over the past minute.
merch_small_amt_cnt_15min	LONG	amount_unified_currency event_type merchant_id	Number of card payments associated with a merchant_id having the amount_unified_currency <=Â small_amount_trn_value over the last 15 minutes.

		payment_method_type	
appr_card_merch_amtstr_cnt_1d	LONG	amount_as_str event_type info_type payment_auth_status payment_card_id merchant_id	
card_merch_dist_email_1d	LONG	customer_email event_type merchant_id payment_card_id	Number of distinct customer_emails associated with payment events involving a given merchant_id and a payment_card_id over the past day.
card_merch_cnt_1d	LONG	event_type merchant_id payment_card_id	Number payment events involving a given merchant_id and a payment_card_id over the past day.
fail_card_cnt_1d	LONG	event_type payment_auth_status payment_card_id	Number of failed attempts involving a given payment_card_id.
cnp_card_dist_merch_5min	LONG	card_presence_flag event_type merchant_id payment_card_id	Number distinct merchant_ids involved in CP payment events performed with the same payment_card_id.
apm_merch_cust_email_sum_24h	LONG	customer_email event_type merchant_id payment_method_id payment_method_type	Sum amount involved in APM payments performed by a given customer_email to a merchant_id using a given payment_method_id over the last 24 hours.
apm_merch_cust_email_cnt_24h	LONG	customer_email event_type merchant_id payment_method_id payment_method_type	Number of APM payments performed by a given customer_email to a merchant_id using a given payment_method_id over the last 24 hours.
merch_cardbin_cur7		merch_cardbin_cntDist_cardExpD	

		merch_cardbin_cntDist_card_10m	

Use Models

This section points you to the Pulse sections that explain how you can build models and AutoML runs.

Build modelsâ

Pulse supports your Data Science loop by enabling you to quickly train your models and evaluate the results over several iterations. A model project aggregates several iterations of a model and provides tools for you to assess whether the features you designed are useful, well constructed, etc. At each iteration, you are able to understand how close you are to reaching your projects goals and what actions you took to reach those results. Building and evaluating your models with Pulse involves:

- Sampling data
- Training a model

- Evaluating a model
- Tuning model parameters
- Comparing models

Standard feature planâ

Transaction Fraud for PSPs delivers an out-of-the-box standard feature plan. It contains a set of features identified by Feedzai from industry knowledge and experimentations with historical data.

The feature model can be used as a placeholder to rapidly create a ML model into the âTFP Scoring Modelâworkflow element, which serves as a placeholder for your ML model.

Feature Planâ

- **Name**: PSP Model Features Plan.
- **Output schema**: Features Plan Schema (Payments Workflow Schema + fields calculated on the PSP Model Features Plan). Only a model should use this schema.
- **Position on the workflow**: After Model Features Plan and before all the rules.

Modelâ

- **Name**: TFP Scoring Model.
- **Output schema**: Uses a simple schema to minimize TFP Scoring Model's processing impact. Note that, to start using a model, you should use the Features Plan Schema instead.
- **Placeholder model**: Always scores events with 0.
- **Position on the workflow**: Before the decision rules block.

Features are grouped into operations / data transformations according to their type, e.g. map operations are grouped into "Feature group 1".

The following table includes the standard features in the PSP Model Features Plan.

info

To see the features in the PSP Rules Metrics Plan, check the tables in the Rule Metrics page.

tip

Use the table's **horizontal scroll bar** to see all its columns.

Feature group 1	Feature group 2	Profiles	Feature group 3
dayOfMonth_event_occurred_at	cos_dayOfWeek_event_occurred_at	merchant_id_avg_amount_unified_currency_1_hour	probability_log_amo
dayOfWeek_event_occurred_at	cos_hour_event_occurred_at	merchant_id_avg_amount_unified_currency_1_day	probability_log_amo

emailName_billing_email		merchant_id_avg_log_amount_unified_currency_1_hour	probability_log_amo
norm_billing_email		merchant_id_avg_log_amount_unified_currency_1_day	ratio_merchant_id_
hour_event_occurred_at		merchant_id_countDistinct_customer_id_1_hour	ratio_merchant_id_
log_amount_unified_currency		merchant_id_countDistinct_merchant_mcc_1_hour	ratio_merchant_id_
month_event_occurred_at		merchant_id_countDistinct_payment_card_id_1_hour	ratio_merchant_id_
		merchant_id_countDistinct_payment_card_bin_1_hour	ratio_merchant_id_
		merchant_id_countDistinct_billing_email_1_hour	ratio_merchant_id_
		merchant_id_countDistinct_customer_id_1_day	ratio_merchant_id_
		merchant_id_countDistinct_merchant_mcc_1_day	ratio_merchant_id_
		merchant_id_countDistinct_payment_card_id_1_day	ratio_merchant_id_
		merchant_id_countDistinct_payment_card_bin_1_day	ratio_merchant_id_
		merchant_id_countDistinct_billing_email_1_day	ratio_merchant_id_
		merchant_id_count_1_hour	ratio_merchant_id_
		merchant_id_count_1_day	ratio_merchant_id_
		merchant_id_max_amount_unified_currency_1_hour	ratio_merchant_id_
		merchant_id_max_amount_unified_currency_1_day	ratio_merchant_id_
		merchant_id_stdDev_amount_unified_currency_1_hour	ratio_merchant_id_
		merchant_id_stdDev_amount_unified_currency_1_day	ratio_merchant_id_
		merchant_id_stdDev_log_amount_unified_currency_1_hour	

		merchant_id_stdDev_log_amount_unified_currency_1_day
		merchant_id_sum_amount_unified_currency_1_hour
		merchant_id_sum_amount_unified_currency_1_day
		merchant_id_timeSincePrev_event_occurred_at_1_hour
		merchant_id_timeSincePrev_event_occurred_at_1_day
		merchant_id_vel_1_hour
		merchant_id_vel_1_day
		merchant_id_avg_amount_unified_currency_1_month
		merchant_id_avg_log_amount_unified_currency_1_month
		merchant_id_count_1_month
		merchant_id_max_amount_unified_currency_1_month
		merchant_id_stdDev_amount_unified_currency_1_month
		merchant_id_stdDev_log_amount_unified_currency_1_month
		merchant_id_sum_amount_unified_currency_1_month
		merchant_id_timeSincePrev_event_occurred_at_1_month
		merchant_id_vel_1_month

The following sections help you build a model using the standard feature plan, and explain how models can improve rule performance.

Use the standard feature plan to build a modelâ

Use the out-of-the-box standard feature plan, i.e. **Model Features Plan**, to build a model and then run the Model Features Plan.

Import a data sourceâ

To train an algorithm and build a model, you need a set of events described through fields and labels. This dataset is used as input, over which the algorithm will iterate until it learns and recognizes patterns. These patterns allow to establish relationships between eventsâfeatures and the labels that the model intends to predict. The labels are typically referred to as classes.

The Transaction Fraud for PSPs use case was designed to handle events labeled as legitimate or fraudulent. In more specific terms, the dataset is expected to have a field `fraud` that indicates whether or not an event is fraudulent (`fraud == False` or `fraud == True`).

To import your dataset into Pulse, you must follow the procedure described in the Importing your Dataset into Pulse guide.

1. When importing your dataset, choose the option **Import from an Existing Schema** on the **Data Source Characterization** tab:

1. A pop up window will allow you to **Select the Pre Omnichannel Metrics Plan Schema** and confirm. Pulse will indicate if your dataset is compatible with the solutionâs input schema:

1. Save and create the data source.

If your raw data schema does not match the input data schema expected by the solution (i.e. **Pre Omnichannel Metrics Plan Schema**), you can modify your raw data. The Validating the Data Schema guide helps you validate and convert your data.

You can learn more in Cleaning up Data.

Run the standard feature planâ

Typically, you want to train your machine learning model on a dataset containing derived fields and historical profiles, or metrics. Historical profiles describe how a particular entity behaves and can be used to identify deviations or anomalies. Certain changes in the behavior may help you detect financial crime. For example, consider the scenario where a customer who only uses their credit card to pay small amounts on monthly streaming services, suddenly starts making daily purchases at much higher amounts than usual.

You can achieve this by building up your feature plan and generating a feature dataset. The Model Features Plan is designed to compute pre-defined derived fields and metrics. In this plan, an input data source is processed, metrics are computed on top of it, and the resulting data source is persisted.

To compute these derived fields and metrics for your dataset, you should execute the Model Features Plan using your dataset as the Plan input data source. The process of running the plan to generate the enhanced data source is described in the Generating the Features Dataset guide.

info

For better model performance, combine the features in the PSP Model Features Plan with the PSP Rules Plan.

Read the [Build and evaluate a model] section to learn how to run the standard feature plan in order to build your ML model section. The same process should be followed for the PSP Rules Plan in case you want to use its features in the model.

Data Science Platform considerationsâ

Within the data science platform there are some considerations to take into account. The following table highlights scalability concerns that are relevant when executing plans.

	Context	Implication
Spark Jobs & data volume	Metrics done in the Data Science Platform make use of Spark to distribute the workload across a Hadoop cluster.	

	Distributing metrics through a Spark job means that data will be moved around in the cluster in large quantities. Due to the nature of profiles, each entity's value (e.g. each card or account) will be processed in the same Spark machine (or executor).	Even though a metric may only need 2 or 3 fields to compute, Spark will still move the records with all their columns to the executors.
Spark Jobs & skewness	Very high skews in grouping entities (e.g. 10% of entities represent 90% of the total records), can make Spark Tuning more complex.	Avoid including highly skewed grouping entities in your metrics. If they cannot be avoided, consider the following: • Randomize special entities (e.g. government-related beneficiaries, gmail, etcâ¦). • Salt the largest entities: added cost of joining the features.

To understand more about the implications of distributed Spark Jobs within the Data Science Platform, check the relevant data partitioning documentation.

tip

Pulse allows you to configure data partitioning using flexible UI options to ensure that Spark performs data re-organization as homogenous and well distributed as possible. Data partitioning is a key step for processing distributed jobs in a Spark cluster spawned by plan executions to enable Pulse to efficiently handle large volumes of data. The data-aware optimized partitioning is enabled by default. You may see this functionality in the **Advanced Options** tab when configuring a Plan Execution (see image below). To learn more about it, read the [Data Partitioning for the Metrics Operator]() section.

Troubleshootingâ

1. Missing fields: The schema of the dataset to be evaluated may not correspond the schema expected by the Model Features Plan (Pre Omnichannel Metrics Plan Schema). Therefore, pay attention to field names, type, and order when importing a dataset.

If you are unable to include a particular field present in the Pre Omnichannel Metrics Plan Schema in the to-be-evaluated dataset, observe the following scenarios and their consequences:

- The missing field is not used by the plan
  - **Impact**: This scenario has little impact on the model creation pipeline, as it only requires the user to change the input schema used by the Plan.
  - **Workaround**: Duplicate the Model Features Plan and just modify the schema used in the Plan Settings, changing the Plan input schema to the one defined by your dataset. Don't forget to set the timestamp field to `event_occurred_at`. No Operator in the Plan should be modified.
- The missing field is used by the plan
  - **Impact**: This scenario may have a higher impact on the model creation pipeline, as it requires you to modify the Plan Operators. Since fields and metrics derived from missing fields cannot be created, they must be removed so that the plan can run. Note that by removing derived fields and metrics, the model performance can be affected.
  - **Workaround**: Duplicate the Model Features Plan to avoid losing the original Plan. In the duplicate plan, you must identify those to-be-created or -updated fields that depend on a given missing field, and consequently, cannot be calculated. As it will not be possible to generate these new fields or metrics, they should be removed from the Plan Operators. Modify the schema used in the Plan Settings, changing the Plan input schema to the one defined by your dataset. Don't forget to set the timestamp field to `event_occurred_at`.

info

When the `customer_email` field is not available in payment events, the PSP Rules Plan will coalesce the missing `customer_email` field to the `billing_email` field.

1. Grouping entities with several missing values: Suppose you want to compute profiles grouped by the `device_ip` field but, in your dataset, about half of the transactions donâ□□t have a device IP address. You start by adding a filter to the metric window, e.g. `device_ip != null`, but this is not enough.

   When computing the profile, all the transactions will still be loaded and distributed to the executors before the filter is applied. There will be a very large partition with all the missing IPs, which will likely crash the Spark job.

    1. Replace that missing value with something else, such as the transaction ID (`event_external_id`) or the event timestamp (`event_occurred_at`).
    2. Create a new field device_ip_salted in a Map Operator before the Metrics Operator.
    3. Assign it a value similar to `device_ip ?? event_external_id`.
    4. Update the metric window filter to `device_ip_salted != event_external_id`. This will still yield the same end result, but without generating a massive partition in the Spark job.

Edit Map Operatorâ

Learn more in:

- [Creating Derived Features]
- [Creating Historical Profiles]
- [Best Practices for Plans]

Build and evaluate a modelâ

As the output of the Model Features Plan, you will have a new dataset with the original features and the additional features created through the Plan. You should use this new dataset to train a model.

If the dataset you are going to use to build your model does not have all the data labeled, you can adopt one of the following strategies:

- Remove the unlabeled events from the dataset.

OR

- Coalesce the unlabeled events to a default value. For example, you may assume that transactions that were not deliberately marked as fraudulent are legitimate, and therefore, should have the fraud `label = False`.

A possible workaround for the missing labels is to run a Pulse Plan using a Map Operator to create a new label field or update an existing label field, as the following image shows:

Typically, data scientists use different data splits to train, validate, and optionally, test a model. There shouldn't be any overlap between the training, validation, and test sets. As a general machine learning guideline, these splits are 80/20/20, i.e., 80% of your data for training, 20% for validation, and 20% for testing (hold-out data set). Here, the split is enforced to be defined timewise, because the transformations are time-sensitive.

The training, validation, and test splits can be generated through JupyterLab or by creating and running a new Pulse Plan. If you choose to partition the data using a Pulse Plan, you will have to create a Filter Operator for each split that you want to generate.

As a usage example, imagine that you want to create three splits, e.g. one for training, one for validation, and one for test:

The condition to be observed for the creation of these splits is a temporal criterion based on the timestamp value of the `event_occurred_at` field.

This means that, for the three filters, you will have the following conditions:

- Training Split: `event_occurred_at < timestamp1`
- Validation Split: `event_occurred_at >= timestamp1 and event_occurred_at < timestamp2`
- Test Split: `event_occurred_at >= timestamp2`

info

The following condition must be observed: `timestamp1 < timestamp2`. If the dataset is extremely unbalanced, the same Pulse Plan can also be used to sample the most frequent class.

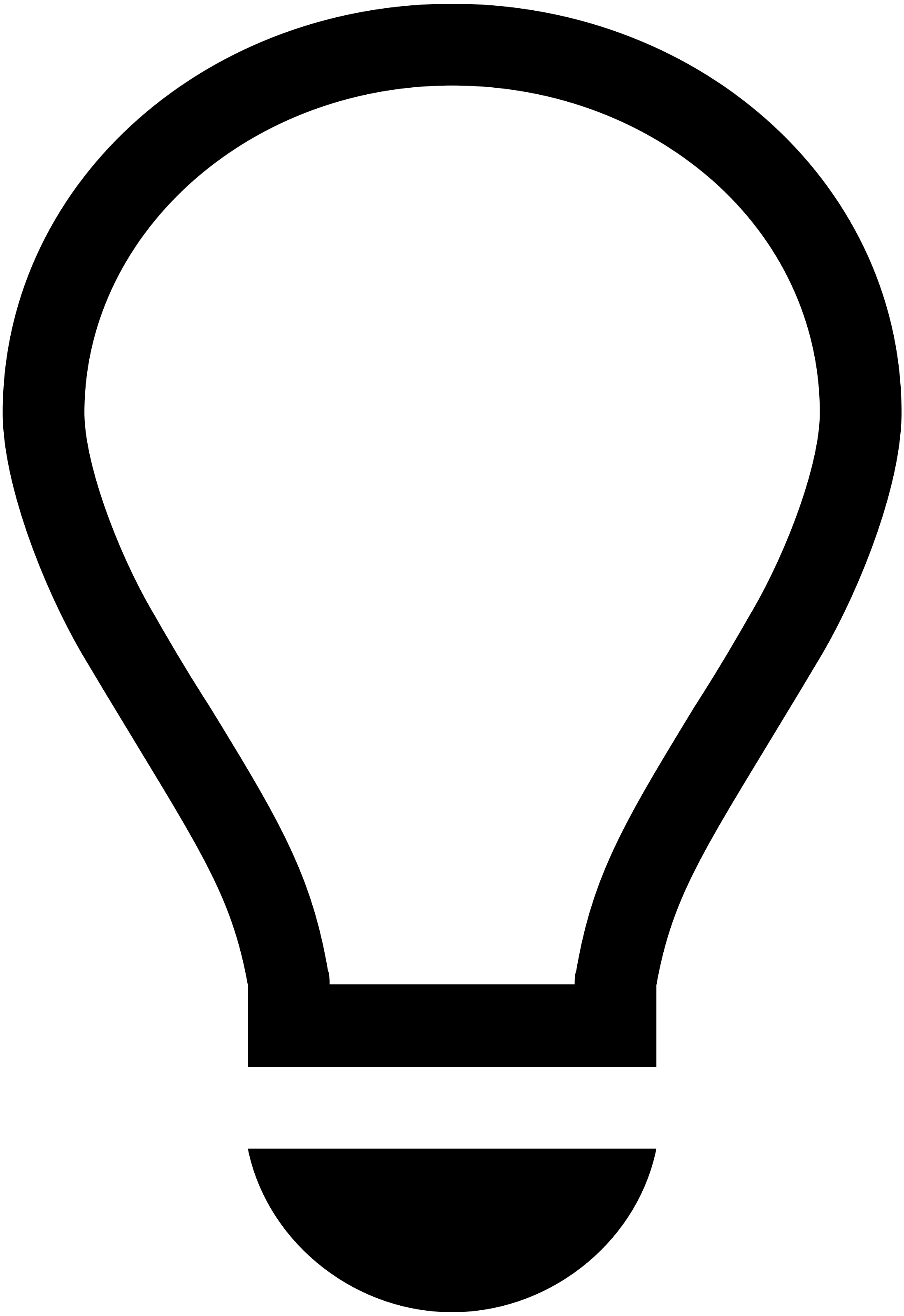
1. Separate the events belonging to each class using a Filter Operator.
2. Use a Sample Operator to reduce the number of events in the most frequent class.
3. The sampled majority class and the instances belonging to the minority class must pass through a Union Operator to compose the new dataset.

After running the splitting Plan in Pulse, the new dataset splits will be listed in the Data Sources section on the **Data Science** tab. The new dataset splits will have the same schemas and can be used in the usual training and evaluation model pipeline.

1. Use the training split to train your model. You may also use the validation split to validate your model. In the **Validation & Reporting** tab, select the Validation **Validation Data Source** type and choose the validation dataset split that you have just created.

Learn more in [Evaluating a Model](#).

1. Use the test split to evaluate your model.



tip

When training an algorithm, the features in the dataset are expected to have a different impact on the model performance. You may find features that contribute to the improvement of the model's performance and others that just add noise to the model. It is, therefore, recommended that you run an analysis on your dataset to [examine the statistics for each field](#). For example, categorical fields with only one category, e.g. empty value, should not be considered when training your model.

You can also [compute the Feature Importance] and remove the features that had a negative impact on the previous model iteration.

Learn more in:

- Tuning Model Parameters
- Comparing Models

How models improve rule performanceâ

What is the model score?â

A machine learning model is used to label new (unseen) events. In a classification scenario, where a binary model is used to predict fraud events, the model assigns a value (score) representing its confidence that a given event belongs to one of the classes (fraud and legit). Typically, a fraudulent event is characterized as having the field `fraud == True`, while legitimate events are represented by events having the field `fraud == False`.

The score generated by Pulse models corresponds to a probability of an event belonging to the positive class (`fraud == True`), thus ranging from 0 to 1. However, to make the interpretation easier, Pulse returns the score of a model multiplied by 1000. Thus, the score assigned by the model to the positive class will always vary between 0 and 1000. The greater the confidence of the model in labeling an event as `fraud`, the higher the score returned.

Where to position the model in the workflow?â

The machine learning model trained on the raw and derived features should be inserted into the **Payments Workflow** using a Scoring operator. Use the TFP Scoring Model operator that is included in the out-of-the-box workflow (see image below). This way, the model can use the derived fields and metrics that are generated by this plan.

When configuring your model, select the Model Project and the Model dropdown fields. Ensure that the target outcome of the model is compatible with the selected outcome. The following image presents an example of a model configuration:

How is the model score used in the workflow?â

In the **Transaction Fraud for PSPs Workflow**, the model score produced will be evaluated by decision rules. When evaluating the model score, the other rule blocks will take precedence over the model's decision. Therefore, if the PosNeg rule block triggers an `approve` for a payment event, the workflow will have the final outcome `approve`. Also, if any rules block triggers a decline for a payment event, the workflow will have the final outcome `decline`, regardless of the model result.

Work with Rules

Learn how to fully manage your rules, including:

- Understanding how rules work.
- Writing rules that are specific to your business.
- Evaluating and testing rules.
- Deploying rules to runtime.

As a data scientist or fraud prevention agent, you can leverage rules to detect and mitigate fraud events at scale in real time. The following user journey shows the different stages required to set up your rules and deploy them to runtime:

Understand rulesâ

This section provides you with an overview of how rules work to detect and prevent fraud with the resources that come out of the box with your application.

API response with Feedzai's recommendationââ

When a PSP invokes Feedzai's API to score a request via the payments endpoint, or merchant provisioning, Feedzai sends an API response containing a decision recommendation. This field helps you decide whether the payment is fraudulent or legitimate:

The event is processed by the risk engine platform, Pulse, which uses data science resources to evaluate the event. If you look at the result generated by rules, the API response contains a decision field with the recommendation values `approve` or `decline`, and the boolean alert field. The following section explains how both decision outcomes and alerts are generated.

How is Feedzai's recommendation generated?ââ

It is important to understand how rules process events and contribute to the API response to detect and prevent fraud efficiently. This section breaks down this life cycle in the following conceptual steps:

1. Mapping API fields to schema fields
2. Rule options in the Pulse application
3. Rule blocks and outcomes in the Pulse workflow
4. Manual revision in Case Manager
5. API response

1. Mapping API fields to Pulse schemaââ

The fields in the API request are mapped to a Pulse schema named Payments API Schema. Learn more about data schemas to better grasp what they are used for.

2. Rule options in the Pulse applicationââ

Before getting into the specific logic behind each event, let's have a closer look at two important rule settings. Each rule is configured with Actions and Mark As:

When a rule is triggered, Pulse sends the event to Case Manager as "Alerted" for manual revision. This also sets the alert field as `true` in the API response, thus allowing fraud prevention agents to focus their investigation only on suspicious activity.

- The **Actions** option includes the rule code (e.g. Device-ID-Review-PN13) so that it is possible to identify, in the API response's **Actions** array, which rules triggered.

- The **Mark As** option lets you select the outcome values that are configured beforehand in the Pulse workflow. When a rule is triggered, the **Mark As** value is written to the schema field that is associated with this rule outcome. For example, in this case, the `decline` value is written in the decisionâs internal field.

Different rules are defined with different outcomes according to the specificities of each event. The position of rules projects in the Pulse workflow, and the additional custom services dictate how the final recommendation is generated. This is further discussed in the next step.

3. Rule blocks and outcomes in the Pulse workflowââ

Each set of APIs contains a Pulse workflow (i.e. *Merchant Provisioning Workflow*, *Payments Workflow*, and *Reference Data Workflow*). The possible rule outcomes and the rules' processing order are defined in each workflow:

Payments workflow

How does the *Payments Workflow* generate the final outcome?

The Payments Workflow is configured with decision and fraud outcomes. The fraud outcome is generally configured to be used by machine learning models. This means that each rule contributes to the decision field, and the first rule triggering sets the final value.

4. Manual revision in Case Managerâ□□â□□

Payment events can generate alerts when one of the rules, configured with the Alert action, is triggered. The event is then displayed in Case Managerâ□□s Alerts page.

Fraud prevention agents investigate alerts in Case Manager and decide whether the event is fraudulent or legitimate. This decision is registered in the `feedback_value` field as `true` or `false` for `fraud` and `not fraud` (`feedback_label`), respectively. Then, the Feedback Rules project evaluates this result and makes a decision on adding entities, such as Card ID, to PosNeg Lists. The following payments can then be automatically `approved` or `declined` .

5. API responseâ□□â□□

At this point, you already understand how Feedzai scores an event. Note that, when you send an API request via Payments stream using one of its scoring endpoints, you will get an API response with the following key fields:

- `decision` : The final outcome generated by rules.
- `alert` : Whether the event generated an alert that is visible on the Alerts page in Case Manager.
- `score` : The numeric value a machine learning model generates to evaluate an event. Possible values range from 0 to 1000.
- `action_codes` : Every rule has an array of Actions including the name of the rule (e.g. [Device-IP-PN1]). The `action_codes` array includes all the actions of triggered rules to further understand which rules have triggered for each event.

Rule configurations in Pulseâ□□â□□

To define a rule in Pulse, you have the following main configurations available:

- **Variables**: Allows you to declare variables that are further used in the rule's condition including thresholds or transformation/conversion of the input value (e.g. variable that converts an amount from cents to US dollar). Variables are useful to avoid having large condition statements, which are error-prone and difficult to maintain.
- **Conditions**: The condition expression must return a boolean value that dictates whether the event triggers a rule.
- **Explanations**: Allows you to understand why a rule has triggered in a human-readable way. If an event triggers, the explanation is displayed in Case Manager to provide additional insights to fraud prevention agents.
- **Actions**: Operations that are performed when the rule is triggered. The standard values that help you meet your custom business logic are:
    - - Sends the event to Case Manager's Alerts page for manual revision.
    - - Flags entities such as cards, accounts, devices, among others, to be added to the PosNeg list with the decline value.
    - - Flags entities such as cards, accounts, devices, among others, to be added to the PosNeg list with the approve value.
- **Decisions**: The rule decision on the outcome of the transaction. The standard values that help you meet your custom business logic are:
    - The decision of this rule is to approve the event.
    - The decision of this rule is to decline the event.
    - This rule does not decide the event's outcome.

Custom business logicâ□□â□□

The combination of rule actions and decisions described in the previous section allows you to easily define custom business logic. For example, imagine that you want to define the transaction responses:

- **Soft decline**: This use case involves rules that decline the transaction but also generate an alert. The transaction should be displayed in Case Managerâ□□s Alert page for fraud prevention agent review.
- **Hard decline**: This use case involves rules that decline the transaction and do not generate an alert. The transaction wonâ□□t be displayed in Case Managerâ□□s Alert page for fraud prevention agent review.

- **Soft approve**: This use case involves rules that approve the transaction but also generate an alert. The transaction should be displayed in Case Managerâs Alert page for fraud prevention agent review.
- **Hard approve**: This use case involves rules that approve the transaction and do not generate an alert. The transaction wonât be displayed in Case Managerâs Alert page for fraud prevention agent review.

Understand rules projectsâ

A rules project is an ordered collection of rules that share similar traits. Grouping your rules in different rules projects allows you to:

- Keep your work organized.
- Set general configurations (e.g. a schema) that apply to all the rules that share similarities.
- Order rules projects in a workflow so that real-time events are first processed by rules with higher priority and then by rules with lower priority.

Standard rules projectsââ

Transaction Fraud for PSPs includes standard rules projects to help you get started:

(*) Fraud Rules are subdivided into several logics, i.e. velocity rules, behavior rules, MCC rules, and mismatch rules.

Conceptually, rules projects are divided into the following main groups according to the event type that arrives to the workflow:

- Feedback events
- Scorable events

Feedback eventsââ

Feedback events are evaluated by the following rules project:

- **Feedback Rules**: Contains rules that classify entities into âdeclineâor âapproveâlists according to the fraud prevention agent's feedback on the event. Examples of rules include:
  - Feedback fraud: Entity is listed to be declined if the fraud prevention agent marks it as "Fraud", or if there is a Feedback event received through the API marking the event as "Fraud".
  - Feedback not fraud: Entity is listed to be approved if the fraud prevention agent marks it as "Not Fraud", or if there is a Feedback event received through the API marking the event as "Not Fraud".

The typical flow of a triggered feedback rule is as follows:

To learn how Feedback rules automatically put an entity into a PosNeg list, check the following sections:

- Create lists
- List management service

Scorable eventsââ

Scorable events are the events that return a decision outcome. For instance, Payment events, or particular Info events, such as Refunds or OCTs, return a decision outcome. Scorable events are evaluated by the following rules projects:

- **APM Rules**, **Card Present Rules** and **Fraud Rules**: Contains rules that look for fraud patterns.
- **By Merchant PosNeg Rules**: Implements decisions based on listed entity values. This means that these rules will check if entities are listed to be approved / declined, and, based on that, they set the respective approve / decline decision. Note that, for PosNeg rules, you should use complex lists.
- **PosNeg Rules**: Implements decisions based on listed entity values. This means that these rules will check if entities are listed to be approved / declined, and, based on that, they set the respective approve / decline decision. Note that, for PosNeg rules, you should use complex lists.
- **Refund/OCT Rules**: Contains rules that look for fraudulent refunds / OCTs.
- **Decision Rules**: Contains rules that have triggered based on the score produced by a data science model.

All eventsââ

The following rules project is evaluated for all event types:

- **Self-service Rules**: Contains rules that are written in Case Manager by Fraud Prevention Managers and are processed in Pulse. Fraud Prevention Managers can use this functionality to write global and âby Merchantâ rules directly in production to quickly act on an emerging fraud pattern.
- **By Merchant Self-service Rules**: Contains rules that are written in Case Manager by Fraud Prevention Agents (merchant) and are processed in Pulse. There rules are shared between the fraud prevention agent (merchant) and the fraud prevention manager (PSP) to write âby Merchantârules directly in production to quickly act on an emerging fraud pattern.

Rules visibility in Case Managerâ

The visibility of rules in Case Manager can be managed in the respective rules project.

By default, the visibility will be public, which means that the triggered rules and respective explanations are available to all users that have generic permissions to see rules.

If the rules of a specific rules project must be visible to the landlord, perform the following configuration:

1. In the sidebar, open the **Data Science** tab.
2. Scroll down to find the Rules Projects widget.
3. Select the **Configure** icon under Actions to edit the respective rules project.
4. In the **Advanced Options**, under Visibility, select Specific Groups and type `rmpsp_landlord_resources`.

In this scenario, only users with access to landlord resources will be able to see the rules and the respective explanation in Case Manager.

info

With this option, you can create scenarios where a user can see an Alert/Decline but the explanation won't be available.

Rules precedenceâ□□â□□

An event can trigger several rules in multiple rules projects. It is important to understand what happens when an event triggers several rules that generate different outcomes. The following sections explain this precedence level for rules that generate decisions and alerts.

Precedence in decisionsâ□□â□□

The following image shows the precedence level for rules that trigger an approve or decline decision. The label *triggered* in this example means that at least one rule in the rules project has been triggered.

Rules precedence regarding decisions works as follows:

- **Scenario 1**: If no rule is triggered, the final outcome is â□□approveâ□□

- **Scenario 2**: When one of the rules with the decision decline is triggered, the final outcome is decline.
- **Scenarios 3 and 4**: When multiple rules with different decisions are triggered, the final outcome is the decision of the rule that has triggered first.

Precedence in alertsââ

The following image shows the precedence level for rules projects that trigger an alert:

Rules precedence regarding alerts works as follows:

- **Scenario 1**: If no rule is triggered, the transaction is not displayed in Case Manager's Alerts page. However, fraud fraud prevention agents can still access the transaction in Case Manager's search page.
- **Scenario 2**: When one of the rules with the action alert is triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why the rule was triggered.
- **Scenario 3**: When multiple rules with alert action are triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why each rule was triggered.

Rules orderââ

The order in which rules are processed can be defined inside each rules project. This allows you to define the order according to the priority of the outcome, i.e. rules with higher priority (e.g. hard decline) can be evaluated before rules with lower priority (e.g. âhard approveâ).

Rule componentsâ

Rules are composed of variables, conditions, and integrations with enrichment functions and environment metrics/ variables. Learn about each of these components to understand how you can use them when building a rule.

Create rulesâ

A rule, when used in a workflow, evaluates if a certain condition is true for the current transaction. According to the result, the rule can trigger actions in the workflow like rejecting or accepting a transaction, and contribute to the score of the transaction.

The following sections help you successfully create rules for your business needs:

- Create rules.
- Operators in rules.
- Data types in rules.
- Default UDFs.

Request a reviewâ

Ensuring every change is reviewed by a second pair of eyes helps prevent unwanted or unplanned changes in production. As such, the Four Eyes Check feature allows you to request reviews from your colleagues.

Learn how to:

- Request peer reviews
- Review rules

Create listsâ

A list is a resource that allows you to store items (e.g. card IDs), which can be referenced in rules.

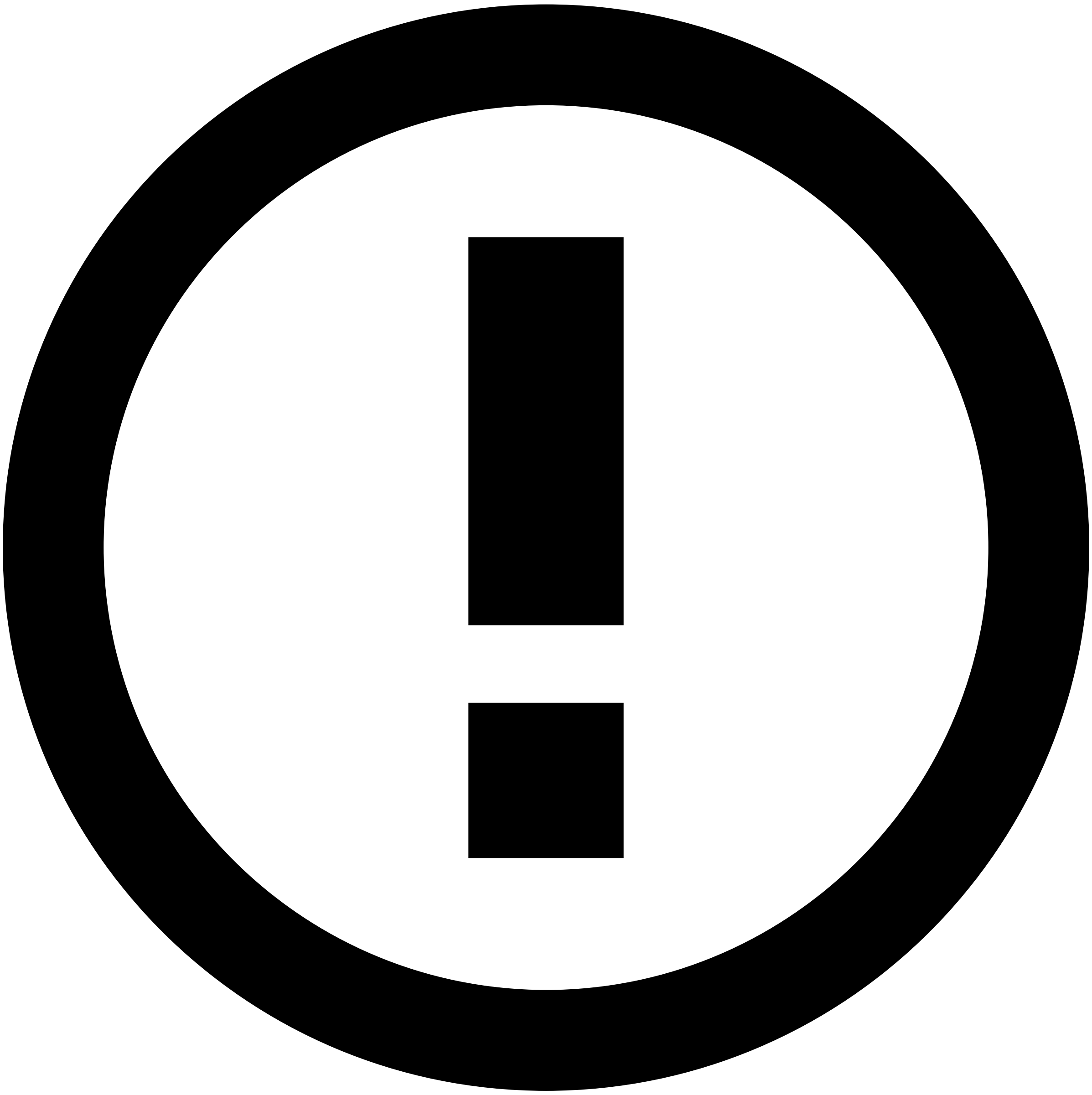
You can create lists in Pulse with different types of information. Use lists when writing rule conditions to confirm if the value of a given item is stored in a certain list. For example, you can create rules that decline or accept transactions performed by credit cards or digital events whose IDs are listed to be declined or approved.

In general, PosNeg lists are typically configured to identify items as "approve" or "decline" using a combination of key-value pair and activation timeframe.

Consider the following examples.

Decline listsâ

The following image illustrates an example of a card listed to be **declined**.



info

Timeframe: As a standard configuration, a card is permanently stored as listed to be declined. Therefore, the **Active from** option only specifies the initial date. However, in the case of a fraud feedback event (e.g. cardholder response via API or fraud prevention agent feedback via Case Manager) identifying the transaction as *Not Fraud*, the card is automatically tagged as `approve` (i.e. listed to be approved), thus overriding the `decline` value.

Approve listsâ

The following image illustrates an example of a card listed to be **approved**:

info

Timeframe: As a standard configuration, a card is listed to be approved within 2 hours (the time is configurable). After that, the card becomes automatically an inactive item (but still on the list). The **Active from** and **To** options are used to define this time interval.

To learn how to create lists, refer to the Pulse documentation - Creating Lists and Adding List Items.

Default listsâ

Transaction Fraud for PSPs includes default lists to speed up your work. Check the Lists page to confirm the out-of-the-box lists.

List management serviceâ

The list management service allows you to configure the automatic list transitions that an entity (e.g. card ID) undergoes when triggered by different rule actions.

Rule actions have standard values to identify decline and approve lists in a PosNeg List. The list management service is responsible for:

1. Checking when a rule triggers one of those actions.
2. Adding the entity to the PosNeg List and tagging it as decline or approve.

You can extend and adapt the list management service to fulfill your use cases.

To open the list management service:

1. In the sidebar, select the **Workflows** tab.
2. In the Diagram, select the pencil to edit the *Workflow List Manager*.

tip

Advanced functionality: Configurations in the list management service should be performed by advanced users to ensure that the transitions between lists work as expected. Note that, if the transitions are not well configured, the rules that depend on these transitions will not yield the targeted results.

How the list management service worksâ

The list management service is configured out of the box to classify card IDs as:

- Listed to be declined

- Listed to be approved

Decline listâ

The entries in the **Configuration** table with the *rules.0* prefix are associated with the same action (i.e. to list a card ID as declined). In this case, this configuration is set up to tag a card as **decline**:

Key	Value	Mandatory	Explanation
rules.0.actions	Auto Decline Card	Yes	The rule action that triggers this service to list a card ID to be declined.
rules.0.entity	payment_card_id	Yes	The entity that is added to the list (i.e. the key of the key-value pair in the list configuration options).
rules.0.listid	ffce7e5ce8f4013a	Yes	The list ID that is used to store the card ID (i.e. PosNeg Listed Cards).
rules.0.appid	c145d72d00d46bab	No	The app ID where the list exists. If not defined, the default is the app ID where the service exists.
rules. 0.listFieldName	value	No	The name of the list's field that is used in the list ID to store the card ID.
rules.0.entries. 0.value	decline	Yes	The value used to list a card ID to be declined (i.e. the value of the key-value pair in the list configuration options).

Approve listâ

The entries in the **Configuration** table with the prefix *rules.1* are used to list card IDs to be approved:

Key	Value	Mandatory	Explanation
rules.1.actions	Auto Approve Card	Yes	The rule action that triggers this service to list a Card ID to be approved.
rules.1.entity	payment_card_id	Yes	The entity that is added to the list (i.e. the key of the key-value pair in the list configuration options).
rules.1.listid	ffce7e5ce8f4013a	Yes	The list ID that is used to store the card ID (i.e. PosNeg Listed Cards).
rules.1.appid	c145d72d00d46bab	No	The app ID where the list exists. If not defined, the default is the app ID where the service exists.
rules. 1.listFieldName	value	No	The name of the list's field that is used in the list ID to store the card ID.
rules.1.entries. 0.value	approve	Yes	The value used to list a card ID to be approved (i.e. the value of the key-value pair in the list configuration options).
rules.1.entries.0.ttl	7200000	No	The duration that a card remains in the PosNeg Listed Cards list as approved in miliseconds.

Evaluate and test rulesâ

You can evaluate your rules to assess their effectiveness by creating and analyzing a rule snapshot evaluation.

Deploy rules to runtimeâ

The deployment process of rules is different depending on whether this is the first deployment, or an iteration over an existing deployment.

Learn more about this topic in Pulseâs Deploying Rules to the Production Environment page.

Pulse Guide

The Pulse guide leads you on the recommended tasks you must perform in Pulse throughout the complete life cycle of your project. It contains instructions on what you should do to perform each task and how you should keep moving forward, often complemented with examples, and always providing more context and conceptual details.

Pulse focuses on a democratization of Data Science, which allows both data scientists and fraud prevention agents to use the platform. All of the the tasks in this journey will be about writing or improving the risk strategy for the merchant monitoring use case, heavily influenced by Data Science and Machine Learning models, providing an understanding on how to work with rules.

This guide helps you work with rules.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/data-scientist/pulse/work-with-rules

- Operational Work
- Data Scientist & Fraud Strategist
- Pulse Guide
- Work with Rules

On this page

Work with Rules

This section provides you with an overview of how rules work to alert risky merchants with the resources that come out of the box with your application. Learn how to fully manage your rules, including:

- Understanding how rules work.
- Writing rules that are specific to your business.
- Evaluating and testing rules.
- Deploying rules to runtime.

As a data scientist or risk strategist, you can leverage rules to detect and mitigate risky events at scale in real time. The following user journey shows the different stages required to set up your rules and deploy them to runtime:

How is the alert generated?ââ

It is important to understand how rules process events and contribute to the API response to detect risky merchants efficiently. This section breaks down this life cycle in the following conceptual steps:

1. Rule options in the Pulse application
2. Rule blocks and outcomes in the Pulse workflow
3. Manual revision in Case Manager
4. API response

1. Rule options in the Pulse applicationââ

Before getting into the specific logic behind each event, let's have a closer look at two important rule settings. Each rule is configured with Actions and Mark As:

When a rule is triggered, Pulse sends the event to Case Manager as "Alerted" for fraud prevention agents to focus their investigation only on suspicious activity.

- The **Actions** option includes the rule code (e.g. Failed-3DS-Count-Rate-FR1) so that it is possible to identify which rules triggered.

- The **Mark As** option lets you select the outcome values that are configured beforehand in the Pulse workflow. When a rule is triggered, the **Mark As** value is written to the schema field that is associated with this rule outcome.

Different rules are defined with different outcomes according to the specificities of each event. The position of rules projects in the Pulse workflow, and the additional custom services dictate how the final alert is generated.

2. Rule blocks and outcomes in the Pulse workflowââ

Each set of APIs contains a Pulse workflow (i.e. *Merchant Monitoring Workflow*). The possible rule outcomes and the rules' processing order are defined in the workflow:

Merchant Monitoring workflow

How does the *Merchant Monitoring Workflow* generate the final outcome?

The Merchant Monitoring Workflow is configured with bankruptcy, chargeback, fraud, money laundering, and unusual activity rules. The `risky` outcome is generally configured to be used by machine learning models.

3. Manual revision in Case Managerââ

Merchant activity can generate alerts when one of the rules, configured with the Alert action, is triggered. The event is then displayed in Case Managerâs Alerts page.

Fraud prevention agents investigate alerts in Case Manager and decide whether the event is `risky` or `not risky` by *actioning* or *dropping* the alert.

4. API responseââ

At this point, you already understand how Feedzai provides PSPs with a response. When you send an API request, you will get an API response with the following key fields:

- `alert`: Whether the event generated an alert that is visible on the Alerts page in Case Manager.
- `action_codes`: Every rule has an array of Actions including the name of the rule (e.g. [Failed-3DS-Count-Rate-FR1]). The `action_codes` array includes all the actions of triggered rules to further understand which rules have triggered for each event.

Rule configurations in Pulse

To define a rule in Pulse, you have the following main configurations available:

- **Variables**: Allows you to declare variables that are further used in the rule's condition including thresholds or transformation/conversion of the input value (e.g. variable that converts an amount from cents to US dollar). Variables are useful to avoid having large condition statements, which are error-prone and difficult to maintain.
- **Conditions**: The condition expression must return a boolean value that dictates whether the event triggers a rule.
- **Explanations**: Allows you to understand why a rule has triggered in a human-readable way. If an event triggers, the explanation is displayed in Case Manager to provide additional insights to fraud prevention agents.
- **Actions**: Operations that are performed when the rule is triggered. The standard value that helps you meet your custom business logic is the following:
    - Sends the event to Case Manager's Alerts page for manual revision.

Understand rules projects

A rules project is an ordered collection of rules that share similar traits. Grouping your rules in different rules projects allows you to:

- Keep your work organized.
- Set general configurations (e.g. a schema) that apply to all the rules that share similarities.

Standard rules projects

Merchant Monitoring includes the following standard rules projects to help you get started:

- **Bankruptcy risk rules**: Account for merchants who are at risk of or are concealing their inability to ensure payment of their debt.
- **Chargeback rules**: Account for merchants with abnormal customer transaction activity, which may lead to higher volume and risk of reported transaction chargebacks.
- **Fraud rules**: Prevent various fraud schemes that are generally associated with collusive/fraudulent merchants.
- **Money laundering rules**: Merchants with abnormal activity indicating potential business fronts used to process transactions for products and/or services that are unknown and/or illegal.
- **Unusual activity rules**: Rules that account for suspicious activity that is not yet associated with a particular fraud scenario but requires close monitoring.

Rules precedence

An event can trigger several rules in multiple rules projects. It is important to understand what happens when an event triggers several rules that generate different outcomes. The following section explain this precedence level for rules that generate alerts.

Precedence in alerts

The following image shows the precedence level for rules projects that trigger an alert:

Rules precedence regarding alerts works as follows:

- **Scenario 1**: If no rule is triggered, the transaction is not displayed in Case Manager's Alerts page. However, fraud fraud prevention agents can still access the transaction in Case Manager's search page.
- **Scenario 2**: When one of the rules with the action alert is triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why the rule was triggered.
- **Scenario 3**: When multiple rules with alert action are triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why each rule was triggered.

Rules order

The order in which rules are processed can be defined inside each rules project. This allows you to define the order according to the priority of the outcome.

Rule componentsâ

Rules are composed of variables, conditions, and integrations with enrichment functions and environment metrics/ variables. Learn about each of these components to understand how you can use them when building a rule.

Create rulesâ

A rule, when used in a workflow, evaluates if a certain condition is true for the current transaction. According to the result, the rule can trigger actions in the workflow like rejecting or accepting a transaction, and contribute to the score of the transaction.

The following sections help you successfully create rules for your business needs:

- Create rules.
- Operators in rules.
- Data types in rules.
- Default UDFs.

Create listsâ

A list is a resource that allows you to store items (e.g. card IDs) which can be then referenced in rules.

You can create lists in Pulse with different types of information. Use lists when writing rule conditions to confirm if the value of a given item is stored in a certain list. For example, you can create rules that mark events as risky or not risky.

Learn how to create lists.

Evaluate and test rulesâ

You can evaluate your rules to assess their effectiveness by creating and analyzing a rule snapshot evaluation.

Deploy rules to runtimeâ

The deployment process of rules is different depending on whether this is the first deployment, or an iteration over an existing deployment.

Learn more about this topic in Pulseâs Deploying Rules to the Production Environment page.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/fraud-prevention-agent/case-manager/case-manager-guide

- Operational Work
- Fraud Prevention Agent
- Case Manager Guide

Case Manager Guide

Case Manager is an alert management tool that enables fraud prevention agents to review events that fall within a risk threshold and raise alerts. Case Manager includes out-of-the-box resources tailored to the merchant monitoring use case, which automate event processing and organize teams' workload efficiently.

Select the most suitable option for your specific use case:

Merchant Monitoring (standard)

Each merchant is assessed on a daily basis (i.e. every 24 hours), with each assessment resulting in a new event. The standard out-of-the-box configuration provides a comprehensive overview of merchant activity and risk within the specified cycle, with optimal performance.

Ad-hoc investigations (with add-on)

Trigger a merchant investigation either upon receiving a request from an external source or on an ad-hoc basis.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/fraud-prevention-manager/analytics-dashboards/analytics-dashboards-guide

-

- Operational Work
- Fraud Prevention Manager
- Analytics Dashboards Guide

Analytics Dashboards Guide

The analytics dashboards include metrics that provide you with real-time charts with information about merchant portfolio, temporal trends, agent performance, and rule monitoring. Based on these insights, you will have visibility on the merchant's business activity, including information about transactions (i.e. processed, alerted, approved, declined, etc), as well as about temporal trends (i.e. volume of orders processed throughout time). You can also improve workload management, or write self-service rules to quickly block a fraudulent behavior.

This guide provides you with the following information:

- Merchant portfolio overview
- Transaction overview dashboard
- Agent performance - alerts dashboard
- Agent performance - cases dashboard
- Rule monitoring dashboard

Merchant Portfolio Overview

The merchant portfolio overview dashboard is an entry point to analyze merchants on an individual or multiple level. It includes, among others, the number of merchants in a PSP's portfolio, the top merchants in relation to the highest transaction amount, and the top regions in relation to the highest volume of transactions.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by fields. For instance, the `fields.merchant_id` field allows you to filter by a specific merchant ID.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An [index pattern] selects the data to use and allows you to define properties of fields.
2. Add the [field] you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. To finalize the creation of your condition, enter a value.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

The **Merchant Portfolio Overview** provides other filters to refine your search such as:

- **Merchant ID**
- **Merchant MCC**
- **Merchant Name**
- **Event Type**: `feedback`, `info`, `merchant monitoring`, `payment`
- **Region**
- **Feedzai Channel**: `mmchannel` or `payments`

Merchant analysisâ

- **Number of merchants**: The total number of merchants.
- **Number of customer regions**: The count of distinct geographic areas into which a merchant divides its customer base.
- **Distribution of transactions over time**: The number of transactions spread across a given period (e.g. over the last 7 days).

- **Top 10 merchants by total amount**: The hierarchical ranking of the top 10 merchants based on their total amounts, where each merchant's portion represents their respective percentage. By selecting a merchant from the graph, you're redirected to the merchant's transaction overview dashboard, which provides a more detailed view of that specific merchant's transactions.

caution

Note that selecting line dots works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Top Merchant by Total Amountâ

This table presents a comprehensive overview of the top merchants, showcasing their total amount. The table consists of the following columns:

- **Merchant ID**: The unique identifier assigned to each merchant.
- **Merchant name**: The names of the merchants.
- **Total amount**: The total transaction amount for each merchant.
- **Feedzai approved amount**: The total transaction amount that has been approved by Feedzai.
- **Feedzai declined amount**: The total transaction amount that has been declined by Feedzai.
- **% Feedzai approved amount**: The transaction amount that has been approved by Feedzai vs. the total amount, in percentage.

- **% Feedzai declined amount**: The transaction amount that has been declined by Feedzai vs. the total amount, in percentage.

tip

- The table is sortable. You can arrange data into ascending or descending order according to a column.
- By hovering over each **merchant name**, you can use the **magnifying glass** icon to either filter out that merchant or filter by that specific merchant only.
- You can also export the table by selecting **Raw** or **Formatted**.

Merchant Geo Locationâ

The following map allows you to hover over any country to see the transaction amount per merchant country code.

- Operational Work
- Fraud Prevention Manager
- Analytics Dashboards Guide
- Transaction Overview

On this page

Transaction Overview

The transaction overview dashboard includes several visualizations that help fraud prevention managers analyze transactions, refunds, chargebacks, and original credit transactions (OCTs) over a defined period for a merchant or subset of merchants. It also includes visualizations for alerts, status codes, geographical and temporal trends.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by fields. For instance, the `fields.merchant_id` field allows you to filter by a specific merchant ID.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the field you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. To finalize the creation of your condition, enter a value.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

The **Transaction Overview** provides other filters to refine your search such as:

- **Merchant ID**
- **Merchant Name**
- **Merchant MCC**
- **Merchant URL**
- **Event Type**: `feedback`, `info`, `merchant monitoring`, `payment`
- **Payment Region**
- **Feedzai Channel**: `mmchannel` or `payments`
- **Lifecycle ID**

Totalâ

- **Total processed**: The aggregate transaction amount and corresponding number of transactions.
- **Total Feedzai approved**: The total transaction amount approved by Feedzai, along with the corresponding number of transactions from merchants, and the percentage of the total number of transactions.

- **Total Feedzai declined**: The total transaction amount declined by Feedzai, along with the corresponding number of transactions from merchants, and the percentage of the total number of transactions.
- **Number of transactions by amount interval**: Total transaction distribution based on different amount intervals (e.g. 2 transactions fall within the $25 - $50 interval).
- **Transaction amount by amount interval**: Total transaction amount distribution based on different amount intervals (e.g. $150 falls within the $100 - $250 interval).

caution

Note that selecting bars works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Refunds, chargebacks and OCTsâ

- **Total refund**: The total transaction amount refunded, along with the corresponding number of refunded transactions, and the percentage of refunded transactions.
- **Total chargeback**: The total amount of chargeback transactions, along with the corresponding number, and the percentage of chargeback transactions.
- **Total OCT**: The total amount of OCTs, along with the corresponding number, and the percentage of OCTs.
- **Total Disputes**: The total amount of disputed transactions, along with the corresponding number, and the percentage of disputed transactions.

Alertsâ

- **Total alerted**: The total transaction amount alerted, along with the corresponding number of alerted transactions, and the percentage of alerted transactions.
- **Total approved and alerted**: The total transaction amount approved and alerted by Feedzai, along with the corresponding number of approved and alerted transactions, and the percentage of approved and alerted transactions.
- **Total declined and alerted**: The total transaction amount declined and alerted by Feedzai, along with the corresponding number of declined and alerted transactions, and the percentage of declined and alerted transactions.
- **Distribution of transactions alerted and declined over the selected time range**: The number of alerted and declined transactions spread across a given period (e.g. over the last 7 days).

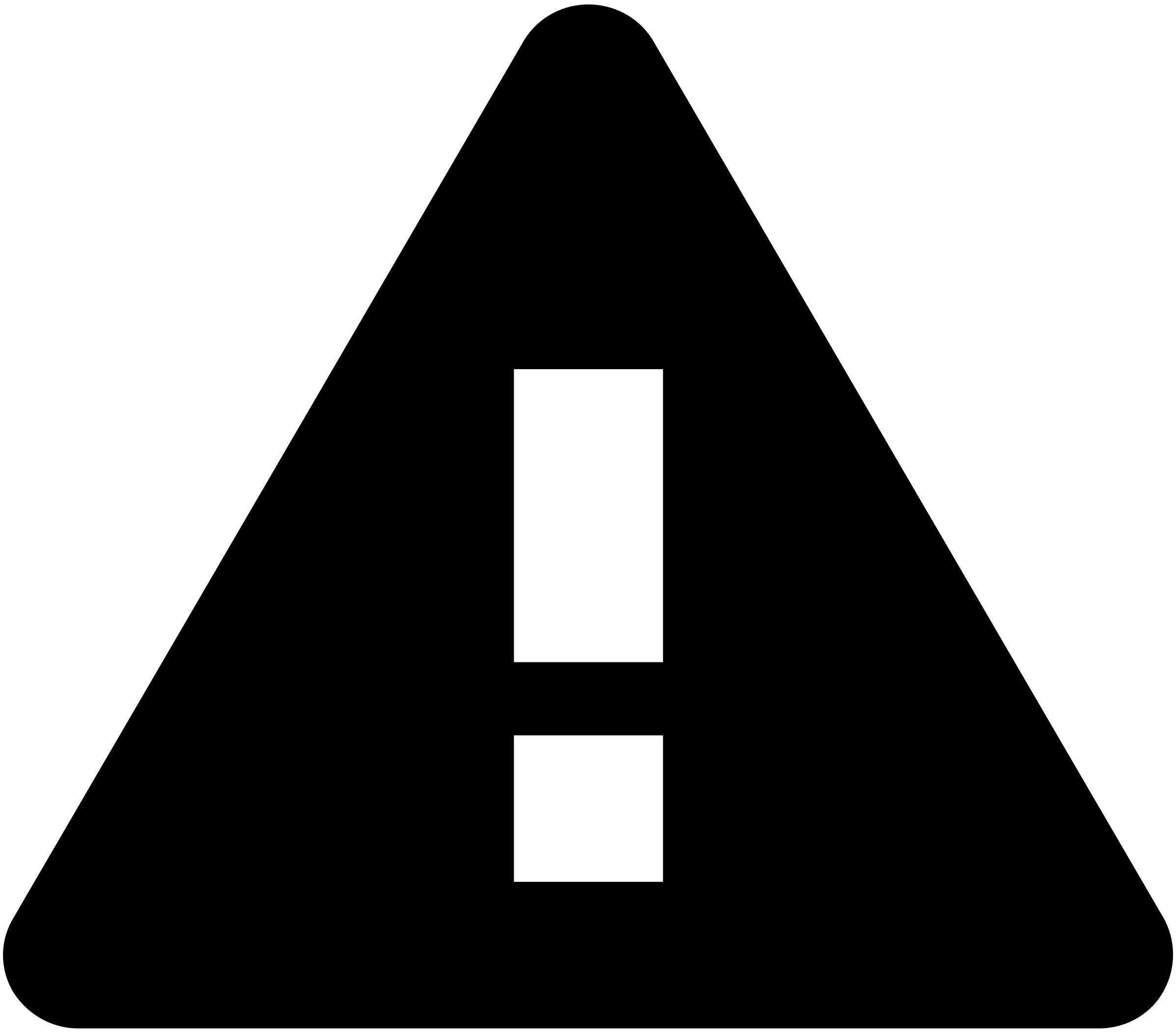
caution

Note that selecting bars works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Status codesâ

- **CNP ratio**: The percentage of card-not-present transactions.

- **Distribution of 3DS authentication**: The percentage of transactions using 3-D Secure (3DS), and whose code used is `Y`.
- **Authorization status code**: The number of orders vs. the number of authorization codes used (e.g. 10000 authorization codes used in 5 orders).



caution

Note that selecting graph slices or bars works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Geographical trendsâ

- **Card country analysis**: A geographical map that allows you to hover over any country to see the number of transactions by terminal IP country, device IP location, device IP country, or payment card country code. A complementary table gives you more detail, including the number of declined transactions, total transaction amount, and declined transaction amount per country.
- **Payment region analysis**: The number and corresponding percentage of payments per region.

tip

- The table is sortable. You can arrange data into ascending or descending order according to a column.
- By hovering over each **terminal IP** or **payment card**, you can use the **magnifying glass** icon to either filter out or filter by those specific countries only.
- You can also export the table by selecting **Raw** or **Formatted**.

caution

Note that selecting graph slices or bars works like a filter, and therefore this action results in changes to all the results of the **Analyst Performance Review - Cases** dashboard.

Temporal trendsâ

- **Processed**: Identifies abnormal behavior by comparing the current value (blue line) against the one-week offset (light blue line).
- **Declined**: Identifies abnormal behavior by comparing the current value (red line) with the one-week offset (light red line).

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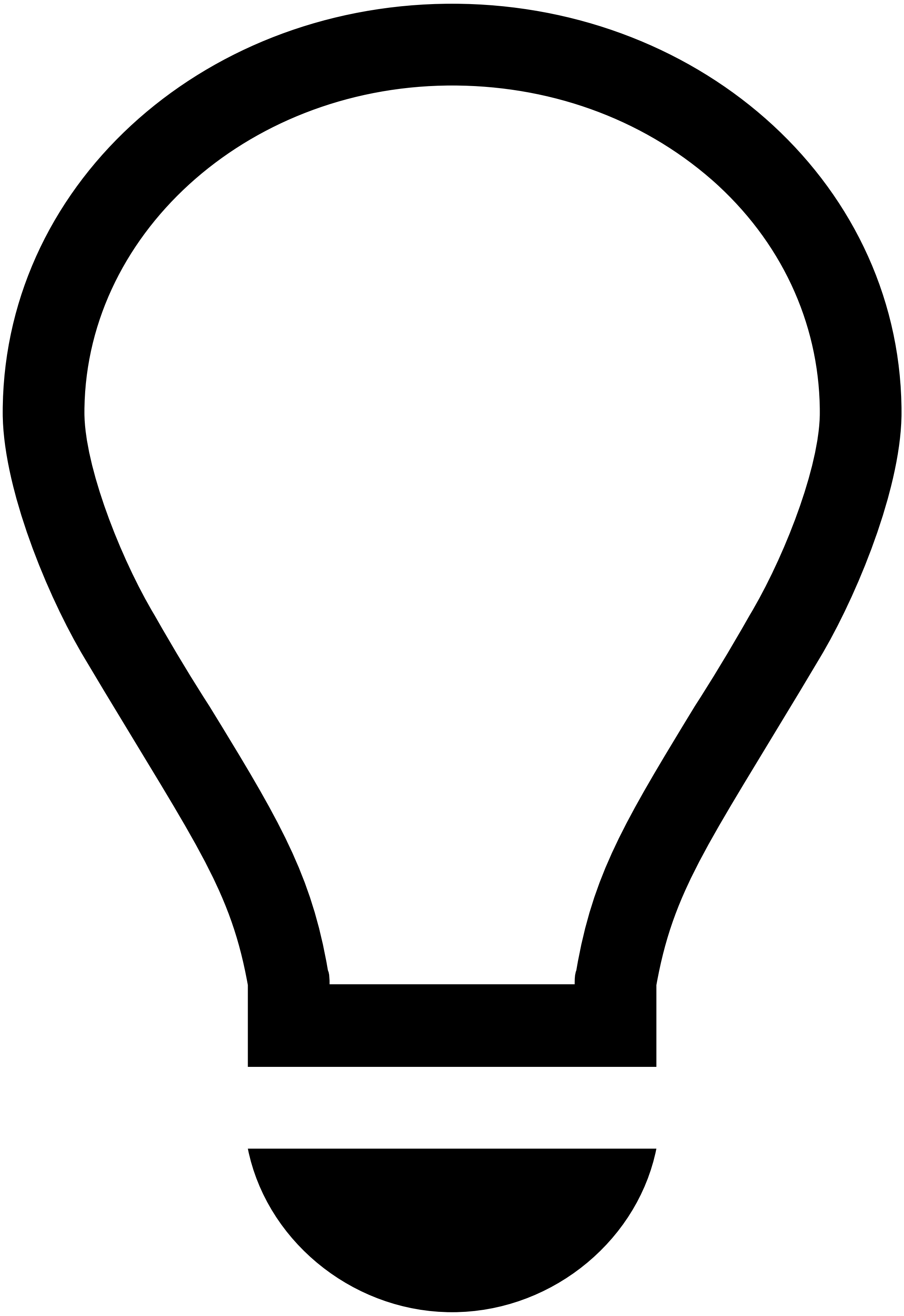
Case Manager Guide

This guide helps you manage your team's workload using:

- **Automation rules**
- **Service-level agreements (SLAs)**
- **Queues**

To improve system configurations available to agents, you will learn about:

- **Reasons**



fraud prevention agent guide

For information on fraud prevention agents' operational flow, refer to the following documentation:

- Case Manager for fraud prevention agents

Automation rulesâ

Automation rules allow you to automatically perform actions on events and cases based on certain conditions. These rules allow to automatically move alerts to queues, assign alerts to agents, link alerts to cases, or even change the alert status.

View automation rulesâ□□

Access the Automation Rules page using the **Settings** option in the main menu.

Filter automation rulesâ□□

Use filters to refine the rules displayed on the list. This can be helpful, for example, to view all rules with a specific name.

The **Search** filter lists rules whose name or description matches the text inserted in this field.

Create automation rulesâ□□

Create rules to automate what should happen when certain actions occur. For example, you can configure a rule that automatically sets a status when an alert arrives to Case Manager.

1. Select the **New Rule** button.

2. Start by adding the rules's basic information:

    1. **Name**: Rule's name. For example, "Set new alerts as unreviewed".
    2. **Description**: Explanation of the rule's purpose. For example, "Set all incoming alerts as unreviewed".
    3. **Channel**: Defines that only events from the selected channel are considered in the rule's conditions.
    4. **Type**: Defines if the rule should be triggered based on **Events** or **Cases**.

3. Specify the rule's conditions according to your needs. In this example, the conditions would be the following:

4. Select the **Enabled** option to indicate if the rule should go into effect immediately or not.

5. **Create** the rule.

info

After creating automation rules, you are still able to edit or delete them from the automation rules list.

tip

For more information on how to manage automation rules, reach out to Feedzai.

SLAsâ

Service Level Agreements (SLAs) allow you to automatically perform actions on events and cases based on a time and schedule.

View SLAsâ

Access the **SLAs** page using the **Settings** option in the main menu.

Filter SLAs Listâ

Use filters to refine the SLAs displayed on the list. This can be helpful, for example, to view all SLAs with a specific name.

The **Search** filter lists only SLAs whose name or description match the text inserted in this field.

Create SLAsâ

Create SLAs to automate what should happen when certain conditions occur. For example, you can configure an SLA that sends an alert to a specific queue depending on the time that alert has been in the system.

1. Select the **New SLA** button to open the SLA creation form.

2. Start by adding the SLA's basic information:

    1. **Name**: SLA's identification. For example: Daily Escalation.
    2. **Description**: Explanation of the SLA's purpose. For example: Move all unreviewed alerts to Urgent queue.
    3. **Channel**: Defines that only events from the selected channel are considered in the SLA's conditions.
    4. **Type**: Defines if the SLA should be triggered based on **Events** or **Cases**.

3. Specify the SLA conditions according to your needs. In this example, the conditions would be the following:

Exclude weekends

When this option is selected, the SLA does not take weekends (Saturdays and Sundays) into consideration. For instance, a 24-hour SLA due on a Saturday at 2:00 PM will be triggered on the following Monday at 2:00 PM. Likewise, a 48-hour SLA due on a Sunday will be triggered on the following Tuesday.

4. **Create** the SLA.

info

After creating SLAs, you are still able to edit or delete them from the SLAs list.

tip

For more information on how to manage SLAs, reach out to Feedzai.

Queuesâ

Queues help you to manage and track alerts and cases. You can think of queues as containers in which you can store alerts or cases and assign them to agents.

View queuesâ

Access the **Queues** page using the **Settings** option in the main menu.

Filter queuesâ

Use filters to refine the queues displayed on the list. This can be helpful, for example, to view all queues from a specific channel.

- **Channels**: Lists only queues from the selected channel.
- **Tenants**: Lists only queues from the selected tenant. Note that this is a multiple choice, so you can discriminate among the queues of various tenants.
- **Type**: Lists only queues from the selected type: Event or Case.
- **Search**: Lists only queues whose name or description match the text inserted in this field.

Create queuesâ

Create queues to store your events and cases in a more organized manner:

1. Select the **New Queue** button.

2. Complete the fields with the information that best describes the new queue.

    1. **Name**: Queue's name (e.g. "Urgent").
    2. **Description**: Brief description of the queue's purpose (e.g. "Alerts to be reviewed in 24h.").
    3. **Type**: Defines if the queue is specific to Events or Cases.
    4. **Channel**: Ensures that only events from the select channel are added to the queue.
    5. **Tenant**: As a landlord in a Multi-Tenancy environment, you can create different queues for different tenants. Adding tenants to queues ensures that only events from the select tenant are added to the queue. In addition, it ensures that the queue is only visible to users with permissions to view that specific tenant.
    6. **Users**: Ensures that the queue is only visible for the selected users.

3. Specify the queue conditions according to your needs. In this example, the conditions would be the following:

4. **Create** the queue.

info

After creating queues, you are still able to edit or delete them from the queues list.

tip

For more information on how to manage queues, reach out to Feedzai.

Reasonsâ

When agents classify an alert, they can select the reason behind their decision. This section explains how you can manage the reasons used by them when classifying alerts.

View reasonsâ

Access the **Reasons** page using the **Settings** option in the main menu.

Filter reasonsâ

Use the filters to refine the reasons displayed on the list. This can be helpful, for example, to view all reasons from a specific state machine.

Create reasonsâ

Create reasons for agents to use when classifying alerts.

1. Select the **New Reason** button.
2. Complete the fields with the information that best describes the new reason.
    1. **Name**: Reasons's name.
    2. **Status**: Select the status associated with the new reason. Reach out to Feedzai for more information on states.
3. **Create** the reason.

info

After creating reasons, you are still able to edit or delete them from the reasons list.

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Integration Guide

This guide provides you with the relevant steps to successfully test rules and test the system.

- Rule Testing: Learn how to test rules using the Postman Collection.
- User Acceptance Testing: Learn how to test rules, and ensure the system is well configured / running properly.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/case-manager/merchant/event-flow

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On this page

Event Flow

As a fraud prevention agent, understanding the event flow is a key step to managing the alert investigation efficiently. This page helps fraud prevention agents understand the operational flow when investigating alerts.

Event statusesâ

Case Manager displays the event's statuses in the **Event**'s page as follows:

- **System Decision**: Indicates the outcome processed by the Feedzai's risk engine. The possible values are:
    - Accept
    - Decline

- **Alert**: Indicates whether the event was alerted (i.e. by one or more self-service rules) or not. This means that you can create your own self-service rules and decide to alert certain payments.
- **Merchant Fraud Label**: Indicates the fraud prevention agent's final decision. The `Fraud` or `Not Fraud` values become available after selecting the **Fraud** or **Not Fraud** button, respectively.
- **Listed Entities**: Indicates if there are entities (e.g. customer, card, or account) associated with a known fraud source.
- **Cases**: Indicates the number of cases linked to the event.

-
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- Investigate Alerts

On this page

Investigate Alerts

As a fraud prevention agent, the first step you should take is to access the workload. Depending on the risk engine's decision, you may find a set of events for review.

This page helps you understand how to access the events and cases that are up for review.

User journeyâ

The event analysis process involves a number of actions each fraud prevention agent (i.e. L1, L2, and L3) can perform, as follows:

Access events and casesâ

When events arrive at Case Manager, they can be displayed on more than one page, depending on the fraud risk and the complexity. On Case Manager's sidebar you can access:

- **Search By Event** : Displays all the events (alerted and non-alerted).
- **Cases** : Events and alerts can be grouped into cases for a more organized action-taking process. This allows you to access events that may be correlated and that require a thorough investigation. Cases offer extra functionalities, such as note attachment.

The following figure gives you an idea of how events, alerts, and cases relate to each other:

tip

Events versus Alerts: The **Search By Event** page includes an **Alert** filter to display only the alerted events.

Filter events and casesâ

The Search by Event and Cases pages allow you to narrow down the workload for your analysis through filters.

Select and analyze eventsâ

As previously mentioned, the key pages to access your workload are Search By Event and Cases pages.

Whether you open an alerted or non-alerted event, you will land on the Event page, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a comprehensive approach, in which you open several events, e.g. from the same customer ID, and search for fraud evidence by looking at multiple widgets.

The Event page displays the following information:

- **Rules triggered**: Shows all the rules triggered by the event.
- **Transaction**: Shows the transaction details, such as date, amount, and payment method. The **Card ID** field allows you to quickly update lists.
- **Customer**: Includes information about the customer, such as the customer's ID, city, and address. To help your investigation, you can check this widget both against the billing and shipping widgets. The **Customer Email** field allows you to quickly update lists.
- **Terminal**: In a card present transaction, it shows information about where the transaction took place. The **Terminal ID** field allows you to quickly update lists.
- **Device**: In a card not present transaction, it shows information about the device used to purchase goods/services. The **Device ID** and **Device IP** fields allow you to quickly update lists.
- **Dispute**: In the event of chargeback, the dispute widget shows information on the reason why the payment is being disputed.
- **Billing**: Includes details about the invoice, such as the billing name, address, and ZIP code.
- **Shipping**: Includes details about the shipping, such as the shipping name, address, and ZIP code.
- **Items**: Shows information such as the name and ID of the purchased goods/services.
- **Payment Method**: Includes information about the payment method used, e.g. card or alternative payment method details. The **Card Country** field allows you to quickly update lists.
- **Transaction History**: Displays all the transactions associated with the event under investigation.
- **Shipping Distance**: Displays the distance between the shipping and device locations.
- **Attached Files**: This widget allows you to add files that are useful for subsequent investigation steps.
- **Notes**: This widget shows any notes made under a given event.
- **Activity Log**: This widget displays the different actions performed in each event, namely regarding linked events, attached files, and notes.
- **Location**: Displays the distance between the shipping and device locations.
- **Refunds and Chargebacks**: Shows the refunds / chargebacks for the payment under investigation.

Add entities to listsâ

Transaction Fraud includes out-of-the-box lists that automatically classify incoming events. Case Manager looks at specific entities in each event (such as card, customer, terminal or device) and validates, through the out-of-the-box lists, if those entities are associated with a known fraud source or if they belong to a secure source.

As a fraud prevention agent L2 or L3, you can manage lists to temporarily or permanently `approve`, `decline`, or `limit/review` certain entities.

To add entities to lists, follow the procedure that applies best to your case:

Add an entity to a list via Settings tabâ

To add an entity to a list, hover over the **Settings** tab and select **Lists Management**:

1. On the far right column, hover over the list you want to update and select the **Update Lists** button.

2. Fill out the mandatory fields. The **Add Note** and **Validity** are optional fields.

Validity field

Note that, by not filling the start and end dates in the *Validity* field, you are not setting an expiration date.

Add or update all entities from an eventâ

To add or update all entities in an event, open the event's page as your starting point:

1. Select the **More Actions** button and choose the **Update Lists** option.

   Tick each box according to your needs (i.e. approve, decline, or limit/review entity).

Validity field

Note that, by not filling the start and end dates in the *Validity* field, you are not setting an expiration date.

Add an entity from an event to a listâ

To add an entity from an event, open the event's page as your starting point:

1. Select the **Update Lists** icon present in the following widgets:
    - **Transaction**, located next to the *Card ID* field.
    - **Customer**, located next to the *Customer Email* field.
    - **Terminal**, located next to the *Terminal ID* field.
    - **Device**, located next to the *Device ID* and *Device IP* fields.
    - **Payment Method**, located next to the *Card Country* field.
2. Select **Add to New** and fill out the mandatory fields. The **Add Note** and **Validity** are optional fields.
3. After updating the necessary entities, the event header updates the **Listed Entities** status accordingly.

Validity field

Note that, by not filling the start and end dates in the *Validity* field, you are not setting an expiration date.

Create Self-Service Rulesâ

Self-Service Rules (SSRs) allow fraud prevention agents L3 to quickly create and publish a rule directly in production.

SSRs are useful to, for instance, restrict the traffic on a specific combination of channels, BIN, Regions, etc. in order to block fraudulent activity.

Well-defined SSRs provide fraud prevention agents with a clear starting point for their analysis. Instead of manually reviewing every payment, they can focus on payments that trigger specific rule-based alerts. This type of approach saves time and allows them to concentrate their efforts on high-risk payments, leading to more efficient and effective investigations.

To create SSRs, open the **Self Service Rules** tab and follow the instructions. The following image, provides you with an example of an SSR that bypasses the **Auto-Decline-Card-FB1** feedback rule to be auto-approved, even if the transaction is marked as `fraud`.

Review casesâ

The main purpose of cases is to link events when there is a common element that suggests a connection. By aggregating events into a case, you save time in your investigation.

To create a case, follow the procedure that applies best to your case:

Create a case via Cases tabâ

To create a new case, hover over the **Cases** tab and select **New Case**:

The case becomes visible in the **Cases / Case Details** page.

Link event to case or create case from the Event pageâ

The **Event** page allows you to take the following actions:

- Link event to an existing case.
- Create a case from the event.

To create a new case from an event or link the event to an existing case, open the event under analysis and select the **More Actions** button:

Fill out the requested fields according to the procedure that applies best to your case (i.e. link the event to an existing case or to create a case from that event).

Access created casesâ

To access created cases, hover over the **Cases** tab and select the **Cases** option. This option takes you to the **Search By Case** page, which lists all created cases.

By selecting a case from the list of created cases, you're taken to the **Case Details** page.

The header provides you with the following useful information:

- **Name**: Displays the case's name.
- **ID**: Provides the case's alphanumerical ID.
- **Status**: Displays the case's state: **To be reviewed**, **In analysis**, **Waiting for info**, and **Closed**.
- **Source Type**: Provides the case's use case, e.g. Suspected Fraud or Suspected AML.
- **Assignee**: Identifies the user assigned to the case, and, if not assigned to anyone, allows you to assign the case to yourself.

There are a number of actions you can take in a case, namely:

- Assign the case to yourself.
- Update the case's status (i.e. **To be reviewed** / **In analysis** / **Waiting for info** / **Closed**).
- Edit the case.
- See the events linked to the case.
- Attach new files or download existing files attached to the case.
- Add or see notes.
- See the activity log, i.e. confirm the different actions performed, namely regarding linked events, attached files, and notes.

-
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- Merchant

Merchant

Case Manager is an alert management tool that enables fraud prevention agents to review transactions that fall within a risk threshold and raise alerts. Case Manager includes out-of-the-box resources tailored to the Transaction Fraud for PSPs use case, which automate event processing and organize teams' workload efficiently.

This guide helps you to:

- Understand the event flow.
- Investigate alerts, which involves accessing, filtering, and reviewing events, alerts and cases.

-
- Operational Work
- Case Manager Guide
- PSP
- Event Flow

On this page

Event Flow

As a fraud prevention agent, understanding the event flow is a key step to managing the alert investigation efficiently. This page helps fraud prevention agents follow the status of alerted events, e.g. investigation is in progress, or on hold.

In this page, you will also learn how the out-of-the-box automation rules play a role in the event flow. These settings integrate an automated process that sorts the alerts by priority for you.

Event statusesâ

Case Manager displays the event's statuses in the **Event**'s page as follows:

- **System Decision**: Indicates the outcome processed by the Feedzai's risk engine. The possible values are:
    - Accept
    - Decline
- **Operational Status**: Indicates the alerted event's investigation stage. The possible values are:
    - Unreviewed
    - In progress
    - On hold
    - Closed
- **Fraud Label**: Indicates the fraud prevention agent's final decision. The possible values are:
    - Fraud
    - Not Fraud

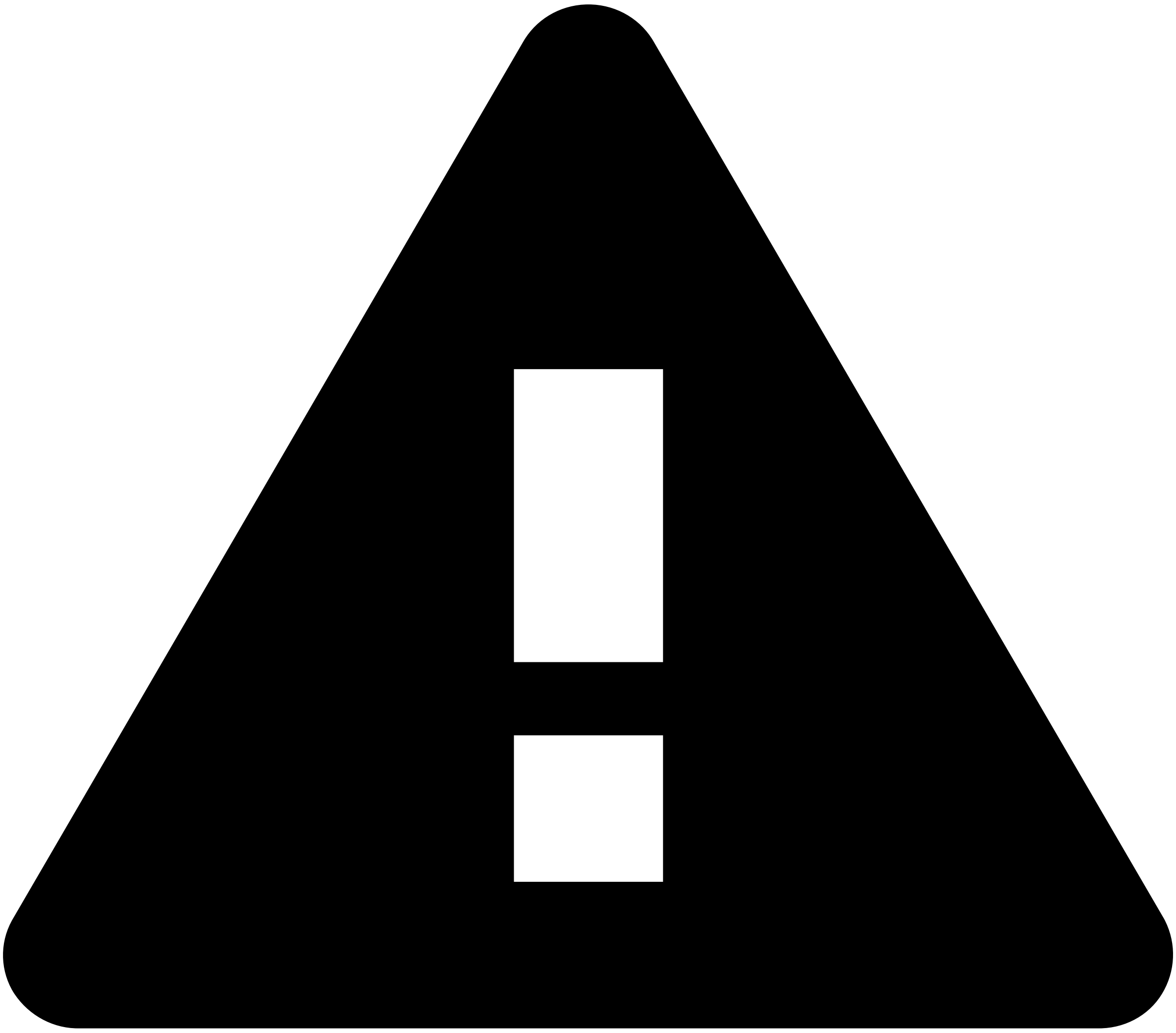
Event status flowâ

The following figure shows the event flow, from the moment it arrives to Case Manager until the fraud prevention agent marks it as `fraud` or `not fraud`.

The system decision is set by the risk engine and won't change during the investigation.

TFP includes out-of-the-box automation rules that fully automate each operational status:

1. Event arrives to Case Manager: a rule sets the operational status to "Unreviewed".
2. Fraud prevention agent is assigned to the event: a rule sets the operational status to "In Progress".
3. Fraud prevention agent takes a `fraud` or `not fraud` decision: a rule sets the operational status to "Closed".



Feedback output for refunds / OCTs / chargebacks

For refunds and OCTs, any decision that the fraud prevention agent takes (i.e. `fraud` or `not fraud`) will reflect in the alerted events as follows:

- All the events associated with a payment's `lifecyle_id` will be updated according to the fraud prevention agent's decision, i.e. `fraud` or `not fraud`.
- For refunds and OCTs with no associated payments, a new feedback event will be generated with the same `lifecycle_id` as the refund or OCT event, and the `fraud` or `not fraud` decision label updated on the refund / OCT.

For chargebacks, which are sent via API, a new feedback event will be generated with the same `lifecycle_id` as the chargeback event, and the `fraud` or `not fraud` decision label updated on all payment / refund / OCT events associated with the same `lifecycle_id`.

-
- Operational Work
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- PSP
- Investigate Alerts

On this page

Investigate Alerts

As a fraud prevention agent, the first step you should take is to access the workload. Depending on the risk engine's decision, you may find a set of events for review.

This page helps you understand how to access the events and cases that are up for review.

Access events, alerts, and casesâ

When events arrive at Case Manager, they can be displayed in more than one page, depending on the fraud risk and the complexity. On Case Manager's sidebar you can access:

- **Search By Event** : Displays all the events (alerted and non-alerted).

- **Alerts** : Events that trigger a risk engine rule with the `alert` action, flagging a high fraud risk. When such rule is triggered, Case Manager displays the event both in the Search by Event and Alerts pages. The Alerts page allows you to focus only on triggered events.
- **Cases** : Events and alerts can be grouped into cases for a more organized action-taking process. This allows you to access events that may be correlated and that require a thorough investigation. Cases offer extra functionalities, such as note attachment.

The following figure gives you an idea of how events, alerts, and cases relate to each other:

tip

Events versus Alerts: The Search By Event page shows all the alerted and non-alerted events. The Alerts page shows only the alerted events.

Since there is a much higher percentage of genuine payments relative to fraudulent ones, the Search By Event page is not an effective way to access your workload.

Filter events, alerts, and casesâ

The Search by Event, Alerts, and Cases pages allow you to narrow down the workload for your analysis through filters.

Alerts* page: Depending on your use case, the Alerts page may group events into more channels.

Select and analyze eventsâ

As previously mentioned, the key pages to access your workload are Alerts and Cases pages.

Whether you open an alerted or non-alerted event, you will land on the Event page, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a comprehensive approach, in which you open several events, e.g. from the same customer ID, and search for fraud evidence by looking at multiple widgets.

The Event page displays the following information:

- **Rules triggered**: Shows all the rules triggered by the event.
- **Transaction**: Shows the transaction details, such as date, amount, and payment method.
- **Genome**: Enables fraud agents to find crime patterns using link analysis and graph technology. Learn more in the Read Genome graphs section.
- **Merchant**: Includes information about the merchant that processed the payment, such as merchant's ID, MCC, and country.
- **Customer**: Includes information about the customer, such as the customer's ID, city, and address. To help your investigation, you can check this widget both against the billing and shipping widgets.
- **Terminal**: In a card present transaction, it shows information about where the transaction took place.
- **Device**: In a card not present transaction, it shows information about the device used to purchase goods/services.
- **Dispute**: In the event of chargeback, the dispute widget shows information on the reason why the payment is being disputed.
- **Billing**: Includes details about the invoice, such as the billing name, address, and ZIP code.
- **Shipping**: Includes details about the shipping, such as the shipping name, address, and ZIP code.
- **Items**: Shows information such as the name and ID of the purchased goods/services.
- **Payment Method**: Includes information about the payment method used, e.g. card or alternative payment method details.
- **Transaction History**: Displays all the transactions associated with the event under investigation.
- **Shipping Distance**: Displays the distance between the shipping and device locations.
- **Attached Files**: This widget allows you to add files that are useful for subsequent investigation steps.
- **Notes**: This widget shows any notes made under a given event.
- **Activity Log**: This widget displays the different actions performed in each event, namely regarding linked events, attached files, and notes.
- **Refunds and Chargebacks**: Shows the refunds / chargebacks for the payment under investigation.

Read Genome graphsâ

Feedzai Genome enables fraud prevention agents to find and visualize complex fraudulent payment patterns using link analysis and graph technology.

The powerful link analysis technology underlying Feedzai Genome, along with the tool's visual capabilities, help fraud prevention agents make decisions faster and with greater accuracy, thus lowering false positives in manual detection. This means that fraud prevention agents can understand the relationship between all the entities participating in a payment, such as the customer, the merchant, and the cards used. This enables fraud prevention agents detect complex fraud scenarios such as Account Takeover.

To better understand how you should analyze links, consider the following example:

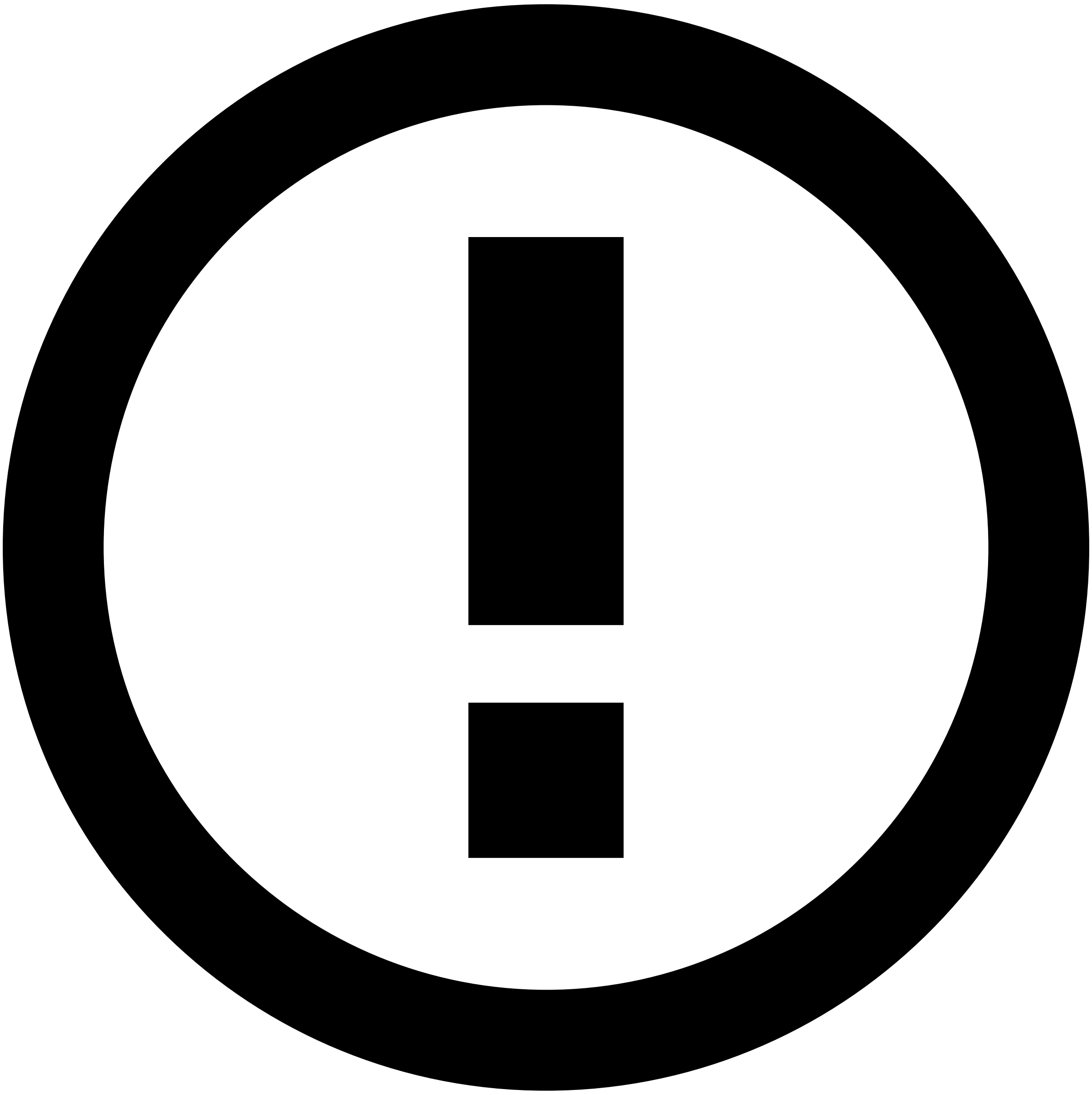
A user performs a payment in a merchant. That transaction typically involves the following entities:

- Customer
- Card
- Merchant

Genome displays each of these entities as nodes, and the relationship between them as the edges that connect the nodes.

tip

Nodes (1) represent entities and **edges (2)** represent the connection between nodes.



info

Genome displays only the nodes that are related to an alert under analysis.

Now that you understand what nodes and edges are, learn how to properly read a Genome graph. Consider the following core concepts:

Node entitiesâ

- Merchant corresponds to the merchant receiving payments.
- Customer corresponds to the customer performing payments.
- Card corresponds to the card of the customer performing payments.

Node and edge colorsâ

The graph's node and edge colors are visible on the **Review** lens and correspond to the state of events associated with those nodes and edges:

- Yellow - at least one event in pending state

- Red - at least one confirmed fraud event
- Purple - at least one declined event
- Green - all events accepted

Edge gradient and money flowsâ

The edge gradient is directly connected to the money flows' functionality. This gradient indicates the money flow direction through the edges' color as follows:

- Grey, with a gradient on the edge's color, indicates who sends and who receives the money. The edge displays a lighter color next to the sender node and darker color next to the receiver node. The money flows from the lightest to the darkest side of the edge.
- Purple is used to indicate that there are transfers in both directions between two entities. For instance, consider two account entities A and B, respectively, where the money would flow from account A to B, as well as from B to A. When there is a history of fraud, the edge displays red instead of purple.
- Red indicates that there is a history of fraud involving at least one transaction.

info

Note that, when opening a graph, the **Review** lens is displayed by default. To see the edge gradients with the aforementioned colors, move to the **Genome** view.

Genuine payment exampleâ

This graph shows a genuine payment use case. Even though there is at least one declined payment from one customer node (purple node), the money flows between these entities are not considered fraudulent (purple gradient).

Fraudulent payment exampleâ

This graph shows a number of fraudulent events:

- (1) In this example, there has been at least one declined payment (purple node). According to the edge gradient color (red gradient), there is also previous history of fraud between these two entities.
- (2) This is a clear example of a fraudulent payment from a customer to a merchant (red nodes). According to the edge gradient color (red gradient), there is also previous history of fraud between these two entities.
- (3) These two examples show at least a confirmed fraudulent payment (yellow nodes). According to the edge gradient color (red gradient), there is also previous history of fraud between these two entities.

To learn more about link analysis, read Genome's documentation.

Review casesâ

The main purpose of cases is to link events when there is a common element that suggests a connection. By aggregating events into a case, you save time in your investigation.

info

Cases can be omnichannel. This means that you can aggregate events from different channels.

The Cases / Case Details page header displays the following information:

- **Name**: Displays the case's name.
- **ID**: Provides the case's alphanumerical ID.
- **Status**: Displays the case's state: 'New', 'Review in progress', 'Closed' and 'Closed with SAR'.
- **Source Type**: Provides the case's use case, e.g. Fraud or AML.
- **Queue**: Specifies the queue the case is stored in.
- **Source type**: Identifies what originated the investigation (e.g. Referrals from customer-facing employees).
- **Assignee**: Identifies the user assigned to the case.

The header also displays several action buttons, according to the case's state: **To be reviewed** / **In analysis** / **Waiting for info** / **Closed**.

Below this header, contextual information is provided as follows:

- **Linked Events**: Presents events associated with the case under investigation.
- **Attached Files**: This widget allows you to add files that support your investigation.
- **Notes**: This widget shows any notes made under a given case.
- **Activity Log**: This widget displays the different actions performed in case, namely regarding linked events, attached files, and notes.

-
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PSP

Case Manager is an alert management tool that enables fraud prevention agents to review transactions that fall within a risk threshold and raise alerts. Case Manager includes out-of-the-box resources tailored to the Transaction Fraud for PSPs use case, which automate event processing and organize teams' workload efficiently.

This guide helps you to:

- Understand the event flow.
- Investigate alerts, which involves accessing, filtering, and reviewing events, alerts and cases.

-
- Operational Work
- Integration Guide
- User Acceptance Testing

On this page

User Acceptance Testing

The User Acceptance Testing (UAT) guide ensures that the Transaction Fraud for PSPs solution is properly tested. The following sections guide you through the necessary steps to test rules, cases, and automation rules.

Requirementsâ

To execute all the tests, make sure you have the customer's QA or Pre-Production environment, as well as the UAT Postman Collection.

The Postman collection includes a set of requests and environment variables. Make sure you import both and change the environment variables into the desired enrivonment.

Test rulesâ

1. Run the `Payment-FR20` folder. Do not run each request individually.

2. Confirm that 4 events show in Case Manager's **Search By Event** page. Note that one payment to merchant `P1 Name (P1 ID)` and three payments to merchant `P2 Name (P2 ID)` occurred.

3. Select the **Payments** channel and confirm if:

    - 1 payment was automatically declined.
    - 3 payments were automatically approved.

Test automation rulesâ

Perform the following steps for the first approved event (i.e. `P1 Name (P1 ID)`):

1. Mark `P1 Name (P1 ID)` event as `Fraud` with the Authorization Dispute reason, and select **Submit and Continue**. In the **Update Lists**, mark the **Terminal ID** as `Decline` and select **Submit**. Confirm if the automation rule has activated and that **Status** has changed to `Closed`.
2. Execute the Postman Collection's last request (`03 - Payment 3`) from the Payment-FR20 folder. Execute it individually and confirm if the response includes the `Terminal-ID-Decline-PN7`.
3. Go to the latest event in Case Manager to confirm if the new event has been declined and triggered the `Terminal-ID-Decline-PN7` rule.

Test queues and casesâ

Perform the following steps for the last declined event (i.e. `P2 Name (P2 ID)`):

1. Go to the **Event Details** page and confirm if the `Cards-Behavior-FR20` has triggered.
2. Select **Assign to me**.
3. Confirm if the automation rule has activated, and if the **Status** changed to **In Progress**.
4. Create a new queue and link `P2 Name (P2 ID)` to a new case.
5. Confirm if `P2 Name (P2 ID)`:
    - Is assigned to the current Case Manager user.
    - Is **In Progress**.
    - Has moved to the new queue.
    - Is linked to the new case.

-

- Operational Work
- Pulse Guide
- Migrate Pulse Resources
- Configure Resources

On this page

Configure Resources

This page helps you configure your resources in the **new environment**. This process includes the following tasks:

- **Resource versioning**: Guides you through the process of changing the name of your resources for versioning.
- **Resource cleanup**: Helps you remove schemas and plans that are not being used, as well as outdated rules projects.
- **Resource checkup**: Helps you make sure that all the resources are configured as expected.

Resource versioningâ

To version plans in Pulse, execute the following steps:

Duplicate plansâ

1. In the **Data Science** tab, duplicate the most recent version of the plans that you want to version (e.g. assume *Custom Plan m1/d1* as an example, following the nomenclature *project-name mm/dd*).

info

Duplicate Plans: Plan executions are not transferred from the original plan to the duplicate version. The next subsection explains how to add executions to duplicate plans.

1. Rename the plans according to your versioning nomenclature (e.g. assume *Custom Plan m1/d2*).

Add executions to plansâ

1. Select the options menu in the Pulse application where you have duplicated the plans.
2. Select the option.
3. Choose the same resource bundle that you have used to import the plans you are working on, and upload it using the button.

4. In the **Choose Resources** step, open the **Plans** tab and add executions as follows:
5. Use the checkbox to choose the plans that you have duplicated (e.g. *Custom Plan m1/d1*).
6. Choose the **Local Plan Name** that you have duplicated (i.e. *Custom Plan m1/d2*).
7. Make sure you have all the plan executions that you want to keep in the new plans by expanding the collapse button under **Plan Executions**.
8. Continue to the **Next** step.
9. Since you are importing the plan executions to the duplicate plan, the schema should always match and the **Schema Conflicts** step should not be prompted. If after duplicating the plan in the Duplicate plans subsection the plan schema is changed, the **Schema Conflicts** step is prompted. Solve the conflicts by choosing **Keep Imported** for each plan execution.
10. Continue to the **Next** step and select the button to complete the import.
11. In the **Data Science** tab, open the executions list using the button in each of the new plans and confirm if they have the correct plan executions.

Create new executionsâ

In your Pulse application, you may have plans that use an existing schema as input and generate a new schema as output. You may also have plans that use the output schema generated from another plan as an input. Start by creating snapshots for the plans that use the existing schema:

1. In the **Data Science** tab, open the plan in which you want to create the new snapshot (e.g. *Custom Plan m1/d2*).
2. In the **More Actions** button, select **Plan Settings** and execute the following steps:
3. In the **Schema** option, choose the schema that applies to the plan (e.g. *Extended Schema [last version]*), and **Confirm** the change using the button in the warning box.
4. In the **Timestamp Field** option, select `event_occurred_at`.
5. **Save** your changes.
6. Select the button and execute the following steps:
7. Rename the execution according to your versioning nomenclature (e.g. *Custom Plan m1/d2 vN*, where N increments the version number).
8. In the **Data Sink** section, tick the export box and rename the new schema (e.g. *Custom Plan m1/d2 vN*). For traceability purposes, Feedzai recommends using the same name for the execution and the schema.
9. the plan execution.
10. Create executions for the plans that input the schema generated in the previous steps (e.g. *Standard Plan m1/d2*):

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Consider the scenario in which you need to edit a Standard Plan. Repeat Steps 2 and 3, and consider the following:

- **Schema**: In this example, the input of the *Standard Plan d1/m2* is the output schema of the Custom Plan generated in Step 3.
- **Execution name**: Example, *Standard Plan m1/d2 vN*, where N increments the version number.
- **Schema name**: Example, *Standard Plan m1/d2 vN*, where N increments the version number.

For traceability purposes, Feedzai recommends using the same name for the execution and the schema. :::

Version rules projects and snapshotsâ

1. In the **Data Science** tab, select the options button in the **Rules Projects** panel to configure each rules project.
2. Name the rules project according to your versioning nomenclature (e.g. *name_of_rules_project mm/dd*).
3. Select the **Schema** created in [Create new executions](#)' step 4.
4. Set the **Target Variable** using the one that applies to your use case (e.g. `decision` in a typical application).
5. **Save** your changes.
6. For each non-self-service rules project (non-SSR):
7. Open the rules project.

8. In the **Snapshots** panel, create a new snapshot.
9. Name the snapshot according to your versioning nomenclature (e.g. *name_of_rules_project mm/dd vN*).
10. the snapshot.

info

The snapshots for self-service rules (SSR) projects will be created in the next subsection.

1. Publish your application using the button.
2. Apply your changes to complete the process using the button.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

Update the workflowâ

1. In the **Workflow** tab, open the workflow that you want to update.
2. In the **Schema**, make sure the correct schema is selected (e.g. *Input Schema vN*, where N increments the version number). If you don't have the correct schema, select it from the list and **Confirm** the change using the button in the warning box.
3. In the diagram, edit each operator (e.g. plans and rules) and execute the following actions:
4. In the **Main** tab, select the resource's and snapshot's latest versions.
5. **Save** your changes.
6. For the operator containing SSR projects, select the rules project's latest version. If you have changed the name of the SSR projects as specified in the previous version's step 2, you need to publish the application to create new snapshots. Pulse will automatically recognize the new snapshot and replace the current value of the snapshot field with the new one. To activate this behavior, execute the following:

7. **Save** and exit the workflow.
8. Publish your SSRs using the button.
9. Apply your changes to complete the process using the button.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

1. Return to the workflow and make sure that the schema of the SSR project has been updated with the new snapshot.

Resource cleanupâ

1. In the **Data Configuration** tab, delete the schemas that are not being used. The final resources are specific to each use case. As a rule of thumb, you should have an Input Schema and Extended Schema.
2. In the **Data Science** tab:
    1. Delete the data sources that were created when importing the resources.

    2. Delete the plans that are not being used.
    3. Delete the outdated rules projects.

Resource checkupâ

1. In the **Data Configuration** tab, under the **Schema** panel, ensure that the new schemas have the new inserted fields.
2. In the **Data Science** tab, under the **Plans** and **Rules Projects** panels, confirm if:
    1. The version is correct.
    2. The input schema is correct.
    3. The **Target Variable** for rules projects is correct.
    4. The new executions and snapshots have been created for plans and rules.
3. In the **Workflow** tab, ensure that each operator has the latest resource and snapshot versions.
4. In the **Lists** tab, make sure that the lists are correctly configured.

Next stepsâ

Now that you finished configuring your resources:

- Make sure you use the Update the Workflow section as reference for publishing further workflow changes.
- Read the Transfer list items page.

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- Migrate Pulse Resources
- Export Resources

On this page

Export Resources

Learn how to export resources (i.e. lists, models projects, plans, and rules projects).

danger

Make sure you back up the environment you want to update:

1. Open Pulse in the new environment.
2. Select the options menu in your Pulse application.
3. **Export Application**.

To export resources:

1. Select the **Options** menu in the Pulse application of your current environment.

2. Select the **Export Resources** option:

1. Each resource has a specific tab. Use each tab to select the resources that you want to export.

info

To make sure that all resources are selected, keep a list of all the model projects, models, plans, plan executions, rules projects, and snapshots that you want to export and insert the name of each resource into the corresponding field.

Model projectsâ

1. In the **Export Resources** pop-up window, the **Model Projects** tab is selected by default. Select the model projects that you want to export.
2. Optionally, if you want to include a file with the model object, select the **Include Binary** option.
3. Make sure that you add each resource to the export bundle.

Plansâ

1. Open the **Plans** tab.
2. Select the plans that you want to export.

info

Since non-deployable plans are not directly deployed in a runtime workflow, you can select only deployable plans.

1. If you want to export a specific configuration of the plan, select the plan executions from the **Executions List**. For traceability and history purposes, Feedzai recommends selecting the last plan execution for each exported plan.

2. Make sure that you add each resource to the export bundle.

Rules projectsâ

1. Open the **Rules Projects** tab.

2. Select the rules projects that you want to export.

3. If you want to export a specific configuration of a rules project, select the snapshots from the **Snapshot List**. For traceability and history purposes, Feedzai recommends selecting all existing snapshots.

4. Make sure that you add each resource to the export bundle.

Once all resources are configured, select the **Export Bundle** button.

Listsâ

Lists are simple resources that can be migrated by exporting the list items from the source environment and importing them in the target environment. The resource transfer process allows you to export and import list metadata, but not list items. List items are exported and imported through a CSV file. To export and import list items refer to the Transfer List Items guide.

Next stepsâ

Now that you can export resources from the current environment, learn how to import resources in the new environment.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/pulse/migrate-pulse-resources/import-resources

-

- [Operational Work]
- [Pulse Guide]
- [Migrate Pulse Resources]
- Import Resources

On this page

Import Resources

Learn how to **import resources** to upload resources to your application in the new environment.

danger

Make sure you perform a backup of the environment that you want to update. If you already performed this backup in the export resources step, ignore the following steps:

1. Open Pulse in the new environment.
2. Select the options menu in your Pulse application.
3. Select the **Export Application** option to download the file with the application.

To import resources:

1. Open the **Options** menu in the Pulse application of your new environment.
2. Select the **Import Resources** option:

1. Execute the sequence of steps.

Upload fileâ

1. **Drag & Drop** the zip file containing the export bundle to the dashed box. Alternatively, **browse for a file** to upload the zip file.
2. Use the button to go to the next step.

Choose resourcesâ

info

To make sure that all resources are selected, keep a list of all the model projects, models, plans, plan executions, rules projects, and snapshots that you want to import and insert the name of each resource on the corresponding field.

You can choose the resources that you want to import by navigating through each resource tab. Note that the **Lists** resource does not have a specific tab because its configurations are automatically imported as follows:

- Lists that do not exist in the new environment are created without items.
- Lists that already existed in the new environment are updated with the new metadata. However, their items are not affected by the import into the new environment.

Modelsâ

1. In the **Import Resources** pop-up window, the **Models** tab is selected by default.
2. Use the checkbox to select the models that you want to import. Note that in Pulse, it is not possible to bulk import models from a model project, and therefore you should import them individually.
3. For each model, choose the **Local Model Project** to which you want to store the model in the new Pulse application. Note that you must have a model project in the new application in order to import the new model.
4. Optionally, if you want to include a file with the model object, select the **Include Binary** option. This option is only available if you include the binary in the export resources step.

info

Dummy Data Source: When you import a model that does not have a compatible data source available in the new environment, Pulse automatically creates a new data source. This new data source only contains the schema with no data associated.

Plansâ

1. Select the **Plans** tab.
2. Use the checkbox to select the plans that you want to import.
3. Choose the **Local Plan Name** for the plan in the new Pulse application.

info

Local Plan Name:

i. You can save the plan as a new plan by choosing the option *project_name* . The *new* option creates a resource named *resource_name (N)*, where N is a number that increments over the highest value of this resource in your new application. Feedzai recommends that you use this approach when you want to version your resources.

ii. You can replace an existing plan by choosing the same name as the one that you are importing. This keeps your environment clean.

4. Choose the plan executions that you want to import by expanding the menu under the **Plan Executions** column. By default, all the plan executions are selected. For traceability and history purposes, Feedzai recommends that you import all the executions.

tip

Executions persistency: Pulse only keeps the executions that you select. Imagine that you choose option 3 and you only select the most recent plan execution. The current plan in the new application will be replaced with the selected execution (i.e. all the executions in the new environment will be deleted).

info

Dummy Data Source: When importing a plan with executions that do not have a compatible data source available in the new environment, Pulse automatically creates a new data source for each execution that you choose to import. This new data source only contains the schema with no data associated.

Rules projectsâ

1. Select the **Rules Projects** tab.
2. Use the checkbox to select the rules projects that you want to import.
3. Choose the **Local Rules Project Name** for the rules project in the new Pulse application:

info

Local Rules Project Name: When you choose the **Local Rules Project Name**, do the same as step 3 for Plans.

4. Select the rule snapshots that you want to import by expanding the menu under the **Snapshots** column. By default, all snapshots are selected. For traceability and history purposes, Feedzai recommends that you import all the executions.

tip

Snapshots persistency: Pulse keeps snapshots as explained in Plans' step 4.

Once you have chosen all the resources that you want to import (i.e. models, plans, and rules projects), go to the **Next** step.

Schema conflictsâ

You are prompted with this step only when the plans' or rules projects' schema previously selected doesn't match the schema of the resources that you currently have in the new environment. Select the schema that you want to keep for each resource:

- **Keep imported**: for the given **Resource Type** and **Resource Name** (i.e. the resource that is being imported), the schema that is selected is kept in the resource that you are importing.
- **Keep local**: for the given **Resource Type** and **Resource Name** (i.e. the resource that is being imported), the schema that is selected is kept in the resource in the new environment.

Once you have chosen the schema for each resource on the list, go to the **Next** step.

Workflowâ

tip

This step is optional since you can also manually update resources in the workflow after importing resources. The workflow's manual update is required in case you decide to perform resource versioning when choosing plans and rules projects in the new environment. This option is recommended if you import a resource that shares the same schema as the one in the new environment (e.g. update a rule that has a new condition).

In this step, associate the resources that you are importing with the new environment's workflow services:

1. Select the **Resource Type** that you want to map to a workflow service.
2. Choose the **Workflow** name that you want to update with the new resource.
3. Choose the **Workflow Service** to which you want to map the resource.
4. After associating the resources with the workflow, select the button.

If the workflow element does not exist in the Workflow yet, you must create the element on the target environment Workflow by taking the following steps:

1. Drag and drop the desired workflow element to match the one from the source environment. Make sure to configure it correctly by selecting the **Edit** icon.

2. Once your workflow element is created, create the connection to the other elements by dragging the **Drag to connect services** icon of the source element to the target element.

   Note that, for data to navigate the workflow from an element to another, schemas have to match. Make sure the schema used in your new element matches the output schema of the previous one. To compare them, select the new element's **Edit** icon and, in the workflow configuration pop-up window, open the **Schemas** tab.

   In case there are schema conflicts, you can adapt the schema of the element -> create a snapshot -> and use it in the new workflow element.

3. The connection between the existing element and the new one can be conditional based on a filter expression. Configure the connection by hovering over the connection arrow and selecting the **Edit** icon.

Next stepsâ

Now that you have imported the new resources in the new application, learn more about the final configurations to make sure that all the resources are working as expected. The next step is also important if you have decided to version your resources.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/tfp/operational-work/pulse/migrate-pulse-resources/overview

-

- Operational Work
- Pulse Guide
- Migrate Pulse Resources

On this page

Migrate Pulse Resources

Rules and Data Science resources such as Plans and Models must be developed, tested, and deployed into Production to score transactions in real time. To keep fraud-fighting tools performant, you need to periodically implement changes that may include:

- Adjusting parameters.
- Incorporating new insights regarding a specific issue.
- Deleting resources because they are no longer relevant.

To ensure that resources that are edited, validated and tested do not affect the runtime in production, the development and testing work are performed in staging environments. This raises the need to transfer your latest developments and combine them with other resources in a new environment. Some of this work is part of the standard product and is not subject to changes whereas, as shown in the following figure, other resources are transferred via a standard procedure named **resource transfer**. You can use resource transfer to autonomously transfer resources between environments.

The Migrate Pulse Resources guide explains how to transfer the following resources:

- Models
- Plans
- Rules
- Lists metadata

Block ownershipâ

While you develop new profiles with plans, re-train models with new data, and/or test new rules using offline environments, your Pulse application (Pulse app or papp) will keep evolving in production. Some resources, such as rules projects are often developed by various teams (i.e. owners). To evolve an application in a collaborative way, you can use the **block ownership**. Each Pulse resource is developed as a block by a single team, within one environment at a time. For example, each rules project should be owned by one team and should be only modified (e.g. adding/removing/editing rules) by the team that owns each rules project.

The block ownership assumes that:

1. Between each deployment, each block evolves in a single environment.
2. For each environment, all the resources evolve using a single Pulse cluster.

By structuring your work with the concept of block ownership, you allow:

1. Multiple teams to work in parallel using different environments.
2. Easy and quick migration of resources between environments.

Flow between environmentsâ

The typical workflow between environments is as follows:

1. **QA**: Run functional tests to verify the effectiveness of the implemented changes. These may include sending transactions to API to validate rules logic or testing profiles with long windows.
2. **Pre-prod**: To ensure the new changes will be successful in Prod, migrate what's in Prod to Pre-prod to run tests with real data.

info

Depending on your service, this environment may not be available out of the box. This is an environment that is commonly used for performance tests.

3. **Prod**: Once tests have successfully run in Pre-prod, update the Prod environment.

This process leverages two Pulse functionalities: exporting and importing resources. The process is completed after you configure the resources in your new environment.

- **Step 1**: Edit resources.
- **Step 2**: Export resources from the current environment.
- **Step 3**: Import resources into the Prod environment.
- **Step 4**: Configure resources in the new environment.

These steps must be followed each time resources are transferred between environments, for instance, between a Pre-prod and a Prod environment.

Next stepsâ

Now that you understand the resource transfer's purpose, you can start exporting resources.

- Operational Work
- Pulse Guide
- Migrate Pulse Resources
- Transfer List Items

On this page

Transfer List Items

Lists are simple resources that can be migrated by exporting the list items from the source environment and importing them in the target environment.

In the process of detecting fraud, it is very common to define lists to block transactions from certain cards, or even to send transactions with common parameters for review (e.g. triggering an alarm every time a transaction occurs). Lists allow you to group and manage these values by storing them in items to further use them with rules and plans.

The resource transfer process allows you to export and import lists metadata, but not list items. List items are exported and imported through a CSV file.

Export list itemsâ

To export list items, take the following steps:

1. Go to the **Lists** tab.
2. Open the list that has the items that you want to export.
3. Export all or only the active list items by selecting the download button. A CSV file is downloaded.

Import list itemsâ

To import list items, take the following steps:

1. Go to the **Lists** tab.

2. Make sure the corresponding list items exist in the target environment. If this is not the case, create a new list by selecting the **New List** button.

tip

You can access the List configuration to make sure the lists from both environments match.

3. Open the list where you want to import the items to.
4. Select the button.
5. **Drag & Drop** or browse the CSV file containing the list items and **Save**.

CSV file formattingâ

When importing CSV files, notice that if you try to upload a list with duplicate entries, Pulse will generate a silent error, and therefore won't notify you.

Consider the following formatting to ensure list items are correctly imported:

Example of a simple listâ

- Card listed to be declined

```
## Header: Key, Description, Start Date (UNIX timestamp), End Date (UNIX timestamp), Enabled

"0000-0000-0000-0000","Stolen card, made multiple transactions in different countries separated by minutes",,,"true"
"1111-1111-1111-1111","Duplicated card, attempted several withdraws trying to determine the maximum amount permitted.","1381708800",,"true"
"2222-2222-2222-2222","Two months ban for breaking rule R102","1381708800","1386633600","true"
```

Example of a complex listâ

- PosNeg list

```
## Header: Key, Value, Description, Start Date (UNIX timestamp), End Date (UNIX timestamp), Enabled

"AAA","{"value":"review"}","Card to be reviewed",,,"true"
"BBB","{"value":"decline}","Card to be declined",,,"true"
```

info

By default, if no dates are set in the **Active from** - **To** fields, new list items become enabled and active.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/fraud-prevention-agent/case-manager/ad-hoc-investigations/event-flow

-
- Operational Work
- Fraud Prevention Agent
- Case Manager Guide
- Add-Hoc Investigations (with add-on)
- Event Flow

On this page

Event Flow

The ad-hoc investigation enables agents with a generic investigation page to log any intelligence gathered from investigations outside the normal merchant monitoring flow. This event allows agents to leave notes and attachments, track the status of the investigation and add it to a case. This page describes the main event life cycles and their associated logic:

- Ad-hoc investigation event header (i.e. sticky bar).
- Ad-hoc investigation event life cycle.

Ad-hoc investigation event's headerâ

Case Manager displays the ad-hoc investigation event header in the **Ad-Hoc Investigation Event Details** page as follows:

- **Alerted**: Indicates whether the ad-hoc event comprises an alert (e.g. triggered one or more alerting rules).
- **Created At**: The date and time the ad-hoc investigation was generated for that merchant.
- **Merchant ID**: The unique ID number of the merchant under investigation.
- **Name**: The name of the merchant under investigation.
- **Status**: Indicates the ad-hoc investigation stage. Possible values are:
    - **Unreviewed**: Has not undergone investigation.
    - **In progress**: Under review by an agent.
    - **On hold**: Temporarily paused while waiting for extra information.
    - **Closed**: Final status after making a decision.
- **Analyst Decision**: The decision to either take action or drop the event. The possible values are:
    - **Action taken**
    - **Dropped**
- **Reasons**: Indicates the reasons why the agent considered the alert irrelevant (**Dropped**) or needed further investigation (**Action taken**).
- **Queue**: Indicates in which queue the ad-hoc investigation event is, based on the investigation priority. Queues can be configured by a fraud prevention manager.
- **Assignee**: The current agent assigned to the ad-hoc investigation event.

info

Note that reasons are self-serviceable and can be configured by a fraud prevention manager or by a fraud prevention agent with such permissions.

Ad-hoc investigation event life cycleâ

ad-hoc investigation event life cycle - step by step

Take merchant Acme Inc as an example:

1. Send a [request] (link-to-endpoint) to create an ad-hoc investigation for merchant Acme Inc, after which the event is created in Case Manager within the Ad-Hoc Investigations channel.
2. Assign the event to yourself and start the investigation.
3. Consider the following status flow after the event has been assigned:

Next stepsâ

Now that you understand the ad-hoc investigation event flow, you are ready to investigate merchants.

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/fraud-prevention-agent/case-manager/ad-hoc-investigations/investigate-merchants

- Operational Work
- Fraud Prevention Agent
- Case Manager Guide
- Add-Hoc Investigations (with add-on)
- Investigate Merchants

On this page

Investigate Merchants

As a fraud prevention agent, the first step you should take is to access the workload. Depending on the risk engine's decision, you may find a set of ad-hoc investigation events for review.

This page helps you understand how to access the events and cases that are up for review.

Access events and casesâ

When events arrive in Case Manager, they can be displayed in more than one page, depending on the fraud risk and the complexity. On Case Manager's sidebar you can access:

- **Search By Event** : Displays all the events (alerted and non-alerted). By default, some filters, such as **Alerted** and **Created Date**, are already populated to always show the latest alerted events.
- **Cases** : Events can be grouped into cases for a more organized action-taking process. This allows you to access events that may be correlated and that require a thorough investigation. Cases offer extra functionalities, such as note attachment.
- **Settings** : This option is only available for certain roles and provides various management tools such as queues, automation rules, SLAs, self-service rules and lists.
- **Dashboards** : This option is only available for certain roles and displays dashboards to help you monitor merchants, agents and rules, as well as analyze transactions, alerts, and cases.

Select and analyze eventsâ

As previously mentioned, the key pages to access your workload are the **Search by Event** and **Cases** pages.

Whether you open an alerted or non-alerted event, you will land on the ad-hoc investigation's **Event** page, which contains information that helps you make your decision.

The **Event** page displays the following information.

Merchant Detailsâ

The **Merchant Details** widget contains information about the merchant under investigation, such as merchant's ID, MCC, and country. To obtain more information about the merchant, open the **Merchant Details** page in a different tab using the **External Link** icon.

Investigation Detailsâ

The **Investigation Details** widget shows where the investigation originated from and the reasons for creating the ad-hoc investigation.

Transaction Historyâ

The **Transaction History** widget shows a complete overview of payments, merchant monitoring, and ad-hoc investigations.

- The **Payments** tab lists in descending chronological order all the alerted payments processed by the merchant under analysis.
- The **Merchant Monitoring** tab lists in descending chronological order all the alerted merchant monitoring events over a specified time period.
- The **Ad-Hoc Investigations** tab lists in descending chronological order all the ad-hoc investigation events created for the merchant under investigation.

Note that, by default, the **Ad-Hoc Investigations** channel is selected and filtered by `Merchant ID` to refine the results by the current merchant under analysis.

You can also narrow down results based on time periods such as "Last week" by selecting the date options filter.

Attached Filesâ

The **Attached Files** widget allows you to add files that support your investigation.

Notesâ

The **Notes** widget shows any notes made under a given ad-hoc investigation, such as "merchant being investigated due to suspicious activity" or "information requested".

Activity Logâ

The **Activity Log** widget includes all the updates made to the event, such as information about the date and time the ad-hoc investigation has been created, or a file has been attached to the event. These updates are listed in descending chronological order. This widget also logs alerts occurring over the course of a day.

Analyze merchantsâ

Fraud prevention agents need to closely monitor merchants to understand the risk they represent. The Merchant Details page displays the merchantâs relevant data and risk insights over the last 30 days.

The **Merchant Details** page displays the following information:

- **Merchant Overview**: Breaks down information into several visualizations, as follows:
  - **Merchant Details**: Contains relevant information about the merchant, such as ID, Merchant Category Code (MCC) or tax identification number, as well as contact details.
  - **Location**: Contains information related to the merchant's location, i.e. registration country and Ultimate Beneficiary Owner (UBO) names.
  - **Account**: Includes information about the merchant's account status (i.e. whether it's active or not) and bank account.
  - **Total Transactions**: Shows the ratio of approved vs. declined transactions over the last 30 days.
  - **Timeline of merchant risk assessments**: Displays a timeline with the number of times the merchant has been alerted in the last 30 days, presenting the evolution of the merchant's risk in a visual way. Other important indicators such as chargebacks, approved transactions, average transaction value, top locations and the number of refunds are key metrics for ongoing management of merchants' risk.
  - **Top location history**: Displays the top transaction distribution (in percentage) per market.
  - **Transaction Amounts**: Shows the minimum, median, and maximum transaction amounts received by the merchant over the last 30 days.
  - **Top Chargebacks**: Displays the total number of chargebacks and chargebacks to sales as well as the total amount of chargebacks over the last 30 days.
  - **Top Refunds**: Displays the total number and amount of refunds over the last 30 days.
- **Merchant Monitoring Assessments**: Shows the previous daily assessments including rules triggered, investigation status, and agents' decisions.
- **Payment History**: Displays all the transactions associated with the merchant under investigation.
- **Notes**: Shows any notes made under a given merchant.
- **Attached Files**: Allows you to add files that are useful for subsequent investigation steps.

Create a caseâ

Cases allow you to group events when there is a common element that suggests a connection between them. This allows you to save time in your investigation and have a more organized and better-structured action-taking process.

info

Cases can be omnichannel. This means that you can aggregate events from **Cards**, **Digital Activity**, and **Transfers**.

You can create cases through the **Cases** page, as well as through the **Event**, **Search**, and **Alerts** pages as follows:

- Cases page
- Search page
- Event page

1. Hover over the **Cases** icon and select **New Case**.
2. Fill out the necessary information.

Review casesâ

Once you know which cases are up for review, you can use a case's details page to start the investigation process. This section helps you to understand the different case details displayed and how they can help you to classify cases.

The **Cases / Case Details** page header displays the following information:

- **Name**: Displays the case's name.
- **ID**: Provides the case's alphanumerical ID.
- **Status**: Displays the case's state: "New", "Review in progress", "Closed" and "Closed with SAR".
- **Source Type**: Provides the case's use case, e.g. Fraud or AML.
- **Queue**: Specifies the queue the case is stored in.
- **Source type**: Identifies what originated the investigation (e.g. Referrals from customer-facing employees).
- **Assignee**: Identifies the user assigned to the case.

The header also displays several action buttons, according to the case's state: **To be reviewed** / **In analysis** / **Waiting for info** / **Closed**.

Below this header, contextual information is provided as follows:

- **Linked Events**: Presents events associated with the case under investigation.
- **Attached Files**: This widget allows you to add files that support your investigation.
- **Notes**: This widget shows any notes made under a given case.
- **Activity Log**: This widget displays the different actions performed in the case, namely regarding linked events, attached files, and notes.

Ad-Hoc Investigations (with add-on)

Ad-hoc investigations allow fraud prevention agents to initiate an investigation for a specific merchant from an external source (e.g. a regulator requesting a specific merchant assessment) or on an ad-hoc basis.

This channel also leverages the Merchant Monitoring workflow by incorporating insights from ad-hoc investigation events. This is accomplished through cross-channel metrics, which allow the Merchant Monitoring workflow score to consider relevant events or data from this channel.

This guide provides you with the following information:

- Event flow
- Investigate merchants on an ad-hoc basis

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/fraud-prevention-agent/case-manager/daily-monitoring/event-flow

- Operational Work
- Fraud Prevention Agent
- Case Manager Guide
- Daily Monitoring (standard)
- Event Flow

On this page

Event Flow

As a fraud prevention agent, understanding the merchant monitoring event flow is a key step to managing the alert investigation efficiently. This page describes the main life cycles and explains the logic behind:

- Merchant monitoring event header (i.e. sticky bar)
- Merchant monitoring event life cycle
- Payment event life cycle

Key conceptsâ

This section clarifies key concepts that will be mentioned throughout this page to ensure your understanding.

Concept	Description
Merchant monitoring assessment	Occurs when the merchant monitoring workflow runs for a particular merchant, updating their profiles (metrics) - e.g. total payments processed over the past 30 days -, and running rules as well as potential machine learning models that look at those profiles to evaluate the risk of that merchant. These assessments occur daily, meaning fraud prevention agents review the previous day's data.
Merchant monitoring event	The event that is generated at the beginning of each day and where the outcome of the merchant monitoring's daily assessment is written.
Related transactions	Transactions that contributed to a merchant monitoring rule, e.g. merchant Acme is alerted if the number of transactions with an amount greater than $500 over the past 24 hours is above a threshold of 10. The related transactions for this rule are the transactions greater than $500 over the past 24 hours processed by merchant Acme.
Risk strategy	Set of rules and machine learning models (when available), which comprise a workflow, as well as agents' actions (e.g. assigning events), state machines (e.g. moving an event to "On Hold"), queues, automation rules and service-level agreements (SLAs). Each channel - payments and merchant monitoring - has their own risk strategy.

Merchant monitoring event headerâ

Case Manager displays the merchant monitoring event's header in the **Merchant Monitoring Event Details** page as follows:

- **Alerted**: Indicates whether the event comprises an alert (e.g. triggered one or more alerting rules).
- **Created at**: The date and time the merchant monitoring event was created with the daily assessment.
- **Status**: Indicates the alerted merchant's investigation stage. Possible values are:
  - **Open**: Has not undergone investigation.
  - **In progress**: Under review by an agent.
  - **On hold**: Temporarily paused while waiting for extra information.
  - **Closed**: Final status after making a decision.
- **Analyst Decision**: The decision to either take action or drop the event. The possible values are:
  - **Action taken**
  - **Dropped**
- **Reasons**: Indicates the reasons why the agent considered the alert irrelevant (**Dropped**) or needed further investigation (**Action taken**).

info

Note that reasons are self-serviceable and can be configured by a fraud prevention manager.

- **Queue**: Indicates in which queue the merchant monitoring event is, based on the investigation priority. Queues can be configured by a fraud prevention manager.
- **Assignee**: The current agent assigned to the alerted merchant monitoring event.

Merchant monitoring event life cycleâ

At the beginning of each day, PSPs and acquirers can review each of their merchants' activity over the past 24 hours to facilitate the decision whether to pay out the funds or not.

Take a close look at the merchant monitoring event life cycle.

merchant monitoring event life cycle - step by step

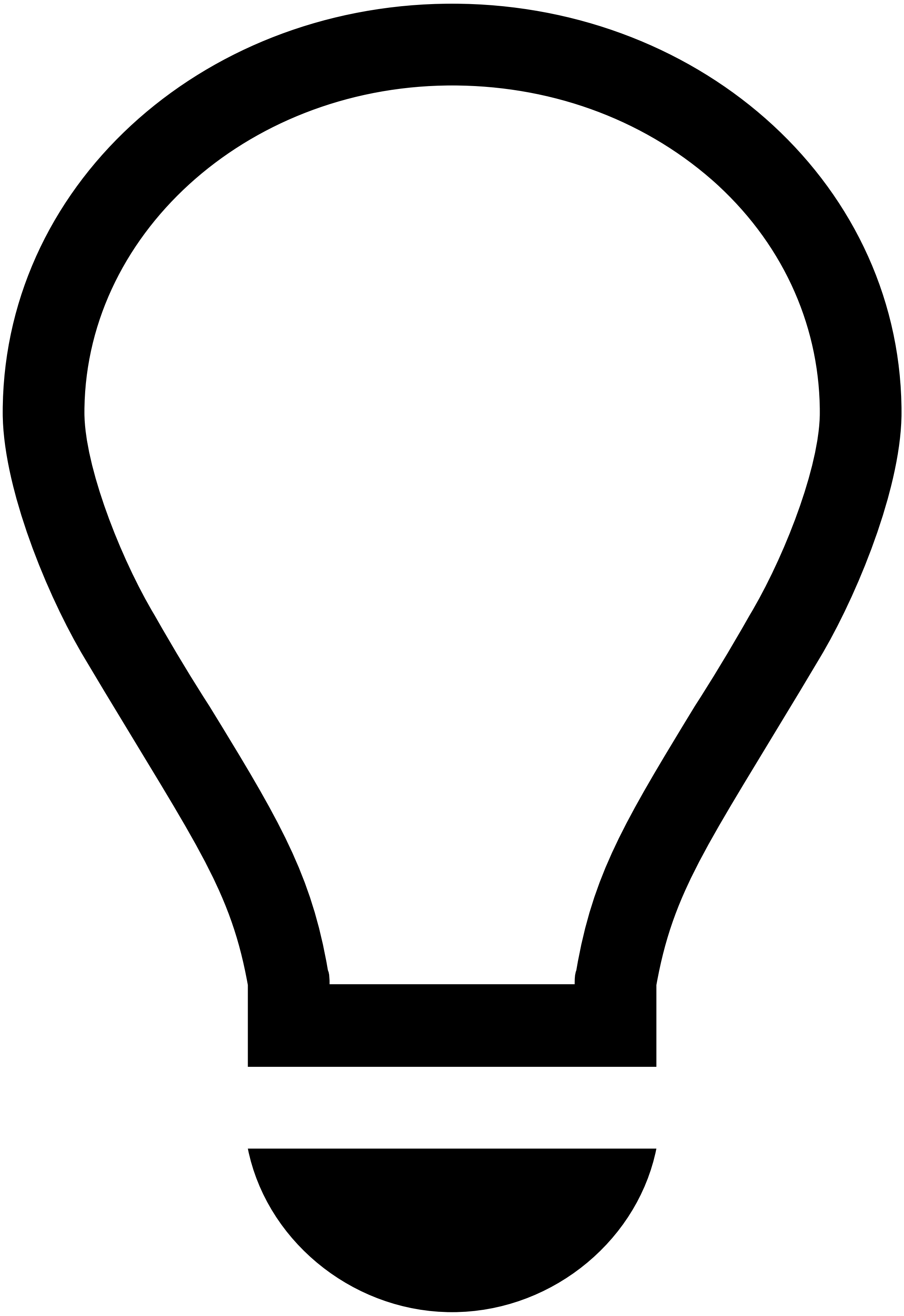
Take merchant Acme Inc as an example:

1. At the start of each working day (e.g. 0:00, 8:00), a daily merchant monitoring event is created for merchant Acme.
2. If the daily assessment does not trigger any rule, Acme's merchant monitoring event remains unalerted for the rest of the day.
3. On the other hand, if the daily assessment triggers a rule, Acme's merchant monitoring event becomes alerted. Consider the following status flow after the event has been assigned:

Payment event life cycleâ

The Payments channel looks at payments performed by merchants' customers. Rules evaluate the legitimacy of these transactions to ensure merchants are not subject to fraudulent payments.

The Payments channel includes **decision-based rules**, which determine whether a payment event is fraudulent or not by either approving or declining, and triggering an alert or not. Note that a daily merchant monitoring event assessment is not impacted by payment rules triggered over the course of the day.



payment event life cycle - step by step

As the diagram shows, this is the typical payment life cycle. Take the following steps to understand how Acme's merchant monitoring event may be impacted by payment events.

1. A payment event, alerted or not, is generated.
2. If the payment event doesn't trigger any rule, it will remain approved and unalerted in the system.
3. If the payment event triggers a decision rule, an alert is generated.

Next stepsâ

Now that you understand the standard merchant monitoring event flow on a daily basis, let's explore an agent's typical investigation flow.

Investigate Alerts

As a fraud prevention agent, the first step you should take is to access the workload. Depending on the risk engine's decision, you may find a set of events for review.

This page helps you understand how to access the events and cases that are up for review.

Access events and casesâ

When events arrive at Case Manager, they can be displayed in more than one page, depending on the fraud risk and the complexity. On Case Manager's sidebar you can access:

- **Search By Event** : Displays all the events (alerted and non-alerted). By default, some filters, such as **Alerted** and **Created Date**, are already populated to always show the latest alerted events.
- **Cases** : Events can be grouped into cases for a more organized action-taking process. This allows you to access events that may be correlated and that require a thorough investigation. Cases offer extra functionalities, such as note attachment.
- **Settings** : This option is only available for certain roles and provides various management tools such as queues, automation rules, SLAs, self-service rules and lists.
- **Dashboards** : This option is only available for certain roles and displays dashboards to help you monitor merchants, agents and rules, as well as analyze transactions, alerts, and cases.

Select and analyze eventsâ

As previously mentioned, the key pages to access your workload are the **Search by Event** and **Cases** pages.

Whether you open an alerted or non-alerted event, you will land on the **Merchant Monitoring Event Details** page, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a specific widget alone. An investigation is a comprehensive approach, in which you open several events, e.g. from the same merchant ID, and search for fraud evidence by looking at multiple widgets.

The **Merchant Monitoring Event Details** page displays the following information.

Merchant Detailsâ

The **Merchant Details** widget contains information about the merchant that processed the payment, such as merchant's ID, MCC, and country. To obtain more information about the merchant, open the **Merchant Details** page in a different tab using the **External Link** icon.

Activity Logâ

The **Activity Log** widget includes all the updates made to the event, such as information about the date and time the system ran a merchant monitoring assessment, or a file has been attached to the event. These updates are listed in descending chronological order.

Merchant Rulesâ

The **Merchant Rules** widget shows all the merchant rules triggered during the assessment window (i.e. last 24 hours).

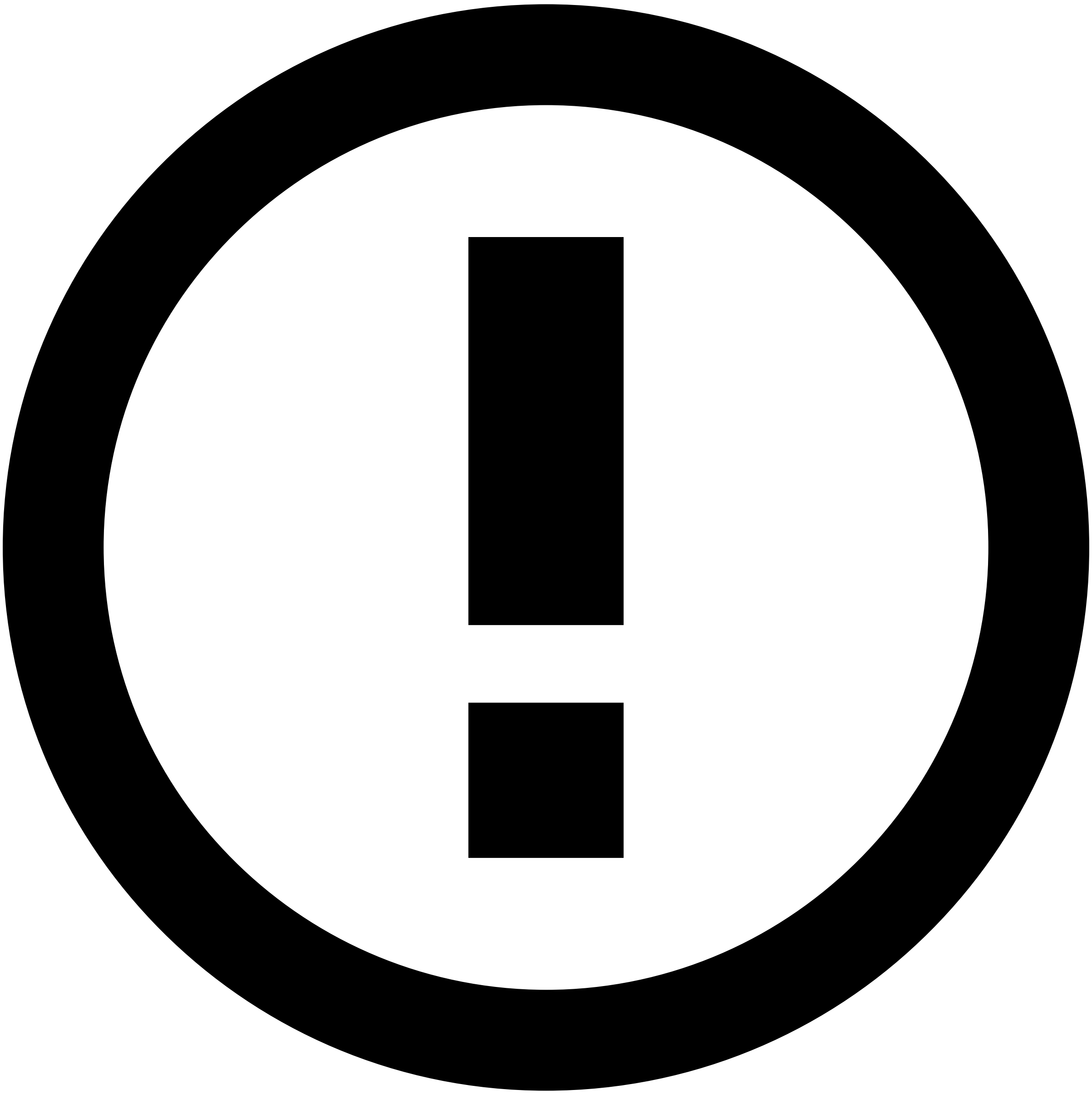
Merchant rules

Merchant rules help acquirers and PSPs detect risky activities occurring on their merchant portfolio, and thus prevent them from being victims of fraudulent merchants.

Learn the risk scenarios in which acquirers and PSPs become victims of their own merchants.

Payment Rulesâ

The **Payment Rules** widget shows all the rules triggered by payments processed by the merchant during the merchant monitoring assessment period (e.g. the previous 24 hours). The widget shows the system decision (`approve` , `review` , `decline`). It is also possible to use the **magnifying glass** icon to search for the payments that triggered a specific rule.



Payment rules

Payment rules help acquirers and PSPs confirm if a particular merchant is being a victim of fraudulent customers.

Learn about the possible fraud scenarios carried out by merchants' customers.

Transaction Historyâ

The **Transaction History** widget shows a complete overview of both merchant monitoring and payment events.

- The **Payments** tab lists in descending chronological order all the alerted payments processed by the merchant under analysis.
- The **Merchant Monitoring** tab lists in descending chronological order all the alerted merchant monitoring events over a specified time period.

Note that, by default, the **Merchant Monitoring** channel is selected and filtered by `Merchant ID` and `Alerted` to refine the results by the current merchant under analysis.

You can also narrow down results based on time periods such as "Last week" by selecting the date options filter.

Related transactionsâ

Related transactions are transactions that contributed to a merchant monitoring rule, e.g. merchant Acme is alerted if the number of transactions with an amount greater than $500 over the past 24 hours is above a threshold of 10. The related transactions for this rule are the transactions greater than $500 over the past 24 hours processed by merchant Acme.

To see related transactions, follow the instructions below.

1. By default, the `Merchant ID` and `Alerted` filters are defined, and no rules are selected.

2. Use the `Related Transactions` filter. Optionally, to refine your results by a specific rule, select a rule in the related transactions options menu.

3. Related transactions filter payments that made a rule trigger. To see them, switch to the **Payments** tab.

4. Check the **Payments** channel to see which payments triggered the selected rule. Optionally, add another rule to the related transactions options menu.

Notesâ

The **Notes** widget shows any notes made under a given merchant, such as "merchant being investigated due to suspicious activity" or "information requested".

You can create notes about:

1. An **Event** in the **Merchant Monitoring Event Details** page.
2. A **Merchant** both in the **Merchant Monitoring Event Details** and **Merchant Details** pages. Note that, in the **Merchant Details** page, you can only create notes about **Merchant**.

Attached Filesâ

The **Attached Files** widget allows you to add files that support your investigation.

Analyze merchantsâ

Fraud prevention agents need to closely monitor merchants to understand the risk they represent. The Merchant Details page displays the merchantâs relevant data and risk insights over the last 30 days.

The **Merchant Details** page displays the following information:

- **Merchant Overview**: Breaks down information into several visualizations, as follows:
  - **Merchant Details**: Contains relevant information about the merchant, such as ID, Merchant Category Code (MCC) or tax identification number, as well as contact details.
  - **Location**: Contains information related to the merchant's location, i.e. registration country and Ultimate Beneficiary Owner (UBO) names.
  - **Account**: Includes information about the merchant's account status (i.e. whether it's active or not) and bank account.
  - **Total Transactions**: Shows the ratio of approved vs. declined transactions over the last 30 days.
  - **Timeline of merchant risk assessments**: Displays a timeline with the number of times the merchant has been alerted in the last 30 days, presenting the evolution of the merchant's risk in a visual way. Other important indicators such as chargebacks, approved transactions, average transaction value, top locations and the number of refunds are key metrics for ongoing management of merchants' risk.
  - **Top location history**: Displays the top transaction distribution (in percentage) per market.
  - **Transaction Amounts**: Shows the minimum, median, and maximum transaction amounts received by the merchant over the last 30 days.
  - **Top Chargebacks**: Displays the total number of chargebacks and chargebacks to sales as well as the total amount of chargebacks over the last 30 days.
  - **Top Refunds**: Displays the total number and amount of refunds over the last 30 days.
- **Merchant Monitoring Assessments**: Shows the previous daily assessments including rules triggered, investigation status, and agents' decisions.
- **Payment History**: Displays all the transactions associated with the merchant under investigation.
- **Notes**: Shows any notes made under a given merchant.

- **Attached Files**: Allows you to add files that are useful for subsequent investigation steps.

Review casesâ

The main purpose of cases is to link events when there is a common element that suggests a connection. By aggregating events into a case, you save time in your investigation.

info

Cases can be omnichannel. This means that you can aggregate events from **Payments** and **Merchant Monitoring**.

The Cases / Case Details page header displays the following information:

- **Name**: Displays the case's name.
- **ID**: Provides the case's alphanumerical ID.
- **Status**: Displays the case's state: "New", "Review in progress", "Closed" and "Closed with SAR".
- **Source Type**: Provides the case's use case, e.g. Fraud or AML.
- **Queue**: Specifies the queue the case is stored in.

- **Source type**: Identifies what originated the investigation (e.g. Referrals from customer-facing employees).
- **Assignee**: Identifies the user assigned to the case.

The header also displays several action buttons, according to the case's state: **To be reviewed** / **In analysis** / **Waiting for info** / **Closed**.

Below this header, contextual information is provided as follows:

- **Linked Events**: Presents events associated with the case under investigation.
- **Attached Files**: This widget allows you to add files that support your investigation.
- **Notes**: This widget shows any notes made under a given case.
- **Activity Log**: This widget displays the different actions performed in the case, namely regarding linked events, attached files, and notes.

Investigation Flow

As an agent, when you log into Case Manager, you land on the **Search by Event** page, which displays all â□□Openâ□□ events in Case Manager. The **Search by Event** page comes preconfigured to display alerted events only, but you can apply additional filters such as date, assignee (e.g. â□□Unassignedâ□□to exclude already assigned events), or system decision (e.g. "Decline").

1. Select the **Merchant Monitoring** channel and open the highest priority event from the list by selecting the event's "Created Date". You will be redirected to the **Merchant Monitoring Event Details** page.

2. In the **Merchant Monitoring Event Details** page, assign the event to yourself. The event is set to â□□In Progressâ□□

3. Open the **Merchant Details** page in a different tab using the **External Link** icon from the **Merchant Details** widget. The **Merchant Details** page offers supplementary historical information for the following purposes:
    - Review merchant KPIs.
    - Review alert and note history.
4. Go back to the **Merchant Monitoring Event Details** page and check the triggered rules in the **Rules Triggered** widget. Look for related transactions (i.e. transactions triggered by the same rules), by scrolling to the **Transaction History** widget. To check for related transactions, apply filters based on triggered rules, if any.
5. If there are related transactions, select one of the payment events. You will be redirected to the **Payment Event Details** page. This page provides the following information relevant to your investigation:
    - Rules triggered.
    - Customer and card details.
    - Notes and attachments.
    - Other event details related to the transaction.

tip

Make sure to keep both the **Merchant Monitoring Event Details** and **Merchant Details** pages open in different tabs to keep reviewing other details.

6. If there aren't any related transactions, you may want to review other merchant monitoring event details. For instance, you can use the **Transaction History** widget to verify if a merchant's previous orders are unusually high, which could be an indication of the merchant selling unauthorized goods.

7. Add notes, as needed.

8. If you need additional information, use the **Hold** button to put the investigation on hold. The event status is set to "On Hold".

9. When you have the necessary information, use the **Resume** button to go back to your investigation. The event status is set to âIn Progressâ
10. Whether you need or not additional information, if there's evidence of fraud, use the button **Take Action**. The agent's decision is set to "Action Taken". After submitting the reasons, the agent's decision is set to âAction Takenâand the event status is set to "Closed".
11. If there isn't any evidence of fraud, use the **Drop** button. After submitting the reasons, the agent's decision is set to âDroppedâand the event status is set to âClosedâ

Automation rules and queues

Note that the out-of-the-box automation rules and queues play a role in the event status transitions. This means that they are responsible for moving an event's status from "In Progress" to "Closed".

Next stepsâ

At this point, you've acquired the essential knowledge to start an alert investigation. Go to Investigate Alerts to initiate your investigations.

Daily Monitoring (standard)

Each merchant is assessed on a daily basis (i.e. every 24 hours), with each assessment resulting in a new event. The standard out-of-the-box configuration provides a comprehensive overview of merchant activity and risk within the specified cycle, with optimal performance.

This guide provides you with the following information:

- Event flow
- Investigation flow
- Investigate alerted events

Source: https://cam.feedzai.com/rmpsp/2.8/docs/common/rmpsp/mm/operational-work/integration-engineer/integration-guide/rule-testing/mm-rules-testing-strategy

Merchant Monitoring Rules Testing Strategy

The goal of these pages is to describe the execution of **Scenario's Rules Testing (E2E)** - where functional end-to-end tests are executed over the rules, with the entire system up and running, to ensure that the system behaves correctly and that merchant monitoring rules only trigger upon the expected conditions.

All different components are covered, namely:

```
 - Event and entity injection
- Calculation of profile metrics
- Rule and Alert triggering
```

The following sections of this document provide you with an overview regarding the different interactions between the system components in order to perform the E2E testing. This aims to guide you during rule testing, by providing you with additional details on how the solution works and how to interact with the different endpoints during the testing process.

- Operational Work
- Integration Engineer
- Integration Guide
- Rule Testing

Rule Testing Guide

Rule testing is the first stage of the rule activation / tuning. This guide explains the relevant steps to successfully complete this stage.

- MM Rules Testing Strategy: contains a brief introduction to the different types of tests that you can perform.
- Scenarios Testing for MM Rules: introduces different system components, with additional explanation on how those components are used during the tests to emulate the system behavior in a production-like scenario.
- Test Scenario Example with Postman Collection: includes a practical example to help you structure a test case to validate a rule, alongside a Postman Collection.
- Testing Endpoints: contains a walkthrough on the different platform endpoints that can be used to reproduce the scenario described above.

Scenario Testing for Merchant Monitoring Rules

In rule testing, the goal is to ensure that merchant monitoring rules are implemented correctly and that trigger alerts as expected. Along with that, this testing activity also ensures that the different data propagates through the different components of the system as expected.

In order to achieve proper testing, the following validations should be met:

- Historical / look-back periods, by ensuring that the metrics are correctly calculated.
- Alerting frequency, by ensuring the events only trigger according to the defined frequency.
- Rule conditions, by ensuring that alerts are triggered correctly.

This set of validations verifies that the rule is well implemented to fulfill its purpose of detecting risky behavior.

Typically, there are two different categories of data processing:

- **Real-time:** Data processed in payment instruments and entity updates used for either real-time checks or profile enrichment.
- **Batch:** Entity-centric checks (e.g. merchant monitoring) driven by an entity activity under the real-time streams that usually requires heavier computations that are not compliant with strict SLAs or that by nature would imply more volumes for agents if built from a transactional perspective.

In the Merchant Monitoring use case, those different data domains are processed by the merchant monitoring workflow.

In the real-time workflows, Payments data is processed. That data is then used to calculate the metrics based on the defined historical and look-back periods.

The metrics aggregations (Historical Profiles) used in the MM rules need to be calculated periodically and that calculation is performed using the real-time data that is processed in the Payments workflow. Those metrics allow you to calculate the historical / look-back periods.

Generic test stepsâ□□

Each test case consists of the following steps:

1. Inject a set of transactions in the Payments workflow - This feeds the system with transactions for the historical and look-back periods of the rule under test.
2. Trigger a Scheduled Metrics Job (SMJ) - SMJs are responsible for calculating long period profiles based on the corresponding historical/look-back period. , the calculated metrics are stored in the Profile Storage. This is the set of metrics and aggregations that the rules rely on.
3. Perform merchant monitoring requests.
4. Confirm the expected results in Case Manager.

To execute the aforementioned steps, some considerations and configurations should be applied to the system. Check the Considerations / Remarks section.

Considerations / remarksâ□□

While testing the system, it is important to take some considerations regarding how the system works in production. Some test modeling techniques are applied to enable the testing of complex scenarios in a matter of minutes, thus keeping the functionalities from the system under test unchanged.

Disabling system automatic triggersâ□□

Scheduled Metrics Jobsâ□□

In a production environment, the calculation of SMJs is triggered automatically by the system, based on the workflow's internal clock. The workflow clock advances when new events are injected in the system, and holds the highest timestamp from the set of events injected until then.

In the RMPSP Solution, the SMJs are configured to trigger every day by default, meaning that every event injected in a new day would lead to SMJs being triggered.

In a system where automated or manual tests are executed, this mechanism should be controlled while executing the tests, because the historical and loop-back periods should be filled with events first, and the SMJs should be triggered to run afterwards.

Other useful configurationsâ□□

Current workflow status resetâ□□

If the environment must restart to its initial state (i.e. a state in which no events are injected in the platform, and the workflow clock is reset to 0), check the Restart Application with Skip Recovery section.

Test Scenario Example with Postman Collection

This section contains an example on how to design test cases for rules. Different data and details are evaluated in order to create a scenario that will lead to the rule being triggered upon the specified conditions.

The test specification example was designed to test the rule `Failed-3DS-Count-Rate-FR1`. This section contains the test description and the test data to be injected into the system for the given merchant monitoring rule. Only the set of fields that contribute to the metric calculation and rule conditions evaluation are indicated. For other fields, default values can be used.

Rule detailsâ

Rule nameâ

`Failed-3DS-Count-Rate-FR1` : Failed 3ds count rate, over the last month, is equal to or above a set threshold.

Rule propertiesâ

Property	Description	Value
Rule Alerting Frequency	The frequency at which the rule will be evaluated and alerts generated for this rule	Everyday
Historical Period	Historical period that applies to the Rule being tested	N/A
Lookback Period	Look-back period that applied to the Rule being tested	1 month

Client risk rating parametersâ

Parameter	Default Value
count_3ds_limit_1m	0.6
min_avg_amount	1
min_trx_count	1

Test specification example for `Failed-3DS-Count-Rate-FR1` â

Test scenarioâ

The fraud rule `Failed-3DS-Count-Rate-FR1` verifies that the failed 3ds payments over the total number of payments, over the last month, is equal to or above a set threshold.

To trigger this rule, the following requests are needed with the same `merchant_id` :

- 1 payment request
- 1 info request with `payment_3ds_status_code` : `F`
- 1 merchant monitoring request

Payment workflow dataâ

Timestamp	Merchant Id	Request
T+0	merchant_id_fr1	payment
T+1	merchant_id_fr1	info
T+2	dummy_event*	payment
T+3	dummy_event*	merchant monitoring
T+4	merchant_id_fr1	merchant monitoring

Dummy events are used to move workflow clocks forward.

Postman Collectionâ

Feedzai offers an out-of-the-box postman collection that enables you to test the `Failed-3DS-Count-Rate-FR1` rule.

For more details on the rules scope, check the official documentation provided by Feedzai in Rules Projects.

The postman collection is available for download here.

General descriptionâ

Postman Execution Order	Description
0 - Setup	Load postman collection
1 - Inject payments	Run requests 1 and 2
2 - Move clocks	Run requests 3 and 4 with dummy data
3 - Run SMJs	Triggers the Scheduled Metric Job that calculates the metrics usin the injected transactions and customers/accounts
4 - Run merchant monitoring	Run request 5 with the merchant monitoring request
5 - Verify data in CM	Run request 5 with the merchant monitoring request

Detailed descriptionâ

1. **Setup**
    - Load both the collection (`RMPSP MM Test Guide.postman_collection.json`) and the environment (`Local RMPSP.postman_environment.json`) in your Postman instance.
2. **Inject payments**
    - Inject payment transactions (requests 1 and 2).
3. **Move workflow clocks**
    - Inject dummy payment and merchant monitoring transactions. These transactions are used to advance the workflow clock to the desired timestamp to be used by the Scheduled Metric Job to calculate the metrics' windows from (requests 3 and 4).
4. **Run SMJs**
    - Log into Pulse.
    - Go to `/pulseviews/management/#apps/mm/jobs` .
    - Run both Scheduled Metrics Jobs from the Registered Jobs section.
    - Wait for the Scheduled Metrics Jobs to finish.
5. **Inject merchant monitoring**
    - Inject merchant monitoring transactions (request 5).
6. **Verify rules triggered in Case Manager**
    - Open the merchant monitoring event in Case Manager, where the `Failed-3DS-Count-Rate-FR1` rule will appear in the Rules Triggered widget.

Testing Endpoints

This section contains a set of REST endpoints that can be used in order to test the system, and the necessary steps to achieve some actions when performing manual interactions with Pulse and Case Manager pages.

This page includes a Postman Collection containing some of the described endpoints. For more details on how to use the collection, check the Test Scenario Example with Postman Collection page.

Paymentsâ

Payment endpointâ

Post	`/api/payments`

Example body *See postman collection*

Info endpointâ

Put	`/api/payments/[lifecycle_id]`

Example body *See postman collection*

Feedback endpointâ

Put	`/api/payments/[lifecycle_id]/feedback/[feedback_label]`

Example body *See postman collection*

Settlement endpointâ

Put	`/api/payments/[lifecycle_id]/settlement`

Example body *See postman collection*

Merchant monitoringâ

Merchant monitoring endpointâ

Post	`/api/merchant`

Example body *See postman collection*

Check registered Schedule Metrics Jobsâ

User interfaceâ

1. Access `/pulseviews/management/#apps/mm/jobs`.
2. Verify the list of jobs that can be triggered for the different workflows under the Registered Jobs table.

Trigger Schedule Metrics Job for profile metric calculationsâ

User interfaceâ

1. Access `/pulseviews/management/#apps/mm/jobs`.
2. Select the Schedule Metrics Job with the name `mm-merchant_monitoring_workflow-Merchant Monitoring Plan-ScheduledMetrics-0 0 0 * * ?-OnDemand`.
3. Select the Schedule Metrics Job `mm-merchant_monitoring_workflow-Merchant Monitoring Plan-ScheduledMetrics-0 0 0 ? * 1-OnDemand`.
4. Confirm that a new job is executed in the Job Execution table.
5. Ensure that the job reached the Finished state.

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The same steps apply to the Schedule Metrics Jobs from the Transfers workflow.

-

- Operational Work
- Integration Engineer
- Integration Guide
- User Acceptance Testing

On this page

User Acceptance Testing

The User Acceptance Testing (UAT) guide ensures that the Merchant Monitoring solution is properly tested. The following sections guide you through the necessary steps to test rules, cases, and automation rules.

Requirementsâ

To execute all the tests, make sure you have the customer's QA or Pre-Production environment, as well as the UAT Postman Collection.

The Postman collection includes a set of requests and environment variables. Make sure you import both and change the environment variables into the desired enrivonment.

Test rulesâ

1. Execute the `MM-BR1` folder from the Postman Collection to trigger the `Monthly-Amount-Decrease-BR1` rule.

   Learn more about this process in the Rule Testing guide.

2. In Pulse, run the Scheduled Metrics Jobs (SMJs) under **Registered Jobs**.

3. Execute the last request in Postman's `MM-BR1` folder.

4. In Case Manager, the last Merchant Monitoring event will display the desired rule under the **Rules Triggered** widget.

5. Perform the following actions:

   - Move the event to queue
   - Link the event to case
   - Update the event's list
   - Add a note to the event

Test casesâ

The **Search By Case** page displays the previous event.

By selecting the case you are redirected to the **Case Details** page, which shows the previously created case.

Test automation rulesâ

1. In the **Automation Rules** page, create the following rule:

   This rule triggers when a new event arrives to Case Manager with a specific **Terminal ID**.

2. Send the Postman Collection's last request (i.e. 03 - Payment 3) from the `Payment-FR20` folder.

3. Go to the **Event Details** page to confirm that the event's queue was automatically set to the previously created queue.